

2 Regular-Sector Large Level Limits from Doubled 3 Chern–Simons Theory to BF Theory

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6 ABSTRACT: We study the large-level limit of doubled Chern–Simons theory toward compact
7 $BF[G]$ theory for G compact, connected, simple, and simply connected, combining regular-
8 sector bulk convergence with boundary and interface analyses. At genus one, lattice-cell
9 embeddings into a common Hilbert space yield elementwise strong-* convergence of the
10 modular representations. Fixed pure-electric fusion operators converge in the same topol-
11 ogy. At genus zero, we prove compact-uniform asymptotic formulas for individual WZW
12 fusion coefficients on open affine–BZ cells. For each fixed regular input pair, the leading
13 cell polynomial represents the group-valued Duistermaat–Heckman density almost every-
14 where in the output; on principal regular chambers, it agrees pointwise with the Liouville
15 volume of the BF pair-of-pants reduction. Combining the two chiral estimates gives a
16 charge-resolved vague limit of rescaled doubled multiplicity measures with density V_{BF}^2 on
17 regular output windows wall-transverse to the limiting input pair. For $G = SU(m)$, $m \geq 3$,
18 the transversality condition is unnecessary, and the compact-uniform estimates, including
19 finitely many fixed compatible shifts, extend across all internal walls of the regular locus.
20 We then investigate the boundary dynamics associated with this bulk limit. Exact wave-
21 functions in the abelian $U(1)$ case exhibit decompactification of the diagonal boundary,
22 while nonabelian WZW and BF wavefunction calculations support a candidate boundary
23 theory of $\dim G$ noncompact free scalars. We also explore flat gauging of its adjoint sym-
24 metry as a continuous-orbifold construction, relating regular twisted sectors to the bulk
25 flux–charge description. To examine transport between different levels, we formulate a
26 refinement criterion requiring charge preservation and flux displacement of order k^{-1} at
27 bounded conformal weight. For adjacent doubled $SU(2)$ levels, we prove that the necessary
28 genus-one consistency conditions obstruct such uniformly refining gapped transport. Ex-
29 act branching and conformal-weight identities for a gapless diagonal-coset candidate relate
30 bounded conformal weight at fixed charge to microscopic flux displacement. These identi-
31 ties motivate a conjectural $c = 1$ Runkel–Watts interface limit and identify its candidate
32 continuous spectral parameter with the rescaled flux displacement.

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84 1 Introduction

85 Three-dimensional Chern–Simons theory with compact gauge group G and positive in-
86 tegral level k has finitely many simple lines and finite-dimensional mapping-class-group
87 representations [1, 2]. For the doubled theory the line category is

$$Z(\mathcal{C}_k(G)) \simeq \mathcal{C}_k(G) \boxtimes \mathcal{C}_k(G)^{\text{rev}}$$

88 [3]. Compact $BF[G]$ theory is described instead by the transformation-group double
89 $\mathfrak{D}(G) = C(G) \rtimes_{\text{Ad}} G$: its regular labels consist of a conjugacy-class coordinate and a
90 character of the corresponding maximal torus [4–6]. The continuous centre also has an
91 operator-algebraic realization in terms of measurable Hilbert fields over the conjugation
92 groupoid [7]. We compare the finite and continuum data through the genus-one modular
93 action and the large-level behavior of genus-zero fusion coefficients.

94 The classical scaling fixes the normalization. If the two connections are defined on the
95 same bundle, set

$$A_{\pm} = A \pm \frac{B}{2k}. \tag{1.1}$$

96 On a closed three-manifold the difference of the level- k Chern–Simons actions equals the
97 BF action plus a cubic term of order k^{-2} . The same change of variables contracts the gauge
98 transformations and the split metric Lie algebra to those of the tangent group $G \rtimes_{\text{Ad}} \mathfrak{g}_{\text{ab}}$.
99 Section 2 records the global assumptions and the normalizations used below.

100 Write $r_k = k + h^\vee$, let Y be the open fundamental alcove, and let $P = X^*(T)$. A
 101 charge-oriented doubled label $a \boxtimes \bar{b}$ is assigned the coordinates

$$x_{a,b} = \frac{a + b + 2\rho}{2r_k}, \quad \eta_{a,b} = a - b. \quad (1.2)$$

102 Thus the average of the shifted chiral labels samples a flux $x \in Y$, while their difference
 103 gives an electric charge $\eta \in P$. These variables turn the finite label sets into progressively
 104 finer samples of $Y \times P$.

105 The first main result concerns the torus. Charles proved that every fixed modular
 106 element acts semiclassically as a Fourier integral operator on the holomorphic quantization
 107 spaces [8, Theorem 9.1]. Continuum convolution and semiclassical $BF+CS$ calculations
 108 give the corresponding regular kernels [9, 10]. Here the doubled labels and explicit lattice
 109 cells give isometries

$$I_k : \mathcal{H}_k \longrightarrow \mathcal{H}_0 = L^2(Y \times P, dx \otimes \#).$$

110 Theorem 3.2 proves that, for every fixed $\gamma \in SL(2, \mathbb{Z})$, the transported operator and
 111 its adjoint converge strongly to the regular BF modular operator. Equal Weyl pairs in
 112 the doubled Kac–Peterson kernel produce the limiting Fourier kernel; Poisson summation
 113 suppresses the unequal pairs. Unitarity and a tightness estimate then control the alcove
 114 boundary and the unbounded charge direction. The theorem is formulated for an admissible
 115 class of cell embeddings, and Lemma 3.11 proves stability within that class.

116 At genus zero, the exact continuum decomposition and the finite fusion-coefficient
 117 asymptotics provide complementary information. Theorem 4.2 specializes the compact-
 118 double tensor-product decomposition of Koornwinder, Bais, and Muller to the principal
 119 regular locus and rewrites it in the present flux–charge normalization [6, Section 6, espe-
 120 cially Theorem 6.10]. It is a direct integral carrying an orbital spectral measure and a
 121 measurable field of multiplicity Hilbert spaces. The finite-level results concern the numer-
 122 ical WZW fusion coefficients and the scalar measures built from them.

123 The support of conjugacy-class multiplication is the multiplicative eigenvalue poly-
 124 tope. Its description by quantum Schubert calculus was obtained for $SU(n)$ by Agnihotri–
 125 Woodward and Belkale, and for compact simply connected groups by Teleman–Woodward
 126 [11–13]. Inside this support, a chosen BZ presentation and common rational refinement
 127 divide Y^3 into open, full-dimensional affine–BZ cells. On each cell the coefficient has a
 128 polynomial leading term, while the resulting density is independent of the chosen refine-
 129 ment.

130 Defosseux proved weak convergence of the quantum-dimension-weighted fusion prob-
 131 abilities to conjugacy-class convolution [14, Proposition 7.1 and Corollary 7.2]. The local
 132 estimate below concerns unweighted individual coefficients. Its combinatorial input is the
 133 Berenstein–Zelevinsky lattice-point model [15], together with chamber quasipolynomiality
 134 and the leading-volume theory for vector partitions [16–18].

135 The occurrence of Duistermaat–Heckman measure as the leading term of a multiplicity
 136 problem is part of a broader theory. Singular quantization relates multiplicities to reduced
 137 spaces [19]; Loizides, Paradan, and Vergne obtain Duistermaat–Heckman measures as the
 138 leading distributions associated with piecewise quasipolynomial multiplicity functions [20];

and McSwiggen studies the volume function governing tensor-product multiplicities [21]. The result proved here is the corresponding local statement for individual, unweighted WZW fusion coefficients. It is compact-uniform on affine-BZ cells and simultaneous for finitely many fixed compatible shifts; the proof retains every contributing affine Kac-Walton summand. The identification with the intrinsic density holds for every fixed input section, and the doubled limit additionally requires control of the moving wall slices. Put $d_G = |\Delta_+| - \text{rank } G$. Main Theorem B, Theorem 4.7, gives on every affine-BZ cell C the compact-uniform estimate

$$r_k^{-d_G} N_{a+\alpha, b+\beta}^{(k)c+\varsigma} = V_{\text{aff}, C}(z_a, z_b; z_c) + O_{K, E}(r_k^{-1}), \quad z_a = \frac{a + \rho}{r_k},$$

for $K \in C$ and any prescribed finite family E of compatible fixed shifts. The proof combines vector-partition asymptotics with the affine Kac-Walton formula and controls all affine Weyl images that can contribute on K . For every fixed regular input pair, the polynomial $V_{\text{aff}, C}$ represents the normalized orbital-convolution density in the output variable. On a principal regular chamber it agrees pointwise with the Liouville volume of the group-valued pair-of-pants reduction.

François and Tarrago prove a pointwise limit for fixed-degree quantum Littlewood-Richardson sequences, $|\lambda_N| + |\mu_N| - |\nu_N| = dN$ with fixed $d \geq 0$ [22, Theorem 3.8]. To obtain the WZW statement used here, Appendix B.6 combines Korff's lift with the simple-current correction, passes to centered $SU(m)$ coordinates, and treats the canonical degree by a subsequence argument. The resulting compact-uniform estimate, valid simultaneously for finitely many compatible fixed shifts and across internal regular walls, is Corollary 4.8.

For chirally compatible labels the two coefficient estimates give the local leading term $V_{\text{aff}, C}^2$. Main Theorem C, Theorem 5.3, passes from that local formula to a charge-resolved measure. If the limiting regular inputs are transverse to the fixed wall arrangement of Eq. (4.46) over a relatively compact output window $U \in Y$, then

$$\mathfrak{N}_{A_k, B_k}^{(k)} \xrightarrow{\vee} V_{\text{BF}}(x, y; z)^2 dz \otimes \sum_{\tau \in \eta + \omega + Q} \delta_\tau$$

vaguely on $U \times P$. The proof combines off-wall Riemann sums with a uniform estimate for lattice points in shrinking wall tubes. In type A , $m \geq 3$, the internal-wall transversality hypothesis drops out. Proposition 5.5 gives a complementary operator statement: fusion by a fixed pure-electric label converges in the strong- $*$ topology on $\mathcal{H}_0$.

Sections 2–5 contain the mathematical arguments. Sections 6 and 7 discuss boundary theories and level-changing interfaces, with the exact abelian calculation in Appendix A. Appendix B collects the detailed Mackey, Duistermaat-Heckman, vector-partition, and lattice-counting proofs.

2 Classical contraction and regular flux-charge scaling

Throughout Sections 2–5, G is compact, connected, simple, and simply connected, and k is a positive integer. We first fix the root and measure normalizations and compute the

classical contraction of the doubled action. We then introduce the regular compact-double representations and the flux-charge coordinates in which the finite labels approach them. The sampling lattice for all doubled states and the output lattice for fixed fusion inputs have different covolumes; both are determined below.

2.1 Root data, alcoves, and measures

Write $\mathfrak{g} = \text{Lie}(G)$ for the compact real Lie algebra and choose a maximal torus $T \subset G$, with Lie algebra $\mathfrak{t}_{\text{Lie}} \subset \mathfrak{g}$. We distinguish it from the normalized root-lattice space used below. We denote the normalized covering vector space by $\mathfrak{t}$ and fix the covering homomorphism

$$\text{Exp}_T : \mathfrak{t} \longrightarrow T, \quad \ker(\text{Exp}_T) = X_*(T). \quad (2.1)$$

Here $X_*(T)$ is the cocharacter lattice. Since G is simply connected, it is the coroot lattice $Q^\vee$ defined below. Set

$$j := d_0 \text{Exp}_T : \mathfrak{t} \longrightarrow \mathfrak{t}_{\text{Lie}}. \quad (2.2)$$

Since Exp_T is a homomorphism of an additive vector group onto a torus, j is a linear isomorphism and $\text{Exp}_T(x) = \exp_G(jx)$. In the usual unitary realization one writes $jx = 2\pi i x$.

We use one invariant form for the level and a second for the root-theoretic coordinates. Let κ_{CS} be the positive Ad-invariant form on the compact real algebra for which the Chern–Weil class

$$\left[\frac{1}{8\pi^2} \kappa_{\text{CS}}(F \wedge F) \right] \in H^4(BG; \mathbb{R}) \quad (2.3)$$

is the image of the basic generator of $H^4(BG; \mathbb{Z})$. In particular, for every closed oriented four-manifold X and every G -bundle on X ,

$$\frac{1}{8\pi^2} \int_X \kappa_{\text{CS}}(F \wedge F) \in \mathbb{Z}. \quad (2.4)$$

The choice of the basic generator in (2.3) fixes the level-one normalization: the integrality condition (2.4) alone would also allow positive integer multiples of this form. The Euclidean form used for roots, alcoves, and Lebesgue measure is

$$\kappa_{\text{geom}} := \frac{1}{(2\pi)^2} \kappa_{\text{CS}}, \quad (x, y) := \kappa_{\text{geom}}(jx, jy), \quad x, y \in \mathfrak{t}. \quad (2.5)$$

With the normalization in (2.3), every long root has squared length two for this form. We use $(\ , \)$ whenever roots, coroots, weights, alcove coordinates, or covolumes occur. Pairings of Lie-algebra-valued gauge fields below use κ_{CS} . Thus

$$\kappa_{\text{CS}}(jx, jy) = (2\pi)^2 (x, y), \quad (2.6)$$

which is the conversion between normalized exponential coordinates and the level form. In Sections 6–7 and Appendix A we continue to write $\langle X, Y \rangle$ for $\mathfrak{g}$ -valued fields; there it means $\kappa_{\text{CS}}(X, Y)$. This use of angle brackets is distinct from the canonical weight–coroot pairing below. When the same physical notation is applied to roots, weights, or lattice variables in

the normalized Cartan space $\mathfrak{t}$, the metric pairing is instead the geometric inner product in Eq. (2.5).

The roots are normalized real character roots: after pullback by Exp_T , a root α satisfies

$$\chi_\alpha(\text{Exp}_T x) = e^{2\pi i \alpha(x)}. \quad (2.7)$$

Thus $\Delta \subset \mathfrak{t}^*$. We use the geometric form in Eq. (2.5) to identify $\mathfrak{t}^* \simeq \mathfrak{t}$ whenever roots occur in metric formulas. Fix simple roots $\alpha_1, \dots, \alpha_\ell$ and the corresponding positive system Δ_+ , where $\ell = \text{rank } G$. For a root α , put $\alpha^\vee = 2\alpha/(\alpha, \alpha)$. The root, coroot, and weight lattices are

$$Q = \mathbb{Z}\Delta, \quad Q^\vee = \mathbb{Z}\{\alpha^\vee : \alpha \in \Delta\}, \quad P = (Q^\vee)^*.$$

In the common Euclidean realization fixed by the geometric form (2.5),

$$Q^\vee \subset Q \subset P. \quad (2.8)$$

Indeed, a long coroot is the corresponding long root, while a short coroot is the corresponding short root multiplied by the lacing number; the second inclusion is the integrality of the Cartan matrix. The first inclusion can be strict in non-simply-laced types, so Q and $Q^\vee$ will not be interchanged below. We also write

$$P_{\mathbb{R}} := P \otimes_{\mathbb{Z}} \mathbb{R}, \quad Q_{\mathbb{R}} := Q \otimes_{\mathbb{Z}} \mathbb{R} = Q^\vee \otimes_{\mathbb{Z}} \mathbb{R} \simeq \mathfrak{t}. \quad (2.9)$$

All these lattices have full rank, so $P_{\mathbb{R}} = Q_{\mathbb{R}} \simeq \mathfrak{t}$. The canonical weight–coroot pairing is $\langle \lambda, \alpha^\vee \rangle = 2(\lambda, \alpha)/(\alpha, \alpha)$. Let

$$P_+ = \{\lambda \in P : \langle \lambda, \alpha_i^\vee \rangle \geq 0 \text{ for } 1 \leq i \leq \ell\}.$$

We write W for the Weyl group, w_0 for its longest element, and $\epsilon(w) = \det(w)$. For a weight λ , its dual highest weight is

$$\lambda^* := -w_0 \lambda. \quad (2.10)$$

Let θ be the highest root, $\rho = \frac{1}{2} \sum_{\alpha \in \Delta_+} \alpha$ the Weyl vector, and $h^\vee$ the dual Coxeter number. Two further abbreviations used in the fusion asymptotics are

$$q_G = |\Delta_+|, \quad d_G = q_G - \ell, \quad r_k = k + h^\vee. \quad (2.11)$$

For simply connected G , restriction to the centre identifies $P/Q \simeq \widehat{Z(G)}$. Thus two highest weights define the same central character precisely when their difference lies in Q . Standard references for the root, coroot, weight, affine-Weyl, and centre conventions used here are [23, 24].

Our exponential convention may therefore be written

$$T = \mathfrak{t}/Q^\vee, \quad x \mapsto g_x := \text{Exp}_T(x) = \exp_G(jx). \quad (2.12)$$

In a unitary realization the final expression is customarily written $\exp_G(2\pi i x)$. In particular,

$$P = X^*(T) = (Q^\vee)^* = \{\eta \in \mathfrak{t} : (\eta, q^\vee) \in \mathbb{Z} \text{ for every } q^\vee \in Q^\vee\}, \quad (2.13)$$

227 and $\eta \in P$ defines the character $\chi_\eta(g_y) = e^{2\pi i(\eta, y)}$. The affine Weyl group $W_{\text{aff}} = W \ltimes Q^\vee$
 228 acts on $\mathfrak{t}$ by $x \mapsto wx + q^\vee$. A closed fundamental alcove and its interior are

$$\bar{Y} = \{x \in \mathfrak{t} : (\alpha_i, x) \geq 0 \ (1 \leq i \leq \ell), \ (\theta, x) \leq 1\}, \quad Y = \text{int } \bar{Y}. \quad (2.14)$$

229 Every conjugacy class has a unique representative g_x with $x \in \bar{Y}$. It is regular precisely
 230 when $x \in Y$, in which case $C_G(g_x) = T$; on $\partial\bar{Y}$ the centralizer is larger. The highest root is
 231 long, so $(\theta, x) = \langle x, \theta^\vee \rangle$ in our normalization; the lower-wall inequalities likewise have the
 232 same sign as $\langle x, \alpha_i^\vee \rangle$. This explains the compatibility between (2.14) and the integrability
 233 inequalities below.

234 The geometric inner product (2.5) fixes Lebesgue measure dx on $\mathfrak{t}$. For a full lattice
 235 $L \subset \mathfrak{t}$, $\text{covol}(L)$ denotes the dx -volume of a fundamental cell. Since P and $Q^\vee$ are dual,

$$\text{covol}(P) \text{covol}(Q^\vee) = 1. \quad (2.15)$$

236 More precisely, if $F \subset \mathfrak{t}$ is any measurable fundamental cell for $Q^\vee$, normalized Haar
 237 measure on T is characterized by

$$\int_T f(t) dm_T(t) = \text{covol}(Q^\vee)^{-1} \int_F f(\text{Exp}_T x) dx \quad (2.16)$$

238 for continuous f . Unless another measure is displayed, an alcove integral uses the flat mea-
 239 sure dx . The push-forward of Haar measure on G to conjugacy classes instead contains the
 240 Weyl-Jacobian factor. The boundary of $\bar{Y}$ is a finite union of lower-dimensional faces and
 241 is therefore dx -null. The Haar push-forward assertion is the Weyl integration formula; see
 242 [25, Ch. 18]. Restriction and extension across this null set give the canonical identification

$$L^2(Y, dx) = L^2(\bar{Y}, dx) \quad \text{as Hilbert spaces of almost-everywhere classes.} \quad (2.17)$$

243 We write $K \Subset U$ when K is compact and contained in the open set U .

244 2.2 The doubled action and its contraction

245 Let M be an oriented three-manifold and let A_+ and A_- be connections on the same
 246 principal G -bundle. Their average is a connection and their difference is an adjoint-valued
 247 one-form. In a common trivialization put

$$F_A = dA + \frac{1}{2}[A, A], \quad \text{CS}_{\kappa_{\text{CS}}}(A) = \int_M \left(\kappa_{\text{CS}}(A \wedge dA) + \frac{1}{3} \kappa_{\text{CS}}(A \wedge [A, A]) \right). \quad (2.18)$$

248 The doubled action on a closed manifold is

$$S_k[A_+, A_-] = \frac{k}{4\pi} (\text{CS}_{\kappa_{\text{CS}}}(A_+) - \text{CS}_{\kappa_{\text{CS}}}(A_-)) \mod 2\pi\mathbb{Z}. \quad (2.19)$$

249 Choose the real-valued lift determined by the relative transgression formula below, and
 250 denote the resulting lifted action by $\tilde{S}_k$. Introduce

$$A = \frac{A_+ + A_-}{2}, \quad B = k(A_+ - A_-), \quad A_\pm = A \pm \frac{B}{2k}. \quad (2.20)$$

251 For an adjoint-valued one-form Ξ , direct transgression gives

$$\begin{aligned} \text{CS}_{\kappa_{\text{CS}}}(A + \Xi) - \text{CS}_{\kappa_{\text{CS}}}(A - \Xi) &= 4 \int_M \kappa_{\text{CS}}(\Xi \wedge F_A) + \frac{2}{3} \int_M \kappa_{\text{CS}}(\Xi \wedge [\Xi, \Xi]) \\ &\quad - 2 \int_{\partial M} \kappa_{\text{CS}}(A \wedge \Xi). \end{aligned} \quad (2.21)$$

252 Taking $\Xi = B/(2k)$ yields

$$\tilde{S}_k[A_+, A_-] = \frac{1}{2\pi} \int_M \kappa_{\text{CS}}(B \wedge F_A) + \frac{1}{48\pi k^2} \int_M \kappa_{\text{CS}}(B \wedge [B, B]) \quad (2.22)$$

253 on a closed manifold. With a fixed boundary trivialization the accompanying boundary
254 term is

$$-\frac{1}{4\pi} \int_{\partial M} \kappa_{\text{CS}}(A \wedge B). \quad (2.23)$$

255 The first term in Eq. (2.22) is the $BF[G]$ action in our normalization. Freed's relative
256 formulation supplies the global bundle and boundary interpretation of these local formulas
257 [26, Secs. 1–3].

258 The infinitesimal gauge transformations contract in the same variables. To distinguish
259 their parameters from the Weyl sign $\epsilon(w)$, write

$$\varepsilon_{\pm} = \varepsilon \pm \frac{\beta}{2k}. \quad (2.24)$$

260 Substitution in $\delta A_{\pm} = d\varepsilon_{\pm} + [A_{\pm}, \varepsilon_{\pm}]$ yields

$$\delta A = D_A \varepsilon + \frac{1}{4k^2} [B, \beta], \quad \delta B = D_A \beta + [B, \varepsilon]. \quad (2.25)$$

261 The $k \rightarrow \infty$ transformations are the ordinary gauge and shift symmetries of $BF[G]$. This
262 is an Inönü–Wigner contraction in the standard sense [27, 28].

263 The same contraction can be read from the metric Lie algebra. For $X \in \mathfrak{g}$, define
264 $J_X = (X, X)$ and $P_X = (X, -X)/(2k)$ in $\mathfrak{g} \oplus \mathfrak{g}$. Then

$$[J_X, J_Y] = J_{[X, Y]}, \quad [J_X, P_Y] = P_{[X, Y]}, \quad [P_X, P_Y] = \frac{1}{4k^2} J_{[X, Y]}. \quad (2.26)$$

265 For the split level form determined by the two Chern–Simons summands, set

$$\langle X_+ \oplus X_-, Y_+ \oplus Y_- \rangle_k^{\text{split}} := k(\kappa_{\text{CS}}(X_+, Y_+) - \kappa_{\text{CS}}(X_-, Y_-)). \quad (2.27)$$

266 In the rescaled basis,

$$\langle J_X, J_Y \rangle_k^{\text{split}} = \langle P_X, P_Y \rangle_k^{\text{split}} = 0, \quad \langle J_X, P_Y \rangle_k^{\text{split}} = \kappa_{\text{CS}}(X, Y). \quad (2.28)$$

267 Here J_X and P_X are the abstract generators in this k -dependent basis; $P_X = (X, -X)/(2k)$
268 specifies their realization in $\mathfrak{g} \oplus \mathfrak{g}$ before the contraction.

269 At $k = \infty$ these formulas give the metric Lie algebra

$$\mathfrak{g} \ltimes_{\text{ad}} \mathfrak{g}_{\text{ab}} \simeq \mathfrak{g} \ltimes_{\text{ad}^*} \mathfrak{g}^*, \quad \langle J_X, P_Y \rangle_{\infty} = \kappa_{\text{CS}}(X, Y),$$

with vanishing J - J and P - P pairings. For the connection $J_A + P_B$, its Chern–Simons functional is

$$2 \int_M \kappa_{\text{CS}}(B \wedge F_A) - \int_{\partial M} \kappa_{\text{CS}}(A \wedge B).$$

Multiplication by $1/(4\pi)$ gives the leading action in Eq. (2.22) and the boundary term in Eq. (2.23). Fourier transformation in the abelian normal factor of the tangent group gives the transformation-group algebra

$$C_0(\mathfrak{g}^*) \rtimes_{\text{Ad}^*} G.$$

This differs from the compact double $\mathfrak{D}(G) = C(G) \rtimes_{\text{Ad}} G$ used below, whose flux parameter is a conjugacy class in G . In an exponential chart near the identity, group-valued conjugacy flux linearizes to the coadjoint variable of the tangent-group contraction. The later constructions take $\mathfrak{D}(G)$ in its independently defined compact form [4, 29].

In the abelian calculation of Section 6 and Appendix A we use lower-case fields. The specialization of (2.20) is

$$a = \frac{1}{2}(a_+ + a_-), \quad b = k(a_+ - a_-). \quad (2.29)$$

The finite-level periodicities are inherited from the two compact fields. In normalized holonomy coordinates, integral shifts $a_{\pm} \mapsto a_{\pm} + \lambda_{\pm}$ give

$$a \mapsto a + \frac{\lambda_+ + \lambda_-}{2}, \quad b \mapsto b + k(\lambda_+ - \lambda_-). \quad (2.30)$$

In particular, transformations which keep the diagonal coordinate fixed have $\lambda_- = -\lambda_+$ and shift b by $2k\lambda_+$. Hence the relative period grows linearly with k ; this is the periodicity used in the boundary wavefunction calculation.

Equations (2.20)–(2.25) concern families for which A and B remain fixed while

$$A_+ - A_- = \frac{B}{k} = O(k^{-1}). \quad (2.31)$$

For such smooth families the cubic term in (2.22) tends to zero. This is a near-diagonal classical contraction on a common bundle. The classical change of variables uses the integer level k , whereas the quantum sampling below naturally involves the shifted level $r_k = k + h^\vee$; boundary comparisons use the polarizations fixed in Section 6.

2.3 Regular compact-double labels

Let $G_{\text{reg}} \subset G$ be the open set of regular elements. We form tensor products using the compact double and use its regular ideal to parametrize the labels in the limiting Hilbert space:

$$\mathfrak{D}(G) = C(G) \rtimes_{\text{Ad}} G, \quad \mathfrak{D}_{\text{reg}}(G) = C_0(G_{\text{reg}}) \rtimes_{\text{Ad}} G. \quad (2.32)$$

Here $\rtimes$ denotes the full crossed product. Since G is compact, the full and reduced crossed products agree [30]. Let

$$i : C(G) \longrightarrow M(\mathfrak{D}(G)), \quad u : G \longrightarrow U(M(\mathfrak{D}(G))), \quad h \longmapsto u_h$$

be the canonical coefficient map and the canonical unitary multipliers. All C^* -tensor products below are minimal. If $m : G \times G \rightarrow G$ is multiplication, let

$$i^{(2)} : C(G \times G) \simeq C(G) \otimes C(G) \longrightarrow M(\mathfrak{D}(G) \otimes \mathfrak{D}(G))$$

be the canonical coefficient map induced by $i \otimes i$. There is a unique nondegenerate coassociative homomorphism

$$\Delta_{\mathfrak{D}} : \mathfrak{D}(G) \longrightarrow M(\mathfrak{D}(G) \otimes \mathfrak{D}(G)) \quad (2.33)$$

whose strict extension $\tilde{\Delta}_{\mathfrak{D}} : M(\mathfrak{D}(G)) \longrightarrow M(\mathfrak{D}(G) \otimes \mathfrak{D}(G))$ is determined on the canonical multipliers by

$$\tilde{\Delta}_{\mathfrak{D}}(i(f)) = i^{(2)}(f \circ m), \quad \tilde{\Delta}_{\mathfrak{D}}(u_h) = u_h \otimes u_h. \quad (2.34)$$

Thus, if π_1 and π_2 are nondegenerate representations, their tensor product is

$$(\pi_1 \boxtimes \pi_2)(a) = \widetilde{\pi_1 \otimes \pi_2}(\Delta_{\mathfrak{D}}(a)), \quad (2.35)$$

where the tilde denotes the strict extension to the multiplier algebra. This is the product-holonomy convention of [6, Secs. 2 and 4].

The open set G_{reg} is invariant under conjugation, so $\mathfrak{D}_{\text{reg}}(G)$ is a C^* -ideal. Every nondegenerate representation of this ideal extends uniquely to a representation of $\mathfrak{D}(G)$ on the same Hilbert space. The full algebra also has representations supported on singular conjugacy classes. The irreducible representations at the identity class are the ordinary irreducible representations of G ; they include the tensor unit and the pure-electric representations used in Subsection 5.3. We therefore retain $\mathfrak{D}(G)$ as the algebra in which tensor products are formed.

For every pair $x, y \in Y$, Lemma 4.1 shows that $g_x q g_y q^{-1}$ is regular for Haar-almost every $q \in G$. The tensor-product decomposition in Theorem 4.2 consequently has regular output holonomy almost everywhere. This measure-theoretic statement is what permits the regular labels to describe that decomposition.

For $x \in Y$ and $\eta \in P$, Mackey–Green induction associates to the orbit of g_x and the stabilizer character χ_{η} the covariant representation

$$\pi_{x,\eta} = \text{Ind}_T^G(\text{ev}_{g_x}, \chi_{\eta}) \quad \text{of} \quad \mathfrak{D}_{\text{reg}}(G). \quad (2.36)$$

Here ev_{g_x} is evaluation at g_x , and $\mathbb{C}_{\eta}$ is the one-dimensional T -module with character χ_{η} . To fix the induction convention, one may realize the carrier as

$$\mathcal{H}_{x,\eta} = \left\{ f : G \rightarrow \mathbb{C} : \begin{array}{l} f \text{ is measurable, modulo a.e. equality,} \\ f(ut) = \chi_{\eta}(t)^{-1} f(u) \quad \text{for every } t \in T \text{ and for a.e. } u \in G, \\ \int_{G/T} |f(u)|^2 d(uT) < \infty \end{array} \right\}. \quad (2.37)$$

Here $d(uT)$ is the normalized G -invariant probability measure on G/T . For $F \in C_0(G_{\text{reg}})$ and $h \in G$, the covariant pair acts by

$$(M_x(F)f)(u) = F(ug_x u^{-1})f(u), \quad (U_h f)(u) = f(h^{-1}u). \quad (2.38)$$

323 The inverse character in (2.37) fixes our induction convention. The same multiplication
 324 formula, now with $F \in C(G)$, gives the extension to the full compact double used for tensor
 325 products in Section 4.1. This is the usual Mackey–Green model [6, 31].

326 Before an alcove representative is chosen, the labels obey

$$(x, \eta) \sim (wx + q^\vee, w\eta), \quad w \in W, \quad q^\vee \in Q^\vee. \quad (2.39)$$

327 Indeed, the translation does not change g_x , while a representative of w conjugates g_x and
 328 transports the stabilizer character from η to $w\eta$. Every regular conjugacy class meets Y
 329 exactly once. Moreover, an affine Weyl element fixing a point of Y must be the identity,
 330 since an interior point lies on no affine reflection hyperplane. Hence choosing $x \in Y$ removes
 331 the ambiguity in both entries, and the regular irreducible classes are parametrized without
 332 repetition by

$$(x, \eta) \in Y \times P. \quad (2.40)$$

333 The common comparison space used at genus one is

$$\mathcal{H}_0 = L^2(Y \times P, dx \otimes \#), \quad \|f\|_{\mathcal{H}_0}^2 = \sum_{\eta \in P} \int_Y |f(x, \eta)|^2 dx, \quad (2.41)$$

334 where P has the discrete topology and $\#$ is counting measure. Since $\partial \bar{Y}$ is null, replacing
 335 Y by $\bar{Y}$ gives the same L^2 space. We write Y because the coordinates (x, η) label regular
 336 fluxes. Thus $\mathcal{H}_0$ is the scalar L^2 space of regular irreducible labels. The carrier spaces $\mathcal{H}_{x, \eta}$
 337 appear separately in the Mackey decomposition of Section 4.1.

338 2.4 Finite labels, sampling meshes, and fusion conventions

339 Let $\mathcal{C}_k = \mathcal{C}_k(G)$ be the level- k WZW fusion category. Its simple objects are indexed by

$$P_k^+ = \{a \in P_+ : \langle a, \theta^\vee \rangle \leq k\}. \quad (2.42)$$

340 The shifted chiral coordinate of $a \in P_k^+$ is

$$z_a = \frac{a + \rho}{r_k} \in Y. \quad (2.43)$$

341 To check that z_a lies in the open alcove, observe that for every simple coroot,

$$\langle z_a, \alpha_i^\vee \rangle = \frac{\langle a, \alpha_i^\vee \rangle + 1}{r_k} > 0, \quad (2.44)$$

342 whereas at the affine wall

$$\langle z_a, \theta^\vee \rangle = \frac{\langle a, \theta^\vee \rangle + h^\vee - 1}{r_k} \leq \frac{k + h^\vee - 1}{k + h^\vee} < 1. \quad (2.45)$$

343 The parametrization of integrable affine highest weights by (2.42), and the shifted level
 344 $k + h^\vee$, are standard; see [24, Chs. 12–13].

For the Kac–Walton formula, parametrize an affine Weyl element by $\widehat{w} = (w, q^\vee) \in W \ltimes Q^\vee$ and define its level- k shifted action on a weight by

$$\widehat{w} \cdot_k \lambda := w(\lambda + \rho) + r_k q^\vee - \rho, \quad \epsilon(\widehat{w}) := \epsilon(w). \quad (2.46)$$

Thus the unscaled coroot $q^\vee$ parametrizes translation by $r_k q^\vee$. We suppress the subscript on the dot action when the level is clear. This shifted affine action is the one entering the Kac–Walton algorithm [24, 32, 33].

We use a charge-oriented convention for doubled simples. If $\bar{b} = -w_0 b$ is the highest weight of the dual representation, the ordered pair (a, b) denotes the actual centre object

$$Z_{a,b}^{\text{ch}} = a \boxtimes \bar{b} \in Z(\mathcal{C}_k) \simeq \mathcal{C}_k \boxtimes \mathcal{C}_k^{\text{rev}}, \quad a, b \in P_k^+. \quad (2.47)$$

For the centre equivalence and the reverse-braiding convention, see [3, 34]. Its flux and charge are respectively the average and the unscaled difference of the two shifted chiral labels:

$$x_{a,b} = \frac{z_a + z_b}{2} = \frac{a + b + 2\rho}{2r_k}, \quad \eta_{a,b} = r_k(z_a - z_b) = a - b. \quad (2.48)$$

The inverse formulas are

$$a = r_k x_{a,b} - \rho + \frac{\eta_{a,b}}{2}, \quad b = r_k x_{a,b} - \rho - \frac{\eta_{a,b}}{2}. \quad (2.49)$$

They show that the map $(a, b) \mapsto (x_{a,b}, \eta_{a,b})$ is injective. Conversely, a pair (x, η) on the ambient affine mesh corresponds to a doubled simple precisely when

$$r_k x - \rho + \frac{\eta}{2} \in P_k^+, \quad r_k x - \rho - \frac{\eta}{2} \in P_k^+. \quad (2.50)$$

The lattice condition applies to the two combinations in Eq. (2.50); it allows $\eta/2$ itself to lie outside P . Equivalently,

$$\frac{a + \rho}{r_k} - \frac{b + \rho}{r_k} = \frac{\eta_{a,b}}{r_k}. \quad (2.51)$$

Since $r_k = k + h^\vee$, this agrees to leading order with the antidiagonal separation in (2.20).

The identities

$$S_{\bar{b}, \bar{d}}^{(k)} = S_{bd}^{(k)}, \quad \theta_{\bar{b}}^{(k)} = \theta_b^{(k)} \quad (2.52)$$

show that the S -entry and categorical-twist entry of $a \boxtimes \bar{b} \in \mathcal{C}_k \boxtimes \mathcal{C}_k^{\text{rev}}$ are $S_{ac}^{(k)} \overline{S_{bd}^{(k)}}$ and $\theta_a^{(k)} (\theta_b^{(k)})^{-1}$, respectively. Thus the convention in Eq. (2.47) agrees with the modular conventions of Section 3, while the endpoint charge is the simple difference $a - b$. Here $\theta_a^{(k)}$ is the categorical twist scalar; it is unrelated to the highest root θ used in the alcove inequalities. The duality identities in (2.52) are the standard unitary ribbon-category identities; see [34, Chs. 2–3].

The finite set sampled at level k is

$$\Lambda_k := \left\{ \left(\frac{a + b + 2\rho}{2r_k}, a - b \right) : a, b \in P_k^+ \right\} \subset Y \times P. \quad (2.53)$$

369 For a fixed charge $\eta \in P$, its exact slice is

$$\Lambda_{k,\eta} = \left\{ \left(\frac{\mu + \rho + \eta/2}{r_k}, \eta \right) : \mu \in P, \mu \in P_k^+, \mu + \eta \in P_k^+ \right\}. \quad (2.54)$$

370 The ambient flux affine lattice in this formula is

$$r_k^{-1}(P + \rho + \eta/2). \quad (2.55)$$

371 The integrability conditions in (2.54) are essential near an alcove wall. On a compact set
372 $K \subseteq Y$, however, they eventually impose no further deletion. To see this, set

$$\delta_K := \min_{x \in K} \min \{ \langle x, \alpha_i^\vee \rangle \ (1 \leq i \leq \ell), 1 - \langle x, \theta^\vee \rangle \} > 0. \quad (2.56)$$

373 If $x = (\mu + \rho + \eta/2)/r_k$ belongs to the ambient mesh, the two recovered weights are $b = \mu$
374 and $a = \mu + \eta$. For each simple coroot,

$$\begin{aligned} \langle \mu, \alpha_i^\vee \rangle &= r_k \langle x, \alpha_i^\vee \rangle - 1 - \frac{1}{2} \langle \eta, \alpha_i^\vee \rangle, \\ \langle \mu + \eta, \alpha_i^\vee \rangle &= r_k \langle x, \alpha_i^\vee \rangle - 1 + \frac{1}{2} \langle \eta, \alpha_i^\vee \rangle. \end{aligned} \quad (2.57)$$

375 Both are bounded below by $r_k \delta_K - 1 - \frac{1}{2} |\langle \eta, \alpha_i^\vee \rangle|$. At the affine wall one similarly has

$$\begin{aligned} k - \langle \mu, \theta^\vee \rangle &= r_k (1 - \langle x, \theta^\vee \rangle) - 1 + \frac{1}{2} \langle \eta, \theta^\vee \rangle, \\ k - \langle \mu + \eta, \theta^\vee \rangle &= r_k (1 - \langle x, \theta^\vee \rangle) - 1 - \frac{1}{2} \langle \eta, \theta^\vee \rangle. \end{aligned} \quad (2.58)$$

376 These quantities are positive once k is large enough, uniformly for $x \in K$. The reverse
377 inclusion is immediate from the definition of $\Lambda_{k,\eta}$, and therefore

$$\{x : (x, \eta) \in \Lambda_{k,\eta}\} \cap K = r_k^{-1}(P + \rho + \eta/2) \cap K \quad (2.59)$$

378 for all sufficiently large k . Thus compact regular flux windows are eventually sampled by
379 the full translated weight lattice, while the charge coordinate remains discrete.

380 We finish with conventions used to state the fusion-coefficient limits. For $a, b, c \in P_k^+$,
381 let

$$N_{ab}^{(k)c} := \dim \text{Hom}_{\mathcal{C}_k}(c, a \otimes b), \quad \mathcal{L}_G := \{(a, b, c) \in P^3 : a + b - c \in Q\}. \quad (2.60)$$

382 Membership in $\mathcal{L}_G$ is the central-character compatibility condition; it is necessary, but not
383 sufficient, for a coefficient to be nonzero. Since $\mathcal{L}_G$ has finite index in P^3 , its real span is

$$\mathcal{L}_{G,\mathbb{R}} := \mathcal{L}_G \otimes_{\mathbb{Z}} \mathbb{R} = P_{\mathbb{R}}^3. \quad (2.61)$$

384 For doubled labels

$$A = (a^+, a^-), \quad B = (b^+, b^-), \quad C = (c^+, c^-),$$

where A means $a^+ \boxtimes \overline{a^-}$ and similarly for B, C , put

$$x_A = \frac{z_{a^+} + z_{a^-}}{2}, \quad \eta_A = a^+ - a^-, \quad (2.62)$$

and define (x_B, η_B) and (x_C, η_C) in the same way. Hereafter A, B, C denote simple-object labels. For the reversed chiral factor, duality followed by the C^* -category adjoint gives a conjugate-linear isomorphism of the relevant multiplicity spaces:

$$\mathrm{Hom}(\overline{c^-}, \overline{a^-} \otimes \overline{b^-}) \simeq \mathrm{Hom}(c^-, b^- \otimes a^-). \quad (2.63)$$

Braided commutativity then gives $N_{b^- a^-}^{(k)c^-} = N_{a^- b^-}^{(k)c^-}$. Hence the doubled fusion coefficient is

$$\mathcal{N}_{AB}^{(k)C} := N_{a^+ b^+}^{(k)c^+} N_{a^- b^-}^{(k)c^-}. \quad (2.64)$$

We call (A, B, C) *chirally compatible* when

$$a^+ + b^+ - c^+ \in Q, \quad a^- + b^- - c^- \in Q. \quad (2.65)$$

Subtracting the two congruences gives the endpoint charge rule

$$\eta_C - \eta_A - \eta_B \in Q. \quad (2.66)$$

We call a charge satisfying (2.66) *centrally allowed*. An actual nonzero channel additionally satisfies the two chiral congruences, the integrability conditions for the output weights, and the fusion inequalities.

There are two different local meshes in use. All doubled states at a fixed charge sample the translated P -mesh of (2.55). Now fix the two inputs, a centrally allowed output charge, and one chirally compatible output $C_0 = (c_0^+, c_0^-)$ at level k . If $C = (c^+, c^-)$ is another such output, equality of the charges gives

$$c^+ - c_0^+ = c^- - c_0^- =: q. \quad (2.67)$$

Either chiral congruence then implies $q \in Q$, and

$$x_C - x_{C_0} = \frac{q}{r_k}. \quad (2.68)$$

Conversely, every $q \in Q$ for which $c_0^+ + q, c_0^- + q \in P_k^+$ gives a chirally compatible output with the same charge: subtracting q preserves both congruences in (2.65). Thus the exact set of chirally compatible output labels at this fixed charge is the affine root-lattice coset intersected with the two integrable alcoves, and

$$x_C - x_{C_0} \in r_k^{-1}Q, \quad \mathrm{covol}(r_k^{-1}Q) = \mathrm{covol}(Q)r_k^{-\ell}. \quad (2.69)$$

On a compact regular output window, the integrability deletion eventually disappears by the estimates (2.57)–(2.58). The factor of two in the definition of x_C cancels because both output weights shift by q ; this is why the mesh is $r_k^{-1}Q$, not $(2r_k)^{-1}Q$. Its covolume is the Riemann-sum factor in the charge-resolved doubled fusion measure. Throughout

Sections 4–5, an *actual doubled triple* means that all six chiral weights lie in P_k^+ . Chiral compatibility by itself imposes only the two root-lattice congruences in Eq. (2.65).

By a regular large- k sequence we mean a sequence for which

$$x_{a_k, b_k} \longrightarrow x \in Y, \quad \eta_{a_k, b_k} = \eta \quad \text{for all sufficiently large } k. \quad (2.70)$$

For operators on a fixed Hilbert space, $A_k \rightarrow A$ in the strong operator topology (abbreviated SOT) means $\|(A_k - A)\psi\| \rightarrow 0$ for every vector ψ . Strong- $*$ convergence means SOT convergence of both A_k to A and A_k^* to A^* . Vague convergence of Radon measures on $Y \times P$ means convergence against every function in $C_c(Y \times P)$; because P is discrete, such a test function meets only finitely many charge fibres.

The Hilbert-space and measure limits in Sections 3–5 are taken on $Y \times P$. Boundary fluxes have larger centralizers and hence different charge labels. In particular, a pure-electric label has identity flux and an irreducible G -representation as its charge. Although these labels are absent from $Y \times P$, their fusion with regular labels defines the bounded operators on $\mathcal{H}_0$ constructed in Subsection 5.3.

3 Strong regular genus-one convergence

Main Theorem A compares the finite doubled genus-one modular operators with operators on the regular label space $\mathcal{H}_0$. The cell isometries I_k fixed below transport the finite operators to $I_k \mathcal{H}_k$; the transported representation acts as the identity on the orthogonal complement.

The proof begins with the doubled Kac–Peterson kernel. Its expansion over Weyl pairs $(u, v) \in W^2$ separates into a Riemann sum for the endpoint Fourier transform when $u = v$, and rapidly decreasing terms when $u \neq v$. This gives convergence on compact flux windows with finitely many charges. Unitarity and a sampling estimate then control the mass outside those windows. Cell approximation places this convergence on $\mathcal{H}_0$; convergence of the two generators and their adjoints gives the result for every fixed mapping class. The last subsection shows that any sampling embeddings satisfying the same approximation properties give this limit.

Recall from Eq. (2.13) that the geometric inner product of Eq. (2.5) identifies the weight lattice with the Euclidean dual lattice, $P = (Q^\vee)^*$. This geometric inner product determines Lebesgue measure on $\mathfrak{t}$ and hence all covolumes below. In the long-root-length-two convention the coroot lattice is integral, so $Q^\vee \subset P$. Consequently,

$$\text{covol}(Q^\vee) = |P/Q^\vee| \text{covol}(P). \quad (3.1)$$

Writing $\ell = \text{rank } G$, set

$$v_k = \text{covol}(P) r_k^{-\ell}, \quad \gamma_G = \text{covol}(Q^\vee)^{-1}. \quad (3.2)$$

Thus v_k is the volume of a fundamental cell of $r_k^{-1}P$. Let $F_{Q^\vee} \subset \mathfrak{t}$ be a measurable fundamental domain for $Q^\vee$. Normalized Haar measure on $T = \mathfrak{t}/Q^\vee$ is characterized by

$$\int_T H(t) dm_T(t) = \gamma_G \int_{F_{Q^\vee}} H([y]) dy \quad (3.3)$$

for every integrable H . We occasionally abbreviate this relation by $dm_T = \gamma_G dy$ when working in a torus coordinate chart.

The finite and limiting Hilbert spaces are

$$\mathcal{H}_k = \ell^2(\Lambda_k), \quad \mathcal{H}_0 = L^2(Y \times P, dx \otimes \#) \cong \bigoplus_{\eta \in P} L^2(Y, dx), \quad (3.4)$$

where $\#$ is counting measure on the discrete lattice P . In particular,

$$\|g\|_{\mathcal{H}_0}^2 = \sum_{\eta \in P} \int_Y |g(x, \eta)|^2 dx.$$

The space $\mathcal{H}_k$ carries the canonical orthonormal basis $\{\delta_{(x, \eta)} : (x, \eta) \in \Lambda_k\}$. If $(x, \eta) \leftrightarrow (a, b)$ and $(y, \omega) \leftrightarrow (c, d)$ under Eq. (2.53), the finite doubled modular S -operator has, in this basis, the matrix

$$(\mathcal{S}_k)_{(x, \eta), (y, \omega)} := S_{ac}^{(k)} \overline{S_{bd}^{(k)}} =: S_{(x, \eta), (y, \omega)}^{(k, -k)}. \quad (3.5)$$

Here the complex conjugate in the second factor reflects the reversed second chiral theory.

For a pointwise-defined function $g : Y \times P \rightarrow \mathbb{C}$, define the restriction and weighted sampling maps

$$Q_k g = g|_{\Lambda_k}, \quad J_k g = \sqrt{v_k} Q_k g. \quad (3.6)$$

We use these maps on smooth test functions; they are not defined on arbitrary L^2 equivalence classes. The factor $\sqrt{v_k}$ ensures that $\|J_k g\|_{\mathcal{H}_k}^2 = v_k \sum_{z \in \Lambda_k} |g(z)|^2$ is the Riemann sum corresponding to the L^2 norm displayed above.

Initially we sample the test-function space

$$C_c^\infty(Y \times P) := C_c^\infty(Y) \otimes_{\text{alg}} c_{00}(P).$$

Here $c_{00}(P)$ denotes the finitely supported complex functions on P . Thus a test function has finitely many nonzero charge fibres, each in $C_c^\infty(Y)$.

Throughout this section, $|\cdot|$ is the norm induced by the geometric inner product of Eq. (2.5), also used on $\mathfrak{t}^*$ through the duality identification. We write $\mathcal{A}(Y \times P)$ for the charge-Schwartz class: every fibre $g(\cdot, \eta)$ extends smoothly to a neighbourhood of $\overline{Y}$, and, for every multi-index α and every $N \geq 0$,

$$\sup_{\eta \in P} (1 + |\eta|)^N \sup_{x \in \overline{Y}} |\partial_x^\alpha g(x, \eta)| < \infty. \quad (3.7)$$

The charge-Schwartz class is unchanged if this norm is replaced by any equivalent norm. Here $C^\infty(\overline{Y})$ means restriction from a neighbourhood of $\overline{Y}$. The class contains $C^\infty(\overline{Y}) \otimes_{\text{alg}} c_{00}(P)$ and is dense in $\mathcal{H}_0$. The same sampling maps are used on $\mathcal{A}(Y \times P)$; Lemma 3.7 provides uniform control of their norms.

We also fix, once and for all, a bounded half-open fundamental parallelotope F_P of P with $0 \in \text{int } F_P$. For $z = (x, \eta) \in \Lambda_k$, put

$$C_{k,z} = Y \cap (x + r_k^{-1} F_P), \quad I_k \delta_z = |C_{k,z}|^{-1/2} 1_{C_{k,z}} \delta_\eta, \quad P_k = I_k I_k^*. \quad (3.8)$$

Since $x \in Y$ and $0 \in \text{int } F_P$, each $C_{k,z}$ has positive measure. The half-open cells are disjoint up to null sets within each charge fibre, and different charge fibres are orthogonal. Hence $I_k : \mathcal{H}_k \rightarrow \mathcal{H}_0$ is an isometry. The estimates proving that these cells fill the limiting Hilbert space, including the cells cut by ∂Y , are given in Lemma 3.9.

Before defining the endpoint operators, we realize $\mathcal{H}_0$ as a quotient–Haar L^2 space on the first-holonomy-regular commuting-pair chart. Let

$$q : T_{\text{reg}} \times T \longrightarrow (T_{\text{reg}} \times T)/W$$

be the quotient map. The diagonal Weyl action is free because the first component is regular. We equip the quotient with

$$m_{\text{quot}} := q_*((dm_T|_{T_{\text{reg}}}) \otimes dm_T), \quad (3.9)$$

the quotient of normalized Haar measure.

For later use, define the diagonal Weyl extension

$$\text{Ext} : L^2(Y \times P) \longrightarrow L^2(T \times P)^W, \quad (\text{Ext } f)(wx, w\eta) = f(x, \eta).$$

Here $x \in Y$, $\eta \in P$, and $w \in W$, and wx is understood as a point of $T = \mathfrak{t}/Q^\vee$. Every regular Weyl orbit meets Y once, so the formula is unambiguous on $T_{\text{reg}} \times P$. Values on the root hypertori may be chosen arbitrarily; those hypertori are Haar-null, and all identities involving Ext are understood almost everywhere.

Proposition 3.1 (Regular quotient–Haar model). *There are natural unitary identifications*

$$L^2((T_{\text{reg}} \times T)/W, m_{\text{quot}}) \simeq L^2(T \times T)^W \simeq L^2(T \times P)^W \simeq L^2(Y \times P, dx \otimes \#) = \mathcal{H}_0. \quad (3.10)$$

The left-hand term is the quotient–Haar L^2 -space on the first-holonomy-regular commuting-pair chart. Its measure is the pushforward m_{quot} defined in Eq. (3.9).

Proof. A commuting pair whose first member is regular is simultaneously conjugate to a pair in $T_{\text{reg}} \times T$, with residual ambiguity W . This gives the Borel identification $\{(g, h) : gh = hg, g \in G_{\text{reg}}\}/G \simeq (T_{\text{reg}} \times T)/W$, as reviewed in [35, Sec. 3]. Because the measure in Eq. (3.9) is a pushforward, pullback along q is an isometry onto the invariant subspace. Root hypertori are Haar-null, so T_{reg} may be replaced by T in L^2 . Fourier transformation in the second torus variable gives the middle identification by Pontryagin–Plancherel theory [36, Ch. 4]. Modulo the root hypertori, the regular torus is the disjoint union of the Weyl translates of the open alcove,

$$T_{\text{reg}} = \bigsqcup_{w \in W} wY \quad \text{up to a Haar-null set.}$$

In each alcove chart, Eq. (2.16) gives $dm_T = \gamma_G dx$. Since $(\text{Ext } f)(wx, w\eta) = f(x, \eta)$,

$$\|\text{Ext } f\|_{L^2(T \times P)}^2 = \gamma_G \sum_{w \in W} \sum_{\eta \in P} \int_Y |f(x, \eta)|^2 dx = |W| \gamma_G \|f\|_{L^2(Y \times P)}^2.$$

Thus

$$U := (|W|\gamma_G)^{-1/2} \text{Ext} : \mathcal{H}_0 \longrightarrow L^2(T \times P)^W$$

is the last unitary in Eq. (3.10). Its normalization is determined by m_{quot} , normalized Haar measure on T , and counting measure on P . This is the regular Fourier–Haar chart appearing in [9]; the cotangent interpretation is described in [37]. $\square$

The two endpoint generators are initially defined on $C_c^\infty(Y \times P)$ by

$$(\mathcal{S}_0 f)(x, \eta) = \gamma_G \sum_{\omega \in P} \int_Y \sum_{w \in W} e^{-2\pi i((wx, \omega) + (w\eta, y))} f(y, \omega) dy, \quad (3.11)$$

$$(\mathcal{T}_0 f)(x, \eta) = e^{2\pi i(\eta, x)} f(x, \eta). \quad (3.12)$$

The sum in the first line is finite on this test space. We shall prove below that $\mathcal{S}_0$ extends to a unitary, that $\mathcal{T}_0$ is unitary, and that together they give the endpoint $SL(2, \mathbb{Z})$ representation.

The S -generator of the doubled modular data is $\mathcal{S}_k$, and its T -generator is the categorical twist

$$\mathcal{T}_k \delta_{(x_{a,b}, \eta_{a,b})} = \theta_a \theta_b^{-1} \delta_{(x_{a,b}, \eta_{a,b})}. \quad (3.13)$$

Indeed, in the reversed factor $\mathcal{S}_{b,d}^{\text{rev}} = \overline{\mathcal{S}_{bd}}$ and $\theta_b^{\text{rev}} = \theta_b^{-1} = \theta_b^{-1}$; this is the matrix convention fixed in Eqs. (2.52) and (3.5). The scalar vacuum phases of the two chiral T -matrices cancel because the chiralities are opposite. More explicitly, the single-chiral RCFT matrix is

$$(T_k^{\text{RCFT}})_{aa} = e^{-2\pi i c_k / 24} \theta_a, \quad (3.14)$$

where

$$c_k = \frac{k \dim G}{k + h^\vee}$$

is the level- k WZW central charge. The reversed chiral factor carries the complex-conjugate scalar. Their product has diagonal entry $\theta_a \theta_b^{-1}$, with no residual central-charge phase. For the reverse braided and ribbon conventions, see [38, Secs. 8.1 and 8.10].

The doubled action is genuine. Indeed, for a modular category $\mathcal{D}$, put

$$\text{Dim } \mathcal{D} := \sum_{X \in \text{Irr}(\mathcal{D})} d_X^2, \quad \Delta_\pm(\mathcal{D}) := \sum_{X \in \text{Irr}(\mathcal{D})} \theta_X^{\pm 1} d_X^2. \quad (3.15)$$

Müger’s Gauss-sum formula [3, Theorem 1.2 and Remark 5.19] gives

$$\Delta_\pm(Z(\mathcal{C}_k)) = \text{Dim } \mathcal{C}_k, \quad \text{Dim } Z(\mathcal{C}_k) = (\text{Dim } \mathcal{C}_k)^2.$$

Thus $Z(\mathcal{C}_k)$ has multiplicative central charge one; the normalized doubled matrices obey $\mathcal{S}_k^4 = 1$ and $(\mathcal{S}_k \mathcal{T}_k)^3 = \mathcal{S}_k^2$. They therefore define a unitary representation $\rho_k : SL(2, \mathbb{Z}) \rightarrow U(\mathcal{H}_k)$.

Let $I_k : \mathcal{H}_k \rightarrow \mathcal{H}_0$ be the cell isometry defined in Eq. (3.8), and let $P_k = I_k I_k^*$ be the orthogonal projection onto its image. We transport ρ_k to $P_k \mathcal{H}_0$ and let it act as the identity on the orthogonal complement:

$$\rho_k^\#(\gamma) := I_k \rho_k(\gamma) I_k^* + (1 - P_k). \quad (3.16)$$

Relative to $\mathcal{H}_0 = I_k \mathcal{H}_k \oplus (I_k \mathcal{H}_k)^\perp$, this operator is unitarily equivalent to $\rho_k(\gamma) \oplus 1$.
 Moreover, $I_k^* I_k = 1$ and $I_k^*(1 - P_k) = (1 - P_k)I_k = 0$, so

$$\rho_k^\#(\gamma_1) \rho_k^\#(\gamma_2) = I_k \rho_k(\gamma_1 \gamma_2) I_k^* + (1 - P_k) = \rho_k^\#(\gamma_1 \gamma_2). \quad (3.17)$$

Since $\rho_k^\#(1) = 1$, this gives a unitary representation on $\mathcal{H}_0$.

Theorem 3.2 (Main Theorem A: regular genus-one modular convergence). *The operators $\mathcal{S}_0$ and $\mathcal{T}_0$ of Eqs. (3.11)–(3.12) extend to unitaries on $\mathcal{H}_0$ and satisfy the $SL(2, \mathbb{Z})$ relations. Denote the resulting representation by ρ_0 , so that $\rho_0(S) = \mathcal{S}_0$ and $\rho_0(T) = \mathcal{T}_0$. Then every $\rho_k^\#$ in Eq. (3.16) is a unitary representation on the same Hilbert space. For each $\gamma \in SL(2, \mathbb{Z})$,*

$$\rho_k^\#(\gamma) \xrightarrow[k \rightarrow \infty]{\text{SOT}} \rho_0(\gamma), \quad \rho_k^\#(\gamma)^* \xrightarrow[k \rightarrow \infty]{\text{SOT}} \rho_0(\gamma)^*. \quad (3.18)$$

Thus the embedded genus-one operators converge pointwise in the strong-* topology on the regular label space $\mathcal{H}_0$. Let $(\tilde{I}_k)$ be an admissible family of sampling isometries in the sense of Eq. (3.63), and put

$$\tilde{P}_k = \tilde{I}_k \tilde{I}_k^*, \quad \tilde{\rho}_k^\#(\gamma) = \tilde{I}_k \rho_k(\gamma) \tilde{I}_k^* + (1 - \tilde{P}_k). \quad (3.19)$$

For every k , this is a unitary representation of $SL(2, \mathbb{Z})$ on $\mathcal{H}_0$. For each fixed $\gamma \in SL(2, \mathbb{Z})$,

$$\tilde{\rho}_k^\#(\gamma) \xrightarrow{\text{SOT}} \rho_0(\gamma), \quad \tilde{\rho}_k^\#(\gamma)^* \xrightarrow{\text{SOT}} \rho_0(\gamma)^*. \quad (3.20)$$

Lemma 3.11 proves these two limits.

3.1 The Kac–Peterson kernel and Poisson decay

In the long-root-length-two convention, the Kac–Peterson matrix [39, 40] is

$$S_{\lambda\mu}^{(k)} = \alpha_k^{\text{KP}} \sum_{w \in W} \epsilon(w) e^{-\frac{2\pi i}{r_k} (w(\lambda + \rho), \mu + \rho)}, \quad |\alpha_k^{\text{KP}}|^2 = \frac{r_k^{-\ell}}{|P/Q^\vee|}. \quad (3.21)$$

The constant can also be read directly from the finite Fourier group. Since $P = (Q^\vee)^*$, the phase is periodic in either weight modulo $r_k Q^\vee$, and

$$|P/r_k Q^\vee| = r_k^\ell |P/Q^\vee|. \quad (3.22)$$

Put $A_k = P/r_k Q^\vee$. The bicharacter

$$([\lambda], [\mu]) \mapsto \exp\left(-\frac{2\pi i}{r_k} (\lambda, \mu)\right)$$

on A_k is perfect. Indeed, if its value is one for every $\mu \in P$, then $\lambda/r_k \in P^* = Q^\vee$, so $\lambda \in r_k Q^\vee$. The same argument applies in the other variable. The unitary discrete Fourier normalization is therefore $|A_k|^{-1/2}$, whose square is the modulus in Eq. (3.21).

To pass from the finite Fourier transform to the chiral matrix, consider its alternating subspace

$$\ell^2(A_k)^{\text{alt}} := \{F \in \ell^2(A_k) : F(w\xi) = \epsilon(w)F(\xi) \text{ for all } w \in W\}.$$

For $\nu \in P$, membership in the dilated open alcove $r_k Y$ is equivalent to

$$\langle \nu, \alpha_i^\vee \rangle > 0 \quad (1 \leq i \leq \ell), \quad \langle \nu, \theta^\vee \rangle < r_k.$$

For the simple walls this follows from $(\alpha_i, \nu) = \frac{(\alpha_i, \alpha_i)}{2} \langle \nu, \alpha_i^\vee \rangle$; for the upper wall we use that the highest root θ is long. All displayed pairings are integral. Writing $\nu = \lambda + \rho$ and using $\langle \rho, \alpha_i^\vee \rangle = 1$ and $\langle \rho, \theta^\vee \rangle = h^\vee - 1$ therefore gives

$$P \cap r_k Y = \rho + P_k^+. \quad (3.23)$$

Every regular orbit of $W \ltimes r_k Q^\vee$ meets $r_k Y$ exactly once. Equivalently, each regular W -orbit in $P/r_k Q^\vee$ has one representative in $\rho + P_k^+$. A class on an affine wall is fixed modulo $r_k Q^\vee$ by an affine reflection, so alternating functions vanish there. Hence

$$\Psi_\lambda = |W|^{-1/2} \sum_{w \in W} \epsilon(w) \delta_{[w(\lambda+\rho)]}, \quad \lambda \in P_k^+,$$

form an orthonormal basis $\{\Psi_\lambda : \lambda \in P_k^+\}$ of $\ell^2(A_k)^{\text{alt}}$. This is the usual shifted-alcove description of integrable affine weights; see [24, Chs. 12–13].

Taking a finite Fourier matrix coefficient between Ψ_λ and Ψ_μ initially gives a double Weyl sum and a factor $|W|^{-1}$. Writing $w = v^{-1}u$ leaves $|W|$ identical contributions, so the $|W|^{-1}$ factor cancels and gives the single Weyl sum and the positive modulus of the prefactor in Eq. (3.21). The remaining multiplier is the conventional global Kac–Peterson phase, giving exactly the normalization in Eq. (3.21). In the present conventions this is Eq. (3.3d) of Ref. [40]; the normalization follows from $P = (Q^\vee)^*$ and Eq. (3.1). The phase of α_k^{KP} will not be needed, since it cancels against its complex conjugate in the doubled matrix. Substitution of the flux–charge coordinates means using all four identities

$$\begin{aligned} a + \rho &= r_k x + \eta/2, & b + \rho &= r_k x - \eta/2, \\ c + \rho &= r_k y + \omega/2, & d + \rho &= r_k y - \omega/2. \end{aligned} \quad (3.24)$$

For a fixed Weyl pair (u, v) , the exponent from the first chiral factor and the conjugate exponent from the second are

$$\begin{aligned} & -\frac{2\pi i}{r_k} (u(r_k x + \eta/2), r_k y + \omega/2) \\ & + \frac{2\pi i}{r_k} (v(r_k x - \eta/2), r_k y - \omega/2). \end{aligned}$$

Expanding these products gives

$$\begin{aligned} S_{(x,\eta),(y,\omega)}^{(k,-k)} &= |\alpha_k^{\text{KP}}|^2 \sum_{u,v \in W} \epsilon(u) \epsilon(v) \exp\{ -2\pi i r_k ((u-v)x, y) - \pi i ((u+v)x, \omega) \\ & \quad - \pi i ((u+v)\eta, y) - \frac{\pi i}{2r_k} ((u-v)\eta, \omega) \}. \end{aligned} \quad (3.25)$$

Summing the diagonal terms $u = v = w$ gives the kernel

$$K_{\text{diag}}((x, \eta), (y, \omega)) = \sum_{w \in W} e^{-2\pi i ((wx, \omega) + (w\eta, y))}. \quad (3.26)$$

Proposition 3.3 (Rapid decay of the off-diagonal Weyl terms). *Let $f \in C_c^\infty(Y \times P)$, $K \Subset Y$, $E \subset P$ finite, and $N \geq 0$. Let $\mathcal{S}_k^{\text{off}}$ be the operator obtained from Eq. (3.25) by retaining the terms with $u \neq v$. Then*

$$\sup_{(x,\eta) \in \Lambda_k \cap (K \times E)} |(\mathcal{S}_k^{\text{off}} Q_k f)(x, \eta)| = O_{f,K,E,N}(r_k^{-N}). \quad (3.27)$$

Proof. For $u \neq v$ and $x \in Y$, suppose for contradiction that $q := (u - v)x \in Q^\vee$. Then $ux = vx + q$, and hence

$$(v^{-1}u, -v^{-1}q) \cdot x = v^{-1}ux - v^{-1}q = x.$$

This is a nonidentity element of the affine Weyl group $W \ltimes Q^\vee$, because $u \neq v$. An interior point of the fundamental alcove has trivial affine-Weyl stabilizer, so the assumed resonance is impossible. Since $K \Subset Y$ and W is finite, these frequencies remain a positive distance from $Q^\vee$, uniformly for $x \in K$. This is the nonresonance needed for Poisson summation.

Fix $u \neq v$ and one of the finitely many input charges ω occurring in f , and put

$$\xi = (u - v)x, \quad b_{u,v,\eta,\omega}(y) = e^{-\pi i((u+v)\eta,y)} f(y, \omega). \quad (3.28)$$

The flux support of f has positive distance from ∂Y , so extension by zero makes $b_{u,v,\eta,\omega}$ a smooth compactly supported function on $\mathfrak{t}$. For a point in the translated input mesh we may write $y = (\mu + \rho + \omega/2)/r_k$; the two reconstructed chiral weights are then $\mu + \omega$ and μ . Since the flux support stays a fixed positive distance from the alcove boundary and ω ranges over a fixed finite set, both weights satisfy all integrability inequalities for sufficiently large k . More explicitly, choose $D_f \Subset Y$ containing every flux support and let E_f be the finite charge support. Put

$$\delta_f = \min_{y \in D_f} \min \left\{ \min_i \langle y, \alpha_i^\vee \rangle, 1 - \langle y, \theta^\vee \rangle \right\} > 0$$

and

$$M_f = \max_{\omega \in E_f} \max \left\{ \max_i |\langle \omega, \alpha_i^\vee \rangle|, |\langle \omega, \theta^\vee \rangle| \right\}.$$

The inequality $r_k \delta_f \geq 1 + M_f/2$ is a single threshold ensuring that both reconstructed weights are integrable for every charge in E_f and every Weyl pair. Beyond this threshold we extend the finite sum to all $\mu \in P$; the extra terms vanish under the zero extension of f .

Apart from the factor $\epsilon(u)\epsilon(v)|\alpha_k^{\text{KP}}|^2$, the contribution of the fixed quadruple (u, v, η, ω) is

$$\exp \left[-2\pi i(\xi, \rho + \omega/2) - \pi i((u+v)x, \omega) - \frac{\pi i}{2r_k}((u-v)\eta, \omega) \right] \Sigma_{r_k}(\xi), \quad (3.29)$$

where

$$\Sigma_{r_k}(\xi) = \sum_{\mu \in P} e^{-2\pi i(\xi, \mu)} b_{u,v,\eta,\omega} \left(\frac{\mu + \rho + \omega/2}{r_k} \right). \quad (3.30)$$

588 The exponential factor in Eq. (3.29) has modulus one. With $\widehat{b}(q) = \int_{\mathfrak{t}} b(y) e^{-2\pi i(q,y)} dy$, our
 589 convention is

$$\sum_{\mu \in P} g(\mu) = \text{covol}(P)^{-1} \sum_{p \in Q^\vee} \widehat{g}(p). \quad (3.31)$$

590 Applying this formula after dilation and translation gives

$$\Sigma_{r_k}(\xi) = \frac{r_k^\ell}{\text{covol}(P)} \sum_{p \in Q^\vee} e^{2\pi i(\xi+p, \rho+\omega/2)} \widehat{b}_{u,v,\eta,\omega}(r_k(\xi+p)). \quad (3.32)$$

591 This is the standard Poisson formula in the normalization of [41, Ch. 5, Sec. 3, and Ch. 6].

592 To estimate the dual-lattice sum uniformly, define

$$B_K := \{(u-v)x : x \in \overline{K}, u, v \in W, u \neq v\}$$

593 and let $\delta_K = \text{dist}(B_K, Q^\vee) > 0$. The positivity follows from the nonresonance above,
 594 together with compactness of K and finiteness of W .

595 For every $L > \ell$,

$$\sup_{\xi \in B_K} \sum_{p \in Q^\vee} |\xi + p|^{-L} < \infty. \quad (3.33)$$

596 Indeed, B_K is bounded. The finitely many terms with $|p|$ in a fixed bounded range are
 597 controlled by δ_K^{-L} , while for all remaining terms $|\xi + p| \geq |p|/2$ uniformly in $\xi \in B_K$; the
 598 lattice series $\sum_{p \in Q^\vee \setminus \{0\}} |p|^{-L}$ converges precisely when $L > \ell$.

599 The functions in Eq. (3.28) range over finite u, v, η, ω sets. For every L , their Fourier
 600 transforms therefore satisfy one uniform Schwartz estimate; the underlying rapid-decay
 601 theorem is reviewed in [41, Chs. 5–6]. Combining that estimate with Eq. (3.33) gives, for
 602 $r_k \geq 1$,

$$\begin{aligned} \sup_{\xi \in B_K} \sum_{p \in Q^\vee} \left| \widehat{b}_{u,v,\eta,\omega}(r_k(\xi+p)) \right| &\leq C_L \sup_{\xi \in B_K} \sum_{p \in Q^\vee} (1 + r_k |\xi + p|)^{-L} \\ &\leq C'_{L,K} r_k^{-L}. \end{aligned} \quad (3.34)$$

603 Thus Eq. (3.32) is $O(r_k^{\ell-L})$. Multiplication by the exact Kac–Peterson factor

$$|\alpha_k^{\text{KP}}|^2 = |P/Q^\vee|^{-1} r_k^{-\ell}$$

604 leaves $O(r_k^{-L})$. Given the exponent N in the statement, choose $L > \max\{\ell, N\}$ and sum
 605 over the finite Weyl-pair and input-charge sets. This proves Eq. (3.27) with all constants
 606 uniform on $K \times E$. $\square$

607 Since $Q^\vee \subset P$ has index $|P/Q^\vee|$,

$$\frac{|\alpha_k^{\text{KP}}|^2}{v_k} = \frac{1}{|P/Q^\vee| \text{covol}(P)} = \gamma_G.$$

608 The diagonal Weyl terms therefore give precisely the endpoint operator $\mathcal{S}_0$ already defined
 609 in Eq. (3.11); Eq. (3.26) is its integral kernel.

Proposition 3.4 (Fourier realization and endpoint modular action). *Under the last unitary in Eq. (3.10), the operator $\mathcal{S}_0$ is the restriction of the Pontryagin transform*

$$(\mathcal{F}F)(x, \eta) = \sum_{\omega \in P} \int_T e^{-2\pi i((x, \omega) + (\eta, y))} F(y, \omega) dm_T(y). \quad (3.35)$$

Consequently $\mathcal{S}_0$ extends uniquely to a unitary on $\mathcal{H}_0$. Together with $\mathcal{T}_0$, it defines a unitary representation ρ_0 of $SL(2, \mathbb{Z})$.

Proof. Write $e_\lambda(x) = e^{2\pi i(\lambda, x)}$. On the Fourier basis $e_\lambda \delta_\nu$ of $L^2(T \times P)$, $\mathcal{F}(e_\lambda \delta_\nu) = e_{-\nu} \delta_\lambda$; hence $\mathcal{F}$ is unitary and commutes with the diagonal Weyl action. Splitting the torus integral into Weyl chambers and changing the charge variable by the same Weyl element gives $\mathcal{F} \text{Ext} = \text{Ext} \mathcal{S}_0$. The unitary identification of Proposition 3.1 therefore intertwines $\mathcal{S}_0$ with $\mathcal{F}|_{L^2(T \times P)^W}$.

For the group action, use the generators

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

of $SL(2, \mathbb{Z})$. For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z})$, define

$$\tilde{\rho}_0(A)(e_\lambda \delta_\nu) = e_{a\lambda + b\nu} \delta_{c\lambda + d\nu}. \quad (3.36)$$

This permutes an orthonormal basis and commutes with the diagonal Weyl action. To fix the representation convention, regard $(\lambda, \nu)^t$ as a column vector. Then

$$\tilde{\rho}_0(A) \tilde{\rho}_0(B)(e_\lambda \delta_\nu) = \tilde{\rho}_0(AB)(e_\lambda \delta_\nu), \quad (3.37)$$

because the rightmost operator first replaces $(\lambda, \nu)^t$ by $B(\lambda, \nu)^t$ and the left operator then applies A . Hence the formula defines a representation. Transporting the restriction of $\tilde{\rho}_0$ through $U = (|W|\gamma_G)^{-1/2} \text{Ext}$ gives the unitary representation

$$\rho_0 : SL(2, \mathbb{Z}) \longrightarrow U(\mathcal{H}_0), \quad \rho_0(S) = \mathcal{S}_0, \quad \rho_0(T) = \mathcal{T}_0.$$

For the first generator, Eq. (3.35) gives $\tilde{\rho}_0(S) = \mathcal{F}$. For the second, the basis formula reads

$$\tilde{\rho}_0(T)(e_\lambda \delta_\nu) = e_{\lambda + \nu} \delta_\nu = e^{2\pi i(\nu, \cdot)} e_\lambda \delta_\nu.$$

Consequently $(\tilde{\rho}_0(T)Uf)(x, \eta) = e^{2\pi i(\eta, x)}(Uf)(x, \eta) = U\mathcal{T}_0 f(x, \eta)$, which proves the two displayed generator identities. In particular $\mathcal{S}_0^4 = 1$ and $(\mathcal{S}_0 \mathcal{T}_0)^3 = \mathcal{S}_0^2$.

629 □

Remark 3.5 (A common conull domain for the modular action). Put $R = T_{\text{reg}} \times T$. For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z}), \quad \Phi_A(x, y) = (ax + cy, bx + dy),$$

the first component of $\Phi_A(x, y)$ is singular only when

$$a\alpha(x) + c\alpha(y) \in \mathbb{Z}$$

for some root α . Since (a, c) is primitive, each such set is the kernel of a nontrivial character of $T \times T$ and has Haar measure zero. It follows that

$$D = \bigcap_{A \in SL(2, \mathbb{Z})} \Phi_A^{-1}(R) \quad (3.38)$$

is conull. Composition by any Φ_B permutes the factors in this intersection, so D is invariant. It is also invariant under the diagonal Weyl action and therefore descends to a common invariant conull subset of $(T \times T)/W$. The modular action consequently defines unitary operators on the quotient–Haar L^2 space of Eq. (3.10).

The equal-Weyl term in (3.25) is a Riemann sum for (3.11); Proposition 3.3 controls the remainder.

Proposition 3.6 (Local weighted convergence). *For $f \in C_c^\infty(Y \times P)$, $K \Subset Y$, and finite $E \subset P$,*

$$v_k \sum_{z \in \Lambda_k \cap (K \times E)} |(\mathcal{S}_k Q_k f - Q_k \mathcal{S}_0 f)(z)|^2 \longrightarrow 0. \quad (3.39)$$

Proof. The support of f meets only finitely many input-charge fibres, say $\omega \in E_f$, and its flux support has positive distance from ∂Y . For the equal-Weyl terms, Eq. (3.21) and $|\alpha_k^{\text{KP}}|^2/v_k = \gamma_G$ give, exactly,

$$\gamma_G v_k \sum_{\omega \in E_f} \sum_{y: (y, \omega) \in \Lambda_k} K_{\text{diag}}((x, \eta), (y, \omega)) f(y, \omega).$$

For each $\omega \in E_f$, extend $f(\cdot, \omega)$ by zero from Y to $\mathfrak{t}$. Because its support is compactly contained in Y , the extension belongs to $C_c^\infty(\mathfrak{t})$. Choose a compact set $D_f \Subset Y$ whose interior contains the supports of all these fibres, and set, for $y \in \mathfrak{t}$,

$$h_{x, \eta, \omega}(y) = K_{\text{diag}}((x, \eta), (y, \omega)) f(y, \omega).$$

The y -gradients of these functions are bounded uniformly for $(x, \eta) \in K \times E$ and $\omega \in E_f$. Let $D_P = \sup_{q \in F_P} |q|$. Comparing the value at the anchor of each translated cell with its integral over that cell gives

$$\left| v_k \sum_{y \in r_k^{-1}(P+s)} h_{x, \eta, \omega}(y) - \int_{\mathfrak{t}} h_{x, \eta, \omega}(y) dy \right| \leq |(D_f)_{2D_P/r_k}| \frac{D_P}{r_k} \sup_{y \in \mathfrak{t}} |\nabla_y h_{x, \eta, \omega}(y)|, \quad (3.40)$$

where $(D_f)_r = \{y : \text{dist}(y, D_f) < r\}$. Only cells meeting the support of $h_{x, \eta, \omega}$ contribute; their union lies in $(D_f)_{2D_P/r_k}$. All volumes and integrals use the fixed geometric Lebesgue measure. The estimate is independent of the translate s . For the mesh with charge ω one takes $s = \rho + \omega/2$; because $D_f \Subset Y$ and E_f is finite, the admitted mesh and the full translated lattice agree on D_f for all sufficiently large k . Thus Eq. (3.40) proves the required uniform convergence to Eq. (3.11). Proposition 3.3 controls the remaining Weyl pairs uniformly. The translated-lattice count $v_k \#(\Lambda_k \cap (K \times E)) = O_{K, E}(1)$ then converts this uniform pointwise convergence into Eq. (3.39). $\square$

659 3.2 No escape of mass

660 To pass from Proposition 3.6 to a global L^2 estimate, we must control samples near ∂Y
 661 and in the large-charge region. We first check that $\mathcal{S}_0 f$ belongs to the charge-Schwartz
 662 class whenever $f \in C_c^\infty(Y \times P)$. Only finitely many input charges ω occur. Differentiating
 663 Eq. (3.11) in x introduces powers of $w^{-1}\omega$, and these are uniformly bounded on that finite
 664 set. To estimate the output charge, let Δ_y be the Euclidean Laplacian for the geometric
 665 inner product. Its Weyl invariance gives

$$(1 - \Delta_y)^N e^{-2\pi i(w\eta, y)} = (1 + 4\pi^2|\eta|^2)^N e^{-2\pi i(w\eta, y)}.$$

666 Integrating by parts moves $(1 - \Delta_y)^N$ onto $f(\cdot, \omega)$. There are no boundary terms because
 667 every input fibre is compactly supported in Y . The resulting integral is bounded uniformly
 668 in $x \in \bar{Y}$, so it is $O_N((1 + |\eta|)^{-2N})$; the same argument applies after any number of
 669 x -derivatives. This is the usual Fourier-transform rapid-decay argument [41, Chs. 5–6].
 670 Hence

$$f \in C_c^\infty(Y \times P) \implies \mathcal{S}_0 f \in \mathcal{A}(Y \times P). \quad (3.41)$$

671 **Lemma 3.7** (Global sampling and charge-tail control). *For $\eta \in P$ write*

$$L_{k,\eta} := \{x \in Y : (x, \eta) \in \Lambda_k\}.$$

672 *There is a constant C_{lat} , independent of k and η , such that*

$$v_k \# L_{k,\eta} \leq C_{\text{lat}}. \quad (3.42)$$

673 *For every fixed $\eta \in P$ and every $\phi \in C(\bar{Y})$,*

$$v_k \sum_{x \in L_{k,\eta}} \phi(x) \longrightarrow \int_Y \phi(x) dx. \quad (3.43)$$

674 *Moreover, if $g \in \mathcal{A}(Y \times P)$ and $E_M := \{\eta \in P : |\eta| \leq M\}$, then*

$$\sup_k v_k \sum_{\eta \notin E_M} \sum_{x \in L_{k,\eta}} |g(x, \eta)|^2 \leq C_{\text{lat}} \sum_{\eta \notin E_M} \sup_{x \in Y} |g(x, \eta)|^2 \xrightarrow{M \rightarrow \infty} 0. \quad (3.44)$$

675 *Consequently*

$$\|J_k g\|_{\mathcal{H}_k}^2 = v_k \sum_{z \in \Lambda_k} |g(z)|^2 \longrightarrow \|g\|_{\mathcal{H}_0}^2. \quad (3.45)$$

676 *The analogous Riemann-sum convergence holds on $K \times E$ for finite $E \subset P$ whenever $K \Subset Y$*
 677 *has Lebesgue-null boundary.*

678 *Proof.* Solving Eq. (2.53) for a, b shows that, at fixed charge, the full translated lattice and
 679 its admitted subset are, respectively,

$$\tilde{L}_{k,\eta} = r_k^{-1}(P + \rho + \eta/2), \quad L_{k,\eta} = \tilde{L}_{k,\eta} \cap Y_{k,\eta}, \quad (3.46)$$

680 where the simultaneous dominance and level conditions on a and b are equivalent to

$$\begin{aligned}
r_k \langle x, \alpha_j^\vee \rangle &\geq 1 + \frac{1}{2} |\langle \eta, \alpha_j^\vee \rangle|, & 1 \leq j \leq \ell, \\
r_k (1 - \langle x, \theta^\vee \rangle) &\geq 1 + \frac{1}{2} |\langle \eta, \theta^\vee \rangle|.
\end{aligned} \tag{3.47}$$

Here we used $\langle \rho, \alpha_j^\vee \rangle = 1$ and $\langle \rho, \theta^\vee \rangle = h^\vee - 1$ in the long-root-length-two convention. Since the alcove has finitely many affine walls, there is a group-dependent constant C such that

$$Y \setminus Y_{k,\eta} \subset \left\{ x \in Y : \text{dist}(x, \partial Y) \leq C \frac{1 + |\eta|}{r_k} \right\}. \tag{3.48}$$

In particular, for fixed η the deleted points lie within an $O_\eta(r_k^{-1})$ neighborhood of ∂Y .

Using the fundamental parallelotope F_P fixed before Main Theorem A, set

$$D := \sup_{q \in F_P} |q|.$$

The cells $x + r_k^{-1} F_P$ based at the full translated lattice in Eq. (3.46) are disjoint up to null boundaries and have volume v_k . Those based at points of $L_{k,\eta}$ lie in the fixed bounded neighborhood $Y + r_k^{-1} F_P$. Hence their total volume is bounded uniformly in the translate, proving Eq. (3.42).

For fixed η , the width in Eq. (3.48) tends to zero. The ordinary translated-lattice Riemann sum over Y converges because the polytope Y is Jordan measurable. Every deleted anchor lies in the strip of Eq. (3.48). Its full cell is therefore contained in the two-sided $(C(1 + |\eta|) + D)/r_k$ neighborhood of ∂Y . These full cells are disjoint up to null boundaries and have volume v_k . The finite-facet slab estimate for the enlarged strip gives

$$v_k \# (\tilde{L}_{k,\eta} \cap (Y \setminus Y_{k,\eta})) = O_\eta(r_k^{-1}).$$

Deleting these points therefore changes the Riemann sum by $o(1)$. This proves Eq. (3.43). Equation (3.42) then gives the first inequality in Eq. (3.44). The charge-Schwartz condition makes its right-hand side summable on the lattice P : explicitly, choose $N > \ell/2$ in Eq. (3.7). Then, for a suitable C_N ,

$$\sup_{x \in Y} |g(x, \eta)|^2 \leq C_N (1 + |\eta|)^{-2N},$$

and the right-hand side is summable because P has rank ℓ . Its tail therefore tends to zero. The continuous charge tail satisfies the analogous estimate

$$\sum_{\eta \notin E_M} \int_Y |g(x, \eta)|^2 dx \leq |Y| \sum_{\eta \notin E_M} \sup_{x \in \bar{Y}} |g(x, \eta)|^2 \xrightarrow{M \rightarrow \infty} 0.$$

Apply Eq. (3.43) to $\phi(x) = |g(x, \eta)|^2$ for $\eta \in E_M$, and then let $M \rightarrow \infty$ using Eq. (3.44) and the preceding continuous-tail estimate. This proves Eq. (3.45). The proof on $K \times E$ is the same finite-charge argument; the null-boundary hypothesis ensures convergence of the Riemann sums for $\mathbf{1}_K |g(\cdot, \eta)|^2$. $\square$

705 Let

$$r_Y = \max_{x \in \bar{Y}} \text{dist}(x, \partial Y) > 0.$$

706 For $0 < \delta < r_Y$ we use the compact polytope

$$K_\delta = \{x \in Y : \text{dist}(x, \partial Y) \geq \delta\}. \quad (3.49)$$

707 It is nonempty and its boundary has Lebesgue measure zero. Moreover $K_\delta \uparrow Y$ as $\delta \downarrow 0$.
 708 For $M > 0$ put $E_M = \{\eta \in P : |\eta| \leq M\}$, and let $\Pi_{k,\delta,M}$ be the coordinate projection in
 709 $\mathcal{H}_k$ onto $\Lambda_k \cap (K_\delta \times E_M)$.

710 **Theorem 3.8** (No escape of mass). *For every $f \in C_c^\infty(Y \times P)$,*

$$\|\mathcal{S}_k J_k f - J_k \mathcal{S}_0 f\|_{\mathcal{H}_k} \longrightarrow 0. \quad (3.50)$$

711 *More explicitly, for every $\varepsilon > 0$ there are $\delta > 0$, $M < \infty$, and k_0 such that, for $k \geq k_0$,*

$$\|(1 - \Pi_{k,\delta,M})\mathcal{S}_k J_k f\| + \|(1 - \Pi_{k,\delta,M})J_k \mathcal{S}_0 f\| < \varepsilon. \quad (3.51)$$

712 *Here f is fixed before the window (δ, M) is chosen.*

713 *Proof.* Set $a_k = \mathcal{S}_k J_k f$ and $b_k = J_k \mathcal{S}_0 f$. Exact unitarity of the finite doubled matrix,
 714 (3.45), Proposition 3.4, and (3.41) give

$$\|a_k\| \rightarrow \|f\|, \quad \|b_k\| \rightarrow \|f\|.$$

715 Given $\varepsilon > 0$, choose $\delta > 0$ and $M < \infty$ so that the $\mathcal{H}_0$ -norm of $\mathcal{S}_0 f$ outside $\Omega_{\delta,M} := K_\delta \times E_M$
 716 is $< \varepsilon/4$. Put $\Pi_k = \Pi_{k,\delta,M}$. Proposition 3.6 gives $\|\Pi_k(a_k - b_k)\| \rightarrow 0$, while the localized
 717 part of Lemma 3.7 gives $\|\Pi_k b_k\|^2 \rightarrow \|1_{\Omega_{\delta,M}} \mathcal{S}_0 f\|^2$. Therefore $\|\Pi_k a_k\| \rightarrow \|1_{\Omega_{\delta,M}} \mathcal{S}_0 f\|$ as
 718 well. Since $\|a_k\|, \|b_k\| \rightarrow \|\mathcal{S}_0 f\|$, Pythagoras gives the explicit limits

$$\begin{aligned} \|(1 - \Pi_k)a_k\|^2 &= \|a_k\|^2 - \|\Pi_k a_k\|^2 \longrightarrow \|1_{\Omega_{\delta,M}^c} \mathcal{S}_0 f\|^2, \\ \|(1 - \Pi_k)b_k\|^2 &= \|b_k\|^2 - \|\Pi_k b_k\|^2 \longrightarrow \|1_{\Omega_{\delta,M}^c} \mathcal{S}_0 f\|^2. \end{aligned} \quad (3.52)$$

719 The limiting norm is $< \varepsilon/4$. For all sufficiently large k their sum is therefore $< \varepsilon$, which
 720 proves (3.51). Together with $\|\Pi_k(a_k - b_k)\| \rightarrow 0$, the decomposition into Π_k and $1 - \Pi_k$
 721 proves (3.50). $\square$

722 3.3 A common Hilbert space and the modular group

723 Return to the cell isometry fixed in Eq. (3.8). The next lemma proves that its image
 724 eventually fills $\mathcal{H}_0$ in the strong sense needed for the operator limit. The proof treats full
 725 cells, cells truncated by ∂Y , and the small boundary strip not covered by any admitted
 726 cell separately.

727 **Lemma 3.9** (Global cell approximation). *With I_k defined by Eq. (3.8),*

$$P_k \xrightarrow{\text{SOT}} 1, \quad \|I_k^* g - J_k g\| \rightarrow 0, \quad \|I_k J_k g - g\| \rightarrow 0 \quad (g \in \mathcal{A}(Y \times P)). \quad (3.53)$$

728 *Proof.* We first prove the two geometric estimates used below. Since Y is a bounded
 729 polytope with finitely many facets, there is a constant C_Y such that, for every $s > 0$,

$$|\{y \in Y : \text{dist}(y, \partial Y) < s\}| \leq C_Y \min\{1, s\}. \quad (3.54)$$

730 Indeed, after increasing the constant for $s \geq 1$, the set on the left is contained in the union
 731 of finitely many slabs of thickness s adjacent to the facets of Y . Orthogonal projection to
 732 a facet and Fubini's theorem bound the volume of each slab by s times a fixed bounded
 733 $(\ell - 1)$ -dimensional cross-sectional volume.

734 Let $D = \sup\{|q| : q \in F_P\}$. For fixed η , the cells of the full translated lattice $\tilde{L}_{k,\eta}$ tile
 735 $\mathfrak{t}$ up to null boundaries. If $y \in Y$ does not belong to a cell anchored at a point of $L_{k,\eta}$,
 736 let $q \in \tilde{L}_{k,\eta}$ be the anchor of the full cell containing y . Then $|y - q| \leq D/r_k$. If $q \notin Y$,
 737 the point y lies within D/r_k of ∂Y . If $q \in Y \setminus Y_{k,\eta}$, Eq. (3.48) places q , and hence y , in a
 738 boundary strip of width $C'(1 + |\eta|)/r_k$. Applying Eq. (3.54) proves

$$\left| Y \setminus \bigcup_{x \in L_{k,\eta}} (Y \cap (x + r_k^{-1} F_P)) \right| \leq C \min\left\{1, \frac{1 + |\eta|}{r_k}\right\}. \quad (3.55)$$

739 For the second estimate, suppose that an admitted cell anchored at x is truncated, so
 740 that $|C_{k,(x,\eta)}| < v_k$. Its full cell meets the complement of Y . Since $x \in Y$ and that cell
 741 has diameter at most $2D/r_k$, both its anchor and every point of the cell lie in the $2D/r_k$ -
 742 neighbourhood of ∂Y . The full cells are pairwise disjoint up to null sets and each has
 743 volume v_k . Their union has volume $O(r_k^{-1})$ by the same facet-slab argument. Comparing
 744 volumes gives

$$v_k \#\{x \in L_{k,\eta} : |C_{k,(x,\eta)}| < v_k\} \leq \frac{C}{r_k}, \quad (3.56)$$

745 with C independent of the translated lattice and hence of η .

746 Now fix $\eta \in P$ and identify $h \in C^1(\bar{Y})$ with $h\delta_\eta \in \mathcal{H}_0$. On a full cell in the η -summand,
 747 $I_k^* h$ is $\sqrt{v_k}$ times the cell average, whereas $J_k h$ is $\sqrt{v_k} h(x)$. If $C = x + r_k^{-1} F_P$ is full, Jensen's
 748 inequality and $|y - x| \leq D/r_k$ give

$$\left| v_k^{-1/2} \int_C h(y) dy - \sqrt{v_k} h(x) \right|^2 \leq \int_C |h(y) - h(x)|^2 dy \leq \frac{D^2}{r_k^2} \|\nabla h\|_\infty^2 |C|.$$

749 Summing this inequality and using Eq. (3.42) gives a total full-cell error $O(r_k^{-2}) \|\nabla h\|_\infty^2$.
 750 On truncated cells, Cauchy-Schwarz gives

$$\left| |C|^{-1/2} \int_C h - \sqrt{v_k} h(x) \right|^2 \leq 4v_k \|h\|_\infty^2.$$

751 After summing over the truncated cells, Eq. (3.56) gives

$$\sum_{C \text{ truncated}} \left| |C|^{-1/2} \int_C h - \sqrt{v_k} h(x_C) \right|^2 \leq \frac{C}{r_k} \|h\|_\infty^2.$$

752 Here x_C denotes the anchor of C . Together with the full-cell estimate, this proves

$$\|I_k^* h - J_k h\|_{\ell^2(L_{k,\eta})} \longrightarrow 0.$$

753 The same decomposition, now also using Eq. (3.55), gives

$$\|I_k J_k h - h\|_{L^2(Y)} \longrightarrow 0. \quad (3.57)$$

754 For the truncated-cell part of the second statement, note that the squared norm of $I_k J_k h$
 755 on such a cell is exactly $v_k |h(x)|^2$, independently of its volume. Its total contribution is
 756 again $O(r_k^{-1}) \|h\|_\infty^2$, while the L^2 mass of h on the union of the truncated and uncovered
 757 regions tends to zero by the two strip estimates.

758 For $g \in \mathcal{A}(Y \times P)$, let $g_M = 1_{E_M} g$. The preceding fixed-charge estimates, summed
 759 over the finite set E_M , apply to g_M . For the remaining tail $g^{>M} = g - g_M$, contractivity
 760 of I_k^* and isometry of I_k give

$$\begin{aligned} \|I_k^* g^{>M} - J_k g^{>M}\| &\leq \|g^{>M}\| + \|J_k g^{>M}\|, \\ \|I_k J_k g^{>M} - g^{>M}\| &\leq \|J_k g^{>M}\| + \|g^{>M}\|. \end{aligned}$$

761 Here $\|g^{>M}\| \rightarrow 0$ as $M \rightarrow \infty$, while Eq. (3.44) gives $\sup_k \|J_k g^{>M}\| \rightarrow 0$. This proves the
 762 last two limits in Eq. (3.53). On the same dense class,

$$\|P_k g - g\| \leq \|I_k^* g - J_k g\| + \|I_k J_k g - g\| \longrightarrow 0.$$

763 Since the orthogonal projections P_k are contractions and $\mathcal{A}(Y \times P)$ is dense in $\mathcal{H}_0$, the first
 764 limit follows on all of $\mathcal{H}_0$. $\square$

765 *Proof of Main Theorem A.* Proposition 3.4 gives the endpoint representation. For the S -
 766 generator, put

$$\mathcal{S}_k^\# = I_k \mathcal{S}_k I_k^* + (1 - P_k). \quad (3.58)$$

767 Thus $\mathcal{S}_k^\# = \rho_k^\#(S)$. For $f \in C_c^\infty(Y \times P)$, insert $J_k f$ and $J_k \mathcal{S}_0 f$. The function $\mathcal{S}_0 f$ belongs
 768 to $\mathcal{A}(Y \times P)$ by Eq. (3.41), so the cell lemma applies to both functions:

$$\|\mathcal{S}_k^\# f - \mathcal{S}_0 f\| \leq \|I_k^* f - J_k f\| + \|\mathcal{S}_k J_k f - J_k \mathcal{S}_0 f\| + \|I_k J_k \mathcal{S}_0 f - \mathcal{S}_0 f\| + \|(1 - P_k) f\|.$$

769 Equations (3.53) and (3.50) make all four terms vanish, and density then gives

$$\mathcal{S}_k^\# \xrightarrow{\text{SOT}} \mathcal{S}_0. \quad (3.59)$$

770 The adjoints converge strongly as well. Indeed, if U_k and U are unitary and $U_k \rightarrow U$
 771 strongly, then, for every ψ ,

$$\|(U_k^* - U^*)\psi\| = \|\psi - U_k U^* \psi\| \longrightarrow 0. \quad (3.60)$$

772 Applying this identity to $\mathcal{S}_k^\#$ and $\mathcal{S}_0$ gives strong-* convergence.

773 For the twist, the WZW formula [40, Eq. (3.3b)] and cancellation of the two vacuum
 774 phases give an exact identity. Substitution of Eq. (2.48) gives $(a, a + 2\rho) - (b, b + 2\rho) =$
 775 $2r_k(\eta, x)$, and hence

$$\begin{aligned} \theta_{(a,b)} &= \exp \left[\frac{2\pi i}{2r_k} ((a, a + 2\rho) - (b, b + 2\rho)) \right] \\ &= e^{2\pi i(\eta, x)} = \chi_\eta(g_x). \end{aligned} \quad (3.61)$$

Thus the endpoint twist is the operator already defined in Eq. (3.12), and the exact sampling identity is $\mathcal{T}_k J_k = J_k \mathcal{T}_0$. Both $\mathcal{T}_0$ and $\mathcal{T}_0^*$ preserve $C_c^\infty(Y \times P)$. Put $\mathcal{T}_k^\# = I_k \mathcal{T}_k I_k^* + (1 - P_k)$; then $\mathcal{T}_k^\# = \rho_k^\#(T)$. For $f \in C_c^\infty(Y \times P)$, exact intertwining on sampled vectors gives

$$\|\mathcal{T}_k^\# f - \mathcal{T}_0 f\| \leq \|I_k^* f - J_k f\| + \|I_k J_k \mathcal{T}_0 f - \mathcal{T}_0 f\| + \|(1 - P_k)f\|. \quad (3.62)$$

All three terms tend to zero by Lemma 3.9. The same calculation with $\mathcal{T}_k^*$ and $\mathcal{T}_0^*$ is valid because $\mathcal{T}_k^* J_k = J_k \mathcal{T}_0^*$. Since the sampling core is dense and all four operators are unitary, these two estimates extend by density and prove $\mathcal{T}_k^\# \rightarrow \mathcal{T}_0$ in the strong-* topology.

The generators $\mathcal{S}_k^\#$ and $\mathcal{T}_k^\#$, together with their inverses, converge strongly to the corresponding operators for ρ_0 . Fix $\gamma \in SL(2, \mathbb{Z})$ and choose a word in $S^{\pm 1}, T^{\pm 1}$ representing it. Products of uniformly bounded strongly convergent operators converge strongly, so $\rho_k^\#(\gamma) \rightarrow \rho_0(\gamma)$. Applying the same argument to γ^{-1} gives convergence of the adjoints. This proves the pointwise strong-* convergence in Eq. (3.18). The assertion for admissible embeddings follows from Lemma 3.11, proved below. $\square$

3.4 Independence of the sampling embeddings

The preceding proof uses two approximation properties of I_k : its adjoint agrees asymptotically with weighted sampling, and sampled smooth functions reconstruct in L^2 . We now show that these properties suffice to determine the operator limit.

Definition 3.10 (Admissible sampling embeddings). A family of isometries $\tilde{I}_k : \mathcal{H}_k \rightarrow \mathcal{H}_0$ is called admissible if, for every $g \in \mathcal{A}(Y \times P)$,

$$\|\tilde{I}_k^* g - J_k g\| \rightarrow 0, \quad \|\tilde{I}_k J_k g - g\| \rightarrow 0. \quad (3.63)$$

Lemma 3.11 (Robustness under admissible sampling embeddings). For an admissible family $(\tilde{I}_k)$, put $\tilde{P}_k = \tilde{I}_k \tilde{I}_k^*$. Then

$$W_k := \tilde{I}_k I_k^* \rightarrow 1, \quad W_k^* = I_k \tilde{I}_k^* \rightarrow 1 \quad \text{strongly on } \mathcal{H}_0. \quad (3.64)$$

Consequently, if $U_k \in U(\mathcal{H}_k)$ and

$$I_k U_k I_k^* + (1 - P_k) \rightarrow U \quad \text{strongly-}, \quad (3.65)$$

then the same limit is obtained with the alternative embeddings:

$$\tilde{I}_k U_k \tilde{I}_k^* + (1 - \tilde{P}_k) \rightarrow U \quad \text{strongly-}. \quad (3.66)$$

Proof. Fix $g \in \mathcal{A}(Y \times P)$. Insert the common sampled vector $J_k g$:

$$\begin{aligned} \|W_k g - g\| &\leq \|I_k^* g - J_k g\| + \|\tilde{I}_k J_k g - g\|, \\ \|W_k^* g - g\| &\leq \|\tilde{I}_k^* g - J_k g\| + \|I_k J_k g - g\|. \end{aligned} \quad (3.67)$$

Both right-hand sides tend to zero by Lemma 3.9 and (3.63). Since W_k and W_k^* are contractions, density extends the two limits to all of $\mathcal{H}_0$. The same two assumptions also give

$$\|\tilde{\mathbf{P}}_k g - g\| \leq \|\tilde{I}_k^* g - J_k g\| + \|\tilde{I}_k J_k g - g\| \longrightarrow 0$$

on $\mathcal{A}(Y \times P)$, and hence $\tilde{\mathbf{P}}_k \rightarrow 1$ strongly by density.

Let $U_k^\# = I_k U_k I_k^* + (1 - \mathbf{P}_k)$. The identity

$$\tilde{I}_k U_k \tilde{I}_k^* = W_k U_k^\# W_k^* \quad (3.68)$$

holds because $W_k(1 - \mathbf{P}_k) = 0$ and $(1 - \mathbf{P}_k)W_k^* = 0$. Products of strongly convergent uniformly bounded operators converge strongly. Equations (3.64) and (3.65) therefore show that the right-hand side of (3.68) converges strongly to U . The complementary projection tends strongly to zero. Repeating the same argument with U_k^* proves the adjoint convergence. $\square$

Corollary 3.12 (Cell criterion for admissible embeddings). *For each $z = (x_z, \eta_z) \in \Lambda_k$, let $\tilde{C}_{k,z} \subset Y$ be measurable with positive measure, with cells in the same charge fibre pairwise disjoint modulo null sets, and define*

$$\tilde{I}_k \delta_z = |\tilde{C}_{k,z}|^{-1/2} 1_{\tilde{C}_{k,z}} \delta_{\eta_z}.$$

Assume that for every finite $E \subset P$,

$$\sup_{\substack{z \in \Lambda_k \\ \eta_z \in E}} (\text{diam } \tilde{C}_{k,z} + \text{dist}(x_z, \tilde{C}_{k,z})) \longrightarrow 0, \quad (3.69)$$

$$\sum_{\substack{z \in \Lambda_k \\ \eta_z \in E}} (|\tilde{C}_{k,z}|^{1/2} - v_k^{1/2})^2 \longrightarrow 0, \quad (3.70)$$

$$\left| Y \setminus \bigcup_{\substack{z \in \Lambda_k \\ \eta_z = \eta}} \tilde{C}_{k,z} \right| \longrightarrow 0 \quad (\eta \in E). \quad (3.71)$$

Then $(\tilde{I}_k)$ is admissible, and hence gives the same strong-* limit as I_k .

Proof. Fix a finite set $E \subset P$ and write

$$\delta_{k,E} := \sup_{\substack{z \in \Lambda_k \\ \eta_z \in E}} (\text{diam } \tilde{C}_{k,z} + \text{dist}(x_z, \tilde{C}_{k,z})).$$

For $g \in \mathcal{A}(Y \times P)$, let

$$\omega_{g,E}(r) := \max_{\eta \in E} \sup_{\substack{x, y \in \bar{Y} \\ |x-y| \leq r}} |g(x, \eta) - g(y, \eta)|.$$

Then $\delta_{k,E} \rightarrow 0$ by Eq. (3.69), while $\omega_{g,E}(r) \rightarrow 0$ as $r \downarrow 0$.

818 Put $\tilde{C}_z = \tilde{C}_{k,z}$. Directly from the definitions,

$$\begin{aligned} (\tilde{I}_k^* g)_z - (J_k g)_z &= \frac{1}{|\tilde{C}_z|^{1/2}} \int_{\tilde{C}_z} (g(y, \eta_z) - g(x_z, \eta_z)) dy \\ &\quad + (|\tilde{C}_z|^{1/2} - v_k^{1/2}) g(x_z, \eta_z). \end{aligned}$$

819 The cells in each charge fibre are disjoint. Hence, with $\|g\|_{\infty, E} := \max_{\eta \in E} \sup_{\bar{Y}} |g(\cdot, \eta)|$,

$$\begin{aligned} \|\tilde{I}_k^*(1_E g) - J_k(1_E g)\|^2 &\leq 2|E| |Y| \omega_{g, E}(\delta_{k, E})^2 \\ &\quad + 2\|g\|_{\infty, E}^2 \sum_{\substack{z \in \Lambda_k \\ \eta_z \in E}} (|\tilde{C}_z|^{1/2} - v_k^{1/2})^2 \longrightarrow 0. \end{aligned} \quad (3.72)$$

820 On $\tilde{C}_z$ the second sampling operator is

$$(\tilde{I}_k J_k g)(y, \eta_z) = \left(\frac{v_k}{|\tilde{C}_z|} \right)^{1/2} g(x_z, \eta_z).$$

821 Integrating the difference from $g(y, \eta_z)$ over the cells gives

$$\begin{aligned} \sum_{\substack{z \in \Lambda_k \\ \eta_z \in E}} \int_{\tilde{C}_z} |(\tilde{I}_k J_k g)(y, \eta_z) - g(y, \eta_z)|^2 dy \\ \leq 2|E| |Y| \omega_{g, E}(\delta_{k, E})^2 + 2\|g\|_{\infty, E}^2 \sum_{\substack{z \in \Lambda_k \\ \eta_z \in E}} (v_k^{1/2} - |\tilde{C}_z|^{1/2})^2. \end{aligned} \quad (3.73)$$

822 The right-hand side tends to zero. On the uncovered part of a charge fibre, $\tilde{I}_k J_k g$ vanishes,
823 and Eq. (3.71) shows that the remaining integral of $|g|^2$ also tends to zero. This proves
824 both sampling limits for $1_E g$.

825 Finally, take $E_M = \{\eta \in P : |\eta| \leq M\}$ and put $g^{>M} = (1 - 1_{E_M})g$. Since $\tilde{I}_k$ is an
826 isometry,

$$\begin{aligned} \|\tilde{I}_k^* g^{>M} - J_k g^{>M}\| &\leq \|g^{>M}\| + \|J_k g^{>M}\|, \\ \|\tilde{I}_k J_k g^{>M} - g^{>M}\| &\leq \|J_k g^{>M}\| + \|g^{>M}\|. \end{aligned}$$

827 The charge-Schwartz condition gives $\|g^{>M}\| \rightarrow 0$, and Lemma 3.7 gives $\sup_k \|J_k g^{>M}\| \rightarrow 0$.

828 First let $k \rightarrow \infty$ on the finite-charge part and then let $M \rightarrow \infty$. The two conditions in
829 Eq. (3.63) follow, and Lemma 3.11 applies. $\square$

830 If the cells satisfy the preceding criterion, their embeddings give ρ_0 itself. Multiplying
831 the embeddings by phases with a strong-* limit conjugates this realization. Suppose that
832 $\tilde{I}_k = M_{\phi_k} I_k$, where $|\phi_k| = 1$ almost everywhere, and that $M_{\phi_k} \rightarrow M_\phi$ strongly-*. The
833 corresponding extensions satisfy the exact identity

$$\tilde{I}_k \rho_k(\gamma) \tilde{I}_k^* + (1 - \tilde{I}_k \tilde{I}_k^*) = M_{\phi_k} \rho_k^\#(\gamma) M_{\phi_k}^*.$$

834 Consequently, for every fixed $\gamma \in SL(2, \mathbb{Z})$,

$$\tilde{I}_k \rho_k(\gamma) \tilde{I}_k^* + (1 - \tilde{I}_k \tilde{I}_k^*) \xrightarrow{\text{SOT}} M_\phi \rho_0(\gamma) M_\phi^*, \quad (3.74)$$

835 and the same convergence holds for the adjoints.

4 Regular fusion: exact endpoint and large-level limits

The compact-double Mackey decomposition of two regular inputs consists of an orbital output measure and a measurable multiplicity Hilbert field. After Weyl-denominator normalization, the orbital measure has the group-valued Duistermaat–Heckman density. On a principal regular-value chamber this density is the Liouville volume of the pair-of-pants reduction. Main Theorem B identifies its fixed-input density class with the leading affine–BZ polynomial of an individual chiral fusion coefficient, and Section 5 combines the two chiral polynomials. We formulate tensor products in the full compact double, which also contains the tensor unit, pure-electric representations, and singular-flux sectors.

Throughout this section,

$$r_k = k + h^\vee, \quad d_G = |\Delta_+| - \text{rank } G, \quad z_a = \frac{a + \rho}{r_k} \in Y.$$

At a principal regular value, the three-holed-sphere reduced space has real dimension $2d_G$.

4.1 Regular tensor products in the full compact double

We use the full compact double and its regular ideal from Eq. (2.32). Their regular induced representations are realized on the carriers in Eq. (2.37), and their tensor product is defined by the multiplier-valued coproduct in Eq. (2.35). In this model, on the dense convolution $*$ -algebra $C(G \times G) \subset C(G) \rtimes_{\text{Ad}} G$, with the first variable the flux variable and the second the group variable, define

$$\begin{aligned} & (\Pi_{(x,\eta) \boxtimes (y,\omega)}(F)\Phi)(u, v) \\ & := \int_G F(ug_x u^{-1} v g_y v^{-1}, h) \Phi(h^{-1}u, h^{-1}v) dh. \end{aligned} \tag{4.1}$$

This is the integrated covariant representation of [6, Eq. (4.3)]; it extends to $\mathfrak{D}(G)$. Equivalently, $C(G)$ acts by evaluating a function on the product holonomy $ug_x u^{-1} v g_y v^{-1}$, while G acts by simultaneous left translation in u and v . All tensor-product symbols below refer to this representation-level action.

For regular inputs, Lemma 4.1 shows that the output spectral measure gives full mass to regular holonomies. Thus the tensor product in Eq. (4.1) restricts nondegenerately to the regular ideal. This statement concerns the spectral measure: particular relative positions can still give a singular product. For later use, if $z \in \bar{Y}$, $g_z = \text{Exp}_T(z)$, $G_z = C_G(g_z)$, and $\sigma \in \widehat{G_z}$, write

$$\mathcal{H}_{z,\sigma} := \text{Ind}_{G_z}^G \sigma. \tag{4.2}$$

For $z = x \in Y$ and $\sigma = \chi_\eta$, this is the regular representation $\mathcal{H}_{x,\eta}$ of Eq. (2.36).

Let $\pi_{\bar{Y}} : G \rightarrow \bar{Y}$ send an element to the unique point of the closed alcove representing its conjugacy class. To fuse two regular lines, use their relative position

$$X = T \backslash G / T, \quad z(q) = \pi_{\bar{Y}}(g_x q g_y q^{-1}), \quad H_q = T \cap q T q^{-1}. \tag{4.3}$$

Here q records the relative orientation of the two input holonomies, $z(q)$ is the resulting output flux, and H_q is the residual stabilizer. The normalized Haar measure induces a probability measure $d\bar{q}$ on X . Its push-forward under $z(q)$ is

$$\nu_{x,y} := (\pi_{\bar{Y}})_*(\nu_x * \nu_y), \quad (4.4)$$

where ν_x is the normalized G -invariant orbital probability measure on the conjugacy class of g_x . Rokhlin disintegration gives

$$d\bar{q} = \int_{\bar{Y}} p_{x,y}^z d\nu_{x,y}(z), \quad (4.5)$$

$$F_{x,y}^z := \{\bar{q} \in X : z(q) = z\}, \quad p_{x,y}^z(F_{x,y}^z) = 1 \quad \text{for } \nu_{x,y}\text{-almost every } z.$$

Here and below, conditional measures and the associated fibre Hilbert spaces are defined only for $\nu_{x,y}$ -almost every z , as usual in Rokhlin disintegration [6, Appendix A, especially Theorem A.5].

Let $X_{\text{pr}} \subset X$ be the principal orbit-type locus $H_q = Z(G)$, and set

$$X_{\text{pr,reg}}(x, y) := X_{\text{pr}} \cap z_{x,y}^{-1}(Y), \quad F_{x,y}^{z,\text{pr}} := F_{x,y}^z \cap X_{\text{pr,reg}}(x, y), \quad (4.6)$$

where $z_{x,y}(q) = \pi_{\bar{Y}}(g_x q g_y q^{-1})$. When the inputs are fixed, we abbreviate $X_{\text{pr,reg}}(x, y)$ to $X_{\text{pr,reg}}$. This abbreviation will also be used in Appendix B.1.

Lemma 4.1 (Generic principal-regular locus). *Let G be a compact, connected, simply connected simple Lie group and let $T \subset G$ be a maximal torus. For normalized Haar-almost every $q \in G$,*

$$T \cap qTq^{-1} = Z(G). \quad (4.7)$$

Consequently, for every $x, y \in Y$,

$$d\bar{q}(X_{\text{pr,reg}}(x, y)) = 1. \quad (4.8)$$

Proof. The centre is contained in every maximal torus. Suppose that $K_q := T \cap qTq^{-1}$ is strictly larger than $Z(G)$. The quotient $K_q/Z(G)$ is then a nontrivial compact abelian Lie group, so it contains a nonidentity element of finite order. Choose a lift $t \in K_q$. If the order of its image is n , then $t^n \in Z(G)$. The centre of a compact connected simple Lie group is finite; hence t is a noncentral torsion element of T .

The torsion subgroup of the finite-dimensional torus T is countable, since $T_{\text{tors}} = \bigcup_{n \geq 1} T[n]$ and every $T[n]$ is finite. For a fixed noncentral $t \in T_{\text{tors}}$, put

$$E_t = \{q \in G : q^{-1}tq \in T\}. \quad (4.9)$$

The intersection of a conjugacy class with T is one Weyl orbit. Choosing representatives $n_w \in N_G(T)$ with $W \cdot t = \{n_w^{-1}tn_w : w \in W\}$ gives

$$E_t = \bigcup_{w \in W} C_G(t)n_w. \quad (4.10)$$

Because t is noncentral, $C_G(t)$ is a proper closed subgroup of the connected Lie group G and therefore has lower dimension and Haar measure zero. Thus every E_t is null, and

$$\{q : T \cap qTq^{-1} \neq Z(G)\} = \bigcup_{t \in T_{\text{tors}} \setminus Z(G)} E_t \quad (4.11)$$

is a countable union of null sets. This proves (4.7). The exceptional set is bi- T -invariant. Since $d\bar{q}$ is the push-forward of normalized Haar measure under $G \rightarrow T \backslash G / T$, its image in the double-coset space is also null.

It remains to exclude singular output holonomy. The conjugacy-class distribution of $g_x q g_y q^{-1}$ is the class-space push-forward of $\nu_x * \nu_y$. For regular x, y , Appendix B.3 proves that this convolution has an $L^2(G)$ density. Since $G \setminus G_{\text{reg}}$ is Haar-null, the output is regular for almost every q . Intersecting the two conull sets proves (4.8). $\square$

For the present regular inputs, the two centralizers called N_A and N_B in [6] are both T . Its set G° consists of those $q \in G$ for which

$$t_1 q t_2^{-1} = q, \quad t_1, t_2 \in T \implies t_1 = t_2 \in Z(G). \quad (4.12)$$

The displayed implication is equivalent to $T \cap qTq^{-1} = Z(G)$: given an element of the intersection, use it as $t_1 = q t_2 q^{-1}$, and conversely rearrange Eq. (4.12). Thus the first part of Lemma 4.1 proves Assumption 6.7 of the cited paper in the form needed here.

The representatives and conjugators entering the L^2 disintegration can be chosen as Borel fields by [6, Corollary A.3]; Appendix B.1 describes the choices explicitly. Lemma 4.1 also gives $\nu_{x,y}(\partial Y) = 0$, so the principal regular integral may be taken over Y . Exceptional orbit types are described by the full stabilizer formula below.

The compact-group Mackey tensor-product theorem first disintegrates over $T \backslash G / T$ and then over representations of the output centralizer [6, Sec. 6, especially Theorem 6.10 and Eqs. (6.55)–(6.62)]. We use that theorem in the following principal-regular form. Lemma 4.1 verifies that the required locus has full measure. Appendix B.1 recalls the unitary disintegration, converts the measures to the normalization used here, and derives the charge condition.

For a weight λ , let $\mathbb{C}_\lambda$ denote the one-dimensional representation of $Z(G)$ obtained by restricting χ_λ . The coefficient line which occurs on the principal locus is

$$\mathcal{L}_{\eta,\omega,\tau} := \text{Hom}_{Z(G)}(\mathbb{C}_\tau, \mathbb{C}_\eta \otimes \mathbb{C}_\omega). \quad (4.13)$$

This coefficient space is one-dimensional when $\tau - \eta - \omega \in Q$, and it is zero otherwise.

Theorem 4.2 (Mackey decomposition on the principal regular locus). *Let G be compact, connected, simple, and simply connected, let $T \subset G$ be the fixed maximal torus, and let $x, y \in Y$ and $\eta, \omega \in P$. Equip $\mathcal{H}_{x,\eta} \hat{\otimes} \mathcal{H}_{y,\omega}$ with the explicit representation-level action $\Pi_{(x,\eta) \boxtimes (y,\omega)}$ of Eq. (4.1). This Hilbert representation is unitarily equivalent, up to $\nu_{x,y}$ -null sets, to*

$$\mathcal{H}_{x,\eta} \hat{\otimes} \mathcal{H}_{y,\omega} \cong \int_Y^\oplus \bigoplus_{\tau \in \eta + \omega + Q} \mathcal{H}_{z,\tau} \hat{\otimes} \mathcal{M}_{x,y}^{z;\eta,\omega,\tau} d\nu_{x,y}(z), \quad (4.14)$$

921 where, for $\nu_{x,y}$ -almost every $z \in Y$ and each $\tau \in \eta + \omega + Q$,

$$p_{x,y}^z(F_{x,y}^{z,\text{pr}}) = 1, \quad (4.15)$$

922 and

$$\mathcal{M}_{x,y}^{z;\eta,\omega,\tau} = L^2(F_{x,y}^{z,\text{pr}}, p_{x,y}^z; \mathcal{L}_{\eta,\omega,\tau}). \quad (4.16)$$

923 The space in Eq. (4.16) is the intrinsic multiplicity space. Choosing a unit vector in the
924 Hom line gives an isomorphism

$$\mathcal{M}_{x,y}^{z;\eta,\omega,\tau} \cong L^2(F_{x,y}^{z,\text{pr}}, p_{x,y}^z),$$

925 but no such choice is preferred. Its unitary-equivalence class is independent of the choice.
926 For a charge outside the displayed coset the corresponding multiplicity space is zero. In-
927 deed, since G is simply connected, equality of central characters is equivalent to congruence
928 modulo the root lattice, so the input and output central characters agree precisely when

$$\tau - \eta - \omega \in Q. \quad (4.17)$$

929 The countable charge sum may be realized on one common $\nu_{x,y}$ -conull subset of Y .
930 Thus the output of Theorem 4.2 consists of the orbital measure and the measurable family of
931 Hilbert spaces in Eq. (4.16). The general stabilizer-restriction formula and the measurable
932 choices entering its proof are given once in Appendix B.1. For later reference, we denote
933 this measure-field datum by

$$\mathfrak{F}_{\text{Mack}}(x, \eta; y, \omega) := (d\nu_{x,y}(z), \{\mathcal{M}_{x,y}^{z;\eta,\omega,\tau}\}_{z \in Y, \tau \in \eta + \omega + Q}). \quad (4.18)$$

934 At a fixed exceptional double coset $\bar{q}$, the orbitwise multiplicity of $\mathcal{H}_{z,\sigma}$ is

$$m_{\bar{q}}(\sigma) = \dim \text{Hom}_{H_q}(\sigma^{s_q}|_{H_q}, \chi_\eta|_{H_q} \otimes {}^q\chi_\omega|_{H_q}), \quad (4.19)$$

935 with the notation of Appendix B.1.

936 4.2 The orbital density and the pair-of-pants volume

937 We first normalize the orbital density for every fixed pair of regular inputs, then identify
938 its geometric representative on principal regular-value chambers. The character formula at
939 the end of the subsection also proves the square integrability needed in the doubled limit.

940 4.2.1 The orbital density for fixed inputs

941 Put

$$\delta_G(x) = \sum_{w \in W} \epsilon(w) e^{2\pi i(w\rho, x)}, \quad \Pi_\rho = \prod_{\alpha \in \Delta_+} 2\pi(\rho, \alpha). \quad (4.20)$$

942 For fixed $x, y \in Y$, the measure $\nu_{x,y}$ of Eq. (4.4) is absolutely continuous with respect to
943 dz ; Appendix B.3 proves the needed $L^2(G)$ orbital-convolution estimate. The assignment
944 $(x, y) \mapsto \nu_{x,y}$ is a Borel probability kernel from Y^2 to Y . Let $m_Y(A) = \int_A dz$. Since

945 Y is a second-countable locally compact space, a countable algebra generating its Borel
 946 sets is dense, for the symmetric-difference pseudometric, with respect to the finite measure
 947 $m_Y + \nu_{x,y}$ for every (x, y) . The hypotheses of the parameterized Radon–Nikodym theorem
 948 are therefore satisfied with the constant kernel m_Y and the kernel $\nu_{x,y}$. Theorem 2.3 of
 949 [42] supplies a jointly Borel function $h : Y^3 \rightarrow [0, \infty)$ such that

$$\nu_{x,y}(A) = \int_A h(x, y, z) dz \quad \text{for every } (x, y) \in Y^2 \text{ and every Borel } A \subset Y. \quad (4.21)$$

950 Fix one such h and define

$$V_{\text{BF}}(x, y; z) := \frac{\text{covol}(Q)}{\Pi_\rho} \frac{|\delta_G(x)\delta_G(y)|}{|\delta_G(z)|} h(x, y, z). \quad (4.22)$$

951 The Weyl denominator is nonzero on Y . For every fixed input pair, $V_{\text{BF}}(x, y; \cdot)$ is there-
 952 fore a representative of the normalized orbital-convolution density. Fixed-input identities
 953 involving this function are identities modulo dz -null sets, and its support is its essential
 954 support. Pointwise geometric formulas use the chamber polynomial introduced below.

955 4.2.2 The reduced space and its volume

956 At a principal regular value the normalized orbital density equals the symplectic volume
 957 of the three-holed-sphere reduction. This scalar is the leading coefficient in the large-level
 958 asymptotic; the exact multiplicity space remains the L^2 space in Eq. (4.16). For $g \in G$,
 959 write $\mathcal{O}(g) = \{hgh^{-1} : h \in G\}$ and, for $x \in Y$, put $\mathcal{O}_x = \mathcal{O}(g_x)$. With the group-valued
 960 moment-map conventions of [43, Secs. 5 and 9, especially Theorem 9.3], the reduced space
 961 for boundary holonomies g_x, g_y, g_z^{-1} is

$$\overline{\mathcal{P}}_{x,y,z}^{\text{BF}} := \{(A, B, C) \in \mathcal{O}_x \times \mathcal{O}_y \times \mathcal{O}(g_z^{-1}) : ABC = 1\} / (G/Z(G)). \quad (4.23)$$

962 We use the following terminology throughout. A boundary label is *regular* when $x, y, z \in Y$;
 963 this concerns the three conjugacy classes. The parameter is a *regular value* when e is a
 964 regular value of the fused group-valued moment map. It is *principal regular* when, in ad-
 965 dition, the fibre meets the ambient principal orbit-type stratum, whose stabilizer is $Z(G)$.
 966 A critical value is called *principal critical* when its fibre nevertheless contains points with
 967 stabilizer $Z(G)$. The pointwise function V_{geom} is the orbifold Liouville volume on principal
 968 regular chambers and the top-stratum extension at the principal critical parameters spec-
 969 ified below. General measure statements use the kernel V_{BF} fixed in Eq. (4.22). Here the
 970 quotient means the quotient of the single fibre $\Phi^{-1}(e)$ by the effective action of $G/Z(G)$,
 971 where $\Phi(A, B, C) = ABC$. Composing Φ with $G \rightarrow G/Z(G)$ would instead produce the
 972 larger fibre $\Phi^{-1}(Z(G))$ and a different reduced space.

973 The inverse on the third boundary can equivalently be recorded in the alcove. Recall
 974 from Eq. (2.10) that $\lambda^* = -w_0\lambda$. Every simple reflection changes a weight by an element
 975 of Q , so the Weyl group acts trivially on P/Q . It follows that

$$[\lambda^*] = -[\lambda] \quad \text{in } P/Q. \quad (4.24)$$

976 In particular, $a + b - c \in Q$ is equivalent to $a + b + c^* \in Q$, as required when the oppositely
 977 oriented third marking is labelled by c^* . For the alcove coordinate, put

$$z^* := -w_0 z, \quad (4.25)$$

978 then $z^* \in Y$ and

$$\mathcal{O}(g_z^{-1}) = \mathcal{O}_{z^*}, \quad \chi_\lambda(g_{z^*}) = \chi_\lambda(g_z^{-1}) = \overline{\chi_\lambda(g_z)}. \quad (4.26)$$

979 We continue to index the output by z and record the inverse through the conjugate third
 980 character. The overline denotes the full reduced space, including every orbit-type stratum.
 981 The quotient by $G/Z(G)$ removes the unavoidable ineffective centre. At a singular value
 982 the quotient is a compact stratified symplectic space, by the singular reduction theorem of
 983 [44] and its group-valued form in Section 2.7 of [45].

984 Put

$$M_{x,y,z} = \mathcal{O}_x \times \mathcal{O}_y \times \mathcal{O}(g_z^{-1}). \quad (4.27)$$

985 This is a compact connected quasi-Hamiltonian G -space. Its principal stabilizer is $Z(G)$.
 986 Indeed, the stabilizer of a triple is the intersection of its three centralizers, and a generic
 987 pair of regular elements has maximal tori meeting precisely in $Z(G)$, as in Lemma 4.1.
 988 Since the centre fixes every triple, no smaller stabilizer can occur. More explicitly, if the
 989 first two elements are $ug_x u^{-1}$ and $vg_y v^{-1}$ and $q = u^{-1}v$, then

$$C_G(ug_x u^{-1}) \cap C_G(vg_y v^{-1}) = u(T \cap qTq^{-1})u^{-1}.$$

990 The connected-fibre theorem for simply connected G [43, Theorem 7.2] shows that
 991 $\Phi^{-1}(e)$ is connected whenever it is nonempty. At a regular value, the moment-map dif-
 992 ferential has full image, so the infinitesimal stabilizer at each point of the fibre vanishes;
 993 the action is locally free modulo the centre. If the fibre also meets the principal orbit-type
 994 stratum of $M_{x,y,z}$, the stabilizer and connectedness hypotheses in the group-valued volume
 995 formula [45, Theorem 4.1] hold. For the effective group the principal stabilizer is trivial,
 996 and

$$\text{Vol}(G/Z(G)) = \frac{\text{Vol}(G)}{|Z(G)|}. \quad (4.28)$$

997 This is the source of the centre factor in the volume formula below. After division by
 998 $G/Z(G)$, a principal regular reduction is a compact effective symplectic orbifold.

999 Write $\Phi_{x,y,z} : M_{x,y,z} \rightarrow G$ for the fused moment map. Define

$$\mathcal{R}_{\text{pr}} := \left\{ (x, y, z) \in Y^3 : \begin{array}{l} e \text{ is a regular value of } \Phi_{x,y,z}, \\ \Phi_{x,y,z}^{-1}(e) \neq \emptyset, \quad \Phi_{x,y,z}^{-1}(e) \cap (M_{x,y,z})_{\text{pr}} \neq \emptyset \end{array} \right\}. \quad (4.29)$$

1000 **Definition 4.3** (Principal regular-value chamber). A principal regular-value chamber is a
 1001 connected component S of $\mathcal{R}_{\text{pr}}$. On such a component the reduced cohomology class varies
 1002 affinely with (x, y, z) and the reduced volume is a single polynomial, denoted by V_S .

The polynomial variation is the group-valued Duistermaat–Heckman variation described in [46, Proposition 3.3 and Theorem A.1].

For (x, y, z) in a principal regular-value chamber write

$$\mathcal{P}_{x,y,z}^{\text{BF},\text{pr}} := (\overline{\mathcal{P}}_{x,y,z}^{\text{BF}})_{\text{pr}} \quad (4.30)$$

for its open principal locus. With the geometric inner product of Section 2 and $g_x = \text{Exp}_T(x) = \exp_G(jx)$ (written as $\exp(2\pi i x)$ in a unitary realization), let $\omega_{x,y,z}$ be the reduced quasi-Hamiltonian form. At a principal regular value define

$$\begin{aligned} V_{\text{geom}}(x, y, z) &:= \int_{\overline{\mathcal{P}}_{x,y,z}^{\text{BF}}} \frac{\omega_{x,y,z}^{d_G}}{d_G!} \\ &= \int_{\mathcal{P}_{x,y,z}^{\text{BF},\text{pr}}} \frac{\omega_{x,y,z}^{d_G}}{d_G!}. \end{aligned} \quad (4.31)$$

The first integral in Eq. (4.31) means orbifold integration in local uniformizing charts. It agrees with the integral over the open principal locus because the other orbit types have positive codimension and therefore zero Liouville measure. The symplectic form determines this intrinsic volume; the choices of Haar probability measures enter only when it is compared with orbital convolution.

Let (x, y, z) be a principal critical parameter in the interior of the moment-map image, and put

$$F_{\text{pr}} := \Phi^{-1}(e) \cap (M_{x,y,z})_{\text{pr}}.$$

Every $m \in F_{\text{pr}}$ has $G_m = Z(G)$ and hence $\mathfrak{g}_m = 0$. After right trivialization of TG , [43, Proposition 4.1(3)] gives

$$\text{im}(d\Phi_m) = \mathfrak{g}_m^\perp = \mathfrak{g}.$$

Thus F_{pr} consists of regular points of Φ , and its quotient by $G/Z(G)$ is a symplectic manifold of real dimension

$$3(\dim G - \ell) - 2 \dim G = 2(|\Delta_+| - \ell) = 2d_G.$$

The fibre $\Phi^{-1}(e)$ is connected by [43, Theorem 7.2]. Local linearization and singular reduction then show that the image of F_{pr} is the unique connected open dense symplectic stratum of $\overline{\mathcal{P}}_{x,y,z}^{\text{BF}}$; see [44, Theorem 5.9] and [45, Section 2.7]. Denote it by $\mathcal{P}_{x,y,z}^{\text{BF},\text{top}}$ and set

$$V_{\text{geom}}(x, y, z) := \int_{\mathcal{P}_{x,y,z}^{\text{BF},\text{top}}} \frac{\omega_{x,y,z}^{d_G}}{d_G!}. \quad (4.32)$$

Under the holonomy description of the three-holed-sphere moduli space, [46, Proposition 3.3 and Appendix A.2] identifies this integral with the continuous piecewise-polynomial extension from neighboring principal regular values. This defines V_{geom} at principal critical parameters. The measure formulas at the remaining singular parameters use the section-wise kernel V_{BF} .

For each principal regular-value chamber S , Theorem 4.1 and Proposition 4.6 of [45] give

$$\begin{aligned} V_{\text{BF}}(x, y; z) &= V_S(x, y; z) \quad \text{for } dz\text{-almost every } z \in S_{x,y}, \\ V_S &= V_{\text{geom}} \quad \text{pointwise on } S, \end{aligned} \quad (4.33)$$

where $S_{x,y} = \{z : (x, y, z) \in S\}$. Thus the general-group density identification is sectionwise almost everywhere, while its geometric representative is pointwise on a principal regular-value chamber. The polynomiality is joint in (x, y, z) ; inversion of the third holonomy is the linear change $z \mapsto -w_0 z$.

The fixed Borel representative gives, for every fixed (x, y) ,

$$\nu_{x,y}(dz) = \frac{\Pi_\rho}{\text{covol}(Q)} \frac{|\delta_G(z)|}{|\delta_G(x)\delta_G(y)|} V_{\text{BF}}(x, y; z) dz. \quad (4.34)$$

Gauge fixing the first two boundary holonomies identifies the underlying principal relative-position fibre with the principal reduced space. The conditional measure $p_{x,y}^z$ has total mass one; the Liouville measure in Eq. (4.31) has total mass $V_{\text{geom}}(x, y; z)$. The orbital density determines this second normalization.

4.2.3 The character formula and square integrability

Haar measures on G and T , the orbital measures ν_x , and their class-space convolutions $\nu_{x,y}$ have total mass one. The measure dz is geometric Lebesgue measure on the alcove: a fundamental cell of Q has volume $\text{covol}(Q)$, and the torus period lattice is $Q^\vee$. We retain these conventions in the character formula, so that all conversion factors are explicit.

With the notation of Eq. (4.20), let $\Phi_G(x, y; \cdot) \in L^2(G)^G$ be the class defined by

$$\Phi_G(x, y; g) = \sum_{\lambda \in P_+} \frac{\chi_\lambda(g_x) \chi_\lambda(g_y) \overline{\chi_\lambda(g)}}{\dim V_\lambda} \quad \text{in } L^2(G). \quad (4.35)$$

Appendix B.3 constructs from these partial sums a jointly measurable central representative whose section has this L^2 class for every fixed $(x, y) \in Y^2$. We use that representative throughout and write $\Phi_G(x, y; z)$ for its value at g_z . The factor $1/\dim V_\lambda$ in this series follows directly from the probability normalization of the orbital measures. For an irreducible unitary representation π_λ , Schur's lemma and the trace give

$$\int_G \pi_\lambda(g) d\nu_x(g) = \frac{\chi_\lambda(g_x)}{\dim V_\lambda} \text{id}_{V_\lambda}. \quad (4.36)$$

Convolution multiplies the two scalar Fourier transforms. Central Peter–Weyl inversion contributes one factor $\dim V_\lambda$, leaving the coefficient

$$\frac{\chi_\lambda(g_x) \chi_\lambda(g_y)}{\dim V_\lambda}$$

in Eq. (4.35). For fixed regular inputs x, y , this L^2 class is the central density of orbital convolution. Its Fourier coefficients and its identification with the group-valued Duistermaat–Heckman convolution measure are given in Section 4.2, Eqs. (29)–(30), and Proposition 4.6

of [45]. Consequently, evaluation at g_z gives an almost-everywhere defined output density. The conjugate on the third character follows from using the inverse third holonomy in (4.23). Polynomial representatives will be used only after a regular-value chamber has been chosen.

Proposition 4.4 (Orbital density and pair-of-pants volume). *For every fixed $x, y \in Y$, the Borel representative fixed in Eq. (4.22) satisfies, for dz -almost every $z \in Y$,*

$$V_{\text{BF}}(x, y; z) = \frac{\text{covol}(Q)}{\text{covol}(Q^\vee)\Pi_\rho} |\delta_G(x)\delta_G(y)\delta_G(z)| \Phi_G(x, y; z). \quad (4.37)$$

The identity is sectionwise: the exceptional z -null set may depend on the fixed pair (x, y) . On a principal regular-value chamber the same expression agrees with V_{geom} for dz -almost every output by Eq. (4.33); the pointwise polynomial identification is Proposition B.10. For every fixed $x, y \in Y$, one also has

$$d\nu_{x,y}(z) = \frac{\Pi_\rho}{\text{covol}(Q)} \frac{|\delta_G(z)|}{|\delta_G(x)\delta_G(y)|} V_{\text{BF}}(x, y; z) dz. \quad (4.38)$$

Proof. For each fixed x, y , Weyl integration applied to the central L^2 density of orbital convolution gives

$$h(x, y, z) = \frac{|\delta_G(z)|^2}{\text{covol}(Q^\vee)} \Phi_G(x, y; z) \quad \text{for } dz\text{-almost every } z, \quad (4.39)$$

and substitution into Eq. (4.22) gives Eq. (4.37). The spectral-density formula is the rearrangement Eq. (4.34).

For the geometric identification, the group-valued Duistermaat–Heckman theorem relates this orbital convolution to the reduced Liouville volume [45, Theorem 4.1, Eq. (30), and Proposition 4.6]. The conversion of the volume formula in [46, Theorem 5.2] to the geometric inner product of Section 2 is carried out in Proposition B.2. It yields Eq. (4.33) with the prefactor displayed above. $\square$

At an interior critical value for which Eq. (4.32) applies, V_{geom} denotes the continuous principal-stratum volume. At points on the boundary of $\text{ess supp}_z V_{\text{BF}}(x, y; \cdot)$, and at singular parameters outside that hypothesis, only the fixed-input density class enters the measure identity.

For every fixed regular input pair, the square of this density is integrable on the whole alcove. Weyl integration and Eq. (4.37) give

$$\int_Y |V_{\text{BF}}(x, y; z)|^2 dz = \left(\frac{\text{covol}(Q)}{\text{covol}(Q^\vee)\Pi_\rho} \right)^2 |\delta_G(x)\delta_G(y)|^2 \text{covol}(Q^\vee) \|\Phi_G(x, y; \cdot)\|_{L^2(G)}^2 < \infty. \quad (4.40)$$

The L^2 convergence on the right is proved in Appendix B.3. Hence $V_{\text{BF}}(x, y; z)^2 dz$ is a finite Radon measure on Y and is independent of how the density is changed on its null set. In coefficient limits we use the appropriate chamber polynomial pointwise; in measure limits we use the square of the almost-everywhere density class.

4.3 One-chiral fusion-coefficient asymptotics

4.3.1 Fusion-hypergroup convergence

Let $d_a^{(k)} = S_{0a}^{(k)} / S_{00}^{(k)}$ and let $N_{ab}^{(k)c}$ denote the level- k WZW fusion coefficient. The normalized one-chiral fusion distribution

$$\sigma_{a,b}^{(k)} := \sum_{c \in P_k^+} N_{ab}^{(k)c} \frac{d_c^{(k)}}{d_a^{(k)} d_b^{(k)}} \delta_{z_c} \quad (4.41)$$

is a probability measure. Indeed, quantum dimension is a ring character of the fusion ring, so $\sum_c N_{ab}^{(k)c} d_c^{(k)} = d_a^{(k)} d_b^{(k)}$. This is the probability normalization of the fusion hypergroup in [14, Eqs. (5)–(7)].

Defossez proves the corresponding fusion-hypergroup convergence in [14, Proposition 7.1 and Corollary 7.2]. Because the present paper uses the shifted points $z_a = (a + \rho)/r_k$, normalized orbital probability measures, and a fixed convention for the third character, Appendix B.3 gives a character-moment proof in these conventions. The cited result is formulated with unshifted scaled weights. Integrability gives $\|a\| = O(k)$, and

$$\frac{a}{k} - z_a = \frac{h^\vee a - k\rho}{k(k + h^\vee)} = O(k^{-1}) \quad (4.42)$$

uniformly over $a \in P_k^+$. Thus $a_k/k \rightarrow x$ if and only if $z_{a_k} \rightarrow x$, and similarly for the second input.

Proposition 4.5 (Defossez fusion-hypergroup limit). *If $z_{a_k} \rightarrow x$ and $z_{b_k} \rightarrow y$ with $x, y \in Y$, then*

$$\sigma_{a_k, b_k}^{(k)} \implies \nu_{x,y} \quad \text{on } \bar{Y}. \quad (4.43)$$

The proof in Appendix B.3 begins with the exact finite-level character identity (B.34). Stone–Weierstrass then gives the stated weak convergence and fixes the normalization used below.

The weak limit identifies averages of the coefficients against continuous test functions. To control an individual coefficient at a moving lattice point, we use the BZ and Kac–Walton formulas.

4.3.2 General G on open affine–BZ chambers

The local argument has two inputs. Tensor multiplicities are described by the Berenstein–Zelevinsky lattice-point model, in the precise form of [15, Theorem 2.2]. Fusion then introduces the affine Weyl images in the Kac–Walton formula [24, Exercise 13.35]; a proof of that formula is given in [33]. Nontrivial affine translations may contribute; Lemma B.8 shows that their number is finite near every fixed compact set.

The BZ decomposition is rational and polyhedral. We keep it separate from the regular-value decomposition of the group-valued Duistermaat–Heckman density, whose critical set is not assumed to be rational.

Recall the compatible lattice $\mathcal{L}_G$ from Eq. (2.60). For an affine Weyl element $\widehat{w} = (\bar{w}, \gamma^\vee) \in W \ltimes Q^\vee$, write

$$\Psi_{\widehat{w}}(x, y, z) = (x, y, \bar{w}z + \gamma^\vee). \quad (4.44)$$

Let $\mathcal{A}_G$ be the finite affine Weyl set of Lemma B.8. Fix a finite family $\mathcal{H}_{\text{BZ}}$ of rational hyperplanes whose union contains the boundary of the real tensor cone, the codimension-one faces of the relative BZ complex, and the faces on which strict feasibility fails. Let $\mathcal{H}_{\text{alc}}$ be the supporting hyperplanes of the closed alcove, let $\pi_j : P_{\mathbb{R}}^3 \rightarrow P_{\mathbb{R}}$ be the three coordinate projections, and put

$$\mathcal{H}_{\text{alc}}^{(3)} := \{\pi_j^{-1}(H) : H \in \mathcal{H}_{\text{alc}}, 1 \leq j \leq 3\}. \quad (4.45)$$

With $\Psi_{\widehat{w}}$ as in Eq. (4.44), set

$$\mathcal{H}_{\text{aff}} := \{\Psi_{\widehat{w}}^{-1}(H) : \widehat{w} \in \mathcal{A}_G, H \in \mathcal{H}_{\text{BZ}}\} \cup \mathcal{H}_{\text{alc}}^{(3)}, \quad \mathcal{C}_{\text{aff}} := \bar{Y}^3 \cap \bigcup_{H \in \mathcal{H}_{\text{aff}}} H. \quad (4.46)$$

The family $\mathcal{H}_{\text{aff}}$ is finite by Eq. (B.120). Its arrangement induces a finite rational polyhedral complex on $\bar{Y}^3$, with wall set $\mathcal{C}_{\text{aff}}$.

Lemma 4.6 (Cells of the affine-BZ refinement). *Let $\mathcal{W}_{\text{BZ}}$ be the union of the boundary of the real tensor cone, the boundaries of the relative BZ chambers, and the proper faces on which strict feasibility fails. For every $\widehat{w} \in \mathcal{A}_G$,*

$$\bar{Y}^3 \cap \Psi_{\widehat{w}}^{-1}(\mathcal{W}_{\text{BZ}}) \subseteq \mathcal{C}_{\text{aff}}, \quad \partial \bar{Y}^3 \subseteq \mathcal{C}_{\text{aff}}. \quad (4.47)$$

Consequently, if C is a connected component of $Y^3 \setminus \mathcal{C}_{\text{aff}}$, then, for every $\widehat{w} \in \mathcal{A}_G$, its image is either disjoint from the closed real tensor cone or contained in one full-dimensional strictly feasible relative BZ cell.

Proof. Every set in the first inclusion lies in the inverse image of a hyperplane from $\mathcal{H}_{\text{BZ}}$, while $\mathcal{H}_{\text{alc}}^{(3)}$ contains every face of $\bar{Y}^3$. This proves Eq. (4.47). The affine image of the connected set C cannot pass into or out of the tensor cone without meeting its boundary. Within the cone it cannot pass between BZ chambers, or meet a face where strict feasibility fails, without meeting $\mathcal{W}_{\text{BZ}}$. Since C avoids every inverse image in Eq. (4.47), the stated alternative follows. $\square$

A *polyhedral affine-BZ cell* is a connected component of $Y^3 \setminus \mathcal{C}_{\text{aff}}$. By Lemma 4.6, the signed sum of the leading BZ volume polynomials has fixed summands on each such cell C . It is therefore a polynomial there; denote it by $V_{\text{aff}, C}$.

The reduced word, relative BZ presentation, intermediate cellwise function V_{add} , supporting hyperplanes, and common refinement are auxiliary choices. Given two such constructions, intersect their open cell decompositions. On each common open cell, Proposition B.10 identifies both leading polynomials, for every fixed input pair, with the same orbital density for dz -almost every output. Since the restrictions are polynomials, they coincide on the common cell.

The exponent d_G first appears as the dimension of a generic reduced BZ fibre. If the parameters remain in a full-dimensional strictly feasible relative BZ cell, the tensor multiplicity is the number of lattice points in that fibre. Lemma B.5 and Theorem B.6 give, uniformly after a bounded displacement of the labels and for integers $r \rightarrow \infty$,

$$m_{a_r b_r}^{c_r} = V_C(a_r, b_r; c_r) + O_{K,B}(r^{d_G-1}), \quad (a_r, b_r, c_r) = ru_r + e_r, \quad (4.48)$$

uniformly when u_r ranges in a compact subset K of that cell and the compatible displacement e_r has norm at most B . The leading homogeneous term is independent of the quasipolynomial residue class. This explains why a fixed compatible shift can change the lower-order terms without changing the leading volume.

Kac–Walton writes the fusion coefficient as an alternating sum of these tensor multiplicities. Only finitely many summands occur, and their cells and signs are constant on a polyhedral affine–BZ cell. Proposition B.9 therefore gives the leading polynomial $V_{\text{aff},C}$. Proposition B.10 then compares its Riemann sums with the independent fusion-hypergroup limit. For every fixed input pair it identifies this polynomial with V_{BF} for dz -almost every output in $C_{x,y} := \{z : (x, y, z) \in C\}$. For every principal regular-value chamber S , joint polynomiality gives

$$V_{\text{aff},C} = V_{\text{geom}} \quad \text{on } S \cap C. \quad (4.49)$$

Theorem 4.7 (Main Theorem B: general- G chamberwise coefficient limit). *Let C be a polyhedral affine–BZ cell, hence an open full-dimensional component of $Y^3 \setminus \mathcal{C}_{\text{aff}}$, and let $K \Subset C$. Let $E \subset P^3$ be finite and suppose that*

$$\alpha + \beta - \varsigma \in Q \quad \text{for every } (\alpha, \beta, \varsigma) \in E. \quad (4.50)$$

As $k \rightarrow \infty$ through positive integers, uniformly for

$$(a, b, c) \in \mathcal{L}_G \cap (P_k^+)^3, \quad (z_a, z_b, z_c) \in K, \quad (4.51)$$

and for all $(\alpha, \beta, \varsigma) \in E$ for which $a + \alpha, b + \beta$, and $c + \varsigma$ belong to P_k^+ , one has

$$r_k^{-d_G} N_{a+\alpha, b+\beta}^{(k)c+\varsigma} = V_{\text{aff},C}(z_a, z_b; z_c) + O_{K,E}(r_k^{-1}). \quad (4.52)$$

The implied constant is independent of k , of the labels in Eq. (4.51), and of the chosen element of E .

In particular, suppose that

$$(a_k, b_k, c_k) \in \mathcal{L}_G \cap (P_k^+)^3, \quad (z_{a_k}, z_{b_k}, z_{c_k}) \longrightarrow (x, y, z) \in C. \quad (4.53)$$

Then

$$r_k^{-d_G} N_{a_k b_k}^{(k)c_k} \longrightarrow V_{\text{aff},C}(x, y; z). \quad (4.54)$$

For every fixed (x, y) , the functions $V_{\text{aff},C}(x, y; \cdot)$ and $V_{\text{BF}}(x, y; \cdot)$ agree dz -almost everywhere on $C_{x,y} := \{z : (x, y, z) \in C\}$. For every principal regular-value chamber S , one has $V_{\text{aff},C} = V_{\text{geom}}$ pointwise on $S \cap C$.

Proof of Main Theorem B. Proposition B.9 proves the uniform estimate in Eq. (4.52); its lattice check includes every shift satisfying Eq. (4.50). A sequence converging to a point of C eventually lies in a compact subset of C . Applying the estimate with the zero shift and using continuity of $V_{\text{aff},C}$ gives Eq. (4.54). Proposition B.10 proves the sectionwise density identity. Fubini's theorem gives equality of the two volume polynomials almost everywhere on $S \cap C$, and polynomiality extends this equality to every point. $\square$

4.3.3 Type A across internal walls

For $G = SU(m)$ with $m \geq 3$, the existence and identification of the pointwise scaled quantum Littlewood–Richardson limit were proved by François and Tarrago [22, Theorem 3.8]. The form needed here also allows finitely many fixed compatible shifts, uses the centered WZW coordinates $z_a = (a + \rho)/r_k$, and is uniform when the unshifted points range over $K \Subset Y^3$. Appendix B.6 proves this strengthened formulation from the centered Kac–Peterson determinant. Its summable majorant is supplied by [22, Lemmas 3.14–3.15]. Korff's inverse lifting formula [47, Corollary 4.4] is used to compare the WZW and quantum-cohomology conventions. The positive toric-hive representation in Corollary 4.9 follows by comparing Proposition 3.1 and Theorem 2.5 of [22]. The quotient conventions are those of [48, Theorem 6.20 and Corollary 8.2].

The determinant series in Appendix B.6 converges absolutely and uniformly on Y^3 and hence defines a continuous function there. For every fixed input pair its output section represents the same dz -density class as V_{BF} , while on each principal regular-value chamber it equals V_{geom} pointwise. We denote this continuous representative by $V_{\text{BF}}^{A,\text{cont}}$. It supplies the common value on internal affine, polyhedral, and multiplicative-Horn walls, including boundary points of the multiplicative-Horn support that remain in Y^3 .

Corollary 4.8 (All-regular type-A fusion-coefficient limit). *Let $G = SU(m)$ with $m \geq 3$ and $d_G = (m-1)(m-2)/2$. Let $K \Subset Y^3$, and let $\mathcal{E} \subset P^3$ be a finite nonempty set of shifts such that $u + v - w \in Q$ for every $(u, v, w) \in \mathcal{E}$. Then*

$$\lim_{k \rightarrow \infty} \sup_{\substack{(u,v,w) \in \mathcal{E}, \ (a,b,c) \in \mathcal{L}_G \cap (P_k^+)^3 \\ (z_a, z_b, z_c) \in K, \ a+u, b+v, c+w \in P_k^+}} \left| r_k^{-d_G} N_{a+u, b+v}^{(k)c+w} - V_{\text{BF}}^{A,\text{cont}}(z_a, z_b; z_c) \right| = 0. \quad (4.55)$$

The supremum is understood to be zero if its indexing set is empty. In particular, every compatible sequence converging to a point of Y^3 satisfies the corresponding pointwise limit; internal affine, polyhedral, and multiplicative-Horn walls are included. For zero shifts, the pointwise identification agrees with [22, Theorem 3.8]. Appendix B.6 proves the compact-uniform statement and its simultaneous stability under the fixed compatible shifts.

Corollary 4.9 (François–Tarrago toric-hive representation). *Let $G = SU(m)$ with $m \geq 3$, let $x, y, z \in Y$, and let $\alpha, \beta, \gamma \in [0, 1]^m$ be the nonincreasing eigenangle lists of g_x, g_y, g_z . Put*

$$|\theta|_1 := \sum_{i=1}^m \theta_i, \quad d = |\alpha|_1 + |\beta|_1 - |\gamma|_1. \quad (4.56)$$

1207 The determinant-one condition makes d an integer. If $d \in \{0, \dots, m\}$, then

$$V_{\text{BF}}^{A, \text{cont}}(x, y; z) = \sum_{g \in \mathfrak{G}_{d, m}} \text{Vol}_g(\mathcal{P}_{\alpha, \beta, \gamma}^g). \quad (4.57)$$

1208 Here $\mathfrak{G}_{d, m}$ is the full set of maps $g : R_{d, m} \rightarrow \mathbb{Z}/3\mathbb{Z}$ satisfying the boundary and lozenge
 1209 conditions of [22, Definition 2.1]; no quotient of this set of labelings is taken. The polytope
 1210 $\mathcal{P}_{\alpha, \beta, \gamma}^g$ is defined in [22, Definitions 2.3–2.4]; the coordinate-volume normalization Vol_g
 1211 is the one used in [22, Theorem 2.5]. In particular, Vol_g is Lebesgue measure in the
 1212 coordinates indexed by $\text{Supp}(g) \setminus \partial R_{d, m}$. Empty polytopes contribute zero.

1213 For general G , the fusion-coefficient limit is pointwise on every open affine-BZ cell, and
 1214 its geometric interpretation is pointwise on principal regular-value chambers. For $SU(m)$,
 1215 $m \geq 3$, the continuous representative $V_{\text{BF}}^{A, \text{cont}}$ gives the limit throughout Y^3 , including the
 1216 internal walls.

1217 The rank-one case is different because $d_G = 0$. If the level- k $SU(2)$ weights are
 1218 identified with the integers $0, \dots, k$, then

$$N_{ab}^{(k)c} = 1 \iff a + b - c \in 2\mathbb{Z}, \quad |a - b| \leq c \leq \min\{a + b, 2k - a - b\}, \quad (4.58)$$

1219 and the coefficient is zero otherwise. A bounded compatible shift can therefore change the
 1220 coefficient at an endpoint of the fusion interval without changing the limiting point. The
 1221 continuum fusion interval, the endpoint measure, and an explicit shifted example are given
 1222 in Appendix B.8.

1223 5 Doubled fusion coefficients and fixed electric operators

1224 For doubled labels the fusion multiplicity is the product of two chiral coefficients. The
 1225 two chiral triples approach the same flux triple from opposite $O(r_k^{-1})$ directions, while the
 1226 output labels lie on a fixed-charge root-lattice mesh. These features determine the local
 1227 error and the normalization of the limiting measure.

1228 5.1 Charge-resolved doubled multiplicity measures

1229 For actual doubled labels in the charge-oriented convention, write

$$A = (a^+, a^-), \quad B = (b^+, b^-), \quad C = (c^+, c^-), \quad (5.1)$$

1230 where all six entries belong to P_k^+ . Thus A denotes $a^+ \boxtimes \overline{a^-}$, and similarly for B, C . Set

$$x_A = \frac{a^+ + a^- + 2\rho}{2r_k}, \quad \eta_A = a^+ - a^-, \quad (5.2)$$

1231 and use the same notation for B and C . The intrinsic doubled fusion multiplicity is the
 1232 integer

$$\mathcal{N}_{AB}^{(k)C} := N_{a^+ b^+}^{(k)c^+} N_{a^- b^-}^{(k)c^-}. \quad (5.3)$$

1233 We call a triple (A, B, C) *chirally compatible* when

$$a^+ + b^+ - c^+ \in Q, \quad a^- + b^- - c^- \in Q. \quad (5.4)$$

1234 Noncompatible triples have zero multiplicity by the central-character rule.

1235 Let $U \Subset Y$ be relatively compact and open, and enumerate the family in Eq. (4.46) as

$$\mathcal{H}_{\text{aff}} = \{H_1, \dots, H_M\}, \quad H_j = \{(x, y, z) : A_j x + B_j y + C_j z = d_j\}. \quad (5.5)$$

1236 For $(x, y) \in Y^2$, define the fixed-input wall slice

$$\mathcal{C}_{\text{aff}}(x, y; \overline{U}) := \{z \in \overline{U} : (x, y, z) \in \mathcal{C}_{\text{aff}}\}. \quad (5.6)$$

1237 We say that (x, y) is *wall-transverse over $\overline{U}$* when this set has zero ℓ -dimensional Lebesgue
1238 measure, where $\ell = \text{rank } G$.

1239 **Proposition 5.1** (Generic wall transversality). *Every pair outside*

$$E_{\text{aff}} := \bigcup_{j: C_j=0} \{(x, y) \in Y^2 : A_j x + B_j y = d_j\} \quad (5.7)$$

1240 *is wall-transverse over every relatively compact open set $U \Subset Y$. The set E_{aff} is a finite
1241 union of proper rational affine hyperplanes and therefore has measure zero.*

1242 *Proof.* If $C_j \neq 0$, the fixed-input z -slice of H_j is empty or a proper affine hyperplane in
1243 $P_{\mathbb{R}}$. If $C_j = 0$, the slice is empty unless $A_j x + B_j y = d_j$. Thus, for $(x, y) \notin E_{\text{aff}}$, the set
1244 $\mathcal{C}_{\text{aff}}(x, y; \overline{U})$ is a finite union of lower-dimensional affine polyhedra and is Lebesgue null. $\square$

1245 **Lemma 5.2** (Uniform lattice count in wall tubes). *Let $L \subset \mathbb{R}^\ell$ be a full lattice, let B' be
1246 compact, and let $B \subset \text{int } B'$ be compact. Suppose that $\mathcal{W}$ is a finite union of polyhedra of
1247 dimension at most $\ell - 1$ meeting B . For any subset $A \subset \mathbb{R}^\ell$ and $\varepsilon > 0$, write*

$$A_\varepsilon := \{u \in \mathbb{R}^\ell : \text{dist}(u, A) < \varepsilon\}.$$

1248 *Put*

$$\delta = \text{dist}(B, \mathbb{R}^\ell \setminus \text{int } B') > 0.$$

1249 *Then, uniformly in the translate $v + r^{-1}L$,*

$$\#((v + r^{-1}L) \cap B \cap \mathcal{W}_\varepsilon) \leq \text{covol}(L)^{-1} r^\ell \text{vol}_\ell(B' \cap \mathcal{W}_{2\varepsilon}). \quad (5.8)$$

1250 *The estimate holds whenever $D/r \leq \min\{\varepsilon, \delta\}$, where D is the radius bound for the funda-
1251 mental cell chosen in the proof.*

1252 *Proof.* Choose a bounded half-open fundamental cell F of L with $0 \in \overline{F}$, put $D =$
1253 $\sup_{u \in F} \|u\|$, and attach $v + r^{-1}(q + F)$ to every point $v + r^{-1}q$ counted on the left. These
1254 cells are disjoint and each has volume $\text{covol}(L)r^{-\ell}$. Their union lies in $B' \cap \mathcal{W}_{\varepsilon+D/r}$ when
1255 $D/r \leq \delta$; the second inequality is what keeps every attached cell inside B' . When also
1256 $D/r \leq \varepsilon$, this set is contained in $B' \cap \mathcal{W}_{2\varepsilon}$. Division by the cell volume proves Eq. (5.8). $\square$

Theorem 5.3 (Main Theorem C: charge-resolved doubled multiplicity-measure limit).

Recall that $d_G = |\Delta_+| - \text{rank } G$.

(i) Local doubled coefficient. Let $\mathcal{C}$ be a polyhedral affine-BZ cell, let $K \subseteq \mathcal{C}$, and let $E \subset P^3$ be finite. Uniformly over chirally compatible actual doubled labels satisfying

$$(x_A, x_B, x_C) \in K, \quad (\eta_A, \eta_B, \eta_C) \in E, \quad (5.9)$$

one has

$$r_k^{-2d_G} \mathcal{N}_{AB}^{(k)C} = V_{\text{aff}, \mathcal{C}}(x_A, x_B; x_C)^2 + O_{K,E}(r_k^{-1}). \quad (5.10)$$

For every fixed input pair, $V_{\text{aff}, \mathcal{C}}$ represents V_{BF} for dz -almost every output in $\mathcal{C}_{x,y} := \{z : (x, y, z) \in \mathcal{C}\}$. For every principal regular-value chamber S , it equals V_{geom} pointwise on $S \cap \mathcal{C}$.

(ii) Charge-resolved measure. Let $U \subseteq Y$ be relatively compact and open. Fix $\eta, \omega \in P$, and let A_k, B_k be actual doubled labels with

$$\eta_{A_k} = \eta, \quad \eta_{B_k} = \omega, \quad x_{A_k} \longrightarrow x, \quad x_{B_k} \longrightarrow y, \quad (5.11)$$

where $x, y \in Y$. Define the positive raw-multiplicity Radon measure on the locally compact space $Y \times P$, with P discrete,

$$\mathfrak{N}_{A_k, B_k}^{(k)} := \text{covol}(Q) r_k^{-(2d_G + \text{rank } G)} \sum_C \mathcal{N}_{A_k B_k}^{(k)C} \delta_{(x_C, \eta_C)}. \quad (5.12)$$

Here the sum runs over all actual doubled simple labels C at level k ; terms outside the two chiral fusion supports are simply zero. Because P has the discrete topology, this finite sum is a Radon measure, and every test function in $C_c(U \times P)$ meets finitely many charge sectors. The limiting measure is locally finite and is supported on $\eta + \omega + Q$ in the charge variable. All Lebesgue measures and covolumes below use the geometric inner product fixed in Section 2. If (x, y) is wall-transverse over $\bar{U}$, then

$$\mathfrak{N}_{A_k, B_k}^{(k)} \xrightarrow{v} V_{\text{BF}}(x, y; z)^2 dz \otimes \sum_{\tau \in \eta + \omega + Q} \delta_\tau \quad \text{on } U \times P. \quad (5.13)$$

Here $V_{\text{BF}}(x, y; \cdot)$ is the fixed-input section of the Borel kernel fixed in Eq. (4.22). Equivalently, for every $f \in C_c(U \times P)$,

$$\begin{aligned} \lim_{k \rightarrow \infty} \text{covol}(Q) r_k^{-(2d_G + \text{rank } G)} \sum_C \mathcal{N}_{A_k B_k}^{(k)C} f(x_C, \eta_C) \\ = \sum_{\tau \in \eta + \omega + Q} \int_U f(z, \tau) V_{\text{BF}}(x, y; z)^2 dz. \end{aligned} \quad (5.14)$$

For $G = SU(m)$, $m \geq 3$, the wall-transversality assumption can be removed: the conclusion holds on every relatively compact regular output window. The limiting Radon measure remains $V_{\text{BF}}(x, y; z)^2 dz$; the continuous function $V_{\text{BF}}^{A, \text{cont}}$ supplies the pointwise wall values in the coefficient estimate and in the Riemann-sum proof.

Proof. We first prove the local estimate and identify the output mesh. To pass to measures, we apply that estimate away from the fixed-input wall slice and bound the remaining lattice points in a shrinking neighbourhood of the slice. The moving input fluxes need only converge; no rate of convergence is assumed.

Midpoint identities and charge conservation. The definitions give the exact identities in the real weight space

$$z_{a^\pm} = x_A \pm \frac{\eta_A}{2r_k}, \quad z_{b^\pm} = x_B \pm \frac{\eta_B}{2r_k}, \quad z_{c^\pm} = x_C \pm \frac{\eta_C}{2r_k}. \quad (5.15)$$

These identities hold in $P_{\mathbb{R}}$. Main Theorem B is applied to the two integral endpoint triples (a^+, b^+, c^+) and (a^-, b^-, c^-) ; the segment between them is used to keep both endpoints in one affine-BZ cell.

If (5.3) is nonzero, each chiral factor obeys the central-character rule (5.4). Subtracting the two congruences gives

$$\eta_C - \eta_A - \eta_B \in Q. \quad (5.16)$$

Thus only the charges $\tau \in \eta + \omega + Q$ can occur in part (ii).

Product of the chiral asymptotics. Put $X_0 = (x_A, x_B, x_C)$ and let $X_\pm$ be the two triples on the right-hand side of (5.15). If $X_0 \in K \subseteq \mathcal{C}$ and the charges range over the finite set E , then X_+ and X_- lie in a single compact subset of $\mathcal{C}$ for all sufficiently large k . Applying Theorem 4.7 separately to the two actual integral triples yields

$$r_k^{-d_G} N_{a^\pm b^\pm}^{(k)c^\pm} = V_{\text{aff}, \mathcal{C}}(X_\pm) + O_{K, E}(r_k^{-1}). \quad (5.17)$$

The polynomial $V_{\text{aff}, \mathcal{C}}$ has bounded first derivatives on a compact neighbourhood of K . The midpoint identities (5.15) therefore give

$$V_{\text{aff}, \mathcal{C}}(X_\pm) = V_{\text{aff}, \mathcal{C}}(X_0) + O_{K, E}(r_k^{-1}).$$

Since $V_{\text{aff}, \mathcal{C}}$ is uniformly bounded on K , multiplying the two uniform chiral estimates proves Eq. (5.10). Proposition B.10 identifies the cell polynomial with V_{BF} almost everywhere on every fixed-input output section. On a principal regular-value chamber, the pointwise identity with V_{geom} is Eq. (4.49). This proves all assertions in part (i).

The fixed-charge root-lattice mesh. Fix an allowed output charge τ and write $c^+ = c^- + \tau$. Then

$$x_C = \frac{c^- + \rho + \tau/2}{r_k}. \quad (5.18)$$

The negative-chirality central rule restricts c^- to $a_k^- + b_k^- + Q$. Conversely, once $\tau - \eta - \omega \in Q$, that rule implies the positive-chirality rule as well. Put

$$v_{k, \tau} := \frac{a_k^- + b_k^- + \rho + \tau/2}{r_k}, \quad Y_{k, \tau} := \left\{ z \in P_{\mathbb{R}} : z + \frac{\tau}{2r_k} \in Y, \quad z - \frac{\tau}{2r_k} \in Y \right\}.$$

The exact set of chirally compatible, integrable candidate output fluxes of charge τ is

$$\Lambda_{k, \tau}^{A_k, B_k} = (v_{k, \tau} + r_k^{-1}Q) \cap Y_{k, \tau}. \quad (5.19)$$

Together with the translated-lattice condition, the two alcove conditions are equivalent to $c^\pm \in P_k^+$. For fixed τ , the relation $c^+ = c^- + \tau$ shows that c^- and the point x_C determine one another. Thus every point of $\Lambda_{k,\tau}^{A_k, B_k}$ represents exactly one candidate output label, with unit mesh multiplicity in the Riemann sum. If $B \Subset Y$ and τ is fixed, then $B \subset Y_{k,\tau}$ for all sufficiently large k . Therefore, on the compact windows used below,

$$\Lambda_{k,\tau}^{A_k, B_k} \cap B = (v_{k,\tau} + r_k^{-1}Q) \cap B. \quad (5.20)$$

Points outside the Horn support contribute zero. Simultaneously replacing c^- and c^+ by $c^- + q$ and $c^+ + q$, with $q \in Q$, changes their average by q/r_k . Hence the mesh cell volume is $\text{covol}(Q)r_k^{-\text{rank } G}$; the simultaneous shift of the two chiral weights removes the factor $1/2$ from their average.

Let $f \in C_c(U \times P)$. Since P is discrete, the set

$$T_{\text{adm}}(f) := \text{pr}_P(\text{supp } f) \cap (\eta + \omega + Q) \quad (5.21)$$

is finite. Every charge sector outside $\eta + \omega + Q$ vanishes exactly. In each remaining sector, the mesh (5.19) enumerates the chirally compatible integrable candidates, including those with zero coefficient. Thus the left-hand side of (5.14) becomes

$$\sum_{\tau \in T_{\text{adm}}(f)} \text{covol}(Q)r_k^{-\text{rank } G} \sum_{x_C \in \Lambda_{k,\tau}^{A_k, B_k}} f(x_C, \tau) r_k^{-2d_G} \mathcal{N}_{A_k B_k}^{(k)C}. \quad (5.22)$$

We shall use the translated-lattice Riemann sum in a form that is uniform in the translate. If $g \in C_c(P_{\mathbb{R}})$ and $v_k \in P_{\mathbb{R}}$ is arbitrary, then

$$\text{covol}(Q)r_k^{-\text{rank } G} \sum_{q \in Q} g(v_k + q/r_k) \longrightarrow \int_{P_{\mathbb{R}}} g(z) dz. \quad (5.23)$$

To see this, tile $P_{\mathbb{R}}$ by the cells $v_k + r_k^{-1}(q + F_Q)$, where F_Q is a bounded fundamental cell of Q . On every cell meeting a fixed compact neighbourhood of $\text{supp } g$, the difference between g at the lattice point and its cell average is bounded by the modulus of continuity of g at $O(r_k^{-1})$. The bound is independent of v_k , and summing the cell errors proves Eq. (5.23).

Localization near the wall slice. The flux projection $K_f := \text{pr}_Y(\text{supp } f)$ is a compact subset of U . Cover K_f by finitely many open sets $O_j \subset P_{\mathbb{R}}$ for which there are compact full-dimensional polytopes B_j, B'_j satisfying

$$\overline{O_j} \subset \text{int } B_j, \quad B_j \subset \text{int } B'_j, \quad B'_j \Subset U.$$

Choose a continuous partition of unity $\{\vartheta_j\}$ on a neighbourhood of K_f , subordinate to the sets O_j . On the support of f we then have

$$f(z, \tau) = \sum_j f(z, \tau) \vartheta_j(z).$$

The sum is finite, and the flux support of each summand lies in $\text{int } B_j$. It is therefore enough, by linearity, to prove the limit for one summand. We fix such a summand, continue to denote it by f , and write B, B' for the corresponding pair B_j, B'_j . Thus

$$\text{pr}_Y(\text{supp } f) \subset \text{int } B, \quad B \subset \text{int } B', \quad B' \Subset U.$$

1335 Put $\delta_B = \text{dist}(B, P_{\mathbb{R}} \setminus \text{int } B') > 0$. Choose a compact neighbourhood $K_B \Subset Y^3$ of $\{(x, y, z) :$
 1336 $z \in B'\}$ which, for all sufficiently large k , contains

$$\left(x_{A_k} + s \frac{\eta}{2r_k}, x_{B_k} + s \frac{\omega}{2r_k}, z + s \frac{\tau}{2r_k} \right) \quad (z \in B', -1 \leq s \leq 1, \tau \in T_{\text{adm}}(f)).$$

1337 This containment is uniform in $\tau \in T_{\text{adm}}(f)$ and $s \in [-1, 1]$. The wall complex is the
 1338 common refinement over the finite affine Weyl set $\mathcal{A}_G$, so one threshold in k works for every
 1339 affine image, admissible charge, and wall meeting K_B . For $H \in \mathcal{H}_{\text{aff}}$, put $W_H := \bar{Y}^3 \cap H$,
 1340 and define

$$\mathcal{W}_B := \{W_H : H \in \mathcal{H}_{\text{aff}}, W_H \cap K_B \neq \emptyset\}. \quad (5.24)$$

1341 The family is finite because $\mathcal{H}_{\text{aff}}$ is finite. Every wall met by a moving segment in K_B
 1342 belongs to $\mathcal{W}_B$. For $W \in \mathcal{W}_B$ set

$$\mathcal{W}_{x,y}(W) := \{z \in B' : (x, y, z) \in W\}, \quad \mathcal{W}_0 := \bigcup_{W \in \mathcal{W}_B} \mathcal{W}_{x,y}(W), \quad (5.25)$$

1343 and, for $-1 \leq s \leq 1$ and $\tau \in T_{\text{adm}}(f)$, set

$$\mathcal{W}_{k,s,\tau}(W) := \left\{ z \in B' : \left(x_{A_k} + s \frac{\eta}{2r_k}, x_{B_k} + s \frac{\omega}{2r_k}, z + s \frac{\tau}{2r_k} \right) \in W \right\}. \quad (5.26)$$

1344 Each $\mathcal{W}_{k,s,\tau}(W)$ is the corresponding polyhedral slice of W . For fixed k , W , and τ , the
 1345 union over $s \in [-1, 1]$ is compact: it is the projection to B' of the closed subset of $B' \times [-1, 1]$
 1346 defined by the displayed incidence condition. For every fixed $W \in \mathcal{W}_B$ and $\tau \in T_{\text{adm}}(f)$,
 1347 compactness and closedness give

$$\limsup_{k \rightarrow \infty} \bigcup_{-1 \leq s \leq 1} \mathcal{W}_{k,s,\tau}(W) \subseteq \mathcal{W}_{x,y}(W). \quad (5.27)$$

1348 Indeed, if z_k belongs to the union on the left, pass to subsequences with $z_k \rightarrow z \in B'$
 1349 and $s_k \rightarrow s \in [-1, 1]$. The corresponding triples converge to (x, y, z) , so closedness of W
 1350 gives $z \in \mathcal{W}_{x,y}(W)$. Since both $\mathcal{W}_B$ and $T_{\text{adm}}(f)$ are finite, the preceding compactness
 1351 turns this sequential limsup statement into eventual containment in every prescribed open
 1352 neighbourhood of $\mathcal{W}_0$. Hence, for every $\varepsilon > 0$ and all sufficiently large k ,

$$\bigcup_{\substack{W \in \mathcal{W}_B, \tau \in T_{\text{adm}}(f) \\ -1 \leq s \leq 1}} \mathcal{W}_{k,s,\tau}(W) \subseteq (\mathcal{W}_0)_{\varepsilon}. \quad (5.28)$$

1353 Wall transversality makes $\mathcal{W}_0$ Lebesgue null. Each of its finitely many constituent slices
 1354 is a polyhedron, so $\mathcal{W}_0$ is a finite union of polyhedra of dimension at most $\text{rank } G - 1$.
 1355 Indeed, a nonempty full-dimensional polyhedron contains a relatively open set and has
 1356 positive Lebesgue measure; since the finite union $\mathcal{W}_0$ is null, none of its constituent slices
 1357 can be full-dimensional.

1358 *Cellwise convergence away from the walls.* Fix $0 < \varepsilon < \delta_B/4$. Choose a continuous
 1359 function $\chi_{\varepsilon} : P_{\mathbb{R}} \rightarrow [0, 1]$ which vanishes on $(\mathcal{W}_0)_{\varepsilon}$ and equals one outside $(\mathcal{W}_0)_{2\varepsilon}$. We first
 1360 treat the test function $f_{\varepsilon}(z, \tau) = f(z, \tau)\chi_{\varepsilon}(z)$.

For all sufficiently large k , Eq. (5.28), with a smaller tube on its right-hand side, puts every moving wall inside $(\mathcal{W}_0)_{\varepsilon/2}$. Hence, for $z \in \text{supp } f_\varepsilon$, the segment

$$s \mapsto \left(x_{A_k} + s \frac{\eta}{2r_k}, x_{B_k} + s \frac{\omega}{2r_k}, z + s \frac{\tau}{2r_k} \right), \quad -1 \leq s \leq 1, \quad (5.29)$$

does not meet a proper affine-BZ wall. The midpoint and both chiral triples therefore belong to the same polyhedral cell.

The affine-BZ cell may vary with z . Because $(\mathcal{W}_0)_\varepsilon$ is open and f_ε vanishes there,

$$K_\varepsilon := \text{pr}_Y(\text{supp } f_\varepsilon) \subset B \setminus (\mathcal{W}_0)_\varepsilon.$$

This inclusion also covers points at distance exactly ε from $\mathcal{W}_0$. The compact set $\Gamma_\varepsilon := \{(x, y, z) : z \in K_\varepsilon\}$ is disjoint from the fixed finite wall arrangement. It can therefore be covered by finitely many sets $\{(x, y, z) : z \in O_j\}$, where each O_j is open in $P_\mathbb{R}$ and has compact closure in $\text{int } B \setminus (\mathcal{W}_0)_{\varepsilon/2}$. After shrinking these sets if necessary, there are compact sets

$$K_j \Subset K'_j \Subset \mathcal{C}_j,$$

with $\mathcal{C}_j$ a polyhedral affine-BZ cell, such that $\{(x, y, z) : z \in \overline{O_j}\} \subset K_j$. For all sufficiently large k , the entire segment (5.29) over $\overline{O_j}$ then lies in K'_j .

Choose continuous compactly supported functions $\psi_1, \dots, \psi_m$ with $\text{supp } \psi_j \subset O_j$ and $\sum_j \psi_j = 1$ on a neighbourhood of K_ε . Thus $f_\varepsilon = \sum_j f_\varepsilon \psi_j$. Every summand is a continuous compactly supported test function attached to a single cell. The partition identity gives the exact decomposition on cell overlaps.

Apply part (i) on the finitely many compact sets K'_j , with the finite charge set

$$E_f = \{(\eta, \omega, \tau) : \tau \in T_{\text{adm}}(f)\}.$$

This gives one threshold in k and one constant for the $O(r_k^{-1})$ remainder. A fixed compact output window contains $O(r_k^{\text{rank } G})$ mesh points, each with Riemann-sum weight $\text{covol}(Q)r_k^{-\text{rank } G}$, so the summed remainder is still $O(r_k^{-1})$. Uniform continuity of the finitely many cell polynomials also gives

$$V_{\text{aff}, \mathcal{C}_j}(x_{A_k}, x_{B_k}; z)^2 \longrightarrow V_{\text{aff}, \mathcal{C}_j}(x, y; z)^2$$

uniformly for $z \in \text{supp } \psi_j$. The translated Riemann-sum formula (5.23) therefore applies to each summand. Finally, Proposition B.10 identifies each limiting cell polynomial with the fixed-input V_{BF} almost everywhere on its output section; their squares define the same measure. Summing over the cells and charges yields

$$\begin{aligned} & \lim_{k \rightarrow \infty} \sum_{\tau \in T_{\text{adm}}(f)} \text{covol}(Q)r_k^{-\text{rank } G} \sum_{x_C \in \Lambda_{k, \tau}^{A_k, B_k}} f_\varepsilon(x_C, \tau) r_k^{-2d_G} \mathcal{N}_{A_k B_k}^{(k)C} \\ &= \sum_{\tau \in T_{\text{adm}}(f)} \int_U f_\varepsilon(z, \tau) V_{\text{BF}}(x, y; z)^2 dz. \end{aligned} \quad (5.30)$$

Vanishing of the wall contribution. Lemma B.8 places every nonzero Kac–Walton summand occurring over K_B , for all sufficiently large k , in the fixed finite set $\mathcal{A}_G$. The global tensor estimate (B.73) then gives a constant C_B such that

$$0 \leq N_{ab}^{(k)c} \leq \sum_{\hat{w} \in \mathcal{A}_G} m_{ab}^{\hat{w} \cdot c} \leq C_B r_k^{d_G} \quad (5.31)$$

for every chiral triple occurring in the moving-wall argument. Consequently,

$$\mathcal{N}_{A_k B_k}^{(k)C} \leq C_B^2 r_k^{2d_G}.$$

Put

$$a(\varepsilon) := \text{vol}_{\text{rank } G}((\mathcal{W}_0)_{4\varepsilon} \cap B'). \quad (5.32)$$

Since $\mathcal{W}_0$ is compact and null, continuity from above of Lebesgue measure gives $a(\varepsilon) \rightarrow 0$ as $\varepsilon \downarrow 0$. Lemma 5.2, applied with tube width 2ε , gives the following bound, uniformly in the translate (5.19), for fixed ε and all sufficiently large k satisfying $D/r_k \leq \min\{2\varepsilon, \delta_B\}$:

$$\#(\Lambda_{k,\tau}^{A_k, B_k} \cap (\mathcal{W}_0)_{2\varepsilon} \cap B) \leq \text{covol}(Q)^{-1} a(\varepsilon) r_k^{\text{rank } G}. \quad (5.33)$$

Here $D = \sup_{u \in F_Q} \|u\|$ for the bounded fundamental cell F_Q used in Eq. (5.23). We first fix ε and then take k large enough for the stated cell-radius condition. Only after passing to the limit in k do we let ε decrease to zero. Let $R_{k,\varepsilon}(f)$ be the difference between the Riemann sums for f and f_ε . Since $1 - \chi_\varepsilon$ vanishes outside $(\mathcal{W}_0)_{2\varepsilon}$, the uniform coefficient bound, the finiteness of $T_{\text{adm}}(f)$, and Eq. (5.33) imply

$$\limsup_{k \rightarrow \infty} |R_{k,\varepsilon}(f)| \leq |T_{\text{adm}}(f)| \|f\|_\infty C_B^2 a(\varepsilon). \quad (5.34)$$

Equation (4.40) shows that the limiting density V_{BF}^2 is integrable. Absolute continuity of its Lebesgue integral therefore makes the sum, over the finitely many admissible charges, of its mass in $(\mathcal{W}_0)_{2\varepsilon} \cap B$ tend to zero with ε . Combining this observation with Eq. (5.30), and sending first $k \rightarrow \infty$ and then $\varepsilon \downarrow 0$, proves (5.13).

Type A. For $SU(m)$ with $m \geq 3$, Corollary 4.8 applies to both chiral triples uniformly through the internal walls. On the fixed compact flux window it gives a uniform $o(1)$ error after each coefficient is divided by $r_k^{d_G}$. Multiplying the two chiral estimates still leaves a uniform $o(1)$ error in the normalized doubled coefficient. There are $O(r_k^{\text{rank } G})$ points of a fixed-charge mesh in the window, while each point carries the Riemann-sum factor $\text{covol}(Q) r_k^{-\text{rank } G}$; hence the total error is $o(1)$. Uniform continuity of $V_{\text{BF}}^{A, \text{cont}}$ handles the moving inputs and turns the leading terms into the translated-lattice Riemann sum. This proves the final assertion without deleting a wall tube. $\square$

5.2 The spherical-trace measure

The spherical trace gives a second normalization of the same doubled multiplicities. The underlying spherical-trace and quantum-dimension identities are reviewed in [34, Chs. 2–3]. Its fixed-charge limit is as follows.

1415 **Corollary 5.4** (Fixed-charge asymptotic of the doubled trace probability). *Let $D_A =$*
 1416 *$d_{a^+}^{(k)} d_{a^-}^{(k)}$, and similarly for B, C , and put*

$$\mathbb{P}_{A,B}^{(k)} := \sum_C \mathcal{N}_{AB}^{(k)C} \frac{D_C}{D_A D_B} \delta_{(x_C, \eta_C)}. \quad (5.35)$$

1417 *This is a probability measure. The rescaled measures $r_k^{\text{rank } G} \mathbb{P}_{A_k, B_k}^{(k)}$ are locally finite on*
 1418 *$U \times P$, and the convergence below is vague. Under the hypotheses of Theorem 5.3(ii),*

$$r_k^{\text{rank } G} \mathbb{P}_{A_k, B_k}^{(k)} \xrightarrow{\text{v}} \frac{\Pi_\rho^2}{\text{covol}(Q)} \frac{|\delta_G(z)|^2}{|\delta_G(x) \delta_G(y)|^2} V_{\text{BF}}(x, y; z)^2 dz \otimes \sum_{\tau \in \eta + \omega + Q} \delta_\tau \quad (5.36)$$

1419 *on $U \times P$, with the same type-A strengthening.*

1420 *Proof.* The spherical dimension identity $\sum_C \mathcal{N}_{AB}^{(k)C} D_C = D_A D_B$ proves the probability
 1421 normalization; it is the product of the one-chiral fusion-hypergroup normalizations in [14,
 1422 Eqs. (6)–(7)]. The sine-product formula in Eq. (B.107) [14, Eq. (5)] gives, uniformly for z_a
 1423 in a compact subset of Y ,

$$d_a^{(k)} = \frac{r_k^{|\Delta_+|}}{\Pi_\rho} |\delta_G(z_a)| (1 + O_K(r_k^{-2})). \quad (5.37)$$

1424 Indeed, $\sin(\pi(\rho, \alpha)/r_k) = \pi(\rho, \alpha) r_k^{-1} (1 + O(r_k^{-2}))$ for each positive root, while $|\delta_G(u)| =$
 1425 $\prod_{\alpha > 0} 2 |\sin \pi(\alpha, u)|$.

1426 Fix $f \in C_c(U \times P)$. Its charge projection is finite. For a doubled label, $z_{a^\pm} =$
 1427 $x_A \pm \eta_A/(2r_k)$, and the analogous identities hold for B and C . Thus all six shifts occurring
 1428 on the support of f are $O(r_k^{-1})$. Choose compact sets $K_{\text{in}} \Subset Y$ and $K_{\text{out}} \Subset U$ which contain,
 1429 respectively, the two moving input fluxes and a neighbourhood of the flux projection of
 1430 $\text{supp } f$. For all large k the four chiral input sampling points lie in K_{in} , and the two chiral
 1431 output sampling points over $\text{supp } f$ lie in K_{out} . Since the Weyl denominator has no zeros
 1432 in Y ,

$$\inf_{u \in K_{\text{in}} \cup K_{\text{out}}} |\delta_G(u)| > 0. \quad (5.38)$$

1433 This is the uniform lower bound needed when the input quantum dimensions are placed
 1434 in the denominator. Applying Eq. (5.37) to the six chiral labels and using smoothness of
 1435 $|\delta_G|$ on compact regular flux windows gives

$$\frac{D_C}{D_A D_B} = r_k^{-2|\Delta_+|} \Pi_\rho^2 \frac{|\delta_G(x_C)|^2}{|\delta_G(x_A) \delta_G(x_B)|^2} (1 + O(r_k^{-1})). \quad (5.39)$$

1436 The definitions of the trace and raw measures give

$$r_k^{\text{rank } G} \int f d\mathbb{P}_{A_k, B_k}^{(k)} = \frac{1}{\text{covol}(Q)} \int r_k^{2|\Delta_+|} \frac{D_C}{D_{A_k} D_{B_k}} f(x_C, \eta_C) d\mathfrak{N}_{A_k, B_k}^{(k)}(x_C, \eta_C), \quad (5.40)$$

1437 because $2|\Delta_+| - (2d_G + \text{rank } G) = \text{rank } G$. On the support of f , the multiplier in the last
 1438 integral converges uniformly, by (5.39), to

$$\Pi_\rho^2 \frac{|\delta_G(z)|^2}{|\delta_G(x) \delta_G(y)|^2}.$$

1439 To apply vague convergence with this k -dependent multiplier, let

$$g(z) = \Pi_\rho^2 \frac{|\delta_G(z)|^2}{|\delta_G(x)\delta_G(y)|^2}.$$

1440 Choose $\chi \in C_c(U \times P)$ with $0 \leq \chi \leq 1$ and $\chi = 1$ on $\text{supp } f$. Positivity and the vague
1441 convergence in Theorem 5.3(ii) give

$$\int \chi d\mathfrak{N}_{A_k, B_k}^{(k)} \longrightarrow \sum_{\tau \in \eta + \omega + Q} \int_U \chi(z, \tau) V_{\text{BF}}(x, y; z)^2 dz;$$

1442 in particular, the masses of the raw measures on $\text{supp } f$ are uniformly bounded. Let g_k
1443 denote the multiplier in Eq. (5.40), viewed as a function on the finite-level output atoms.
1444 Then Eq. (5.39) gives

$$\varepsilon_k := \sup \left\{ |g_k(x_C, \eta_C) - g(x_C)| : (x_C, \eta_C) \in \text{supp } f, \mathcal{N}_{A_k B_k}^{(k)C} > 0 \right\} \longrightarrow 0.$$

1445 The supremum is understood to be zero when the displayed set is empty. It follows that

$$\left| \int f(g_k - g) d\mathfrak{N}_{A_k, B_k}^{(k)} \right| \leq \|f\|_\infty \varepsilon_k \int \chi d\mathfrak{N}_{A_k, B_k}^{(k)} \longrightarrow 0.$$

1446 The fixed function fg is compactly supported, so vague convergence also gives

$$\int fg d\mathfrak{N}_{A_k, B_k}^{(k)} \longrightarrow \sum_{\tau \in \eta + \omega + Q} \int_U f(z, \tau) g(z) V_{\text{BF}}(x, y; z)^2 dz.$$

1447 Substituting these two limits into Eq. (5.40) and dividing by $\text{covol}(Q)$ proves Eq. (5.36),
1448 including its constant. $\square$

1449 5.3 Fixed pure-electric fusion operators

1450 Fusion by a fixed pure-electric line gives an operator comparison on the same space $\mathcal{H}_0$.
1451 Its convergence follows by combining Verlinde diagonalization with the strong-* modular
1452 limit.

1453 We use the induced-representation model and the same representation-level tensor
1454 action as in Eq. (4.1). In the notation of [6], their equations (3.1), (3.3), (4.1), and (4.3)
1455 identify $([e], V_\lambda) \otimes ([g_x], \chi_\eta)$ with $([g_x], V_\lambda|_T \otimes \chi_\eta)$, since regularity gives $C_G(g_x) = T$. Using
1456 $V_\lambda|_T \simeq \bigoplus_{\nu \in P} m_\lambda(\nu) \chi_\nu$ then gives

$$([e], V_\lambda) \otimes ([g_x], \chi_\eta) \simeq \bigoplus_{\nu \in P} m_\lambda(\nu) ([g_x], \chi_{\eta+\nu}). \quad (5.41)$$

1457 Only finitely many weight multiplicities $m_\lambda(\nu)$ are nonzero. Thus, on the space $\mathcal{H}_0$ of
1458 Eq. (3.4), fusion with this line acts by the finite charge-shift sum

$$(\mathbf{N}_\lambda^{BF} f)(x, \eta) := \sum_{\nu \in P} m_\lambda(\nu) f(x, \eta - \nu), \quad (5.42)$$

representing the pure-electric line $([e], V_\lambda)$. Since translations in the discrete charge variable are unitary,

$$\|\mathbf{N}_\lambda^{BF}\| \leq \sum_{\nu \in P} m_\lambda(\nu) = \dim V_\lambda.$$

Identity flux lies on the boundary of the regular parametrization, while the induced finite charge-shift action preserves $\mathcal{H}_0$ and is bounded.

For fixed $\lambda \in P_+$, the doubled label $(\lambda, 0)$ exists for all sufficiently large k . Let $\mathbf{N}_{\lambda,k}$ denote fusion by this label in $\mathcal{C}_k \boxtimes \mathcal{C}_k^{\text{rev}}$. Under the parametrization in Eq. (2.53), its matrix element in the canonical basis of $\mathcal{H}_k$ from Eq. (3.4) is

$$(\mathbf{N}_{\lambda,k})_{(c,b),(a,b')} = N_{\lambda a}^{(k)c} \delta_{b,b'}. \quad (5.43)$$

Its flux-charge coordinates are

$$x_{(\lambda,0)} = \frac{\lambda + 2\rho}{2r_k} \longrightarrow 0, \quad \eta_{(\lambda,0)} = \lambda. \quad (5.44)$$

Proposition 5.5 (Strong-* pure-electric fusion-operator limit). *For fixed $\lambda \in P_+$, the embedded fusion operators on $\mathcal{H}_0$ satisfy*

$$\mathbf{N}_{\lambda,k}^\# := I_k \mathbf{N}_{\lambda,k} I_k^* \xrightarrow[k \rightarrow \infty]{\text{SOT}} \mathbf{N}_\lambda^{BF}, \quad (\mathbf{N}_{\lambda,k}^\#)^* \xrightarrow[k \rightarrow \infty]{\text{SOT}} (\mathbf{N}_\lambda^{BF})^*. \quad (5.45)$$

Proof. Fusion matrices are diagonal in the modular S basis by the Verlinde formula [49]. We therefore first take the limit of their diagonal eigenvalue functions and then conjugate by the strong-* limit of $\mathcal{S}_k^\#$. At the endpoint, Fourier transform turns multiplication by a character into the finite sum of charge shifts in Eq. (5.42).

Our convention is $(\mathbf{N}_{\lambda,k})_{c,a} = N_{\lambda a}^{(k)c}$, with output rows and input columns. Hence the chiral eigenvectors are the columns of $(\mathcal{S}^{(k)})^*$: the column indexed by a has eigenvalue $S_{\lambda a}^{(k)}/S_{0a}^{(k)}$. For the doubled operator the corresponding eigenvectors are the columns of $(\mathcal{S}_k)^*$, which fixes the conjugation order in the displayed diagonalization below.

For the label (a, b) in the parametrization (2.53), Verlinde diagonalization [49] and the Kac–Peterson character identity [39, 40] give the eigenvalue

$$\Phi_{\lambda,k}(x, \eta) := \frac{S_{\lambda a}^{(k)}}{S_{0a}^{(k)}} = \chi_\lambda \left(\exp \left[-2\pi i \left(x + \frac{\eta}{2r_k} \right) \right] \right). \quad (5.46)$$

With the column convention in (5.43), let $M_{\Phi_{\lambda,k}}$ be the diagonal multiplication operator on $\ell^2(\Lambda_k)$. Verlinde diagonalization on $\mathbf{P}_k \mathcal{H}_0$ reads

$$\mathbf{N}_{\lambda,k}^\# = (\mathcal{S}_k^\#)^* M_{\Phi_{\lambda,k}}^\# \mathcal{S}_k^\#, \quad M_{\Phi_{\lambda,k}}^\# := I_k M_{\Phi_{\lambda,k}} I_k^*. \quad (5.47)$$

Recall the three extension conventions:

$$\mathcal{S}_k^\# = I_k \mathcal{S}_k I_k^* + (1 - \mathbf{P}_k), \quad \mathbf{N}_{\lambda,k}^\# = I_k \mathbf{N}_{\lambda,k} I_k^*, \quad M_{\Phi_{\lambda,k}}^\# = I_k M_{\Phi_{\lambda,k}} I_k^*. \quad (5.48)$$

Thus $\mathcal{S}_k^\#$ is the identity on $(1 - \mathbf{P}_k)\mathcal{H}_0$, whereas the fusion and multiplier operators vanish there. Verlinde diagonalization proves the preceding identity on $\mathbf{P}_k\mathcal{H}_0$, and these conventions extend it to all of $\mathcal{H}_0$. Put

$$\Phi_\lambda(x, \eta) := \chi_\lambda(e^{-2\pi i x}) = \sum_{\nu \in P} m_\lambda(\nu) e^{-2\pi i(\nu, x)}.$$

The function is independent of η and Weyl invariant in x , so multiplication by it preserves $\mathcal{H}_0$; denote the resulting operator by M_{Φ_λ} . To make the comparison with a continuum multiplier explicit, define $\tilde{\Phi}_{\lambda,k}$ to be constant with value $\Phi_{\lambda,k}(x, \eta)$ on $C_{k,(x,\eta)} \times \{\eta\}$ for every $(x, \eta) \in \Lambda_k$, and set it equal to zero off the union of these cells. Since multiplication by a cell-constant function preserves $\mathbf{P}_k\mathcal{H}_0$, one has the exact identity

$$M_{\Phi_{\lambda,k}}^\# = M_{\tilde{\Phi}_{\lambda,k}} \mathbf{P}_k. \quad (5.49)$$

Put $\mathcal{D} = C_c^\infty(Y) \otimes_{\text{alg}} c_{00}(P)$. If $\psi \in \mathcal{D}$, its flux support is contained in a compact set $K \Subset Y$, so $\text{dist}(K, \partial Y) > 0$, and its charge support is finite. On this set $\eta/(2r_k) \rightarrow 0$ uniformly, and the cell diameters tend to zero. For all large k these finitely many compact flux supports lie in the union of the corresponding cells. Uniform continuity of the fixed character χ_λ , together with Eq. (5.49), therefore gives

$$\|(M_{\Phi_{\lambda,k}}^\# - M_{\Phi_\lambda})\psi\| \rightarrow 0.$$

Here one also uses $\mathbf{P}_k\psi \rightarrow \psi$, as proved in Eq. (3.53), to remove the cell projection. Both multiplication operators have norm at most $\dim V_\lambda$. Since $\mathcal{D}$ is dense, the usual three-term approximation argument extends the convergence from $\mathcal{D}$ to every vector in $\mathcal{H}_0$:

$$M_{\Phi_{\lambda,k}}^\# \xrightarrow{\text{SOT}} M_{\Phi_\lambda}. \quad (5.50)$$

We next conjugate by the modular operators. The families $\mathcal{S}_k^\#$ and $M_{\Phi_{\lambda,k}}^\#$ are uniformly bounded, and Section 3 gives strong convergence of both $\mathcal{S}_k^\#$ and $(\mathcal{S}_k^\#)^*$. Applying these facts one factor at a time to a fixed vector yields

$$\mathbf{N}_{\lambda,k}^\# \xrightarrow{\text{SOT}} \mathcal{S}_0^* M_{\Phi_\lambda} \mathcal{S}_0. \quad (5.51)$$

To identify the limit, use the covering realization $U\mathcal{S}_0 = \mathcal{F}U$ on $L^2(T \times P)^W$. The weight expansion and the Fourier convention in Eq. (3.35) give

$$(\mathcal{F}^* M_{\Phi_\lambda} \mathcal{F})(x, \eta) = \sum_{\nu \in P} m_\lambda(\nu) F(x, \eta - \nu). \quad (5.52)$$

The multiplicity function m_λ is Weyl invariant, so the weighted sum preserves the diagonal Weyl-invariant subspace and descends to $\mathcal{H}_0$. By Eq. (5.42), it is precisely $\mathbf{N}_\lambda^{BF}$.

Let $\lambda^* = -w_0\lambda$ be the highest weight of the contragredient representation. The fusion adjoint and the charge-shift adjoint satisfy

$$(\mathbf{N}_{\lambda,k}^\#)^* = \mathbf{N}_{\lambda^*,k}^\#, \quad (\mathbf{N}_\lambda^{BF})^* = \mathbf{N}_{\lambda^*}^{BF}. \quad (5.53)$$

Applying the already proved strong convergence to λ^* gives strong convergence of the adjoints and completes the strong-* statement. $\square$

The genus-one operators and the genus-zero numerical observables thus have complementary limits on $Y \times P$. After the chosen admissible embeddings, the modular actions converge pointwise in the strong-* topology. The one-chiral coefficients have the chamber-wise limits of Main Theorem B, and the charge-resolved doubled measures have the vague limits of Main Theorem C. A fixed electric representation acts at the endpoint by the finite charge-shift operator in Eq. (5.42).

6 The large- k limit of the boundary theories

This section records boundary evidence compatible with the regular-sector bulk limit, but it is not used in the proofs of Main Theorems A–C. Viewing the three-dimensional TQFT as the SymTFT of a boundary suggests a corresponding sequence of boundary theories $\mathcal{B}_k[G] \rightarrow \mathcal{B}[G]$, associated respectively with boundary conditions $\mathcal{B}_k$ of $\text{CS}_k[G] \times \text{CS}_{-k}[G]$ and $\mathcal{B}$ of $BF[G]$; neither the boundary condition nor the comparison across levels is canonical. We focus on the CS–WZW correspondence [1, 2] and on a pair of WZW factors. For $G = U(1)$, subsection 6.1 gives an exact comparison of the wavefunctions and momentum lattices in the stated polarizations, including the decompactifying radius. For compact simple G under the hypotheses stated below, subsection 6.2 instead gives four compatible checks for a sigma-model candidate with target $\mathfrak{g}$. We do not prove convergence of the full boundary CFT, its local fields, OPE coefficients, sewing data, or correlation functions.

6.1 Boundary wavefunction for the $U(1)$ case

We begin with $G = U(1)$, where the wavefunction calculation can be carried out in closed form and the relevant limit is visible in the momentum lattice of a compact boson. The abelian Lie algebra is of course not semisimple, so this case falls outside the hypotheses used elsewhere in this paper. Here and below, “exact” refers to the wavefunctions, lattice identities, and radii obtained after the stated choices of polarization and symmetry boundary. It does not assert a canonical identification of the complete local-field spaces at different levels or convergence of OPE coefficients and correlators.

Throughout this subsection and appendix A we use the Euclidean action and the weight e^{-S} . This is the Wick-rotated form of the topological normalization in section 2; the diagonal and rescaled antidiagonal fields themselves will be normalized exactly as in (2.20). For radii we use the convention

$$p_{L,R} = \frac{1}{\sqrt{2}} \left(\frac{n}{R_c} \pm w R_c \right), \quad (h, \bar{h}) = \left(\frac{p_L^2}{2}, \frac{p_R^2}{2} \right). \quad (6.1)$$

$U(1)$ Chern-Simons theory $\text{CS}_k[U(1)]$ As a warm up, we review the wavefunction for the $U(1)$ Chern-Simons theory following [2]. The method can be generalized to noncompact abelian BF theory.

Our starting point is $\text{CS}_k[U(1)]$ on a three-manifold $\mathcal{M}$ whose boundary is a torus of modular parameter τ , parametrized by $z = \sigma^1 + \tau \sigma^2$ with $\sigma^i \in [0, 1)$, with the action

$$S = \frac{ik}{4\pi} \int_{\mathcal{M}} A \wedge dA. \quad (6.2)$$

1544 Varying (6.2) leaves a boundary term

$$\delta S = -\frac{ik}{4\pi} \int_{\mathcal{M}} A \wedge \delta A = -\frac{k}{2\pi} \int_{\partial\mathcal{M}} d^2 z (A_z \delta A_{\bar{z}} - A_{\bar{z}} \delta A_z), \quad (6.3)$$

1545 where the holomorphic and anti-holomorphic components are

$$A_z = \frac{A_2 - \bar{\tau} A_1}{2i\tau_2}, \quad A_{\bar{z}} = \frac{\tau A_1 - A_2}{2i\tau_2}, \quad (6.4)$$

1546 and we fix the conventions

$$dz \wedge d\bar{z} = -2i\tau_2 d\sigma^1 \wedge d\sigma^2, \quad d^2 z \equiv \frac{i}{2} dz \wedge d\bar{z}, \quad \int d^2 z = \tau_2. \quad (6.5)$$

1547 We add the standard counterterm at the boundary [2] and consider the action

$$S' = S - \frac{k}{2\pi} \int_{\partial\mathcal{M}} d^2 z A_z A_{\bar{z}}. \quad (6.6)$$

1548 The equation of motion $\delta S'/\delta A_z$ is satisfied without imposing any boundary condition for
 1549 δA_z , so that the path integral can be evaluated with boundary conditions of fixed $A_{\bar{z}}$ and
 1550 arbitrary A_z . Introduce the wavefunction $\Psi(A_{\bar{z}})$ as

$$\Psi(A_{\bar{z}}) = \int_{A_z|_{\partial\mathcal{M}} \text{ fixed}} \mathcal{D}A e^{-S'}, \quad (6.7)$$

1551 which depends on the boundary value of $A_{\bar{z}}$. A large gauge transformation $A \rightarrow A + \omega$ at
 1552 the boundary multiplies the wavefunction by a factor assembled from the bulk action

$$\Psi(A_{\bar{z}}) \rightarrow \Psi(A_{\bar{z}}) e^{\frac{ik}{4\pi} \int_{\partial\mathcal{M}} \omega \wedge A} e^{\frac{k\tau_2}{2\pi} (A_z \omega_{\bar{z}} + A_{\bar{z}} \omega_z + |\omega_z|^2)} e^{i\varphi(\omega)}, \quad (6.8)$$

1553 where the first term is from S , the second term is from the boundary term S_{∂} , and the last
 1554 term is an additional cocycle to assure associativity of large gauge transformations.

1555 We will briefly comment on the first term. In general, a large gauge transformation
 1556 on the boundary cannot be extended as a gauge transformation in the bulk. Instead, one
 1557 might consider the deformed gauge field $\tilde{A} = A + \tilde{\omega}$, where $\tilde{\omega}$ is an arbitrary extension
 1558 whose restriction along the boundary is $\tilde{\omega}|_{\partial\mathcal{M}} = \omega$. Denote $r_{\leq 0}$ as the normal direction
 1559 and put the boundary $\partial\mathcal{M}$ at $r = 0$. The path integral does not depend on the detail
 1560 of the extension, and we can choose $\tilde{\omega} = \omega \rho_{\epsilon}(r)$ where $\rho_{-\epsilon}(r)$ is a smooth cutoff function
 1561 satisfying $\rho_{\epsilon}(0) = 1$ and $\rho_{\epsilon}(\epsilon) = 0$. Therefore the bulk action satisfies

$$\frac{ik}{4\pi} \int_{\mathcal{M}} \tilde{A} \wedge d\tilde{A} = \frac{ik}{4\pi} \int_{\mathcal{M}} A \wedge dA + \frac{ik}{4\pi} \int_{\partial\mathcal{M}} \omega \wedge A + \frac{ik}{2\pi} \int_{\mathcal{M}} \tilde{\omega} \wedge dA + \frac{ik}{4\pi} \int_{\mathcal{M}} \tilde{\omega} \wedge d\tilde{\omega}. \quad (6.9)$$

1562 On the RHS, the last term vanishes since $\tilde{\omega} \wedge \tilde{\omega} = 0$, and the third term has order $\mathcal{O}(\epsilon)$.
 1563 Since the path integral does not depend on ϵ , we can send $\epsilon \rightarrow 0$ so that

$$\frac{ik}{4\pi} \int_{\mathcal{M}} \tilde{A} \wedge d\tilde{A} = \frac{ik}{4\pi} \int_{\mathcal{M}} A \wedge dA + \frac{ik}{4\pi} \int_{\partial\mathcal{M}} \omega \wedge A, \quad (6.10)$$

1564 which gives the first term in the transformation.

1565 Parametrize the value of $A_{\bar{z}}$ at the boundary as

$$A_{\bar{z}} = \frac{i\pi}{\tau_2} \xi, \quad (6.11)$$

1566 and the large gauge transformation shifts ξ according to

$$\xi \rightarrow \xi + n + m\tau, \quad n, m \in \mathbb{Z}. \quad (6.12)$$

1567 Then we can write

$$\Psi(\xi + n + m\tau) = \Psi(\xi) \exp \left[\frac{k\pi}{\tau_2} (n + m\bar{\tau}) \xi + \frac{k\pi}{2\tau_2} |n + m\tau|^2 + i\pi k n m \right], \quad (6.13)$$

1568 where we choose the cocycle as $\varphi = \pi k n m$. It is useful to introduce $f = \Psi e^{-k\pi\xi^2/2\tau_2}$,
1569 which reduces the multiplier to

$$f(\xi + 1) = f(\xi), \quad f(\xi + \tau) = f(\xi) e^{-2\pi i k \xi - i\pi k \tau}, \quad (6.14)$$

1570 which is the quasi-periodicity of a level- k theta function. Since $f(\xi)$ is periodic along the
1571 real axis, we can do Fourier expansion

$$f = \sum_{\lambda \in \mathbb{Z}} c_{\lambda} e^{2\pi i \lambda \xi}, \quad (6.15)$$

1572 and it turns the second relation in (6.14) into the recursion relation

$$c_{\lambda+k} = c_{\lambda} e^{2\pi i \lambda \tau + \pi i k \tau}, \quad (6.16)$$

1573 which connects c_{λ} to $c_{\lambda+k}$, therefore the solution space is k -dimensional. Solving the
1574 recursion relation using the ansatz $c_{\lambda} = e^{i\pi\lambda^2/k}$, and setting $\lambda = \sqrt{k} p$ with $p = \sqrt{k} (n+j/k)$
1575 and $u = \sqrt{k} \xi$, the k independent wavefunctions read

$$\Psi_j(\xi) = \frac{1}{\eta(\tau)} \sum_{n \in \mathbb{Z}} \exp \left[i\pi \tau p^2 + 2\pi i p u + \frac{\pi}{2\tau_2} u^2 \right], \quad j = 0, \dots, k-1, \quad (6.17)$$

1576 with $1/\eta(\tau)$ the one-loop determinant of the topologically trivial fluctuations. We recover
1577 the standard count of k states, and j is the anyon label of $U(1)_k$.

1578 **Non-compact abelian BF theory** The non-compact BF theory runs through the same
1579 steps with a single change that turns out to control everything. The action is

$$S = \frac{i}{2\pi} \int_{\mathcal{M}} b \wedge da = \frac{i}{4\pi} \int_{\mathcal{M}} b \wedge da + \frac{i}{4\pi} \int_{\mathcal{M}} a \wedge db, \quad (6.18)$$

1580 in which a remains a compact $U(1)$ gauge field while b is valued in $\mathbb{R}$, so that b admits no
1581 large gauge transformations. Here we write the BF action as a a - b symmetric way up to a
1582 boundary terms $\int_{\partial\mathcal{M}} a \wedge b$, and we will remove it by adding a local boundary counterterm
1583 to the action.

1584

Introduce the combinations

$$A_{\pm} = \frac{1}{\sqrt{2}} \left(\frac{a}{r} \pm rb \right), \quad (6.19)$$

1585

so that the BF action (6.18) can be written as a pair of Chern–Simons theories at opposite

1586

unit level,

$$S = \frac{i}{4\pi} \int (A_+ \wedge dA_+ - A_- \wedge dA_-), \quad (6.20)$$

1587

where r is a real and positive parameter. Consider the two *independent* holomorphic

1588

coordinates parametrized by

$$\xi = \frac{\tau_2}{i\pi} (A_+)_{\bar{z}} = \frac{1}{\sqrt{2}} \left(\frac{u}{r} + vr \right), \quad \bar{\xi} = -\frac{\tau_2}{i\pi} (A_-)_z = \frac{1}{\sqrt{2}} \left(\frac{\bar{u}}{r} - \bar{v}r \right), \quad (6.21)$$

1589

with

$$u = \frac{\tau_2}{i\pi} a_{\bar{z}}, \quad v = \frac{\tau_2}{i\pi} b_{\bar{z}}. \quad (6.22)$$

1590

Deform the action with the proper boundary term

$$S' = S - \frac{1}{2\pi} \int_{\partial\mathcal{M}} d^2z (|A_+|^2 + |A_-|^2), \quad (6.23)$$

1591

and we can repeat the computation for the $U(1)$ Chern–Simons case with the two indepen-

1592

dent holomorphic coordinates ξ and $\bar{\xi}$.

1593

The large gauge transformation is generated purely by a , and we have

$$\xi \rightarrow \xi + \frac{1}{\sqrt{2}r} (n + m\tau), \quad \bar{\xi} \rightarrow \bar{\xi} + \frac{1}{\sqrt{2}r} (n + m\bar{\tau}). \quad (6.24)$$

1594

Denote $\delta = \frac{1}{\sqrt{2}r} (n + m\tau)$, the wavefunction transform according to[]

$$\Psi(\xi + \delta, \bar{\xi} + \delta^*) = \Psi(\xi, \bar{\xi}) \exp \left[\frac{\pi}{\tau_2} (\xi \delta^* + \bar{\xi} \delta + |\delta|^2) \right], \quad (6.25)$$

1595

which is associative under the large gauge transformation. Similarly, let us introduce

1596

 $f = \Psi e^{-\frac{\pi}{2\tau_2} (\xi^2 + \bar{\xi}^2)}$, and one can check the large gauge transformation is generated by

$$\begin{aligned} f \left(\xi + \frac{1}{\sqrt{2}r}, \bar{\xi} + \frac{1}{\sqrt{2}r} \right) &= f(\xi, \bar{\xi}), \\ f \left(\xi + \frac{\tau}{\sqrt{2}r}, \bar{\xi} + \frac{\bar{\tau}}{\sqrt{2}r} \right) &= f \exp \left[-\frac{2\pi i}{\sqrt{2}r} (\xi - \bar{\xi}) + \frac{\pi \tau_2}{r^2} \right]. \end{aligned} \quad (6.26)$$

1597

Then let us consider Fourier expansion and assume

$$f = \sum_{p_L, p_R} c_{p_L, p_R} e^{2\pi i p_L \xi - 2\pi i p_R \bar{\xi}}, \quad (6.27)$$

1598

where the summation runs over all possible values of p_L, p_R . The first relation implies

$$e^{\frac{2\pi i (p_L - p_R)}{\sqrt{2}r}} = 1 \quad \Rightarrow \quad p_L - p_R \in \sqrt{2}r\mathbb{Z}, \quad (6.28)$$

1599 and the other relation gives the recursion relation

$$c_{p_L + \frac{1}{\sqrt{2}r}, p_R + \frac{1}{\sqrt{2}r}} = c_{p_L, p_R} e^{\frac{2\pi i(p_L \tau - p_R \bar{\tau})}{\sqrt{2}r} - \frac{\pi \tau_2}{r^2}}, \quad (6.29)$$

1600 and we take the ansatz

$$c_{p_L, p_R} = \exp [i\pi \tau p_L^2 - i\pi \bar{\tau} p_R^2]. \quad (6.30)$$

1601 Based on the above discussion, we can parametrize

$$p_L = \frac{1}{\sqrt{2}} \left(\frac{\ell + \varphi}{r} + rN \right), \quad p_R = \frac{1}{\sqrt{2}} \left(\frac{\ell + \varphi}{r} - rN \right), \quad (6.31)$$

1602 with $\varphi \in [0, 1)$ and $N \in \mathbb{Z}$, and they satisfy $p_L - p_R = \sqrt{2}rN$ as required. Then the
1603 independent wavefunctions are

$$\Psi_{\varphi, N}(\xi, \bar{\xi}) = \frac{1}{|\eta|^2} \sum_{\ell \in \mathbb{Z}} \exp \left[i\pi \tau p_L^2 - i\pi \bar{\tau} p_R^2 + 2\pi i(p_L \xi - p_R \bar{\xi}) + \frac{\pi}{2\tau_2}(\xi^2 + \bar{\xi}^2) \right]. \quad (6.32)$$

1604 Notice that the labels (φ, N) of the wavefunction are exactly the same to the labels of the
1605 bulk line $W_{\varphi, N}$. In the bulk BF theory, φ parametrize the continuous family of the Wilson
1606 line of b , and N is the discrete electric charge associated to the Wilson line of a . While in
1607 the boundary momentum lattice, the same N occupies the winding slot and $\ell + \varphi$ occupies
1608 the momentum slot.

1609 In the SymTFT picture, the wavefunctions $\Psi_{\varphi, N}(\xi, \bar{\xi})$ actually gives the data of phys-
1610 ical boundary. To retrieve the partition function, one need to include the topological
1611 boundary determined by the Lagrangian algebra $\mathcal{L}$. Formally, suppose the wavefunction is
1612 labeled by Ψ_i , and the Lagrangian algebra is $\mathcal{L} = \bigoplus_i n_i W_i$, where i labels the anyons in
1613 the bulk. Then the partition function associated with $\mathcal{L}$ is written as

$$Z_{\mathcal{L}} = \sum_i n_i \Psi_i. \quad (6.33)$$

1614 In the present case, the bulk anyon is labeled by $W_{\varphi, N}$. Let us first consider

$$\mathcal{L}_1^{BF} = \bigoplus_N W_{0, N}, \quad (6.34)$$

1615 so that the Wilson line of a can end on the topological boundary, and we can turn on the
1616 a -background on it. Setting $v = 0$ we have

$$\xi = \frac{1}{\sqrt{2}} \frac{u}{r}, \quad \bar{\xi} = \frac{1}{\sqrt{2}} \frac{\bar{u}}{r}, \quad (6.35)$$

1617 which gives

$$Z_{\mathcal{L}_1^{BF}} = \frac{1}{|\eta|^2} \sum_{\ell, N \in \mathbb{Z}} \exp \left[i\pi \tau p_L^2 - i\pi \bar{\tau} p_R^2 + 2\pi i(p_L \xi - p_R \bar{\xi}) + \frac{\pi}{2\tau_2}(\xi^2 + \bar{\xi}^2) \right], \quad (6.36)$$

1618 with

$$p_L = \frac{1}{\sqrt{2}} \left(\frac{\ell}{r} + rN \right), \quad p_R = \frac{1}{\sqrt{2}} \left(\frac{\ell}{r} - rN \right). \quad (6.37)$$

1619 Set $R = r/\sqrt{2}$ and rewrite

$$e^{-\frac{\pi}{2\tau_2}(|\xi|^2+|\bar{\xi}|^2)} Z_{\mathcal{L}_1^{BF}} = \frac{1}{|\eta|^2} \sum_{\ell, N \in \mathbb{Z}} e^{2\pi i u_2 N} q^{\frac{1}{2}\left(\frac{\ell+u_1}{2R}+RN\right)^2} \bar{q}^{\frac{1}{2}\left(\frac{\ell+u_1}{2R}-RN\right)^2}, \quad (6.38)$$

1620 where we set $u = u_2 + \tau u_1$. The RHS is exactly the partition function of a compact boson
 1621 with a generic $U(1)_W$ background parametrized by the spatial and temporal holonomies
 1622 (u_1, u_2) , up to an additional phase factor $e^{-\frac{\pi}{2\tau_2}(|\xi|^2+|\bar{\xi}|^2)}$ which can be canceled by adding
 1623 a local counterterm on the boundary[].

1624 Alternatively, we can consider another Lagrangian algebra

$$\mathcal{L}_2^{BF} = \int_0^1 d\varphi W_{\varphi,0}, \quad (6.39)$$

1625 so that the Wilson line of b can end on the topological boundary, and we can turn on the
 1626 b -background on it. Setting $u = 0$ we have

$$\xi = \frac{r}{\sqrt{2}}v, \quad \bar{\xi} = -\frac{r}{\sqrt{2}}\bar{v}, \quad (6.40)$$

1627 and also

$$p_L = p_R = \frac{1}{\sqrt{2}r}(\ell + \varphi). \quad (6.41)$$

1628 The integral is evaluated as

$$\begin{aligned} Z_{\mathcal{L}_2^{BF}} &= \frac{1}{|\eta|^2} \int_{-\infty}^{+\infty} \sqrt{2}r d\lambda \exp(-2\pi\tau_2\lambda^2 + 2\pi i\lambda(\xi - \bar{\xi})) \exp\left(+\frac{\pi}{2\tau_2}(\xi^2 + \bar{\xi}^2)\right) \\ &= \frac{\sqrt{2}R}{|\eta|^2\sqrt{\tau_2}} e^{\frac{\pi}{\tau_2}\xi\bar{\xi}}, \end{aligned} \quad (6.42)$$

1629 so that one has

$$e^{-\frac{\pi}{2\tau_2}(|\xi|^2+|\bar{\xi}|^2)} Z_{\mathcal{L}_2^{BF}} = \frac{\sqrt{2}R}{|\eta|^2\sqrt{\tau_2}} e^{-\frac{2\pi R^2}{\tau_2}|v_2+\tau v_1|^2}, \quad (6.43)$$

1630 where we set $v = v_2 + \tau v_1$. This is the partition function of a non-compact boson with
 1631 a generic background of the translation symmetry.

1632 **Double $U(1)$ Chern-Simons theory in the large k limit** Then we turn to the double
 1633 copy of Chern-Simons theory $U(1)_k \times U(1)_{-k}$, and consider the large k limit. We will show
 1634 that, with a proper choice of boundary condition, the partition function in the large k limit
 1635 agrees with those of $U(1)$ non-compact BF theory we just discussed.

1636 Begin with the double Chern-Simons theory

$$S = \frac{ik}{4\pi} \int a_+ \wedge da_+ - \frac{ik}{4\pi} \int a_- \wedge da_-, \quad (6.44)$$

1637 and we can introduce the BF fields

$$a = \frac{1}{2}(a_+ + a_-), \quad b = k(a_+ - a_-), \quad (6.45)$$

where the period of b goes to infinity as $k \rightarrow +\infty$. To compare with the previous result, we redefine the Chern-Simons fields according to (6.19) by setting

$$A_{\pm} = \frac{1}{\sqrt{2}} \left(\frac{a}{r} \pm rb \right) = \frac{1}{\sqrt{2}} \left(\frac{a_+ + a_-}{2r} \pm kr(a_+ - a_-) \right), \quad (6.46)$$

and introduce the same holomorphic coordinates as

$$\xi = \frac{\tau_2}{i\pi} (A_+)_{\bar{z}}, \quad \bar{\xi} = -\frac{\tau_2}{i\pi} (A_-)_z. \quad (6.47)$$

We have two independent large gauge transformations for a_+ and a_- . Denote

$$\mu = \frac{1}{\sqrt{2}} \left(\frac{1}{2r} + kr \right), \quad \nu = \frac{1}{\sqrt{2}} \left(\frac{1}{2r} - kr \right), \quad (6.48)$$

which satisfies

$$\mu + \nu = \frac{1}{\sqrt{2}r}, \quad \mu - \nu = \sqrt{2}kr, \quad \mu^2 - \nu^2 = k. \quad (6.49)$$

The large gauge transformation reads

$$\delta\xi = \mu D_+ + \nu D_- \equiv E, \quad \delta\bar{\xi} = \nu \bar{D}_+ + \mu \bar{D}_- \equiv F^*, \quad (6.50)$$

where

$$D_{\pm} = n_{\pm} + m_{\pm}\tau, \quad \bar{D}_{\pm} = n_{\pm} + m_{\pm}\bar{\tau}. \quad (6.51)$$

The cocycle φ is defined by

$$\varphi(\omega + \omega') = \varphi(\omega) + \varphi(\omega') - \frac{1}{4\pi} \int_{\partial\mathcal{M}} (\omega_{A_+} \wedge \omega'_{A_+} - \omega_{A_-} \wedge \omega'_{A_-}), \quad (6.52)$$

where

$$\omega_{A_+} = \mu\omega_+ + \nu\omega_-, \quad \omega_{A_-} = \nu\omega_+ + \mu\omega_-, \quad (6.53)$$

so that

$$\omega_{A_+} \wedge \omega'_{A_+} - \omega_{A_-} \wedge \omega'_{A_-} = (\mu^2 - \nu^2)(\omega_+ \wedge \omega'_+ - \omega_- \wedge \omega'_-), \quad (6.54)$$

and is solved by

$$\varphi = \pi k(m_+ n_+ - m_- n_-). \quad (6.55)$$

The wavefunction transforms under the large gauge transformations according to

$$\Psi(\xi + E, \bar{\xi} + F^*) = \Psi(\xi, \bar{\xi}) \exp \left[\frac{\pi}{2\tau_2} (2\xi E^* + E E^*) + \frac{\pi}{2\tau_2} (2\bar{\xi} F + F F^*) + i\varphi \right]. \quad (6.56)$$

Set $\Psi = f e^{\frac{\pi}{2\tau_2}(\xi^2 + \bar{\xi}^2)}$, one can check that $f(\xi + E, \bar{\xi} + F^*) = f(\xi, \bar{\xi}) \times \delta f$ transforms under the four generators as the following table

(n_+, m_+, n_-, m_-)	$(\delta\xi, \delta\bar{\xi})$	δf
$(1, 0, 0, 0)$	(μ, ν)	1
$(0, 0, 1, 0)$	(ν, μ)	1
$(0, 1, 0, 0)$	$(\mu\tau, \nu\bar{\tau})$	$e^{-2\pi i\mu\xi + 2\pi i\nu\bar{\xi} - i\pi\mu^2\tau + i\pi\nu^2\bar{\tau}}$
$(0, 0, 0, 1)$	$(\nu\tau, \mu\bar{\tau})$	$e^{-2\pi i\nu\xi + 2\pi i\mu\bar{\xi} - i\pi\nu^2\tau + i\pi\mu^2\bar{\tau}}$

1653 Consider the Fourier expansion

$$f(\xi, \bar{\xi}) = \sum_{p_L, p_R} c_{p_L, p_R} e^{2\pi i(p_L \xi - p_R \bar{\xi})}, \quad (6.57)$$

1654 and the first two generators give

$$p_L \mu - p_R \nu \in \mathbb{Z}, \quad p_L \nu - p_R \mu \in \mathbb{Z}. \quad (6.58)$$

1655 Parametrize p_L, p_R according to

$$p_L = \mu \hat{n}_+ + \nu \hat{n}_-, \quad p_R = \nu \hat{n}_+ + \mu \hat{n}_-, \quad (6.59)$$

1656 we have

$$p_L \mu - p_R \nu = (\mu^2 - \nu^2) \hat{n}_+ = k \hat{n}_+, \quad p_L \nu - p_R \mu = (\nu^2 - \mu^2) \hat{n}_- = -k \hat{n}_-, \quad (6.60)$$

1657 so one can further parametrize

$$\hat{n}_\pm = \frac{j_\pm}{k}, \quad j_\pm \in \mathbb{Z}. \quad (6.61)$$

1658 Then consider the other two generators. The third one shifts $(p_L, p_R) \rightarrow (p_L - \mu, p_R - \nu)$,
1659 which is effectively

$$\hat{n}_+ \rightarrow \hat{n}_+ - 1, \quad \text{or} \quad j_+ \rightarrow j_+ - k, \quad (6.62)$$

1660 and it gives the recursion relation

$$c_{p_L + \mu, p_R + \nu} = c_{p_L, p_R} e^{2\pi i(\mu \tau p_L - \nu \bar{\tau} p_R) + i\pi \mu^2 \tau - i\pi \nu^2 \bar{\tau}}. \quad (6.63)$$

1661 Similarly, the fourth one shifts $(p_L, p_R) \rightarrow (p_L - \nu, p_R - \mu)$, which is effectively

$$\hat{n}_- \rightarrow \hat{n}_- - 1, \quad \text{or} \quad j_- \rightarrow j_- - k, \quad (6.64)$$

1662 and it gives the recursion relation

$$c_{p_L + \nu, p_R + \mu} = c_{p_L, p_R} e^{2\pi i(\nu \tau p_L - \mu \bar{\tau} p_R) + i\pi \nu^2 \tau - i\pi \mu^2 \bar{\tau}}. \quad (6.65)$$

1663 Both recursion relations are solved by the ansatz

$$c_{p_L, p_R} = \exp \left[i\pi \tau p_L^2 - i\pi \bar{\tau} p_R^2 \right], \quad (6.66)$$

1664 and there are exactly k^2 solution.

1665 Based on the above discussion, let us write

$$j_\pm = k n_\pm + \lambda_\pm, \quad \lambda_\pm \in \{0, \dots, k-1\}, \quad n_\pm \in \mathbb{Z}, \quad (6.67)$$

1666 and the k^2 independent wavefunctions are

$$\Psi_{\lambda_+, \lambda_-}(\xi, \bar{\xi}, \tau) = \frac{1}{|\eta|^2} \sum_{n_+, n_-} \exp \left[i\pi \tau p_L^2 - i\pi \bar{\tau} p_R^2 + 2\pi i(p_L \xi - p_R \bar{\xi}) + \frac{\pi}{2\tau_2}(\xi^2 + \bar{\xi}^2) \right], \quad (6.68)$$

1667 with

$$p_L = \frac{1}{\sqrt{2}} \left[\frac{j_+ + j_-}{2kr} + r(j_+ - j_-) \right], \quad p_R = \frac{1}{\sqrt{2}} \left[\frac{j_+ + j_-}{2kr} - r(j_+ - j_-) \right]. \quad (6.69)$$

1668 We then consider the partition function built by choosing a Lagrangian algebra. Denote
1669 the anyons as W_{λ_+, λ_-} , we first consider the diagonal one

$$\mathcal{L}_1^{CS} = \bigoplus_{\lambda} W_{\lambda, \lambda}, \quad (6.70)$$

1670 which set $\lambda_+ = \lambda_- = \lambda$. We have

$$\frac{j_+ + j_-}{2kr} = \frac{k(n_+ + n_-) + 2\lambda}{2kr}, \quad r(j_+ - j_-) = kr(n_+ - n_-), \quad (6.71)$$

1671 and we label

$$w = n_+ - n_-, \quad m = \frac{k}{2}(n_+ + n_-) + \lambda = \frac{k}{2}w + kn_- + \lambda. \quad (6.72)$$

1672 For the Chern-Simons to be non-spin, we choose k even, then w, m can be chosen as
1673 independent integers. Thus we have

$$p_L = \frac{m}{kr} + krw, \quad p_R = \frac{m}{kr} - krw, \quad (6.73)$$

1674 which matches the momenta of a compact boson with radius kr .

1675 On the other hand, if we choose the off-diagonal one

$$\mathcal{L}_2^{CS} = \bigoplus_i W_{\lambda, k-\lambda}, \quad (6.74)$$

1676 and we set $\lambda_+ = \lambda, \lambda_- = k - \lambda$. Then one has

$$\frac{j_+ + j_-}{2kr} = \frac{n_+ + n_- + 1}{2r}, \quad r(j_+ - j_-) = kr(n_+ - n_- - 1) + 2\lambda r. \quad (6.75)$$

1677 We then set

$$m = n_+ + n_- + 1, \quad (6.76)$$

1678 and

$$w = \frac{(n_+ - n_- - 1)k}{2} + \lambda = \frac{(m - 2n_- - 2)k}{2} + \lambda = \frac{k}{2}m + \lambda - (n_- + 1)k, \quad (6.77)$$

1679 and both m, n can take arbitrary integers when k is even. Therefore we have

$$p_L = \frac{1}{\sqrt{2}} \left(\frac{m}{2r} + 2rw \right), \quad p_R = \frac{1}{\sqrt{2}} \left(\frac{m}{2r} - 2rw \right), \quad (6.78)$$

1680 which is independent of k and correspondent to the compact boson with radius $2r$.

1681 Sending $k \rightarrow \infty$, the compact boson corresponding to the diagonal boundary conse-
1682 quently decompactifies, while that for the off-diagonal boundary remains compact since
1683 the radius is independent of k . We see that the two choices of Lagrangian algebras recover
1684 the results in the BF case. Roughly speaking, we have

$$\mathcal{L}_1^{CS} \rightarrow \mathcal{L}_2^{BF}, \quad \mathcal{L}_2^{CS} \rightarrow \mathcal{L}_1^{BF}. \quad (6.79)$$

6.2 The limiting theory for general G

For G compact, connected, simple, and simply connected, we give four complementary checks of the candidate boundary theory: a sigma model into $\mathfrak{g}$, or $\dim G$ non-compact free scalars. They test central charge, a local action expansion, global symmetry, and current algebra, respectively. Agreement of these checks is evidence for the candidate, not a uniqueness statement or a proof of convergence of the full CFT.

The first is the central charge. The Sugawara construction gives the level- k WZW model [50] central charge

$$c_k = \frac{k \dim G}{k + h^\vee}, \quad (6.80)$$

so that $\lim_{k \rightarrow \infty} c_k = \dim G$. A boundary theory for $\text{CS}_k[G] \times \text{CS}_{-k}[G]$ built from a chiral and an antichiral WZW factor therefore approaches central charge $(\dim G, \dim G)$, consistent with $\dim G$ real non-compact non-chiral scalars. Central charge alone is not diagnostic: many inequivalent CFTs can share it. It therefore supplies only the first necessary check on the candidate tested below.

The second piece of evidence comes from the action itself. The sigma-model metric on the target is proportional to $k\kappa$, so the characteristic radius of the target grows like $\sqrt{k}$ while its curvature falls like $1/k$. To keep fluctuations canonically normalized we zoom into a finite region of the tangent space at the identity by writing

$$g = \exp\left(\frac{X}{\sqrt{k}}\right), \quad X = X^a t_a \in \mathfrak{g}, \quad (6.81)$$

hence the Maurer–Cartan form can be expanded as

$$g^{-1}dg = \frac{1}{\sqrt{k}} dX - \frac{1}{2k} [X, dX] + \frac{1}{6k^{3/2}} [X, [X, dX]] + O(k^{-2}). \quad (6.82)$$

Substituting into the kinetic term $\frac{k}{4\pi} \int d^2z \kappa(g^{-1}\partial g, g^{-1}\bar{\partial}g)$, one would expect a cubic vertex at order $k^{-1/2}$, whose coefficient is proportional to

$$\kappa(\partial X, [X, \bar{\partial} X]) + \kappa([X, \partial X], \bar{\partial} X) = 0, \quad (6.83)$$

and this vanishes by invariance of κ . The kinetic term is therefore free through order $k^{-1/2}$, and its first correction is a quartic vertex at order $1/k$ coming from the curvature of the group metric. The Wess–Zumino term does survive at the intermediate order, since $\kappa(\theta, [\theta, \theta])$ scales as $k^{-3/2}$ against a prefactor of k , leaving a cubic interaction suppressed by $k^{-1/2}$. Assembling the two pieces, we have

$$S_{\text{WZW},k}[e^{X/\sqrt{k}}] = \frac{1}{4\pi} \int_{\Sigma} d^2z \kappa_{ab} \partial X^a \bar{\partial} X^b + \frac{1}{\sqrt{k}} S_{\text{WZ}}^{(3)} + \frac{1}{k} S^{(4)} + \dots, \quad (6.84)$$

so that in an orthonormal basis, $\kappa_{ab} = \delta_{ab}$, the leading local action is that of $\dim G$ real non-compact free scalars. Geometrically this means that $(G, k\kappa)$ flattens onto $(T_e G, \kappa) \simeq (\mathfrak{g}, \kappa) \simeq \mathbb{R}^{\dim G}$, which is the same rescaling that produces the tangent group in the bulk. Using $X/\sqrt{k+h^\vee}$ in place of (6.81) changes nothing at leading order. This tangent-space

expansion controls perturbative fluctuations in a fixed coordinate neighborhood; by itself it does not control global field configurations, normalized operators, or correlation functions.

Third, the candidate theory carries the global symmetry we should expect on a boundary of $CS[TG]$. Take

$$S = \int_{\Sigma} \langle dX, dX \rangle, \quad X : \Sigma \rightarrow \mathfrak{g}_{\text{ab}}, \quad (6.85)$$

with $\langle \cdot, \cdot \rangle$ an Ad-invariant form on $\mathfrak{g}$. It is invariant under

$$X \mapsto \text{Ad}_g X + \alpha, \quad g \in G, \quad \alpha \in \mathfrak{g}_{\text{ab}}, \quad (6.86)$$

and the group of such transformations is exactly $TG \simeq G \ltimes_{\text{Ad}} \mathfrak{g}_{\text{ab}}$, realized without any extra ingredient. One point here deserves care. The fields X^a are real, so the manifest linear symmetry of (6.85) is $O(\dim G)$, with constant translations supplying the affine part, and the TG action enters through the homomorphism $\text{Ad} : G \rightarrow SO(\dim G)$ supplied by invariance of κ . For $G = SU(n)$ this is the embedding $PSU(n) \rightarrow SO(n^2 - 1)$ since $\ker(\text{Ad} : G \rightarrow SO(\dim G)) = Z(G)$ for compact connected Lie group with an Ad-invariant positive-definite inner product on $\mathfrak{g}$, and the TG symmetry is a proper subgroup of the full symmetry of the free theory. Thus the symmetry is a compatibility check, not a characterization of the candidate.

The sharpest of the four checks is at the level of the current algebra, where the limit is an İnönü–Wigner contraction [27, 28] with rescaling. Let J_m^a and $\bar{J}_m^a$ be the standard holomorphic and antiholomorphic level- k modes. After folding the antiholomorphic sector we set $\tilde{J}_m^a := \bar{J}_{-m}^a$; the reversal of the mode number changes the sign of its central term. Thus the two commuting affine algebras are written as $\hat{\mathfrak{g}}_k \oplus \hat{\mathfrak{g}}_{-k}$ with $\langle t^a, t^b \rangle = \delta^{ab}$,

$$[J_m^a, J_n^b] = i f^{abc} J_{m+n}^c + km \delta^{ab} \delta_{m+n,0}, \quad [\tilde{J}_m^a, \tilde{J}_n^b] = i f^{abc} \tilde{J}_{m+n}^c - km \delta^{ab} \delta_{m+n,0}, \quad (6.87)$$

with $[J_m^a, \tilde{J}_n^b] = 0$. Recombining into a diagonal sum and a rescaled antidiagonal difference,

$$\mathcal{J}_m^a := J_m^a + \tilde{J}_m^a, \quad \mathcal{P}_m^a := \frac{1}{2k} (J_m^a - \tilde{J}_m^a), \quad (6.88)$$

leads to

$$\begin{aligned} [\mathcal{J}_m^a, \mathcal{J}_n^b] &= i f^{abc} \mathcal{J}_{m+n}^c, \\ [\mathcal{J}_m^a, \mathcal{P}_n^b] &= i f^{abc} \mathcal{P}_{m+n}^c + m \delta^{ab} \delta_{m+n,0}, \\ [\mathcal{P}_m^a, \mathcal{P}_n^b] &= \frac{1}{4k^2} i f^{abc} \mathcal{J}_{m+n}^c. \end{aligned} \quad (6.89)$$

There are three features of (6.89) that are worth stating explicitly. The level cancels exactly between the two factors in the first bracket, so $\mathcal{J}$ generates a loop algebra at level zero. The central term survives only in the mixed bracket, where it is normalized to one at every k . Lastly, the power k^{-1} is forced if the mixed central term is to have a finite nonzero limit while the third bracket vanishes; the constant $1/2$ in (6.88) normalizes that mixed term to one at every k . At $k \rightarrow \infty$ we obtain $\hat{\mathfrak{t}}\mathfrak{g}$, and restricting to zero modes removes the central term and leaves $\mathfrak{t}\mathfrak{g} = \mathfrak{g} \ltimes \mathfrak{g}_{\text{ab}} = \text{Lie}(TG)$. This is the sharpest algebraic check, but contraction of the brackets alone does not establish convergence of their Hilbert-space representations or of the associated vertex-operator data.

At the classical current-algebra level, the phase space of $\dim G$ free scalars realizes the same brackets. With fields $X^a(\sigma)$, conjugate momenta $\Pi_a(\sigma)$ and $\{X^a(\sigma), \Pi_b(\sigma')\} = \delta_b^a \delta(\sigma - \sigma')$, set

$$J_a(\sigma) = f_{ab}^c X^b(\sigma) \Pi_c(\sigma) + \kappa_{ab} \partial_\sigma X^b(\sigma), \quad (6.90)$$

and a direct Poisson-bracket calculation gives

$$\begin{aligned} \{J_a(\sigma), J_b(\sigma')\} &= f_{ab}^c J_c(\sigma) \delta(\sigma - \sigma'), \\ \{J_a(\sigma), \Pi_b(\sigma')\} &= f_{ab}^c \Pi_c(\sigma) \delta(\sigma - \sigma') + \kappa_{ab} \partial_\sigma \delta(\sigma - \sigma'), \\ \{\Pi_a(\sigma), \Pi_b(\sigma')\} &= 0, \end{aligned} \quad (6.91)$$

which is (6.89) at $k = \infty$ under $\mathcal{J} \leftrightarrow J$ and $\mathcal{P} \leftrightarrow \Pi$. The only step requiring care is the cross term, where the two orderings of the orbital piece against $\partial_\sigma X$ produce derivative-of-delta contributions proportional to $X^e(\sigma') - X^e(\sigma)$, and smearing converts these into $f_{ab}^c \kappa_{cd} \partial_\sigma X^d$. The would-be central term in $\{J, J\}$ cancels for the same reason, surviving only in $\{J, \Pi\}$, exactly as in (6.89). Thus the classical free-scalar phase space has the correct contracted Poisson algebra and the required $2 \dim G$ phase-space directions. A quantum realization additionally requires choices of normal ordering and representation, whose convergence is not established by this calculation.

Before turning to the bulk we note a caveat about chirality, since the four checks above all concern the non-chiral boundary theory in which one WZW factor supplies the holomorphic sector and the other the antiholomorphic one, so that each chirality carries $\hat{\mathfrak{g}}_{\pm k}$ separately and the limiting central charge is $(\dim G, \dim G)$. A different and equally natural reading of $\text{CS}_k[G] \times \text{CS}_{-k}[G]$ treats it as a Chern–Simons theory for $G \times G$ at a level of split signature, whose boundary carries $\hat{\mathfrak{g}}_k \oplus \hat{\mathfrak{g}}_{-k}$ within a *single* chirality. Contracting that theory produces the WZW model on the cotangent group $T^*G = G \ltimes_{\text{Ad}^*} \mathfrak{g}^*$, a non-reductive model of the kind introduced by Nappi and Witten and developed in [51–54]. Its Sugawara central charge is $c = \dim \mathfrak{d} = 2 \dim G$ exactly, with no level shift [29], and it differs from $2 \dim G$ free scalars. Its invariant pairing has split signature $(\dim G, \dim G)$ [51, 54]. We mention this because the two readings are easy to confuse. They correspond to genuinely different boundary conditions on the same bulk, and only the non-chiral one is the subject of this subsection.

As a separate geometric consistency check, the tangent group also recovers the line labels of the limiting bulk. This explains why the boundary candidate and the regular bulk endpoint use compatible variables; it is not an independent proof of boundary-CFT convergence. Put BF theory on $\Sigma \times \mathbb{R}$ and split $S = \int \langle B \wedge F_A \rangle$ into temporal and spatial parts. The time components carry no derivatives and act as Lagrange multipliers, B_0 imposing $F_A = 0$ and A_0 imposing $D_A B = 0$, while the spatial components inherit the symplectic form

$$\omega = \int_\Sigma \epsilon^{ij} \langle \delta B_i \wedge \delta A_j \rangle. \quad (6.92)$$

The field B is therefore the momentum conjugate to A , and on the smooth regular locus the reduced phase space is the cotangent bundle $T^* \mathcal{M}_{\text{flat}}(\Sigma, G)$. Taking Σ to be an annulus, so that $\pi_1 = \mathbb{Z}$ and $\text{Hom}(\mathbb{Z}, TG) = TG \cong T^*G$, the variables before the residual quotient

are the holonomy $g = \text{Hol}_{S^1}(A) \in G$ and the flux $p = \oint_{S^1} B \in \mathfrak{g}^*$, with the residual gauge group acting by conjugation. More generally, on the smooth locus the moduli of flat TG -bundles is the tangent bundle of the moduli of flat G -bundles, which is the geometric content behind the identification $\text{CS}[TG] = BF[G]$.

Quantization now proceeds as it does for a particle on a group manifold. The symplectic potential is the one-form $\theta = \langle p, g^{-1}dg \rangle$, and choosing the vertical polarization, the fibres of $T^*G \rightarrow G$, selects wavefunctions independent of p ,

$$\mathcal{H} = L^2(G, dg). \quad (6.93)$$

Positions act by multiplication as $C_0(G)$ and momenta act as the left-invariant vector fields, and by Peter–Weyl, $L^2(G) = \bigoplus_{\lambda} V_{\lambda} \otimes V_{\lambda}^*$, the momentum labels are discrete because G is compact. This is the origin of the fact that the BF charge takes values in the weight lattice while the flux takes values in a conjugacy class, since the holonomy of A is a group-valued position, the flux of B is its conjugate momentum, and momenta on a compact group are discrete weights.

Imposing the residual conjugation symmetry gives the algebra of observables by the crossed product

$$D(G) = C_0(G) \rtimes_{\text{Ad}} G, \quad (6.94)$$

whose irreducible representations are classified by Mackey induction [5] as pairs

$$([g], R), \quad [g] \subset G \text{ a conjugacy class}, \quad R \in \text{Irr } C_G(g), \quad (6.95)$$

the representation attached to $([g], R)$ being the space of L^2 -sections of the bundle $G \times_{C_G(g)} V_R \rightarrow G/C_G(g) \cong [g]$. On the regular stratum the centralizer is the maximal torus, so R runs over the weight lattice and the labels reduce to a point of the open alcove together with a weight, which is the regular flux–charge label set of $BF[G]$. That this is genuinely the quantization of $\text{CS}[TG]$ is established as a theorem in [4], where the authors quantize Chern–Simons theory with gauge group $G \ltimes \mathfrak{g}^*$ and obtain the quantum double $D(G)$ of G as its quantized observable algebra. For $G = SU(2)$ this is the double familiar from Euclidean three-dimensional gravity at vanishing cosmological constant [55].

We close this subsection by two remarks. First, the naive guess that the line data should be representations of functions on TG overcounts by a square root, because $TG \cong T^*G$ is already a phase space and half of its directions are momenta, and the polarization is what removes the redundancy. Second, the choice of polarization is the same choice as the boundary condition. Polarizing along the fibers gives functions of the holonomy and polarizing along the base gives functions of the charge, and Peter–Weyl is the unitary relating them, so the two boundary conditions we compared in section 6.1 present one Hilbert space in two ways.

Scope of the boundary analysis. The $U(1)$ result is exact at the wavefunction and momentum-lattice level for the specified polarizations. For general G , the central-charge limit, tangent-space action, TG symmetry, affine contraction, and classical Poisson realization are mutually compatible checks. We have not proved convergence of the complete

boundary Hilbert spaces, quantum current-algebra representations, local fields, OPE or sewing coefficients, or correlators, and no statement here covers the singular bulk strata.

6.3 Boundary wavefunction for the non-abelian case

In this section, we will turn to the boundary wavefunction for non-abelian BF theory

$$S = \frac{i}{2\pi} \int \text{Tr} b \wedge f = \frac{i}{4\pi} \int \text{Tr} b \wedge da + \frac{i}{4\pi} \int \text{Tr} a \wedge db + \frac{i}{2\pi} \int \text{Tr} b \wedge a \wedge a, \quad (6.96)$$

where $f = da + a \wedge a$ and the kinetic terms of a and b are treated in equal footing. The variation of the action gives the boundary term up to equation of motion

$$\delta S = -\frac{i}{4\pi} \int_{\partial\mathcal{M}} \text{Tr}(b \wedge \delta a) - \frac{i}{4\pi} \int_{\partial\mathcal{M}} \text{Tr}(a \wedge \delta b). \quad (6.97)$$

Following the abelian case, we similarly introduce the combinations

$$A_{\pm} = \frac{1}{\sqrt{2}} \left(\frac{a}{r} \pm rb \right), \quad (6.98)$$

and define the holomorphic and anti-holomorphic components for both $A_{\pm}$ as before, and redefine the action by

$$S' = S - \frac{1}{2\pi} \int_{\partial\mathcal{M}} d^2z \text{Tr} (|A_+|^2 + |A_-|^2). \quad (6.99)$$

We also introduce the complex variables as

$$\xi = \frac{\tau_2}{i\pi} (A_+)_{\bar{z}}, \quad \bar{\xi} = -\frac{\tau_2}{i\pi} (A_-)_z. \quad (6.100)$$

Consider a constant flat background field on the boundary. First of all, via gauge transformation of a , one can always rotate a to a constant in the complexified Cartan direction, and we denote

$$a_{\bar{z}}|_{\partial\mathcal{M}} = \frac{i\pi}{\tau_2} u, \quad u \in \mathfrak{t}_{\mathbb{C}}. \quad (6.101)$$

For generic $a_{\bar{z}}$, we can further use the gauge transformation of $b \rightarrow b + D_a \beta$ and kill all non-Cartan component of b , so that

$$b_{\bar{z}}|_{\partial\mathcal{M}} = \frac{i\pi}{\tau_2} v, \quad v \in \mathfrak{t}_{\mathbb{C}}. \quad (6.102)$$

Then we have the parametrization

$$\xi = \frac{1}{\sqrt{2}} \left(\frac{u}{r} + vr \right), \quad \bar{\xi} = \frac{1}{\sqrt{2}} \left(\frac{\bar{u}}{r} - \bar{v}r \right), \quad (6.103)$$

and they are independent.

After we fixed both $a_{\bar{z}}$ and $b_{\bar{z}}$ along the Cartan direction, there exists two residue gauge transformations preserving the gauge choice. The first one is the single-valued lattice gauge

1834 transformation $g_{m,n} = \exp[2\pi i(n\sigma^1 + m\sigma^2)]$ with $n, m \in Q^\vee \subset \mathfrak{t}$, where $Q^\vee$ is the co-root
 1835 lattice, which induces

$$\xi \rightarrow \xi + \delta, \quad \bar{\xi} \rightarrow \bar{\xi} + \delta^*, \quad \delta = \frac{1}{\sqrt{2}r} (n + m\tau), \quad n, m \in Q^\vee, \quad (6.104)$$

1836 and one has $\delta\bar{\xi} = \delta\xi^*$. The second is the Weyl group $W \subset N(T)$ with N the normalizer of
 1837 T , which acts on both a and b and we write

$$(\xi, \bar{\xi}) \rightarrow (w\xi, w\bar{\xi}), \quad w \in W. \quad (6.105)$$

1838 The wavefunction will similarly satisfies

$$\Psi(\xi + \delta, \bar{\xi} + \delta^*) = \Psi(\xi, \bar{\xi}) \exp \left[\frac{\pi}{\tau_2} (\langle \xi, \delta^* \rangle + \langle \bar{\xi}, \delta \rangle + \langle \delta, \delta^* \rangle) \right], \quad (6.106)$$

1839 where the inner product $\langle \cdot, \cdot \rangle : \mathfrak{t} \times \mathfrak{t} \rightarrow \mathbb{R}$ is contracted using Killing metric. The the abelian
 1840 derivation goes through verbatim if we replace the products by $\langle \cdot, \cdot \rangle$ and the integer lattice
 1841 $\mathbb{Z}$ by co-root lattice $Q^\vee$. Introduce $f = \Psi e^{-\frac{\pi}{2\tau_2}(\langle \xi, \xi \rangle + \langle \bar{\xi}, \bar{\xi} \rangle)}$ and using $\delta - \delta^* = i\frac{\sqrt{2}}{2}m\tau_2$, we
 1842 have

$$\begin{aligned} f \left(\xi + \frac{n}{\sqrt{2}r}, \bar{\xi} + \frac{n}{\sqrt{2}r} \right) &= f(\xi, \bar{\xi}), \\ f \left(\xi + \frac{m\tau}{\sqrt{2}r}, \bar{\xi} + \frac{m\bar{\tau}}{\sqrt{2}r} \right) &= f(\xi, \bar{\xi}) e^{-\frac{2\pi i}{\sqrt{2}r} \langle m, \xi - \bar{\xi} \rangle + \frac{\pi\tau_2}{r^2} \langle m, m \rangle}, \end{aligned} \quad (6.107)$$

1843 for all $n, m \in Q^\vee$.

1844 Consider the Fourier expansion

$$f = \sum_{p_L, p_R \in \mathfrak{t}} c_{p_L, p_R} e^{2\pi i(\langle p_L, \xi \rangle - \langle p_R, \bar{\xi} \rangle)}, \quad (6.108)$$

1845 the first relation require $\frac{1}{\sqrt{2}r} \langle p_L - p_R, n \rangle \in \mathbb{Z}$ for all $n \in Q^\vee$, and hence

$$p_L - p_R = \sqrt{2}r\eta^\vee, \quad \eta \in P, \quad (6.109)$$

1846 where P is the weight lattice, and $\eta^\vee \in \mathfrak{t}$ is the dual of $\eta \in \mathfrak{t}^*$, raising by the Killing metric.

1847 The second relation gives the recursion relation

$$c_{p_L + \frac{m}{\sqrt{2}r}, p_R + \frac{m}{\sqrt{2}r}} = c_{p_L, p_R} \exp \left[\frac{2\pi i}{\sqrt{2}r} (\tau \langle p_L, m \rangle - \bar{\tau} \langle p_R, n \rangle) - \frac{\pi\tau_2}{r^2} \langle m, m \rangle \right], \quad (6.110)$$

1848 which is solved by the ansatz

$$c_{p_L, p_R} = \exp [i\pi\tau \langle p_L, p_L \rangle - i\pi\bar{\tau} \langle p_R, p_R \rangle]. \quad (6.111)$$

1849 Due to the constraint $p_L - p_R = \sqrt{2}r\eta^\vee$ with $\eta \in P$, the independent wavefunctions are
 1850 labeled by a discrete label $\eta \in P$ and also a continuous label $\varphi \in \mathfrak{t}/Q^\vee$, parametrized by

$$p_L = \frac{1}{\sqrt{2}} \left(\frac{\mu + \varphi}{r} + r\eta^\vee \right), \quad p_R = \frac{1}{\sqrt{2}} \left(\frac{\mu + \varphi}{r} - r\eta^\vee \right), \quad \mu \in Q^\vee, \quad (6.112)$$

1851 and we define the theta functions

$$\Theta_{\varphi,\eta}(\xi, \bar{\xi}|\tau) = \sum_{\mu \in Q^\vee} \exp \left[i\pi\tau \langle p_L, p_L \rangle - i\pi\bar{\tau} \langle p_R, p_R \rangle + 2\pi i (\langle p_L, \xi \rangle - \langle p_R, \bar{\xi} \rangle) \right]. \quad (6.113)$$

1852 Then let us consider the action of Weyl group W . The inner product $\langle \cdot, \cdot \rangle$, the co-root
1853 lattice $Q^\vee$ and the weight lattice P are all Weyl invariant, and one has

$$\Theta_{\varphi,\eta}(w\xi, w\bar{\xi}|\tau) = \Theta_{w^{-1}\varphi, w^{-1}\eta}(\xi, \bar{\xi}|\tau), \quad w \in W. \quad (6.114)$$

1854 Physical wavefunction should be Weyl invariant, therefore the labels are reduced to the
1855 diagonal Weyl orbits

$$[\varphi, \eta] \in (\mathfrak{t}/Q^\vee \times P)/W. \quad (6.115)$$

1856 We can use the Weyl rotation to choose a unique representative φ in the open fundamental
1857 alcove $Y = (\mathfrak{t}/Q^\vee)/W$ and leaves $\eta \in P$ unconstrained:

$$[\varphi, \eta] \in Y \times P, \quad (6.116)$$

1858 which is precisely the regular part of the simple object set of the quantum double $D(G)$.

1859 Assembling the pieces, the boundary wavefunctions of non-abelian BF theory over the
1860 regular sector are

$$\Psi_{[\varphi,\eta]}(\xi, \bar{\xi}|\tau) = \frac{1}{C(\xi, \bar{\xi}|\tau)} \frac{e^{\frac{\pi}{2\tau_2}(\langle \xi, \xi \rangle + \langle \bar{\xi}, \bar{\xi} \rangle)}}{|\eta(\tau)|^{2\text{rk}(G)}} \sum_{w \in W} \Theta_{w\varphi, w\eta}(\xi, \bar{\xi}|\tau), \quad (6.117)$$

1861 where the denominator comes from the 1-loop contribution of the path-integral. Then
1862 $|\eta(\tau)|^{2\text{rk}(G)}$ comes from the Cartan oscillators, which are the same to the abelian case. The
1863 $C(\xi, \bar{\xi}|\tau)$ is from the root part.

1864 Before determine the $C(\xi, \bar{\xi}|\tau)$ factor, it is illuminating to consider the modular prop-
1865 erty of the wavefunctions. For that let us assume $C(\xi, \bar{\xi}|\tau)$ is modular invariant, which
1866 turns out to be true at the end of day. The Θ -function (6.113) transform under the T/S -
1867 transformation as

$$\begin{aligned} \Theta_{\varphi,\eta}(\xi, \bar{\xi}|\tau+1) &= e^{2\pi i \langle \varphi, \eta^\vee \rangle} \Theta_{\varphi,\eta}(\xi, \bar{\xi}|\tau), \\ \Theta_{\varphi,\eta} \left(\frac{\xi}{\tau}, \frac{\bar{\xi}}{\bar{\tau}} \middle| -\frac{1}{\tau} \right) &= |\tau|^{\text{rk}(G)} e^{\frac{\pi i}{\tau} \langle \xi, \xi \rangle - \frac{\pi i}{\bar{\tau}} \langle \bar{\xi}, \bar{\xi} \rangle} \int_{\mathfrak{t}/L} d\varphi' \sum_{\eta' \in P} e^{-2\pi i (\langle \varphi, \eta'^* \rangle + \langle \varphi', \eta^* \rangle)} \Theta_{\varphi', \eta'}(\xi, \bar{\xi}|\tau), \end{aligned} \quad (6.118)$$

1868 and the Gaussian factor $G(\xi, \bar{\xi}|\tau) = e^{\frac{\pi}{2\tau_2}(\langle \xi, \xi \rangle + \langle \bar{\xi}, \bar{\xi} \rangle)}$ transform as

$$\begin{aligned} G(\xi, \bar{\xi}|\tau+1) &= G(\xi, \bar{\xi}|\tau), \\ G \left(\frac{\xi}{\tau}, \frac{\bar{\xi}}{\bar{\tau}} \middle| -\frac{1}{\tau} \right) &= G(\xi, \bar{\xi}|\tau) \exp \left[-\frac{\pi i}{\tau} \langle \xi, \xi \rangle + \frac{i\pi}{\bar{\tau}} \langle \bar{\xi}, \bar{\xi} \rangle \right], \end{aligned} \quad (6.119)$$

1869 and the η -function

$$\eta(\tau+1) = e^{\frac{i\pi}{12}} \eta(\tau), \quad \eta \left(-\frac{1}{\tau} \right) = \sqrt{-i\tau} \eta(\tau). \quad (6.120)$$

1870 Combined all together, we find the wavefunction transform under T

$$\begin{aligned} \Psi_{[\varphi, \eta]}(\xi, \bar{\xi}|\tau + 1) &= e^{2\pi i \langle \varphi, \eta^\vee \rangle} \Psi_{[\varphi, \eta]}(\xi, \bar{\xi}|\tau), \\ \Psi_{[\varphi, \eta]} \left(\frac{\xi}{\tau}, \frac{\bar{\xi}}{\bar{\tau}} \middle| -\frac{1}{\tau} \right) &= \int_Y d\varphi' \sum_{\eta' \in P} \sum_{w \in W} e^{-2\pi i (\langle w\varphi, \eta'^* \rangle + \langle \varphi', w\eta^* \rangle)} \Psi_{[\varphi', \eta']}(\xi, \bar{\xi}|\tau), \end{aligned} \quad (6.121)$$

1871 and they exactly match the S/T -kernel in the regular sector derived in []. It also implies
1872 that $C(\xi, \bar{\xi}|\tau)$ should be invariant under modular transformation.

1873 We then turn to the $C(\xi, \bar{\xi}|\tau)$ factor. Instead of evaluating the one-loop integral
1874 directly, we will determine $C(\xi, \bar{\xi}|\tau)$ using a trick based on the following assumption

- 1875 • $C(\xi, \bar{\xi}|\tau)$ is the generalization of the $\eta(\tau)$ coupling to the background fields ξ and $\bar{\xi}$.
- 1876 • $C(\xi, \bar{\xi}|\tau)$ can be read from the one-loop contribution of root parts of the parent
1877 Chern-Simons theory $G_k \times G_{-k}$ in the large k limit.
- 1878 • $C(\xi, \bar{\xi}|\tau)$ is modular invariant.

1879 Recall that the action of the non-abelian BF theory can be written as large k limit of the
1880 parent Chern-Simons theory $G_k \times G_{-k}$ with field redefinition

$$a = \frac{\mathcal{A}_+ + \mathcal{A}_-}{2}, \quad b = k(\mathcal{A}_+ + \mathcal{A}_-), \quad \mathcal{A}_\pm = a \pm \frac{b}{2k}, \quad (6.122)$$

1881 where we use $\mathcal{A}_\pm$ for level- k Chern-Simons theory to distinguish with $A_\pm$. The paired
1882 Chern-Simons gauge fields and their holonomies on the boundary have the form

$$\mathcal{A}_\pm|_{\partial\mathcal{M}} = a|_{\partial\mathcal{M}} + \frac{1}{2k}b|_{\partial\mathcal{M}} \quad \Rightarrow \quad u_\pm = u \pm \frac{v}{2k}, \quad (6.123)$$

1883 and in the large k limit, we have both $u_\pm \rightarrow u$. One then expect $C(\xi, \bar{\xi}|\tau)$ is the gener-
1884 alization of the η -functions coupling to the background u , and the oscillator contributions
1885 for a given root α is therefore proportional to

$$\prod_{n \geq 1} \frac{1}{|1 - e^{2\pi i \alpha(u)} q^n|^2}, \quad (6.124)$$

1886 and we propose the following expression of $C(\xi, \bar{\xi}|\tau)$

$$C(u|\tau) = \exp \left[-\frac{2\pi h^\vee}{\tau_2} \langle \text{Im}(u), \text{Im}(u) \rangle \right] \prod_{\alpha > 0} \left| \frac{i\vartheta_1(\alpha(u)|\tau)}{\eta(\tau)} \right|^2, \quad (6.125)$$

1887 where the prefactor $\exp \left[-\frac{2\pi h^\vee}{\tau_2} \langle \text{Im}(u), \text{Im}(u) \rangle \right]$ is inserted so that $C(\xi, \bar{\xi}|\tau)$ is a modular
1888 invariant.

1889 Before moving on, we will comment on the meaning of u and v . Parametrize $g =$
1890 $\exp \left(iX/\sqrt{k} \right)$, and consider the left and right actions as

$$g \rightarrow h_L g h_R^{-1}, \quad h_L = e^{i\vartheta_L}, h_R = e^{i\vartheta_R}, \quad \vartheta_L, \vartheta_R \in \mathfrak{t}, \quad (6.126)$$

then it is easy to see that, the translation P along the Cartan direction corresponds to $\vartheta_L = -\vartheta_R$, and the rotation R corresponds to $\vartheta_L = \vartheta_R$. And we can write

$$g \rightarrow e^{\frac{i}{2}(\vartheta_L + \vartheta_R)} e^{\frac{i}{2}(\vartheta_L - \vartheta_R)} g e^{\frac{i}{2}(\vartheta_L - \vartheta_R)} e^{-\frac{i}{2}(\vartheta_L + \vartheta_R)} = R(\vartheta_L + \vartheta_R) \circ P(\vartheta_L - \vartheta_R) \circ g. \quad (6.127)$$

In the large k limit, if we want a finite translation of X , one should have $\frac{1}{2}(\vartheta_L - \vartheta_R) \sim 1/\sqrt{k} \rightarrow 0$, which indicates $\vartheta_L = \vartheta_R$. Recall that the left and right $U(1)^{\text{rk}G}$ backgrounds are given by $\xi(\tau)$ and $\bar{\xi}(\bar{\tau})$, then we can identify the background for translation and rotation separately as $(\xi - \bar{\xi}^*)/2$ and $(\xi + \bar{\xi}^*)/2$, which gives

$$\frac{1}{2}(\xi - \bar{\xi}^*) = \frac{r}{\sqrt{2}}v, \quad \frac{1}{2}(\xi + \bar{\xi}^*) = \frac{1}{\sqrt{2}r}u. \quad (6.128)$$

Therefore u, v can be understood as the background of the rotation and translation.

We then consider the symmetry boundary where all b -lines condense. Setting $\eta = 0$ and $u = 0$, we have

$$\lambda \equiv \lambda_L = \lambda_R = \frac{1}{\sqrt{2}} \left(\frac{\mu + \varphi}{r} \right), \quad \mu \in Q^\vee, \quad (6.129)$$

and

$$\xi = \frac{r}{\sqrt{2}}v, \quad \bar{\xi} = -\frac{r}{\sqrt{2}}\bar{v}, \quad (6.130)$$

so that

$$\Theta_{\varphi, \eta}(\xi, \bar{\xi}|\tau) = \sum_{\mu \in Q^\vee} \exp \left[i\pi(\tau - \bar{\tau})\langle \lambda, \lambda \rangle + 2\pi i\langle \lambda, (\xi - \bar{\xi}) \rangle \right], \quad (6.131)$$

and we have

$$\int_{\mathfrak{t}/Q^\vee} d\varphi \Theta_{\varphi, \eta}(\xi, \bar{\xi}|\tau) = \int_{\mathfrak{t}/Q^\vee} d\varphi \sum_{\mu \in Q^\vee} \exp \left[i\pi(\tau - \bar{\tau})\langle \lambda, \lambda \rangle + 2\pi i\langle \lambda, (\xi - \bar{\xi}) \rangle \right]. \quad (6.132)$$

and we can write

$$\int_{\mathfrak{t}/Q^\vee} d\varphi \sum_{\mu \in Q^\vee} = \int_{\mathfrak{t}} d\varphi, \quad (6.133)$$

so that the integral becomes a Gauss integral

$$\int_{\mathfrak{t}} d\varphi \exp \left[-\frac{\pi\tau_2}{r^2} \langle \varphi, \varphi \rangle + \frac{\sqrt{2}\pi i}{r} \langle \varphi, \xi - \bar{\xi} \rangle \right] = \left(\frac{2R^2}{\tau_2} \right)^{\frac{1}{2}\text{rk}G} \exp \left[-\frac{\pi}{2\tau_2} \langle \xi - \bar{\xi}, \xi - \bar{\xi} \rangle \right], \quad (6.134)$$

where $R = r/\sqrt{2}$. Combined with the prefactor $e^{\frac{\pi}{2\tau_2}(\langle \xi, \xi \rangle + \langle \bar{\xi}, \bar{\xi} \rangle)}$, we obtain

$$e^{-\frac{\pi}{2\tau_2}(|\xi|^2 + |\bar{\xi}|^2)} Z_{\mathcal{L}_2^{BF}} = \frac{1}{C(u \rightarrow 0|\tau)} \left(\frac{\sqrt{2}R}{|\eta(\tau)|^2 \sqrt{\tau_2}} \right)^{\text{rk}G} e^{-\frac{2\pi R^2}{\tau_2} \langle v_2 + \tau v_1, v_2 + \bar{\tau} v_1 \rangle}, \quad (6.135)$$

where $v = v_2 + \tau v_1$ is the background of the translation symmetry along the Cartan directions.

We then focus on the $u \rightarrow 0$ limit of $C(u|\tau)$. Recall that the ϑ_1 function satisfy

$$\vartheta_1(z|\tau) \sim 2\eta(\tau)^3 \sin \pi z + \cdots, \quad (6.136)$$

1909 in the $z \rightarrow 0$ limit, therefore one has

$$C(u|\tau) \sim \prod_{\alpha>0} |\eta(\tau)|^4 |\sin \pi \alpha(u)|^2 \rightarrow 0. \quad (6.137)$$

1910 To regularize that, notice that u is the background of the rotation symmetry. Let $X_\alpha \in \mathbb{C}$
 1911 denote the complex coordinate associated with the pair $(\alpha, -\alpha)$, and the rotation through
 1912 the angle $\theta = 2\pi z$ acts by $g_\theta X_\alpha = e^{i\theta} X_\alpha$. Denote the momentum on the complex plane as
 1913 p^2 , and consider the two dimensional heat kernel

$$K_s(x, y) = \langle x | e^{-sp^2} | y \rangle = \frac{1}{4\pi s} \exp \left[-\frac{|x-y|^2}{4s} \right], \quad s = 2\pi\tau_2 \quad (6.138)$$

1914 and consider the partition function

$$\text{Tr}_{L^2(\mathbb{C})} \left(g_\theta e^{-sp^2} \right) = \int_{\mathbb{C}} d^2x K_s(x, g_\theta x) = \frac{1}{4\pi s} \int_{\mathbb{C}} d^2x \exp \left[-\frac{|1 - e^{i\theta}|^2}{4s} |x|^2 \right]. \quad (6.139)$$

1915 For $\theta \neq 0$, it gives

$$\text{Tr}_{L^2(\mathbb{C})} \left(g_\theta e^{-sp^2} \right) = \frac{1}{|1 - e^{i\theta}|^2} = \frac{1}{4 \sin^2 \pi z}. \quad (6.140)$$

1916 Combining all X_α , they give the the divergent part. On the other hand, if we set $\theta = 0$,
 1917 we have instead

$$\text{Tr}_{L^2(\mathbb{C})} \left(e^{-sp^2} \right) = \frac{\text{Vol}(\mathbb{C})}{4\pi s} = \frac{\text{Vol}(\mathbb{C})}{8\pi^2 \tau_2}. \quad (6.141)$$

1918 Therefore at $u \rightarrow 0$, the $C(u|\tau)$ should behave as

$$C(u \rightarrow 0|\tau) = \prod_{\alpha>0} \frac{8\pi^2 |\eta(\tau)|^4 \tau_2}{\text{Vol}(\mathbb{C}_\alpha)}. \quad (6.142)$$

1919 Finally, we have the partition function

$$e^{-\frac{\pi}{2\tau_2}(|\xi|^2 + |\bar{\xi}|^2)} Z_{\mathcal{L}_2^{BF}} = \prod_{\alpha>0} \left(\frac{\text{Vol}(\mathbb{C}_\alpha)}{8\pi^2 |\eta(\tau)|^4 \tau_2} \right) \left(\frac{\sqrt{2}R}{|\eta(\tau)|^2 \sqrt{\tau_2}} \right)^{\text{rk}G} e^{-\frac{2\pi R^2}{\tau_2} \langle v_2 + \tau v_1, v_2 + \bar{\tau} v_1 \rangle}. \quad (6.143)$$

1920 6.4 2d Irrational CFT from flat gauging G

1921 The candidate boundary theory of section 6.2 is a sigma model into $\mathfrak{g}$ whose global sym-
 1922 metry contains the adjoint action of G , $X \mapsto \text{Ad}_g X$, the compact factor of the TG symme-
 1923 try (6.86). In this subsection we gauge this compact factor by *flat* gauge fields: the back-
 1924 ground connection is made dynamical without a kinetic term, so the path integral sums
 1925 over flat G -bundles only, a continuous orbifold in the sense of Gaberdiel and Suchanek [56],
 1926 and the result is an irrational conformal field theory of Runkel–Watts type [57]: a contin-
 1927 uum of spinless primaries supported on the open alcove, with no normalizable vacuum and
 1928 with the degenerate values excluded at the walls.

Flat gauging on the torus. Coupling the scalars to a flat background computes partition functions twisted by its holonomies, so the gauged partition function is the integral of the twisted partition functions over the moduli of flat G -bundles on T^2 : commuting pairs of holonomies modulo conjugation. For G compact, connected, simple, and simply connected every commuting pair lies in a common maximal torus, and the residual identification is the diagonal Weyl group, so the moduli space is $((\mathfrak{t}/Q^\vee) \times (\mathfrak{t}/Q^\vee))/W$. This is not a new arena: the pair (u_1, u_2) , with $e^{2\pi i u_1}$ the spatial and $e^{2\pi i u_2}$ the temporal holonomy, is the rotation background $u = u_2 + \tau u_1$ of section 6.3, its $Q^\vee$ -periodicities being the lattice gauge transformations there and its Weyl-reduced spatial factor being the alcove Y that carries the flux label. We define

$$Z_{\mathfrak{g}/G}(\tau) = \frac{\mathcal{N}}{|W|} \int_{(\mathfrak{t}/Q^\vee)^2} du_1 du_2 Z_{\mathfrak{g}}[u_1, u_2](\tau). \quad (6.144)$$

The measure is flat on the regular stratum: reducing the class integral over G to T costs the Weyl density $\prod_{\alpha>0} |2 \sin \pi \alpha(u_1)|^2 = \prod_{\alpha>0} |1 - e^{2\pi i \alpha(u_1)}|^2$, which cancels against the Jacobian $\prod_{\alpha>0} |1 - e^{2\pi i \alpha(u_1)}|^{-2}$ from solving the commuting constraint for the second holonomy, per root plane the same determinant as the twisted heat-kernel trace (6.140). Modular invariance forces the same conclusion, since a modular transformation mixes the pair, $(u_1, u_2) \mapsto (-u_2, u_1)$ under S , preserving $du_1 du_2$ but no nonconstant density in u_1 alone. The constant $\mathcal{N}$, a ratio of gauge-group volumes, is a counterterm-type normalization that we do not fix; and since Ad kills the center (cf. the remark below (6.86)), the difference between gauging G and $G/Z(G)$ is a decoupled center sector that plays no role in what follows.

Twisted blocks from the boundary wavefunctions. Under $\text{Ad}(T)$ the target splits into $\text{rk} G$ neutral Cartan directions and one complex plane $\mathbb{C}_\alpha$ per positive root, rotated by the angle $2\pi \alpha(u)$. The Cartan scalars are untwisted and neutral, so they contribute the factor $(\sqrt{2}R/(|\eta(\tau)|^2 \sqrt{\tau_2}))^{\text{rk} G}$ of (6.143), independently of u . Each root plane contributes the twisted complex-boson block: writing $u = u_2 + \tau u_1$, $s = \alpha(u_1) \in (0, 1)$ and $t = \alpha(u_2)$, the product representation of ϑ_1 gives

$$e^{2\pi \tau_2 s^2} \left| \frac{\eta(\tau)}{\vartheta_1((t + s\tau) | \tau)} \right|^2 = (q\bar{q})^{\frac{s(1-s)}{2} - \frac{1}{12}} \prod_{n \geq 0} |1 - q^{n+s} e^{2\pi i t}|^{-2} \prod_{n \geq 1} |1 - q^{n-s} e^{-2\pi i t}|^{-2}, \quad (6.145)$$

the trace over a twisted Fock space with modings shifted by $\pm s$, the temporal twist grading the states by their $\mathfrak{t}$ -weight; the ground state is spinless, $h = \bar{h} = \frac{1}{2}s(1-s)$, because the twist is spatial and shifts left and right movers symmetrically. Multiplying over positive roots and using $\sum_{\alpha>0} \alpha \otimes \alpha = h^\vee \langle \cdot, \cdot \rangle$, under which the prefactors assemble into that of (6.125),

$$Z_{\mathfrak{g}}[u_1, u_2](\tau) = \left(\frac{\sqrt{2}R}{|\eta(\tau)|^2 \sqrt{\tau_2}} \right)^{\text{rk} G} \frac{1}{C(u_2 + \tau u_1 | \tau)}, \quad (6.146)$$

with C exactly as proposed in (6.125). Read from right to left, the one-loop factor of the boundary wavefunctions (6.117) is the inverse adjoint-twisted block of the candidate

boundary theory of section 6.2, the bulk flux entering as the spatial twist in accordance with the identification of u as the rotation background in (6.128); read from left to right, the free-field computation derives the three assumptions of section 6.3, including the modular-invariance prefactor. Two further checks against that section. First, at the untwisted point the regularization (6.142) reduces the integrand to the flux-condensed partition function (6.143) at vanishing background,

$$Z_{\mathfrak{g}}[0,0] = \left(\frac{\sqrt{2}R}{|\eta(\tau)|^2 \sqrt{\tau_2}} \right)^{\text{rk}G} \prod_{\alpha>0} \frac{\text{Vol}(\mathbb{C}_\alpha)}{|\eta(\tau)|^4 \tau_2} = Z_{\mathcal{L}_2^{BF}} \Big|_{u=v=0}, \quad (6.147)$$

the ungauged free-scalar partition function computed there; its volume divergences $\text{Vol}(\mathbb{C}_\alpha)$ are confined to the measure-zero untwisted locus, the twist lifting the root zero modes exactly as in (6.140). Second, modular invariance of (6.144) follows from that of C , established through the S/T kernel (6.121), together with the invariance of $du_1 du_2$ and of $\sqrt{\tau_2} |\eta(\tau)|^2$. In SymTFT terms, the gauging replaces the flux-condensing symmetry boundary behind (6.143) by its charge-condensing counterpart; we do not present the latter as a sum over wavefunctions, because the charge algebra is supported at the identity flux, on the singular stratum outside the regular family (6.117), and the background-integral definition (6.144) uses only the u -dependence of the regular data (for gauging of continuous symmetry categories see [7, 9]).

The gauged partition function and its spectrum. On Weyl-invariant integrands $\frac{1}{|W|} \int_{(\mathfrak{t}/Q^\vee)^2} du_1 du_2 = \int_Y d\varphi \int_{\mathfrak{t}/Q^\vee} du_2$ up to the measure-zero singular twists, and we rename the spatial twist $\varphi \in Y$ to match the bulk flux label. The u_2 -integral projects each twisted sector onto the states of vanishing $\mathfrak{t}$ -weight, the invariants of the residual centralizer T of the regular twist, and this projection is level matching: from the charge assignments in (6.145), a state of total weight λ has

$$h - \bar{h} \equiv \lambda(\varphi) \pmod{1}, \quad (6.148)$$

so the retained states carry integer spin, while the continuous-spin states, such as $a_{-s_\alpha}^{(\alpha)} |0\rangle_{\text{tw}}$ with $h - \bar{h} = s_\alpha$, are removed. Carrying out the projection in (6.145) gives

$$Z_{\mathfrak{g}/G}(\tau) = \mathcal{N} \left(\frac{\sqrt{2}R}{|\eta(\tau)|^2 \sqrt{\tau_2}} \right)^{\text{rk}G} \int_Y d\varphi (q\bar{q})^{h_{\text{tw}}(\varphi) - \frac{\dim G - \text{rk}G}{24}} \mathcal{F}_\varphi(q, \bar{q}), \quad (6.149)$$

with $s_\alpha := \alpha(\varphi) \in (0,1)$ for $\varphi \in Y$ and

$$\begin{aligned} \mathcal{F}_\varphi(q, \bar{q}) &= \int_{\mathfrak{t}/Q^\vee} du_2 \prod_{\alpha>0} \prod_{n \geq 0} |1 - q^{n+s_\alpha} e^{2\pi i \alpha(u_2)}|^{-2} \prod_{n \geq 1} |1 - q^{n-s_\alpha} e^{-2\pi i \alpha(u_2)}|^{-2} \\ &= 1 + \sum_{\alpha>0} \left[(q\bar{q})^{s_\alpha} + (q\bar{q})^{1-s_\alpha} \right] + \frac{\dim G - \text{rk}G}{2} (q + \bar{q}) + \dots \end{aligned} \quad (6.150)$$

The $(q + \bar{q})$ term collects, for each positive root, the holomorphic invariant $a_{-s_\alpha}^{(\alpha)} a_{-(1-s_\alpha)}^{(-\alpha)}$ at integer level, a twisted-sector shadow of the invariant chiral algebra: the G -invariant

subalgebra of the $\mathfrak{g}$ -valued Heisenberg algebra retains no spin-one currents but contains the Casimir currents of spins $e_i + 1$, acting at central charge $\dim G$.

The twisted ground-state weight has the closed form

$$h_{\text{tw}}(\varphi) = \bar{h}_{\text{tw}}(\varphi) = \frac{1}{2} \sum_{\alpha > 0} \alpha(\varphi) (1 - \alpha(\varphi)) = \rho(\varphi) - \frac{h^\vee}{2} \langle \varphi, \varphi \rangle, \quad (6.151)$$

strictly concave on the closed alcove, positive on Y , and vanishing precisely at the vertices at which $e^{2\pi i \varphi}$ is central, where the adjoint twist trivializes. Its unique maximum sits at the interior point φ_* determined by $h^\vee \langle \varphi_*, x \rangle = \rho(x)$ for all $x \in \mathfrak{t}$, where the Freudenthal–de Vries strange formula $\langle \rho, \rho \rangle = h^\vee \dim G / 12$ gives

$$0 < h_{\text{tw}}(\varphi) \leq h_{\text{tw}}(\varphi_*) = \frac{\langle \rho, \rho \rangle}{2h^\vee} = \frac{\dim G}{24}; \quad (6.152)$$

one checks $\theta(\varphi_*) = (h^\vee - 1)/h^\vee < 1$ for the highest root θ , so $\varphi_* \in Y$. The total ground-state exponent in (6.149) is $h_{\text{tw}}(\varphi) - \dim G / 24$, consistent with the limiting central charge (6.80), and it vanishes at φ_* , the nonabelian analogue of the antiperiodic point. Restoring the Cartan momentum continuum from the factor $\tau_2^{-\text{rk} G / 2} \propto \int_{\mathfrak{t}^*} dp (q\bar{q})^{\frac{1}{2} \langle p, p \rangle}$, the primaries of the gauged theory are

$$h = \bar{h} = h_{\text{tw}}(\varphi) + \frac{1}{2} \langle p, p \rangle, \quad (\varphi, p) \in Y \times \mathfrak{t}^*, \quad (6.153)$$

in the normalization fixed by (6.84); regular alcove points have trivial stabilizer, so $\varphi \in Y$ exhausts the Weyl quotient and p is unconstrained.

This is a Runkel–Watts-type spectrum. As in (7.25), every primary is spinless and belongs to a continuum attached to the open alcove with its walls removed, and the spinlessness has the same origin, a symmetric twist of a nonchiral free field. The differences are quantitative: the central charge is $\dim G$ rather than 1, and the Runkel–Watts continuum $h = x^2/4$ is one-dimensional and unbounded, whereas the twist part of (6.153) is $\text{rk} G$ -dimensional and bounded by $\dim G / 24$, the unbounded direction being restored by the Cartan momenta. For $G = SU(2)$ the alcove is an interval, $s = \alpha(\varphi) \in (0, 1)$, and $h_{\text{tw}} = \frac{1}{2} s(1-s) \in (0, \frac{1}{8}]$ with $\frac{1}{8} = \frac{3}{24} = \dim SU(2) / 24$ attained at $s = \frac{1}{2}$. Before projection, the φ -twisted sector is graded by the $\mathfrak{t}$ -weight, $\mathcal{H}_\varphi = \bigoplus_\eta \mathcal{H}_{\varphi, \eta}$, the weight- η states being candidate junction sectors of the bulk line $W_{[\varphi, \eta]}$, in parallel with the fixed-charge sectors of section 7.3: the defect sectors of the flat-gauged boundary are organized by the regular flux–charge label set (1.2) underlying Theorem 3.2, realized for η in the root lattice $Q \subseteq P$, with $P/Q \cong \widehat{Z(G)}$ labeling center sectors invisible to the adjoint fields (cf. section 6.2). The gauged theory also carries the dual noninvertible lines $\text{Rep}(G)$, the Wilson lines of the flat gauge field, acting on the twisted continuum by $\chi_R(e^{2\pi i \varphi})$; characters separate regular conjugacy classes, so distinct alcove points are genuinely superselected.

Irrationality. Three features of (6.149) place the gauged theory outside the rational and quasi-rational classes, in the sense in which Runkel and Watts characterize $\mathcal{M}_\infty$ [57, Sections 1 and 2.1]. (i) The spectral measure is absolutely continuous, the pushforward of $d\varphi dp$ under (6.153), so the character decomposition of $Z_{\mathfrak{g}/G}$, with respect to the Virasoro algebra

or to the invariant chiral algebra, is an integral rather than a countable sum. (ii) There is no normalizable vacuum: the untwisted sector, the central values of the spatial holonomy, is a measure-zero locus at which the integrand degenerates to (6.147), and its T -invariant slice alone is not modular invariant, S exchanging projection and twist as in (6.121); the twisted continuum is its modular completion, and the identity survives only as its zero-weight limit, the continuous-orbifold mechanism of [56, Sections 1 and 4]. (iii) The excluded set is the alcove boundary, where root planes develop zero modes, the centralizer jumps, and the blocks (6.145) degenerate, matching the integers removed from (7.25) and reached at the walls through (7.29); this is the boundary incarnation of the regular-sector restriction that governs the bulk statements of this paper.

7 The large- k limit of interface theories

So far we hold the level fixed inside each comparison and treat the family

$$\mathsf{T}_k := \mathrm{CS}_k[G] \times \mathrm{CS}_{-k}[G], \quad k \in \mathbb{Z}_{>0}, \quad (7.1)$$

as an externally parametrized sequence. A physical realization of the limit should place different members of (7.1) in one spacetime and specify what joins them. This section is an extension of, rather than an input to, the bulk convergence theorems. Subsection 7.1 introduces a finite-weight refinement criterion for this problem. Proposition 7.2 then proves an exact obstruction for $SU(2)$, an adjacent level jump, and nonnegative matrices obeying the necessary genus-one S - and θ -conditions. The coset branching rules and weight-displacement identities are exact properties of the proposed gapless chiral algebra, whereas construction of the physical doubled bulk-coupled interface and its Runkel–Watts limit remain conjectural.

7.1 The interface chain and refinement

The category of the doubled theory is the Drinfeld center $Z(\mathcal{C}_k) \simeq \mathcal{C}_k \boxtimes \mathcal{C}_k^{\mathrm{rev}}$. Divide a three-manifold M into parallel slabs $R_K, \dots, R_{K+n}$ meeting along two-dimensional surfaces $R_{K+j} \cap R_{K+j+1} = \Sigma_{K+j+1/2}$, and place T_{K+j} in R_{K+j} . This gives the chain

$$\mathsf{T}_K \big|_{\Sigma_{K+1/2}} \mathsf{T}_{K+1} \big|_{\Sigma_{K+3/2}} \cdots \big|_{\Sigma_{K+n-1/2}} \mathsf{T}_{K+n}, \quad K \geq 1, \quad (7.2)$$

which is illustrated in figure 1.

We seek a two-dimensional theory $\mathfrak{J}_k$ on $\Sigma_{k+1/2}$ coupled to both adjacent bulks. The coupling must specify how a line of T_k continues as a line of T_{k+1} and cancel the anomaly localized by the level jump. An intrinsic two-dimensional theory without these bulk couplings is therefore not the complete interface datum, and the detailed construction of the doubled bulk-coupled interfaces $\mathfrak{J}_k$ remains open. For a gapped interface, only topological transmission data survive in the infrared, described categorically by a bimodule over the two line categories [58–60]. At a gapless conformal fixed point, the interface also carries local operators with nonzero conformal weights [61].

We use the regular coordinates in (2.53), with $a = \lambda_+$ and $b = \lambda_-$. We write $I \in \mathrm{Irr}Z(\mathcal{C}_{k'})$ with coordinates (x_I, η_I) and $J \in \mathrm{Irr}Z(\mathcal{C}_k)$ with coordinates (y_J, ω_J) , keeping

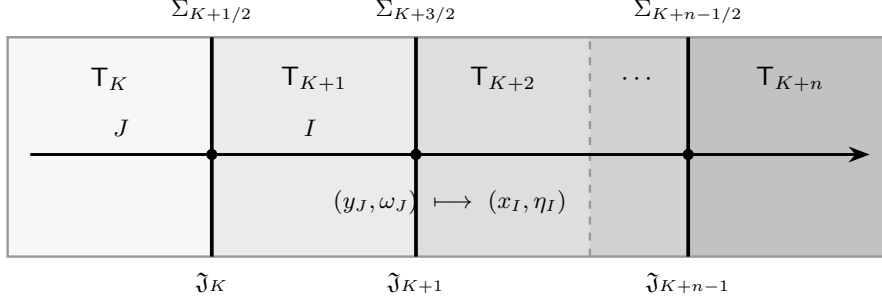


Figure 1. The chain (7.2). Each slab carries one member of the family (7.1), and each thick vertical line carries an interface theory $\mathfrak{J}_k$ coupled to the two slabs it separates. The horizontal arrow follows a line of T_K across the chain. At every fixed conformal-weight cutoff, the refinement requirement (7.3) asks that such crossings preserve the charge η and move the flux x by a bounded multiple of the mesh $(k + h^\vee)^{-1}$, with a bound independent of k .

I in the outgoing role and J in the incoming role throughout. For an interface $\mathfrak{J}_k$ between neighboring levels, let $\mathcal{H}_{IJ}^{\mathfrak{J}_k}$ denote the space of local interface operators at which an incoming line J continues as an outgoing line I . When $\mathfrak{J}_k$ is conformal, this space is graded by the scaling dimension $\Delta = h + \bar{h}$, and we write $\mathcal{H}_{IJ}^{\mathfrak{J}_k}[\Delta \leq \Delta_0]$ for its subspace of states of dimension at most Δ_0 .

Definition 7.1 (Finite-weight refinement). For the convergence problem considered here, we call the neighboring interfaces $\mathfrak{J}_k$, $k \geq 1$, *uniformly refining at finite conformal weight* if, for every fixed $0 \leq \Delta_0 < \infty$, there is a constant C_{Δ_0} , independent of k , such that

$$\mathcal{H}_{IJ}^{\mathfrak{J}_k}[\Delta \leq \Delta_0] \neq 0 \implies \eta_I = \omega_J \text{ and } |x_I - y_J| \leq \frac{C_{\Delta_0}}{k + h^\vee} \quad (7.3)$$

holds for every $k \geq 1$. Here $\mathcal{H}_{IJ}^{\mathfrak{J}_k}[\Delta \leq \Delta_0] \neq 0$ means that a local interface operator of total scaling dimension at most Δ_0 joins the incoming line J to the outgoing line I . The pairs (y_J, ω_J) and (x_I, η_I) are their respective flux–charge coordinates, $h^\vee$ is the dual Coxeter number of G , and C_{Δ_0} may depend on the fixed cutoff Δ_0 but not on k . The scale $(k + h^\vee)^{-1}$ is the mesh of the flux coordinate in (2.53), since moving an integrable weight by one lattice step moves x by $O((k + h^\vee)^{-1})$. Thus, at every fixed conformal-weight cutoff, (7.3) allows a number of lattice steps bounded independently of k , while the charges agree exactly. This criterion models identity-like transport in the regular $BF[G]$ limit; it is not a universal consistency condition on interfaces.

For a gapped interface, the surviving junction spaces are topological and ungraded. Regarding them as concentrated in degree zero and writing their multiplicities as a tunneling matrix $W^{k \rightarrow k+1}$, condition (7.3) specializes to

$$(W^{k \rightarrow k+1})_{IJ} \neq 0 \implies \eta_I = \omega_J, \quad |x_I - y_J| \leq \frac{C_0}{k + h^\vee}. \quad (7.4)$$

In the next subsection we test this gapped specialization and relate the general finite-weight condition to the conformal spectrum of the unit-jump coset candidate.

7.2 Gapped obstruction and the gapless coset candidate

We note that gapped interfaces exist in this setting. Every Drinfeld center has trivial Witt class [62], so $[Z(\mathcal{C}_k)] = [Z(\mathcal{C}_{k+1})] = 0$ and the Witt obstruction to a topological surface defect between the two doubled phases vanishes [59]. Placing a gapped boundary of each phase back to back produces one, with a transport map that factorizes as $W = uv^\top$, so that the admissible outputs depend on u alone and the admissible inputs on v alone. Such an interface is reflecting, namely for such W , the allowed outgoing labels are independent of the incoming label, so this interface leaves the near-diagonal transport required by (7.3) unrealized. More general gapped interfaces can have nonfactorized tunneling matrices. We therefore test arbitrary nonnegative tunneling matrices against the necessary genus-one consistency conditions. To be more precise, recall that W_{IJ} counts the multiplicity with which an incoming line J from the level- k side crosses as an outgoing line I on the level- k' side. Compatibility with the two generators of the torus mapping-class-group action requires

$$S_{k'}W = WS_k, \quad \theta_I^{(k')}W_{IJ} = \theta_J^{(k)}W_{IJ}, \quad W_{IJ} \in \mathbb{Z}_{\geq 0}, \quad (7.5)$$

tested on $T^2 \times I$ with an interface at $t_0 \in I$, where S and θ are the modular S -matrix and the twists of the two doubled phases [58, 63]. These conditions are necessary, so they serve as an obstruction test, while lifting a given intertwiner to a bimodule category remains a further question. For $G = SU(2)$ and neighboring levels they are restrictive in the following sense.

The physical tunnelling condition in (7.5) requires nonnegative integer entries. The following stronger obstruction uses only nonnegative real entries and therefore applies a fortiori to physical tunnelling matrices.

Proposition 7.2 ($SU(2)$ obstruction to uniformly refining gapped transport). *Let $G = SU(2)$, let $k \geq 1$, and let $W \neq 0$ be a nonnegative real matrix obeying the two linear conditions in (7.5) with $k' = k + 1$. Then W carries a nonzero entry W_{IJ} with $\eta_I \neq \omega_J$, or a nonzero charge-preserving entry with*

$$\eta_I = \omega_J \quad \text{and} \quad |x_I - y_J| \geq \frac{1}{2} - \frac{1}{k+3}. \quad (7.6)$$

Consequently, no sequence of nonzero matrices satisfying the necessary genus-one conditions can be the tunneling data of a uniformly refining gapped interface in the sense of (7.3).

Proof. Label the simple objects of $Z(\mathcal{C}_k)$ by pairs (a, b) with $0 \leq a, b \leq k$, and put $p = k+2$. We identify the $SU(2)$ alcove with $[0, 1]$ by writing $x = t\varpi_1$ where ϖ_1 is the fundamental weight of $SU(2)$ and use the scalar coordinate t in the absolute value in (7.3), namely we measure distance in the alcove by $d(x, y) = |t(x) - t(y)|$. In these labels the chiral and doubled S -matrices and the doubled twist are

$$S_{ac}^{(k)} = \sqrt{\frac{2}{p}} \sin\left(\frac{\pi(a+1)(c+1)}{p}\right), \quad \mathcal{S}_{k,(a,b),(c,d)} = S_{ac}^{(k)} S_{bd}^{(k)}, \quad (7.7)$$

$$\theta_{(a,b)}^{(k)} = \exp\left[2\pi i \frac{a(a+2) - b(b+2)}{4p}\right]. \quad (7.8)$$

2114 Suppose first that every nonzero entry of W preserves the charge. Consider the incoming
 2115 level- k label $J_* = (1, 0)$. Every charge-preserving outgoing label has the form $I_q = (q+1, q)$,
 2116 where $0 \leq q \leq k$. For $W_{I_q, J_*} \neq 0$, the θ -equation of (7.5) would require

$$\frac{2q+3}{4(p+1)} \equiv \frac{3}{4p} \pmod{1}. \quad (7.9)$$

2117 Both representatives in (7.9) lie strictly between 0 and $1/2$, so the congruence is an equality.
 2118 It would give $2pq = 3$, which is impossible. The entire J_* -column of W therefore vanishes.

2119 The row of the outgoing vacuum $I_{\text{vac}} = (0, 0)$ must be nonzero. Indeed, if it vanished,
 2120 then the I_{vac} -row of $\mathcal{S}_{k+1}W = W\mathcal{S}_k$ would give, for every incoming label J ,

$$0 = \sum_I \mathcal{S}_{k+1, I_{\text{vac}}} I W_{IJ}. \quad (7.10)$$

2121 The vacuum row of the doubled $SU(2)$ matrix is strictly positive, and W is nonnegative, so
 2122 this would force $W = 0$. Charge preservation now allows us to write the nonzero vacuum
 2123 row as

$$u_a := W_{(0,0),(a,a)} \geq 0, \quad 0 \leq a \leq k, \quad (7.11)$$

2124 with not all u_a zero. Since the J_* -column of W vanishes, $(\mathcal{S}_{k+1}W)_{(0,0), J_*} = 0$. The S -
 2125 intertwining equation, charge preservation in the outgoing-vacuum row, and (7.7) therefore
 2126 give

$$\begin{aligned} 0 &= (\mathcal{S}_{k+1}W)_{(0,0), J_*} = (W\mathcal{S}_k)_{(0,0), J_*} \\ &= \sum_{c,d=0}^k W_{(0,0),(c,d)} \mathcal{S}_{k,(c,d),(1,0)} = \sum_{a=0}^k u_a \mathcal{S}_{k,(a,a),(1,0)} = \sum_{a=0}^k u_a S_{a1}^{(k)} S_{a0}^{(k)} \\ &= \frac{2}{p} \sum_{a=0}^{p-2} u_a \sin\left(\frac{2\pi(a+1)}{p}\right) \sin\left(\frac{\pi(a+1)}{p}\right). \end{aligned} \quad (7.12)$$

2127 The second sine is strictly positive, while the first is strictly positive for $a+1 < p/2$.
 2128 Since the coefficients u_a are nonnegative and not all zero, (7.12) forces $u_a > 0$ for some
 2129 $a+1 \geq p/2$. For the corresponding nonzero entry the charges vanish and

$$x_{I_{\text{vac}}} = \frac{1}{p+1}, \quad y_{(a,a)} = \frac{a+1}{p} \geq \frac{1}{2}, \quad (7.13)$$

2130 which proves (7.6). If the support of W was not charge-preserving to begin with, the
 2131 first alternative of the proposition already holds. Thus every nonzero solution violates the
 2132 gapped specialization of (7.3) unless its degree-zero refinement constant obeys

$$C_0 \geq p \left(\frac{1}{2} - \frac{1}{p+1} \right), \quad (7.14)$$

2133 whose right-hand side diverges as $p \rightarrow \infty$. □

Scope of the obstruction. Only special gapped interfaces preserve the charge. We impose it in (7.3) because the regular $BF[G]$ limit keeps η as an unscaled discrete weight, since a flux displacement of order k^{-1} vanishes in the limit, whereas $\eta_I - \omega_J \neq 0$ remains a nontrivial action on the limiting charge sectors. This selection rule would follow from an appropriate symmetry preserved by an interface, and it lies beyond what gappedness or a bound on the interface central charge supplies. A charge-changing gapped interface may therefore be physically admissible, but it would realize a nontrivial $BF[G]$ defect in place of the uniformly refining transport required here. The proposition rules out gapped interfaces only as realizations of this special kind of transport, namely their genus-one data either change the charge or contain a charge-preserving transmission separated by at least $1/2 - 1/(k+3)$ in the flux coordinate.

Within the scope just stated, the obstruction to a gapped interface satisfying (7.3) motivates a gapless candidate whose local sectors correlate the line labels on the two sides. For an $SU(2)$ level- r jump, a natural chiral algebra is the diagonal coset

$$\begin{aligned} \mathcal{V}_k^{(r)} &:= \frac{SU(2)_k \times SU(2)_r}{SU(2)_{k+r}}, \\ c(\mathcal{V}_k^{(r)}) &= \frac{3k}{k+2} + \frac{3r}{r+2} - \frac{3(k+r)}{k+r+2}, \end{aligned} \quad (7.15)$$

with the denominator embedded diagonally. This coset construction is standard [64], and a related unit-jump construction for $SU(N)$ appears in [65]. For the unit jump the coset is the unitary Virasoro minimal model of [66, 67],

$$\mathcal{V}_k^{(1)} \simeq \mathcal{M}(p, p+1), \quad c(\mathcal{V}_k^{(1)}) = 1 - \frac{6}{(k+2)(k+3)}, \quad (7.16)$$

and, writing $L_\ell(a)$ for the integrable affine $SU(2)_\ell$ module with highest-weight label a , the affine branching rule reads [68, Eq. (2.20)]

$$L_k(a) \otimes L_1(s) = \bigoplus_{\substack{0 \leq b \leq k+1 \\ a+s+b \equiv 0 \pmod{2}}} B_{a,s}^b \otimes L_{k+1}(b), \quad s \in \{0, 1\}, \quad (7.17)$$

up to the usual coset field identification, with $B_{a,s}^b$ the corresponding $\mathcal{M}(p, p+1)$ branching module. In this decomposition, $B_{a,s}^b$ pairs the incoming label a at level k and the auxiliary label s at level one with the outgoing label b at level $k+1$.

To test compatibility with the bulk level jump, we fold the interface by reversing the orientation of the level- $(k+r)$ bulk. The resulting boundary of $SU(2)_k \times SU(2)_{-(k+r)}$ Chern–Simons theory has relative gauge anomaly $k - (k+r) = -r$, so we introduce a boundary $\widehat{su}(2)_r$ current algebra with anomaly $+r$. Its current combines with the level- k current to form $J_{\text{diag}} = J^{(k)} + J^{(r)}$ at level $k+r$, matching the gauge symmetry of the folded outgoing bulk. Gauging this diagonal symmetry leaves the coset (7.15) as the gauge-invariant chiral algebra. Repeating the construction for the negative-level factors and choosing the diagonal modular invariant gives a standalone nonchiral completion. This anomaly-matching argument identifies a compatible candidate; it does not construct the required interface couplings to both bulks.

2166 In the doubled theory, we write the incoming and outgoing labels as $J = (a_+, a_-) \in$
 2167 $\text{Irr}(\mathcal{C}_k) \times \text{Irr}(\mathcal{C}_k^{\text{rev}}) \cong \{0, \dots, k\}^2$ at level k and $I = (b_+, b_-) \in \text{Irr}(\mathcal{C}_{k+1}) \times \text{Irr}(\mathcal{C}_{k+1}^{\text{rev}}) \cong$
 2168 $\{0, \dots, k+1\}^2$ at level $k+1$, and apply (7.17) to the two chiralities with auxiliary labels
 2169 $s_{\pm} \in \{0, 1\}$. To exhibit refining channels, we choose a fixed integer δ , independent of k ,
 2170 and set $b_{\pm} = a_{\pm} + \delta$ and $s_{\pm} \equiv \delta \pmod{2}$ provided $0 \leq a_{\pm} + \delta \leq k+1$. Then the branching
 2171 rule $a_{\pm} + s_{\pm} + b_{\pm} \equiv 0 \pmod{2}$ is satisfied and the common shift preserves the charge

$$\eta_I = b_+ - b_- = a_+ - a_- = \omega_J \quad \text{in units of } \varpi_1, \quad (7.18)$$

2172 and $|x_I - y_J| \leq (|\delta| + 1)/(k + 3)$. Thus a fixed δ moves the line by only a bounded number
 2173 of microscopic label steps. The same branching rule also permits shifts of order k , but
 2174 the calculation below places them at conformal weights of order k^2 , so they stay outside
 2175 every fixed-weight cutoff. Hence the coset contains the required finite-weight near-diagonal
 2176 sectors. Establishing (7.3) for a physical interface nevertheless requires its doubled bulk
 2177 coupling to select charge-preserving left–right combinations in every bounded-weight sector,
 2178 which is beyond the scope of this work.

2179 We can further relate this displacement to conformal weight without restricting to zero
 2180 charge. Consider a charge-preserving channel in (7.17) with common charge $\eta := \eta_I = \omega_J$.
 2181 Charge preservation forces the common shift $b_+ - a_+ = b_- - a_- = \delta$ introduced above.
 2182 Writing $h_{\pm}$ for the lowest weights of the two chiral coset modules, we have [68, Eq. (1.7)]

$$h_{\pm} = \frac{((p+1)(a_{\pm}+1) - p(b_{\pm}+1))^2 - 1}{4p(p+1)} = \frac{(a_{\pm}+1 - p\delta)^2 - 1}{4p(p+1)}. \quad (7.19)$$

2183 The definitions of the flux and charge give

$$a_{\pm} + 1 = p y_J \pm \frac{\eta}{2}, \quad x_I - y_J = \frac{\delta - y_J}{p+1}. \quad (7.20)$$

2184 Substituting these identities into (7.19) yields the exact relations

$$\begin{aligned} \Delta := h_+ + h_- &= \frac{p(p+1)}{2} (x_I - y_J)^2 + \frac{\eta^2/4 - 1}{2p(p+1)}, \\ h_+ - h_- &= -\frac{\eta}{2} (x_I - y_J). \end{aligned} \quad (7.21)$$

2185 For fixed η , the flux part of (7.3) reads $|x_I - y_J| \leq C_{\Delta_0}/p$ at a fixed cutoff Δ_0 , and (7.21)
 2186 gives

$$\Delta \leq \frac{p+1}{2p} C_{\Delta_0}^2 + \frac{\eta^2/4 - 1}{2p(p+1)} = O(1). \quad (7.22)$$

2187 Conversely, for fixed η , a uniform bound on Δ implies $p|x_I - y_J| = O(1)$ and hence the
 2188 flux part of (7.3). On a cylinder of circumference L , the corresponding excitation above
 2189 the vacuum satisfies

$$\frac{L}{2\pi} (E - E_{\text{vac}}) = \Delta. \quad (7.23)$$

2190 Thus, in every fixed-charge channel, bounded displacement in lattice units is equivalent to
 2191 uniform boundedness of the dimensionless cylinder energy. Moreover, (7.21) gives $h_+ -$

$h_- = O(p^{-1})$ under refinement, so these sectors become spinless as $k \rightarrow \infty$. For $\eta \neq 0$ the two finite-level chiral modules are paired off-diagonally and describe candidate junction operators attached to bulk lines, distinct from the intrinsic primaries of the standalone diagonal minimal model.

In the charge-preserving alternative of Proposition 7.2, the forced transition has $\eta = 0$, and (7.21) gives $\Delta \gtrsim p^2/8$. It therefore leaves every fixed conformal-weight sector as $k \rightarrow \infty$, so a gapped interface is unsuitable for the finite-weight refining transport sought here.

Proposition 7.2 constrains only the gapped specialization of (7.3). Its hypotheses describe the transmission by a single ungraded nonnegative matrix W satisfying the topological S - and θ -intertwining conditions. For a gapless interface, $\mathcal{H}_{IJ}^{\mathfrak{J}_k}$ retains its conformal grading, so the ungraded matrix W alone leaves the transmission underdetermined. In the charge-preserving coset sectors above, fixed shifts $\delta = O(1)$ and fixed charges $\eta = O(1)$ occur at bounded conformal weight, whereas an $O(1)$ flux displacement has $\Delta = O(k^2)$ by (7.21). Refinement (7.3) is therefore a condition on the uniformly bounded-weight part of this graded transmission, in place of the support obtained after forgetting the grading. The gapless candidates thus lie exactly outside the hypotheses of the proposition.

7.3 The large-level limit of the unit-jump interface

Let $\mathfrak{J}_k^{(1)} : \mathbb{T}_k \rightarrow \mathbb{T}_{k+1}$ denote a prospective elementary unit-jump interface with candidate chiral algebra $\mathcal{V}_k^{(1)}$. Holding the jump r fixed in (7.15), we find that the first and third terms cancel in the limit, so the exact chiral central charge obeys

$$c(\mathcal{V}_k^{(r)}) \xrightarrow[k \rightarrow \infty]{} \frac{3r}{r+2}, \quad \text{so} \quad c(\mathcal{V}_k^{(1)}) \rightarrow 1. \quad (7.24)$$

The nonzero limiting central charge shows that the coset candidate does not become topological. If this candidate is realized by a physical bulk-coupled interface, conformal degrees of freedom survive on each interface of the chain (7.2). For fixed n , its large- K limit would then suggest BF regions separated by nontrivial limiting interface theories. We now ask which $c = 1$ theory could occupy an elementary interface.

Using (7.16) we recognize the chiral family as exactly the unitary Virasoro minimal series $\mathcal{M}(p, p+1)$, taken with the diagonal modular invariant. Runkel and Watts study that same limit and propose a nonrational $c = 1$ theory $\mathcal{M}_\infty$ whose nonchiral primary spectrum is a continuum of spinless representations with weights

$$h = \bar{h} = \frac{x^2}{4}, \quad x \in S := \mathbb{R}_{>0} \setminus \mathbb{Z}_{>0}, \quad (7.25)$$

with the removed integers labeling modules with null vectors [57, Sections 1 and 2.1]. The proposal is neither rational nor quasi-rational. Its later interpretation in terms of the twisted sector of a continuous orbifold provides additional structural support [56, Sections 1 and 4]; neither construction supplies the bulk couplings required of the interface considered here.

For $G = SU(2)$, we first compare neutral diagonal primaries across levels by selecting sequences in different finite-level Hilbert spaces. For every sufficiently large k , choose an

incoming label $J = (a_k, a_k)$ at level k and an outgoing label $I = (b_k, b_k)$ at level $k + 1$ such that, for a fixed $\delta \in \mathbb{Z}$,

$$\frac{a_k + 1}{k + 2} \longrightarrow t \in (0, 1), \quad b_k - a_k = \delta \implies \frac{b_k + 1}{k + 3} \longrightarrow t, \quad (7.26)$$

so the incoming and outgoing fluxes approach the same macroscopic alcove point t , while δ encodes their fixed microscopic label shift. Substitution in (7.19) gives the Runkel–Watts spectrum (7.25):

$$h \xrightarrow[k \rightarrow \infty]{} \frac{(t - \delta)^2}{4}, \quad \text{that is} \quad x = |t - \delta|. \quad (7.27)$$

The limiting primary is therefore specified by both the macroscopic flux t and the microscopic shift δ , while fixing t alone leaves x undetermined. As t runs over the open alcove and δ over the integers, $|t - \delta|$ fills S exactly, and the integers are missed precisely because t stays regular. Holding the weights fixed in place of the flux gives $(a_k + 1)/(k + 2) \rightarrow 0$ and returns the removed values $x = |\delta|$.

The same limiting weights occur at every fixed charge. Choose a charge-preserving sequence with $y_J \rightarrow t$, $\eta_I = a_+ - a_- = \omega_J = b_+ - b_- = \eta$ fixed, and $b_{\pm} - a_{\pm} = \delta$. Equation (7.19) gives

$$h_{\pm} \longrightarrow \frac{(t - \delta)^2}{4}, \quad h_+ - h_- = \frac{\eta(y_J - \delta)}{2(k + 3)} \longrightarrow 0. \quad (7.28)$$

Thus, along every refining sequence with fixed charge η , the conformal spin $h_+ - h_-$ tends to zero as $k \rightarrow \infty$. As t and δ vary, the charge-preserving branching sequences at every fixed η reproduce the same set S of limiting conformal weights. Comparing (7.21) with (7.25), we identify the Runkel–Watts parameter with the flux displacement across the interface measured in mesh units,

$$x = \lim_{k \rightarrow \infty} (k + 2) |x_I - y_J|, \quad (7.29)$$

so, for the branching sequences above, the candidate Runkel–Watts spectral label equals the rescaled flux displacement, and the integers removed from S are reached at the alcove walls. This prescription selects sequences of primaries in place of a canonical identification of the finite-level Hilbert spaces. At nonzero η , the finite-level pairs in (7.28) are candidate bulk-line-attached junction sectors, distinct from the intrinsic fields of the standalone diagonal minimal model. The calculation fixes their limiting weights, and establishing that they occur in the physical bulk-coupled interface, together with convergence of their normalized fields, OPE coefficients, and correlation functions, remains open.

Conjecture 7.3 (Large-level limit of the unit-jump interface). *For $G = SU(2)$, let $\mathfrak{J}_k^{(1)}$ be a unit-jump interface theory between $\mathbb{T}_k$ and $\mathbb{T}_{k+1}$ whose chiral algebra is $\mathcal{V}_k^{(1)}$ and whose intrinsic nonchiral completion is diagonal. Suppose its neutral diagonal primaries are compared across levels along the sequences specified in (7.26). We conjecture that, after the continuum averaging and normalization required to form the limiting fields, this sector converges as $k \rightarrow \infty$ to the Runkel–Watts theory $\mathcal{M}_{\infty}$ of [57]. We further conjecture that the physical bulk-coupled interface contains the fixed-charge junction sectors of (7.28),*

whose limiting conformal-weight set is S in every fixed regular $BF[SU(2)]$ charge sector. For each fixed n , we conjecture that the $K \rightarrow \infty$ limit of the finite chain (7.2) consists of $n + 1$ $BF[SU(2)]$ regions separated by n such bulk-enriched two-dimensional interfaces.

Evidence for Conjecture 7.3. The central charge matches by (7.24), the chiral family and the modular invariant are the ones Runkel and Watts take, the spectrum produced by (7.26) reproduces S through (7.27), fixed charge leaves this limiting spectrum unchanged by (7.28), and by (7.29) the continuum label is the flux displacement across the interface. These are spectral and algebraic matches; they do not establish convergence of normalized fields, OPE coefficients, or correlators.

Nonuniqueness of the limiting prescription. A competing option is the degeneration of the same minimal series studied by Roggenkamp and Wendland, whose limit acquires a geometric interpretation on an interval together with a commutative algebra of vanishing-energy zero modes [69, Section 4], from which both the limiting metric and the limiting brane geometry can be read off [70]. That option has its own hook here, because the degenerating interval could match the alcove carrying x in (2.53). Runkel and Watts already flag the ambiguity and attribute the alternative to Roggenkamp and Wendland [57, Section 1], and the two limits are distinct as the $\mathcal{N} = 2$ minimal series already shows, where distinct scalings of one coset family reach either a free theory or a continuous orbifold of it [71–73]. They arise from different prescriptions for comparing primaries across levels. Conjecture 7.3 adopts (7.26); a construction of the doubled bulk-coupled interface must determine which sequences remain at finite weight relative to the bulk gap. We also note that $\mathcal{M}_\infty$ was proposed conjecturally in [57], with analytic and numerical crossing tests, so our conjecture rests on that limiting proposal as well as on the unconstructed bulk coupling.

Scope of the interface analysis. The proved ingredients of this section are Proposition 7.2 under its stated $SU(2)$ and genus-one hypotheses, together with the exact coset central-charge, branching, and weight identities. The existence of a physical doubled bulk-coupled coset interface, the selection of its bounded-weight line-attached sectors, their Runkel–Watts convergence, and the limiting chain remain open as stated in Conjecture 7.3. None of these interface claims is used in the proofs of Main Theorems A–C.

8 Discussion and outlook

The three main results compare complementary parts of the regular large-level theory. Main Theorem A places the doubled torus representations in one Hilbert space and proves strong-* convergence. Main Theorem B identifies the leading chamber polynomial of individual WZW fusion coefficients with the normalized BF pair-of-pants volume, and Main Theorem C turns the two chiral estimates into a charge-resolved locally finite measure. The exact Mackey decomposition contains further information: besides its orbital measure, it carries a measurable field of multiplicity spaces.

An intertwiner-level limit is the natural continuation of the genus-zero analysis. It would compare finite Clebsch–Gordan maps with the measurable multiplicity field and

would make associativity and braiding accessible. Higher-genus convergence presents a related problem, since sewing must be controlled in a model that also accommodates the changing centralizers on singular orbit-type strata.

The boundary and interface calculations point to additional structures at the endpoint. The abelian wavefunctions are exact in the chosen polarizations, while the general- G analysis identifies the expected tangent symmetry and current algebra. For adjacent $SU(2)$ levels, Proposition 7.2 gives the genus-one obstruction stated in Section 7.3; the coset construction then suggests a gapless interface and its proposed Runkel–Watts limit. A construction of the bulk coupling would also provide the setting in which such interfaces and their junctions can be composed.

A Boundary theories

In this appendix, we review the boundary theory of $U(1)$ Chern–Simons theory $\text{CS}_k[U(1)]$ following [2]. We then give parallel derivations for noncompact abelian BF theory and for $\text{CS}_k[U(1)] \times \text{CS}_{-k}[U(1)]$.

A.1 $U(1)$ Chern–Simons theory $\text{CS}_k[U(1)]$

Consider the Chern-Simons theory

$$S = \frac{ik}{4\pi} \int_{\mathcal{M}} A \wedge dA, \quad (\text{A.1})$$

and suppose $\mathcal{M}$ has a boundary $\partial\mathcal{M}$, which is a torus parametrized by

$$z = \sigma_1 + \tau\sigma_2, \quad \bar{z} = \sigma_1 + \bar{\tau}\sigma_2, \quad \sigma_i \in [0, 1), \quad (\text{A.2})$$

and introducing the holomorphic and antiholomorphic components

$$A_z = \frac{A_2 - \bar{\tau}A_1}{2i\tau_2}, \quad A_{\bar{z}} = \frac{\tau A_1 - A_2}{2i\tau_2}. \quad (\text{A.3})$$

With the boundary, the variation of the action gives the boundary term

$$\delta S|_{dA=0} = -\frac{ik}{4\pi} \int_{\partial\mathcal{M}} A \wedge \delta A = -\frac{k}{2\pi} \int_{\partial\mathcal{M}} d^2z (A_z \delta A_{\bar{z}} - A_{\bar{z}} \delta A_z), \quad (\text{A.4})$$

where we fix the conventions

$$dz \wedge d\bar{z} = -2i\tau_2 d\sigma^1 \wedge d\sigma^2, \quad d^2z \equiv \frac{i}{2} dz \wedge d\bar{z}, \quad \int d^2z = \tau_2. \quad (\text{A.5})$$

Choosing the boundary condition, introduce the boundary term and define the action

$$S' = S - \frac{k}{2\pi} \int_{\partial\mathcal{M}} d^2z |A|^2, \quad |A|^2 = A_z A_{\bar{z}}, \quad k > 0, \quad (\text{A.6})$$

Then the boundary variation with respect to A_z vanishes when $A_{\bar{z}}$ is fixed, and the path integral can be evaluated with fixed $A_{\bar{z}}$ and arbitrary A_z .

2325 Parametrize the value of $A_{\bar{z}}$ at the boundary as

$$A_{\bar{z}} = \frac{i\pi}{\tau_2} \xi, \quad (\text{A.7})$$

2326 and the large gauge transformation $A \rightarrow A + \omega$ shifts ξ according to

$$\xi \rightarrow \xi + n + m\tau. \quad (\text{A.8})$$

2327 Define the wavefunction

$$\Psi(A_{\bar{z}}) = \int_{A_{\bar{z}}|_{\partial\mathcal{M}} \text{ fixed}} \mathcal{D}A e^{-S'}. \quad (\text{A.9})$$

2328 One can then check that the wavefunction transforms under the large gauge transformation
2329 according to

$$\Psi(A_{\bar{z}}) \rightarrow \Psi(A_{\bar{z}}) e^{\frac{ik}{4\pi} \int_{\partial\mathcal{M}} \omega \wedge A} e^{\frac{k\tau_2}{2\pi} (A_z \omega_{\bar{z}} + A_{\bar{z}} \omega_z + |\omega_z|^2)} e^{i\varphi(\omega)}, \quad (\text{A.10})$$

2330 where the first term comes from S , the second term comes from the boundary term, and
2331 the last term is an additional cocycle to assure associativity of large gauge transformations.

2332 Writing

$$\frac{ik}{4\pi} \int_{\partial\mathcal{M}} \omega \wedge A = \frac{k\tau_2}{2\pi} (A_{\bar{z}} \omega_z - A_z \omega_{\bar{z}}), \quad (\text{A.11})$$

2333 and using

$$A_{\bar{z}} \omega_z = \frac{\pi^2}{\tau_2^2} \xi (\delta\xi)^*, \quad \omega_z \omega_{\bar{z}} = \frac{\pi^2}{\tau_2^2} |\delta\xi|^2, \quad (\text{A.12})$$

2334 we can write

$$\Psi(\xi + n + m\tau) = \Psi(\xi) \exp \left[\frac{k\pi}{\tau_2} (n + m\bar{\tau}) \xi + \frac{k\pi}{2\tau_2} |n + m\tau|^2 + i\pi k n m \right], \quad (\text{A.13})$$

2335 where we choose $\varphi = \pi k n m$. Since one has

$$|\xi + n + m\tau|^2 = |\xi|^2 + (n + m\tau)\bar{\xi} + (n + m\bar{\tau})\xi + |n + m\tau|^2, \quad (\text{A.14})$$

2336 one can check that $|\Psi|^2 e^{-\frac{k\pi}{\tau_2} |\xi|^2}$ is invariant under the large gauge transformation.

2337 Define $f = \Psi e^{-\frac{k\pi}{2\tau_2} \xi^2}$, and one has

$$f(\xi + n + m\tau) = f(\xi) \exp \left\{ \frac{k\pi}{\tau_2} [(n + m\bar{\tau}) - (n + m\tau)] \xi + \frac{k\pi}{2\tau_2} [|n + m\tau|^2 - (n + m\tau)^2] + i\pi k n m \right\}, \quad (\text{A.15})$$

2338 where one has

$$\begin{aligned} \frac{k\pi}{\tau_2} [(n + m\bar{\tau}) - (n + m\tau)] \xi &= \frac{k\pi m}{\tau_2} (\bar{\tau} - \tau) \xi = -2\pi i k m \xi, \\ \frac{k\pi}{2\tau_2} [|n + m\tau|^2 - (n + m\tau)^2] &= \frac{k\pi m}{2\tau_2} (n + m\tau)(\bar{\tau} - \tau) = -i\pi k m (n + m\tau), \end{aligned} \quad (\text{A.16})$$

2339 and one can check

$$f(\xi + n + m\tau) = f(\xi) e^{-\pi i k m^2 \tau - 2\pi i k m \xi}, \quad (\text{A.17})$$

2340 or in other words

$$f(\xi + 1) = f(\xi), \quad f(\xi + \tau) = f(\xi)e^{-2\pi i k \xi - i \pi k \tau}. \quad (\text{A.18})$$

2341 Since $f(\xi)$ is periodic along the real axis, we may Fourier-expand it as

$$f = \sum_{\lambda \in \mathbb{Z}} c_\lambda e^{2\pi i \lambda \xi}, \quad (\text{A.19})$$

2342 The two sides of the second quasi-periodicity relation are therefore

$$\sum_{\lambda \in \mathbb{Z}} c_\lambda e^{2\pi i \lambda \tau} e^{2\pi i \lambda \xi} = e^{-i \pi k \tau} \sum_{\lambda \in \mathbb{Z}} c_{\lambda+k} e^{2\pi i \lambda \xi}. \quad (\text{A.20})$$

2343 This implies the recursion relation

$$c_{\lambda+k} = c_\lambda e^{2\pi i \lambda \tau + \pi i k \tau}, \quad (\text{A.21})$$

2344 This connects c_λ to $c_{\lambda+k}$, so the solution space is k -dimensional, and we can take the ansatz

$$c_\lambda = e^{i \pi \tau \lambda^2 / k}. \quad (\text{A.22})$$

2345 Introduce

$$\lambda = \sqrt{k} p, \quad p = \sqrt{k} \left(n + \frac{r}{k} \right), \quad u = \sqrt{k} \xi \quad (\text{A.23})$$

2346 with $r = 0, \dots, k-1$. Therefore the independent wavefunctions can be written as

$$\Psi_r(\xi) = \frac{1}{\eta(\tau)} \sum_{n \in \mathbb{Z}} \exp \left[i \pi \tau p^2 + 2 \pi i p u + \frac{\pi}{2 \tau_2} u^2 \right], \quad (\text{A.24})$$

2347 where the factor $1/\eta(\tau)$ is the one-loop determinant of the topologically trivial fluctuation,
2348 which is determined independently.

2349 **A.2 Non-compact abelian BF theory**

2350 Now consider the non-compact BF theory

$$S = \frac{i}{2\pi} \int_{\mathcal{M}} b \wedge da = \frac{i}{4\pi} \int_{\mathcal{M}} b \wedge da + \frac{i}{4\pi} \int_{\mathcal{M}} a \wedge db + \text{boundary terms}, \quad (\text{A.25})$$

2351 where a is still a compact $U(1)$ gauge field, but b is non-compact. Therefore b has no large
2352 gauge transformations.

2353 Introducing

$$A_\pm = \frac{1}{\sqrt{2}} \left(\frac{a}{R} \pm R b \right), \quad S = \frac{i}{4\pi} \int (A_+ \wedge dA_+ - A_- \wedge dA_-), \quad (\text{A.26})$$

2354 so that the action is written as a double copy of Chern-Simons theory with opposite level.

2355 Introducing the two *independent* holomorphic coordinates

$$\xi = \frac{\tau_2}{i\pi} (A_+)_z, \quad \bar{\xi} = -\frac{\tau_2}{i\pi} (A_-)_z, \quad (\text{A.27})$$

2356 and with the proper boundary term

$$S' = S - \frac{1}{2\pi} \int_{\partial\mathcal{M}} d^2z (|A_+|^2 + |A_-|^2) , \quad (\text{A.28})$$

2357 we can adopt the same method to evaluate the wave functions.

2358 Since b has no large gauge transformations, only those generated by the compact field
2359 a remain:

$$\xi \rightarrow \xi + \frac{1}{\sqrt{2}R}(n + m\tau), \quad \bar{\xi} \rightarrow \bar{\xi} + \frac{1}{\sqrt{2}R}(n + m\bar{\tau}). \quad (\text{A.29})$$

2360 Here $\delta\bar{\xi} = \delta\xi^*$. Writing $\delta\xi \equiv \delta = \frac{1}{\sqrt{2}R}(n + m\tau)$, the wavefunction transforms according to

$$\Psi(\xi + \delta, \bar{\xi} + \delta^*) = \Psi(\xi, \bar{\xi}) \exp \left[\frac{\pi}{\tau_2} (\xi\delta^* + \bar{\xi}\delta + |\delta|^2) \right]. \quad (\text{A.30})$$

2361 Similarly, let us introduce

$$F = \Psi e^{-\frac{\pi}{2\tau_2}(\xi^2 + \bar{\xi}^2)}, \quad (\text{A.31})$$

2362 and it satisfies the boundary condition

$$F(\xi + \delta, \bar{\xi} + \delta^*) = F(\xi, \bar{\xi}) \exp \left[\frac{\pi}{2\tau_2} (2\xi\delta^* + 2\bar{\xi}\delta + 2|\delta|^2 - 2\xi\delta - 2\bar{\xi}\delta^* - \delta^2 - (\delta^*)^2) \right]. \quad (\text{A.32})$$

2363 Using $\delta - \delta^* = i\frac{\sqrt{2}}{R}m\tau_2$, one has

$$\begin{aligned} 2\xi\delta^* - 2\xi\delta &= -i\frac{2\sqrt{2}}{R}m\tau_2\xi, \\ 2\bar{\xi}\delta - 2\bar{\xi}\delta^* &= i\frac{2\sqrt{2}}{R}m\tau_2\bar{\xi}, \\ 2|\delta|^2 - \delta^2 - (\delta^*)^2 &= -(\delta - \delta^*)^2 = \frac{2m^2\tau_2^2}{R^2}. \end{aligned} \quad (\text{A.33})$$

2364 Denote $P \equiv \frac{1}{\sqrt{2}R}$, one has

$$F(\xi + \delta, \bar{\xi} + \delta^*) = F(\xi, \bar{\xi}) \exp [-2\pi imP(\xi - \bar{\xi}) + 2\pi\tau_2m^2P^2], \quad (\text{A.34})$$

2365 or in other words

$$F(\xi + P, \bar{\xi} + P) = F(\xi, \bar{\xi}), \quad F(\xi + P\tau, \bar{\xi} + P\bar{\tau}) = F e^{-2\pi iP(\xi - \bar{\xi}) + 2\pi\tau_2P^2}. \quad (\text{A.35})$$

2366 Then let us consider Fourier expansion. We assume

$$F = \sum_{\lambda_L, \lambda_R} c_{\lambda_L, \lambda_R} e^{2\pi i \lambda_L \xi - 2\pi i \lambda_R \bar{\xi}}, \quad (\text{A.36})$$

2367 where the first relation implies

$$e^{2\pi i(\lambda_L - \lambda_R)P} = 1, \quad (\text{A.37})$$

2368 which constrains

$$\lambda_L - \lambda_R = \sqrt{2}RN, \quad N \in \mathbb{Z}, \quad (\text{A.38})$$

for some fixed integer N . Therefore there is only one summation. For the other relation

$$\begin{aligned} F(\xi + P\tau, \bar{\xi} + P\bar{\tau}) &= \sum_{\lambda_L, \lambda_R} c_{\lambda_L, \lambda_R} e^{2\pi i \lambda_L (\xi + P\tau) - 2\pi i \lambda_R (\bar{\xi} + P\bar{\tau})} \\ &= \sum_{\lambda_L, \lambda_R} c_{\lambda_L, \lambda_R} e^{2\pi i \lambda_L \xi - 2\pi i \lambda_R \bar{\xi} - 2\pi i P(\xi - \bar{\xi}) + 2\pi \tau_2 P^2}. \end{aligned} \quad (\text{A.39})$$

and shifting $\lambda_L \rightarrow \lambda_L + P, \lambda_R \rightarrow \lambda_R + P$ in the last equation, one has

$$\begin{aligned} \sum_{\lambda_L, \lambda_R} c_{\lambda_L, \lambda_R} e^{2\pi i \lambda_L (\xi + P\tau) - 2\pi i \lambda_R (\bar{\xi} + P\bar{\tau})} \\ = \sum_{\lambda_L, \lambda_R} c_{\lambda_L + P, \lambda_R + P} e^{2\pi i \lambda_L \xi - 2\pi i \lambda_R \bar{\xi} + 2\pi \tau_2 P^2}. \end{aligned} \quad (\text{A.40})$$

so that we have the recursion relation

$$c_{\lambda_L + P, \lambda_R + P} = c_{\lambda_L, \lambda_R} e^{2\pi i P(\lambda_L \tau - \lambda_R \bar{\tau}) - 2\pi \tau_2 P^2}, \quad (\text{A.41})$$

and we can take the ansatz

$$c_{\lambda_L, \lambda_R} = \exp [i\pi \tau \lambda_L^2 - i\pi \bar{\tau} \lambda_R^2]. \quad (\text{A.42})$$

The constraint gives $\lambda_L - \lambda_R = \sqrt{2}RN$, but $\lambda_L + \lambda_R$ is completely free. Therefore the independent wavefunctions are labeled by $N \in \mathbb{Z}$ and λ_L . Due to the recursion, the λ_L is parametrized by $[0, P) = [0, \frac{1}{\sqrt{2}R})$.

Based on the above discussion, we can introduce

$$\lambda_L = \frac{1}{\sqrt{2}} \left(\frac{\ell + \varphi}{R} + RN \right), \quad \lambda_R = \frac{1}{\sqrt{2}} \left(\frac{\ell + \varphi}{R} - RN \right), \quad (\text{A.43})$$

which satisfies $\lambda_L - \lambda_R = \sqrt{2}RN$ and $\varphi \in [0, 1)$. Here φ is the bulk magnetic flux and N the bulk electric charge; in this boundary polarization they occupy the momentum-offset and winding slots, respectively. Then we have the independent wavefunctions

$$\Psi_{\varphi, N}(\xi, \bar{\xi}) = \frac{1}{|\eta(\tau)|^2} \sum_{\ell \in \mathbb{Z}} \exp \left[i\pi \tau \lambda_L^2 - i\pi \bar{\tau} \lambda_R^2 + 2\pi i(\lambda_L \xi - \lambda_R \bar{\xi}) + \frac{\pi}{2\tau_2}(\xi^2 + \bar{\xi}^2) \right]. \quad (\text{A.44})$$

A.3 Double $U(1)$ Chern-Simons theory

We next turn to the doubled Chern-Simons theory $U(1)_k \times U(1)_{-k}$, whose wavefunctions are

$$\Psi_{a_+, a_-}(\xi, \bar{\xi}) = \frac{1}{|\eta(\tau)|^2} \sum_{n_+, n_- \in \mathbb{Z}} \exp \left[i\pi \tau p_+^2 - i\pi \bar{\tau} p_-^2 + 2\pi i(p_+ u - p_- \bar{u}) + \frac{\pi}{2\tau_2}(u^2 + \bar{u}^2) \right], \quad (\text{A.45})$$

with

$$p_{\pm} = \sqrt{k} \left(n_{\pm} + \frac{a_{\pm}}{k} \right), \quad a_{\pm} = 0, \dots, k-1. \quad (\text{A.46})$$

In the following, we shall compare the boundary theory of $k \rightarrow \infty$ limit of the double Chern-Simons theory with the non-compact BF theory.

2386 To relate them, one should be careful with the field redefinition. Begin with the double
2387 Chern-Simons theory

$$S = \frac{ik}{4\pi} \int a_+ \wedge da_+ - \frac{ik}{4\pi} \int a_- \wedge da_-, \quad (\text{A.47})$$

2388 and let us introduce the BF fields

$$a = \frac{1}{2}(a_+ + a_-), \quad b = k(a_+ - a_-), \quad (\text{A.48})$$

2389 where the period of b goes to infinity as $k \rightarrow +\infty$. This is the same field pair as in (2.20)
2390 and (2.29). Using the same parameter R as in the preceding BF calculation, let

$$A_{\pm} = \frac{1}{\sqrt{2}} \left(\frac{a}{R} \pm Rb \right) = \frac{1}{\sqrt{2}} \left(\frac{a_+ + a_-}{2R} \pm kR(a_+ - a_-) \right), \quad (\text{A.49})$$

2391 so that the theory is rescaled to unit level

$$S = \frac{i}{4\pi} \int (A_+ \wedge dA_+ - A_- \wedge dA_-), \quad (\text{A.50})$$

2392 and we have

$$\begin{aligned} A_+ &= \frac{1}{\sqrt{2}} \left(\frac{1}{2R} + kR \right) a_+ + \frac{1}{\sqrt{2}} \left(\frac{1}{2R} - kR \right) a_-, \\ A_- &= \frac{1}{\sqrt{2}} \left(\frac{1}{2R} - kR \right) a_+ + \frac{1}{\sqrt{2}} \left(\frac{1}{2R} + kR \right) a_-. \end{aligned} \quad (\text{A.51})$$

2393 Therefore one should introduce the two *independent* holomorphic coordinates

$$\xi = \frac{\tau_2}{i\pi} (A_+)_{\bar{z}}, \quad \bar{\xi} = -\frac{\tau_2}{i\pi} (A_-)_z, \quad (\text{A.52})$$

2394 and run the calculation again.

2395 Denote

$$\mu = \frac{1}{\sqrt{2}} \left(\frac{1}{2R} + kR \right), \quad \nu = \frac{1}{\sqrt{2}} \left(\frac{1}{2R} - kR \right), \quad (\text{A.53})$$

2396 which satisfies

$$\mu + \nu = \frac{1}{\sqrt{2}R}, \quad \mu - \nu = \sqrt{2}kR, \quad \mu^2 - \nu^2 = k. \quad (\text{A.54})$$

2397 The large gauge transformation reads

$$\delta\xi = \mu D_+ + \nu D_- \equiv E, \quad \delta\bar{\xi} = \nu \bar{D}_+ + \mu \bar{D}_- \equiv F^*, \quad (\text{A.55})$$

2398 where

$$D_{\pm} = n_{\pm} + m_{\pm}\tau, \quad \bar{D}_{\pm} = n_{\pm} + m_{\pm}\bar{\tau}. \quad (\text{A.56})$$

2399 Let us consider the cocycle φ in this case. The cocycle is defined by

$$\varphi(\omega + \omega') = \varphi(\omega) + \varphi(\omega') - \frac{1}{4\pi} \int_{\partial\mathcal{M}} (\omega_{A_+} \wedge \omega'_{A_+} - \omega_{A_-} \wedge \omega'_{A_-}), \quad (\text{A.57})$$

2400 where

$$\omega_{A_+} = \mu\omega_+ + \nu\omega_-, \quad \omega_{A_-} = \nu\omega_+ + \mu\omega_-, \quad (\text{A.58})$$

2401 so that

$$\omega_{A_+} \wedge \omega'_{A_+} - \omega_{A_-} \wedge \omega'_{A_-} = (\mu^2 - \nu^2)(\omega_+ \wedge \omega'_+ - \omega_- \wedge \omega'_-), \quad (\text{A.59})$$

2402 and is solved by

$$\varphi = \pi k(m_+ n_+ - m_- n_-), \quad (\text{A.60})$$

2403 just as before.

2404 Then let us solve the wavefunction. Put $\Psi = f e^{\frac{\pi}{2\tau_2}(\xi^2 + \bar{\xi}^2)}$, one has

$$\begin{aligned} \Psi(\xi + E, \bar{\xi} + F^*) &= \Psi(\xi, \bar{\xi}) \exp \left[\frac{\pi}{2\tau_2} (2\xi E^* + E E^*) + \frac{\pi}{2\tau_2} (2\bar{\xi} F + F F^*) \right] \\ e^{-\frac{\pi}{2\tau_2}(\xi^2 + \bar{\xi}^2)} &\rightarrow e^{-\frac{\pi}{2\tau_2}(\xi^2 + \bar{\xi}^2)} \exp \left[-\frac{\pi}{2\tau_2} (2\xi E + E^2 + 2\bar{\xi} F^* + (F^*)^2) \right] \end{aligned} \quad (\text{A.61})$$

2405 which implies

$$f(\xi + E, \bar{\xi} + F^*) = f(\xi, \bar{\xi}) \exp \left[\frac{\pi}{\tau_2} (\xi(E^* - E) + \bar{\xi}(F - F^*)) + \frac{\pi}{2\tau_2} (E E^* - E^2 + F F^* - (F^*)^2) + i\varphi \right]. \quad (\text{A.62})$$

2406 Introduce

$$M_+ = \mu m_+ + \nu m_-, \quad M_- = \nu m_+ + \mu m_-, \quad (\text{A.63})$$

2407 so that

$$E^* - E = -2i\tau_2 M_+, \quad F - F^* = 2i\tau_2 M_-, \quad (\text{A.64})$$

2408 we then have

$$f(\xi + E, \bar{\xi} + F^*) = f(\xi, \bar{\xi}) \exp [-2\pi i M_+ \xi + 2\pi i M_- \bar{\xi} - i\pi M_+ E + i\pi M_- F^* + i\varphi]. \quad (\text{A.65})$$

2409 For the constant part, one has

$$\begin{aligned} &-i\pi M_+ E + i\pi M_- F^* \\ &= -i\pi(\mu m_+ + \nu m_-)(\mu D_+ + \nu D_-) + i\pi(\nu m_+ + \mu m_-)(\nu \bar{D}_+ + \mu \bar{D}_-) \\ &= i\pi m_+(\nu^2 \bar{D}_+ - \mu^2 D_+) + i\pi m_-(\mu^2 \bar{D}_- - \nu^2 D_-) \\ &\quad + i\pi\mu\nu [m_+(\bar{D}_- - D_-) + m_-(\bar{D}_+ - D_+)], \end{aligned} \quad (\text{A.66})$$

2410 since

$$\bar{D}_\pm - D_\pm = -2i\tau_2 m_\pm, \quad (\text{A.67})$$

2411 the second term gives

$$4\pi\tau_2\mu\nu m_+ m_- \in \mathbb{R}. \quad (\text{A.68})$$

2412 The first term gives

$$\begin{aligned} &i\pi m_+ [(\nu^2 - \mu^2)n_+ + m_+(\nu^2 \bar{\tau} - \mu^2 \tau)] \\ &\quad + i\pi m_- [(\mu^2 - \nu^2)n_- + m_-(\mu^2 \bar{\tau} - \nu^2 \tau)] \end{aligned} \quad (\text{A.69})$$

2413 where one uses $D_\pm = n_\pm + m_\pm \tau$, $\bar{D}_\pm = n_\pm + m_\pm \bar{\tau}$. Using $\mu^2 - \nu^2 = k$, the first and third
2414 terms give

$$-i\pi k m_+ n_+ + i\pi k m_- n_- = -i\pi k(m_+ n_+ - m_- n_-), \quad (\text{A.70})$$

which exactly cancel the cocycle φ .

Therefore we have

$$\begin{aligned} & f(\xi + E, \bar{\xi} + F^*) \\ &= f(\xi, \bar{\xi}) \exp \left[\begin{aligned} & -2\pi i M_+ \xi + 2\pi i M_- \bar{\xi} + i\pi m_+^2 (\nu^2 \bar{\tau} - \mu^2 \tau) \\ & + i\pi m_-^2 (\mu^2 \bar{\tau} - \nu^2 \tau) + 4\pi \tau_2 \mu \nu m_+ m_- \end{aligned} \right], \end{aligned} \quad (\text{A.71})$$

which does not depend on $n_{\pm}$. The four generators act as shown in table 1.

generator	shift $(\delta\xi, \delta\bar{\xi})$	φ	multiplier on f
$(n_+, m_+) = (1, 0)$	(μ, ν)	0	1
$(n_-, m_-) = (1, 0)$	(ν, μ)	0	1
$(n_+, m_+) = (0, 1)$	$(\mu\tau, \nu\bar{\tau})$	0	$e^{-2\pi i \mu \xi + 2\pi i \nu \bar{\xi} - i\pi \mu^2 \tau + i\pi \nu^2 \bar{\tau}}$
$(n_-, m_-) = (0, 1)$	$(\nu\tau, \mu\bar{\tau})$	0	$e^{-2\pi i \nu \xi + 2\pi i \mu \bar{\xi} - i\pi \nu^2 \tau + i\pi \mu^2 \bar{\tau}}$

Table 1. Large-gauge generators and their multipliers on the holomorphic factor f .

Now let us consider the Fourier expansion

$$f(\xi, \bar{\xi}) = \sum_{p_L, p_R} c_{p_L, p_R} e^{2\pi i (p_L \xi - p_R \bar{\xi})}, \quad (\text{A.72})$$

and the first two generators give

$$p_L \mu - p_R \nu \in \mathbb{Z}, \quad p_L \nu - p_R \mu \in \mathbb{Z}. \quad (\text{A.73})$$

Parametrize p_L, p_R according to

$$p_L = \mu \hat{n}_+ + \nu \hat{n}_-, \quad p_R = \nu \hat{n}_+ + \mu \hat{n}_-, \quad (\text{A.74})$$

then we have

$$p_L \mu - p_R \nu = (\mu^2 - \nu^2) \hat{n}_+ = k \hat{n}_+, \quad p_L \nu - p_R \mu = (\nu^2 - \mu^2) \hat{n}_- = -k \hat{n}_-, \quad (\text{A.75})$$

so one can further parametrize

$$\hat{n}_{\pm} = \frac{j_{\pm}}{k}, \quad j_{\pm} \in \mathbb{Z}. \quad (\text{A.76})$$

Then let us consider the other two generators. The third one shifts $(p_L, p_R) \rightarrow (p_L - \mu, p_R - \nu)$, which is effectively

$$\hat{n}_+ \rightarrow \hat{n}_+ - 1, \quad \text{or} \quad j_+ \rightarrow j_+ - k, \quad (\text{A.77})$$

and it gives the recursion relation

$$c_{p_L + \mu, p_R + \nu} = c_{p_L, p_R} e^{2\pi i (\mu \tau p_L - \nu \bar{\tau} p_R) + i\pi \mu^2 \tau - i\pi \nu^2 \bar{\tau}}. \quad (\text{A.78})$$

Similarly, the fourth one shifts $(p_L, p_R) \rightarrow (p_L - \nu, p_R - \mu)$, which is effectively

$$\hat{n}_- \rightarrow \hat{n}_- - 1, \quad \text{or} \quad j_- \rightarrow j_- - k, \quad (\text{A.79})$$

and it gives the recursion relation

$$c_{p_L+\nu, p_R+\mu} = c_{p_L, p_R} e^{2\pi i(\nu\tau p_L - \mu\bar{\tau} p_R) + i\pi\nu^2\tau - i\pi\mu^2\bar{\tau}}. \quad (\text{A.80})$$

Both recursion relations are solved by

$$c_{p_L, p_R} = \exp \left[i\pi\tau p_L^2 - i\pi\bar{\tau} p_R^2 \right], \quad (\text{A.81})$$

and there are exactly k^2 solutions.

Based on the above discussion, let us write

$$j_{\pm} = kn_{\pm} + \lambda_{\pm}, \quad \lambda_{\pm} \in \{0, \dots, k-1\}, \quad n_{\pm} \in \mathbb{Z}, \quad (\text{A.82})$$

and write down the wavefunctions as

$$\Psi_{\lambda_+, \lambda_-}(\xi, \bar{\xi}, \tau) = \frac{1}{|\eta(\tau)|^2} \sum_{n_+, n_- \in \mathbb{Z}} \exp \left[i\pi\tau p_L^2 - i\pi\bar{\tau} p_R^2 + 2\pi i(p_L\xi - p_R\bar{\xi}) + \frac{\pi}{2\tau_2}(\xi^2 + \bar{\xi}^2) \right], \quad (\text{A.83})$$

with

$$p_L = \frac{1}{\sqrt{2}} \left[\frac{\hat{n}_+ + \hat{n}_-}{2R} + kR(\hat{n}_+ - \hat{n}_-) \right], \quad p_R = \frac{1}{\sqrt{2}} \left[\frac{\hat{n}_+ + \hat{n}_-}{2R} - kR(\hat{n}_+ - \hat{n}_-) \right], \quad (\text{A.84})$$

or

$$p_L = \frac{1}{\sqrt{2}} \left[\frac{j_+ + j_-}{2kR} + R(j_+ - j_-) \right], \quad p_R = \frac{1}{\sqrt{2}} \left[\frac{j_+ + j_-}{2kR} - R(j_+ - j_-) \right]. \quad (\text{A.85})$$

These are the momenta used in section 6.1. With the same fields and the same R , the diagonal and charge-conjugation sums at even level have compact radii kR and $2R$, respectively; the associated parity conditions are stated there.

B BF fusion and proofs of the fusion-coefficient limits

This appendix proves the fusion statements of Sections 4 and 5. The first three subsections give the Mackey decomposition, the normalization of the pair-of-pants volume, and the fusion-hypergroup limit. The Mackey construction follows [6], and the geometric normalization uses the quasi-Hamiltonian volume formulas of [43, 46]. The subsequent coefficient proofs combine the relative BZ lattice model with Kac–Walton, identify the leading polynomial through the weighted weak limit, and establish the estimates needed for the doubled root-lattice sums.

B.1 Mackey decomposition for regular BF inputs

We give a proof of Theorem 4.2 and then record the pointwise rule at an exceptional relative position. For $x \in Y$ and $\eta \in P$, let $\mathcal{H}_{x, \eta}$ be the Hilbert space of equivalence classes, modulo equality almost everywhere, of measurable functions $f : G \rightarrow \mathbb{C}$ satisfying

$$f(gt) = \chi_{\eta}(t)^{-1} f(g) \quad (t \in T, \text{ for almost every } g \in G), \quad \|f\|^2 = \int_{G/T} |f(g)|^2 d\dot{g} < \infty, \quad (\text{B.1})$$

where $d\dot{g}$ is the normalized G -invariant measure on G/T . This space carries the regular irreducible representation of $\mathfrak{D}(G) = C(G) \rtimes_{\text{Ad}} G$. This is the standard induced model for irreducible representations of the compact double [6, Definition 3.1 and Theorem 3.2]. The group part acts by left translation and $F \in C(G)$ acts by

$$(Ff)(g) = F(gg_xg^{-1})f(g). \quad (\text{B.2})$$

This is the regular irreducible transformation-group representation labelled by (x, η) in Eq. (2.40). For two such representations, the tensor-product construction of [6] acts on the Hilbert tensor product

$$\mathcal{H}_{x,\eta} \widehat{\otimes} \mathcal{H}_{y,\omega} \simeq \left\{ \Phi : G^2 \rightarrow \mathbb{C} : \begin{array}{l} \Phi(g_1t_1, g_2t_2) = \chi_\eta(t_1)^{-1} \chi_\omega(t_2)^{-1} \Phi(g_1, g_2), \\ \int_{G/T \times G/T} |\Phi(g_1, g_2)|^2 d\dot{g}_1 d\dot{g}_2 < \infty \end{array} \right\}. \quad (\text{B.3})$$

The right-hand side of (B.3) consists of measurable equivariant functions modulo equality almost everywhere. On this Hilbert space define

$$(U_h^{x,y} \Phi)(g_1, g_2) = \Phi(h^{-1}g_1, h^{-1}g_2), \quad (\text{B.4})$$

$$(M_F^{x,y} \Phi)(g_1, g_2) = F(g_1g_xg_1^{-1}g_2g_yg_2^{-1}) \Phi(g_1, g_2), \quad F \in C(G). \quad (\text{B.5})$$

These operators realize the full-double tensor product denoted by (4.1); compare the product-holonomy action in [6, Sec. 4, especially Eq. (4.3)]. The full compact double also contains the tensor unit and the electric and singular-flux representations. For the regular input pairs considered here, Lemma 4.1 shows that the output spectral measure gives full mass to the regular locus; restriction of this tensor product to the regular ideal is consequently nondegenerate.

Give $G/T \times G/T$ the product probability measure. Its quotient by the diagonal G -action is $T \backslash G/T$, and the quotient measure is the probability measure $d\bar{q}$ used in Eq. (4.3). For a representative $q \in G$, recall that

$$H_q = T \cap qTq^{-1}, \quad z(q) = \pi_{\bar{Y}}(g_x q g_y q^{-1}).$$

On the conull set $X_0 = X_{\text{pr,reg}}$, choose Borel representatives $q = q(\bar{q})$ and Borel conjugators $s_q \in G$ satisfying

$$s_q g_{z(q)} s_q^{-1} = g_x q g_y q^{-1}. \quad (\text{B.6})$$

For a fixed exceptional q , the same notation will mean an arbitrary choice of representative and conjugator. Write $G_{z(q)} = C_G(g_{z(q)})$. For $\sigma \in \widehat{G_{z(q)}}$, set

$$\mathcal{H}_{z(q),\sigma} := \text{Ind}_{G_{z(q)}}^G \sigma, \quad (\text{B.7})$$

with the $C(G)$ action determined by the orbit of $g_{z(q)}$. Every $h \in H_q$ commutes with both g_x and qg_yq^{-1} , and hence with their product. It follows that $s_q^{-1}H_q s_q \subset G_{z(q)}$, so the restriction in the following formula is defined. Set

$$m_{\bar{q}}(\sigma) = \dim \text{Hom}_{H_q}(\sigma^{s_q}|_{H_q}, \chi_\eta|_{H_q} \otimes {}^q\chi_\omega|_{H_q}), \quad (\text{B.8})$$

where $\sigma^{s_q}(h) = \sigma(s_q^{-1}hs_q)$ and ${}^q\chi_\omega(h) = \chi_\omega(q^{-1}hq)$. The notation by $\bar{q}$ is justified as follows. If $q' = t_1qt_2$ with $t_1, t_2 \in T$, then $H_{q'} = H_q$, the twisted input character is unchanged, and one may take $s_{q'} = t_1s_q$. If s'_q is a second conjugator for the same output representative, then $s'_q = s_qc$ for some $c \in G_{z(q)}$; replacing s_q by s'_q conjugates the first restricted representation by $\sigma(c)$ and leaves the Hom dimension unchanged. Thus $m_{\bar{q}}(\sigma)$ depends only on the double coset and the equivalence class of σ .

On X_0 one has $H_q = Z(G)$ and $G_{z(q)} = T$. Hence all fibres below are built from fixed subgroup models; only their $C(G)$ actions, through the output holonomy $g_{z(q)}$, vary with q .

Proposition B.1 (Principal regular Mackey disintegration). *For $\bar{q} \in X_0$, let $\mathcal{K}_{\bar{q}}$ be the covariant representation on*

$$\text{Ind}_{Z(G)}^G((\chi_\eta\chi_\omega)|_{Z(G)})$$

in which G acts by left translation and $C(G)$ acts by

$$(M_F^{\bar{q}}\psi)(g) = F(gg_{z(q)}g^{-1})\psi(g), \quad F \in C(G). \quad (\text{B.9})$$

Since $d\bar{q}(X_0) = 1$, there is an equivalence of full-double representations

$$\mathcal{H}_{x,\eta} \widehat{\otimes} \mathcal{H}_{y,\omega} \cong \int_{X_0}^{\oplus} \mathcal{K}_{\bar{q}} d\bar{q}. \quad (\text{B.10})$$

Fourier decomposition through $T/Z(G)$ followed by Rokhlin disintegration along $q \mapsto z(q)$ turns (B.10) into (4.14)–(4.16).

Proof. The orbit-coordinate form of the compact-group tensor-product theorem identifies the measure normalization and the stabilizer character. The measurable representative and conjugator fields used below exist by [6, Corollary A.3].

Put $X_0 = X_{\text{pr,reg}}$. Choose Borel representatives $q = q(\bar{q}) \in G$ for $\bar{q} \in X_0$ and Borel conjugators s_q as in Eq. (B.6), and put

$$\Omega_0 = \{(g_1T, g_2T) \in G/T \times G/T : Tg_1^{-1}g_2T \in X_0\}.$$

Define

$$\begin{aligned} \Theta : (G/Z(G)) \times X_0 &\longrightarrow \Omega_0, \\ (gZ(G), \bar{q}) &\longmapsto (gs_q^{-1}T, gs_q^{-1}qT). \end{aligned} \quad (\text{B.11})$$

This is a Borel bijection. Indeed, the diagonal orbit through $(s_q^{-1}T, s_q^{-1}qT)$ has stabilizer $s_q^{-1}H_qs_q = Z(G)$, which gives bijectivity, and the chosen fields are Borel. Both spaces are standard Borel, so Θ^{-1} is Borel. Let

$$\pi : \Omega_0 \longrightarrow X_0, \quad \pi(g_1T, g_2T) = Tg_1^{-1}g_2T.$$

By definition, $d\bar{q}$ is the push-forward of the product probability measure under π . Disintegrate $(d\dot{g}_1 d\dot{g}_2)|_{\Omega_0}$ over X_0 . The product measure and π are invariant under the diagonal

G -action. For each fixed $h \in G$, uniqueness of Rokhlin disintegration therefore makes the conditional probabilities h -invariant outside a null set. Choose a countable dense subgroup of G and intersect these conull sets. On the resulting common conull set, invariance extends to every $h \in G$: for a continuous function on the compact space $G/T \times G/T$, its integral against a translated conditional measure depends continuously on h . Thus almost every conditional probability is G -invariant. For $\bar{q} \in X_0$, that fibre is the diagonal orbit through $(s_q^{-1}T, s_q^{-1}qT)$, with stabilizer $s_q^{-1}H_q s_q = Z(G)$. Its invariant probability measure is therefore the push-forward of normalized Haar measure on $G/Z(G)$. Consequently,

$$\Theta_*(d\dot{g}_{G/Z(G)} d\bar{q}) = (d\dot{g}_1 d\dot{g}_2)|_{\Omega_0}. \quad (\text{B.12})$$

Both sides have total mass one by Eq. (4.8).

Begin with bounded Borel equivariant functions in Eq. (B.3); their classes form a dense subspace of the Hilbert tensor product. Such representatives can be obtained from bounded Borel sections over $G/T \times G/T$, using Borel sections of the compact-subgroup quotient maps. On this subspace set

$$(\mathcal{U}\Phi)_{\bar{q}}(g) := \Phi(g s_q^{-1}, g s_q^{-1} q). \quad (\text{B.13})$$

If $c \in Z(G)$, then c commutes with both q and s_q , and the two input equivariance relations give

$$(\mathcal{U}\Phi)_{\bar{q}}(gc) = \chi_\eta(c)^{-1} \chi_\omega(c)^{-1} (\mathcal{U}\Phi)_{\bar{q}}(g). \quad (\text{B.14})$$

Thus the $\bar{q}$ -section belongs to $\text{Ind}_{Z(G)}^G((\chi_\eta \chi_\omega)|_{Z(G)})$. The product holonomy at the two arguments of Eq. (B.13) is

$$g s_q^{-1} (g_x q g_y q^{-1}) s_q g^{-1} = g g_{z(q)} g^{-1}. \quad (\text{B.15})$$

It follows directly that $\mathcal{U}$ intertwines the group action and the $C(G)$ action in Eqs. (B.4)–(B.5) with the fibre actions in Eq. (B.9).

Equation (B.12) gives

$$\|\Phi\|^2 = \int_{X_0} \int_{G/Z(G)} |(\mathcal{U}\Phi)_{\bar{q}}(g)|^2 d\dot{g} d\bar{q}. \quad (\text{B.16})$$

The same norm disintegration appears in [6, Lemmas 6.5–6.6 and Corollary 6.8]. Consequently $\mathcal{U}$ extends uniquely to an isometry on the Hilbert completion. To construct its inverse, take a bounded Borel field $\psi_{\bar{q}}$ in the target. The Borel inverse of Θ supplies the orbit coordinate of a pair in Ω_0 . A Borel section of $G \rightarrow G/Z(G)$ then gives $g \in G$, and the equations

$$g_1 = g s_q^{-1} t_1, \quad g_2 = g s_q^{-1} q t_2, \quad (\text{B.17})$$

determine $t_1, t_2 \in T$ measurably. Set

$$\Phi(g_1, g_2) = \chi_\eta(t_1)^{-1} \chi_\omega(t_2)^{-1} \psi_{\bar{q}}(g). \quad (\text{B.18})$$

Equation (B.18) is well defined. If the same pair is represented by (g', t'_1, t'_2) , then $g' = gc$ for some $c \in H_q = Z(G)$ and $t'_i = c^{-1} t_i$. The two character factors contributed by t'_1, t'_2

cancel the factor in (B.14) relating $\psi_{\bar{q}}(g')$ to $\psi_{\bar{q}}(g)$. Thus the right-hand side of (B.18) is unchanged. The two formulas are inverse on the dense bounded Borel subspaces, and the inverse satisfies the same norm identity. They therefore extend to inverse maps on the Hilbert-space completions. Thus the extended isometry is unitary. A different Borel choice changes this unitary by a measurable family of right translations and character factors and therefore gives an equivalent direct integral.

Induction in stages and Fourier series on the compact torus $T/Z(G)$ now give the canonical isotypic decomposition

$$\begin{aligned} \text{Ind}_{Z(G)}^G((\chi_\eta \chi_\omega)|_{Z(G)}) &\cong \text{Ind}_T^G \text{Ind}_{Z(G)}^T((\chi_\eta \chi_\omega)|_{Z(G)}) \\ &\cong \bigoplus_{\substack{\tau \in P \\ \chi_\tau|_{Z(G)} = (\chi_\eta \chi_\omega)|_{Z(G)}}} \text{Ind}_T^G \chi_\tau \hat{\otimes} \mathcal{L}_{\eta, \omega, \tau}. \end{aligned} \quad (\text{B.19})$$

Here $\mathcal{L}_{\eta, \omega, \tau}$ is the intrinsic Hom line of Eq. (4.13). Retaining it avoids making a basis choice in a one-dimensional multiplicity space. For simply connected G , two weights restrict to the same central character if and only if their difference lies in Q . The summation condition is therefore $\tau - \eta - \omega \in Q$.

Since $d\bar{q}(X_0) = 1$, the Rokhlin disintegration satisfies

$$0 = d\bar{q}(X \setminus X_0) = \int_Y p_{x,y}^z(X \setminus X_0) d\nu_{x,y}(z). \quad (\text{B.20})$$

The integrand is nonnegative, and therefore

$$p_{x,y}^z(F_{x,y}^{z, \text{pr}}) = 1 \quad \text{for } \nu_{x,y}\text{-almost every } z. \quad (\text{B.21})$$

The disintegration also furnishes the standard measurable Hilbert field

$$z \longmapsto L^2(F_{x,y}^{z, \text{pr}}, p_{x,y}^z),$$

defined up to a $\nu_{x,y}$ -null set. For each $\tau \in \eta + \omega + Q$, tensoring this field with the fixed line $\mathcal{L}_{\eta, \omega, \tau}$ preserves measurability. Fubini's theorem then gives

$$\int_{X_0}^{\oplus} \mathcal{H}_{z(q), \tau} \hat{\otimes} \mathcal{L}_{\eta, \omega, \tau} d\bar{q} \cong \int_Y^{\oplus} \mathcal{H}_{z, \tau} \hat{\otimes} L^2(F_{x,y}^{z, \text{pr}}, p_{x,y}^z; \mathcal{L}_{\eta, \omega, \tau}) d\nu_{x,y}(z). \quad (\text{B.22})$$

Choosing a unit vector in $\mathcal{L}_{\eta, \omega, \tau}$ recovers the scalar L^2 model, but that optional trivialization is not part of the intrinsic decomposition. The allowed charge set is countable. We may therefore intersect the conull sets on which its summand fields and the Rokhlin rearrangements are defined. On the resulting single $\nu_{x,y}$ -conull subset of Y , Eq. (B.22) holds simultaneously for every $\tau \in \eta + \omega + Q$. Summing over this countable set proves (4.14)–(4.16), with the stated probability normalization and the common conull convention just described. This is the principal-regular specialization of the compact-double Clebsch–Gordan decomposition in [6, Sec. 6, especially Theorem 6.10]. $\square$

This proves the measure and multiplicity-space decomposition (4.18). We next compute the scalar volume of the reduction associated with the same relative-position fibre; it is this volume that occurs in the coefficient asymptotic.

B.2 Intrinsic pair-of-pants volume and its normalization

We prove the volume formula in Proposition 4.4. For regular x, y, z , let

$$\overline{\mathcal{P}}_{x,y,z}^{\text{BF}} := \{(A, B, C) \in \mathcal{O}_x \times \mathcal{O}_y \times \mathcal{O}(g_z^{-1}) : ABC = 1\} / (G/Z(G)). \quad (\text{B.23})$$

This is the reduction of Eq. (4.23), with all orbit-type strata retained. We use the regular-value and principal-value conventions of Subsection 4.2. On a principal regular-value chamber it is a compact effective orbifold, with open principal stratum

$$\mathcal{P}_{x,y,z}^{\text{BF,pr}} := (\overline{\mathcal{P}}_{x,y,z}^{\text{BF}})_{\text{pr}}. \quad (\text{B.24})$$

This agrees with (4.30). The other orbit-type strata have lower dimension. They consequently have zero Liouville measure, and

$$V_{\text{geom}}(x, y; z) = \int_{\overline{\mathcal{P}}_{x,y,z}^{\text{BF}}} \frac{\omega_{x,y,z}^{d_G}}{d_G!} = \int_{\mathcal{P}_{x,y,z}^{\text{BF,pr}}} \frac{\omega_{x,y,z}^{d_G}}{d_G!}. \quad (\text{B.25})$$

The Borel function V_{BF} of Eq. (4.22) represents the orbital density for every fixed input pair. On a principal regular-value chamber, the geometric integral above agrees almost everywhere with that density; the pointwise polynomial identity is established in Proposition B.10.

As computed in Subsection 4.2, a regular conjugacy class has dimension $2|\Delta_+|$; hence on a principal regular-value chamber the reduced dimension is

$$\dim_{\mathbb{R}} \overline{\mathcal{P}}_{x,y,z}^{\text{BF}} = 6|\Delta_+| - 2 \dim G = 2(|\Delta_+| - \ell) = 2d_G. \quad (\text{B.26})$$

The principal-stratum extension at critical parameters is the construction in Eq. (4.32). Replacing the ineffective G -action by the effective $G/Z(G)$ -action leaves the dimension unchanged and fixes the orbifold integration convention.

We first translate the lattice conventions in the volume theorem of Meinrenken and Woodward. They denote the integral lattice by $\Lambda = \ker(\text{Exp}_T)$, use its dual Λ^* , and normalize the invariant inner product so that long roots have length $\sqrt{2}$ [46, Sec. 2 and Theorem 5.2]. Their exponential and Riemannian volume conventions are the normalized ones fixed in Eq. (2.1). Consequently

$$\Lambda = Q^\vee, \quad \Lambda^* = P, \quad \text{Vol}(T) = \text{covol}(Q^\vee). \quad (\text{B.27})$$

The normalized exponential coordinate in Eq. (2.1) incorporates the corresponding $(2\pi)^\ell$ conversion, giving the last identity exactly as written. In the next formula, $\text{Vol}(G)$ and $\text{Vol}(T)$ are the Riemannian volumes induced by the geometric inner product of Section 2. The normalized Haar measures used for orbital probabilities in Section 4 have total mass one and are denoted separately. Their Liouville volume is the degree- $2d_G$ part of $\int e^\omega$, namely $\int \omega^{d_G} / d_G!$. For a three-holed sphere, Theorem 5.2 of the cited paper contributes the factor $|Z(G)| \text{Vol}(G) / \text{Vol}(T)^3$. The inverse holonomy on the third boundary replaces its character by the complex conjugate.

Proposition B.2 (Normalization of the BF volume). *Let S be a principal regular-value chamber. Use the geometric inner product, normalize long roots to have squared length two, and put $g_x = \exp(2\pi i x)$. For fixed $x, y \in Y$, set*

$$S_{x,y} := \{z \in Y : (x, y, z) \in S\}.$$

Then, for every fixed pair (x, y) and almost every $z \in S_{x,y}$,

$$V_{\text{geom}}(x, y; z) = |Z(G)| \frac{\text{Vol}(G)}{\text{Vol}(T)^3} |\delta_G(x) \delta_G(y) \delta_G(z)| \Phi_G(x, y; z). \quad (\text{B.28})$$

Here Φ_G is the jointly measurable class-space representative constructed in Appendix B.3 from a fixed diagonal subsequence of the Peter–Weyl partial sums; we use this representative throughout. The right-hand side of Eq. (B.28) has V_{geom} as its continuous polynomial representative on S , by the volume–density relation proved below. The prefactor is

$$|Z(G)| \frac{\text{Vol}(G)}{\text{Vol}(T)^3} = \frac{\text{covol}(Q)}{\text{covol}(Q^\vee) \Pi_\rho}. \quad (\text{B.29})$$

Proof. After the translation of conventions in (B.27), Theorems 5.2 and 5.7 of [46] give (B.28). Their displayed factor $|Z(G)|$ is the stabilizer factor for the central kernel, while inversion of the third holonomy conjugates the third character.

It remains to translate the Riemannian prefactor into lattice conventions. The exponential kernel is $Q^\vee$, so $\text{Vol}(T) = \text{covol}(Q^\vee)$. The flag-manifold specialization of Macdonald’s volume formula [74] gives

$$\text{Vol}(G/T) = \Pi_\rho^{-1}, \quad \text{Vol}(G) = \frac{\text{covol}(Q^\vee)}{\Pi_\rho}. \quad (\text{B.30})$$

Moreover $P = (Q^\vee)^*$, hence $\text{covol}(P) = \text{covol}(Q^\vee)^{-1}$, and $[P : Q] = |Z(G)|$. Therefore

$$\text{covol}(Q) = |Z(G)| \text{covol}(P) = \frac{|Z(G)|}{\text{covol}(Q^\vee)}, \quad (\text{B.31})$$

which proves (B.29). The group-valued Duistermaat–Heckman result, specifically Theorem 4.1 of [45], supplies the volume–density relation. After local linearization [45, Secs. 2.4 and 4.1], the existence of a polynomial representative on each regular-value chamber follows from the ordinary Duistermaat–Heckman theorem [75]. $\square$

Remark B.3 (Rank-two centre normalization). For $G = SU(3)$ one has $Q = Q^\vee$, $\text{covol}(Q) = \sqrt{3}$, and

$$\Pi_\rho = (2\pi)(2\pi)(4\pi) = 16\pi^3. \quad (\text{B.32})$$

Hence both the lattice and group-volume forms give the prefactor $1/(16\pi^3)$.

The orbital-convolution density follows independently from Weyl integration. Indeed, Peter–Weyl inversion gives the central density $\Phi_G(x, y; z)$ with respect to Haar measure, while the push-forward of normalized Haar measure to Y is

$$\frac{|\delta_G(z)|^2}{\text{covol}(Q^\vee)} dz. \quad (\text{B.33})$$

Substitution of (4.37) proves (4.38). This gives an independent probabilistic interpretation of the same normalization.

B.3 Fusion hypergroups and orbital convolution

Let $a = a_k$ and $b = b_k$ satisfy $z_{a_k} \rightarrow x$ and $z_{b_k} \rightarrow y$, with $x, y \in Y$, as in Proposition 4.5. Fix $\lambda \in P_+$; then $\lambda \in P_k^+$ for all sufficiently large k . The Kac–Peterson character formula, unitarity of $S^{(k)}$, and the Verlinde formula give the following hypergroup identity [39, 76]:

$$\sum_{c \in P_k^+} N_{ab}^{(k)c} \frac{d_c^{(k)}}{d_a^{(k)} d_b^{(k)}} \chi_\lambda(g_{z_c}) = \frac{\chi_\lambda(g_{z_a}) \chi_\lambda(g_{z_b})}{d_\lambda^{(k)}}. \quad (\text{B.34})$$

To fix the conjugation convention, put $p_c = N_{ab}^{(k)c} d_c^{(k)} / (d_a^{(k)} d_b^{(k)})$; this is the fusion hypergroup probability in Eq. (6) of [14]. The corresponding normalized-character identity is Eq. (7) of the same reference. Symmetry of $S^{(k)}$, Verlinde diagonalization, and

$$\frac{S_{\lambda c}^{(k)}}{S_{0c}^{(k)}} = \chi_\lambda(e^{-2\pi i z_c}) = \overline{\chi_\lambda(g_{z_c})} \quad (\text{B.35})$$

give the pre-conjugated identity

$$\begin{aligned} \sum_c p_c \overline{\chi_\lambda(g_{z_c})} &= \frac{1}{S_{00}^{(k)} d_a^{(k)} d_b^{(k)}} \sum_c N_{ab}^{(k)c} S_{c\lambda}^{(k)} \\ &= \frac{\chi_\lambda(g_{z_a}) \chi_\lambda(g_{z_b})}{d_\lambda^{(k)}}. \end{aligned} \quad (\text{B.36})$$

The numbers p_c and $d_\lambda^{(k)}$ are real. Complex conjugation of (B.36) therefore gives (B.34) with output point z_c . Taking $\lambda = 0$ proves that (4.41) has mass one.

For fixed λ , one has $d_\lambda^{(k)} \rightarrow \dim V_\lambda$ and $\chi_\lambda(g_{z_{a_k}}) \rightarrow \chi_\lambda(g_x)$, with the analogous statement for b_k . Thus the right-hand side of (B.34) converges to

$$\frac{\chi_\lambda(g_x) \chi_\lambda(g_y)}{\dim V_\lambda} = \int_G \chi_\lambda(g) d(\nu_x * \nu_y)(g). \quad (\text{B.37})$$

The linear span of the restrictions of irreducible characters to $\bar{Y}$ is a unital self-adjoint algebra: products decompose into characters, and complex conjugation sends χ_λ to χ_λ^* . Characters separate conjugacy classes and hence points of the closed alcove. By the Stone–Weierstrass theorem and the common probability normalization, convergence of all character moments implies the weak convergence in Proposition 4.5. This is the character-moment proof of the limit established in [14, Proposition 7.1 and Corollary 7.2].

The same convention fixes the density without relabelling. Its Peter–Weyl coefficients give the following expansion in $L^2(G)$; convergence is verified below:

$$D_{x,y}(g) = \sum_{\lambda \in P_+} \frac{\chi_\lambda(g_x) \chi_\lambda(g_y)}{\dim V_\lambda} \chi_\lambda(g^{-1}) = \sum_{\lambda \in P_+} \frac{\chi_\lambda(g_x) \chi_\lambda(g_y) \overline{\chi_\lambda(g)}}{\dim V_\lambda}. \quad (\text{B.38})$$

Thus an L^2 representative of the class-space density agrees with $\Phi_G(x, y; z)$ for almost every z , in the convention of (4.35). Using an unconjugated third character would instead describe $D_{x,y}(g^{-1})$.

It remains to check that (B.38) is an L^2 series for regular inputs. The Weyl character and dimension formulas give

$$|\chi_\lambda(g_x)| \leq \frac{|W|}{|\delta_G(x)|}, \quad \dim V_\lambda \geq \prod_{j=1}^{\ell} (\lambda_j + 1), \quad \lambda_j = \langle \lambda, \alpha_j^\vee \rangle. \quad (\text{B.39})$$

In the dimension formula the factors associated with the simple roots give the displayed product, and each remaining positive-root factor is at least one. For fixed regular x, y , the square of the coefficient of $\overline{\chi}_\lambda$ is therefore bounded by a constant times $\prod_{j=1}^{\ell} (\lambda_j + 1)^{-2}$. Its sum over P_+ is at most a constant multiple of $\zeta(2)^\ell$. Since irreducible characters are orthonormal among central functions, the series converges unconditionally in $L^2(G)$. Its limit has the character moments of $\nu_x * \nu_y$. Both define finite central measures, since $L^2(G) \subset L^1(G)$ for Haar probability. Density of the character algebra on the class space identifies these measures. In particular, the L^2 limit is nonnegative almost everywhere and has integral one. Weyl integration together with (B.33) now gives (4.38).

For varying boundary parameters, construct a jointly measurable representative whose section gives the orbital-convolution density for each fixed regular pair.

Choose increasing finite subsets $F_N \subset P_+$ whose union is P_+ , and let $S_N(x, y; g)$ be the corresponding partial sum in Eq. (B.38). If $K \Subset Y^2$, the preceding coefficient bound is uniform on K , because the Weyl denominators stay bounded away from zero. The tail estimate therefore gives

$$\sup_{(x,y) \in K} \|S_N(x, y; \cdot) - D_{x,y}\|_{L^2(G)} \longrightarrow 0. \quad (\text{B.40})$$

Exhaust Y^2 by compact sets $K_1 \subset K_2 \subset \dots$ and choose a diagonal subsequence N_j such that, whenever $j \geq m$,

$$\sup_{(x,y) \in K_m} \|S_{N_{j+1}}(x, y; \cdot) - S_{N_j}(x, y; \cdot)\|_{L^2(G)} \leq 2^{-j}. \quad (\text{B.41})$$

Haar measure is a probability measure, so the same bound holds in $L^1(G)$. For each fixed (x, y) , summing the L^1 bounds and applying Tonelli's theorem shows that $S_{N_j}(x, y; g)$ converges for almost every g . Define the function to be $\lim_j S_{N_j}(x, y; g)$ where this limit exists and zero otherwise. The result is jointly measurable. For each fixed regular pair the same subsequence converges in $L^2(G)$, so its almost-everywhere limit represents the L^2 class of $D_{x,y}$. Each partial sum is central, so the set on which the subsequence converges is conjugation invariant. Weyl integration therefore converts its Haar-null exceptional set into a z -null subset of the alcove for each fixed (x, y) . Evaluating the resulting central representative at g_z defines a function on the alcove. We denote this jointly measurable class-space representative by $\Phi_G(x, y; z)$. It is the representative used in the main text and in the Fubini argument below.

B.4 Tensor multiplicities in the non-affine region

For dominant weights a, b, c , let

$$m_{ab}^c = \dim \text{Hom}_G(V_c, V_a \otimes V_b) \quad (\text{B.42})$$

be the ordinary tensor-product multiplicity, and extend this notation by $m_{ab}^\lambda = 0$ when $\lambda \notin P_+$. We first treat a region in which only the identity affine Weyl element contributes. Put

$$m_{Q^\vee} = \min\{\|\beta^\vee\| : \beta^\vee \in Q^\vee \setminus \{0\}\}, \quad \mathcal{R}_G = \{(x, y, z) \in Y^3 : \|x + y\| + \|z\| < m_{Q^\vee}\}. \quad (\text{B.43})$$

Proposition B.4 (Uniform Kac–Walton stabilization). *For every compact $K \in \mathcal{R}_G$, there is $k_0(K)$ such that*

$$N_{ab}^{(k)c} = m_{ab}^c \quad (\text{B.44})$$

for $k \geq k_0(K)$ and all $a, b, c \in P_k^+$ with $(z_a, z_b, z_c) \in K$.

Proof. The Kac–Walton formula, as stated in [24, Exercise 13.35] and proved in [32, 33], is

$$N_{ab}^{(k)c} = \sum_{(\bar{w}, \beta^\vee) \in W \ltimes Q^\vee} \epsilon(\bar{w}) m_{ab}^{\bar{w}(c+\rho) + r_k \beta^\vee - \rho}. \quad (\text{B.45})$$

Here $\beta^\vee$ is the unscaled coroot: the corresponding affine translation is by $r_k \beta^\vee$. This agrees with the convention in Eq. (2.46). Suppose first that $\beta^\vee = 0$. If $\bar{w} \neq 1$, then $\bar{w}(c + \rho)$ lies in a Weyl chamber different from the strictly dominant chamber containing $c + \rho$. It therefore cannot equal $\lambda + \rho$ for a dominant weight λ , so the corresponding tensor multiplicity vanishes.

Now let $\beta^\vee \neq 0$ and put

$$\lambda = \bar{w}(c + \rho) + r_k \beta^\vee - \rho.$$

If $m_{ab}^\lambda \neq 0$, then λ is dominant and $\lambda \leq a + b$ in the dominance order. For the norm estimate, write $a + b - \lambda = \sum_i n_i \alpha_i$ with $n_i \geq 0$. Since both $a + b$ and λ are dominant,

$$\|a + b\|^2 - \|\lambda\|^2 = \sum_i n_i (\alpha_i, a + b + \lambda) \geq 0.$$

Thus $\|\lambda\| \leq \|a + b\|$. On the other hand,

$$\|\lambda\| \geq r_k \|\beta^\vee\| - \|c + \rho\| - \|\rho\|, \quad \|a + b\| \leq r_k \|z_a + z_b\| + 2\|\rho\|. \quad (\text{B.46})$$

Since $\|c + \rho\| = r_k \|z_c\|$, these inequalities give

$$\|\beta^\vee\| \leq \|z_a + z_b\| + \|z_c\| + \frac{3\|\rho\|}{r_k}. \quad (\text{B.47})$$

Compact containment in (B.43) leaves a positive gap between the right-hand side and $m_{Q^\vee}$. Hence no nonzero $\beta^\vee$ can occur once k is sufficiently large, uniformly on the chosen compact set. Only $(\bar{w}, \beta^\vee) = (1, 0)$ remains, and this term is m_{ab}^c . $\square$

We obtain the tensor-multiplicity asymptotic from a vector-partition formula. Its parameter map may have image contained in a wall of the ambient chamber fan. We therefore first prove a relative chamber statement that includes these walls and their intersections, as well as the proper faces on which the active column configuration changes.

2691 **Lemma B.5** (Relative vector-partition chambers). *Let $A : \mathbb{Z}^n \rightarrow \Gamma$ be a surjective ho-*
 2692 *momorphism to a lattice of rank s . Assume $\ker A_{\mathbb{R}} \cap \mathbb{R}_{\geq 0}^n = \{0\}$, and let $b : L \rightarrow \Gamma$ be a*
 2693 *homomorphism from another lattice. Put $d = n - s$ and*

$$\varphi_A(u) = \#\{v \in \mathbb{Z}_{\geq 0}^n : Av = u\}. \quad (\text{B.48})$$

2694 *There is a finite rational conical polyhedral complex $\mathcal{C}(A, b)$ whose support is $b_{\mathbb{R}}^{-1}(\text{cone}(A))$.*
 2695 *Its cells are understood relative to $L_{\mathbb{R}}$: every cell is a relatively open rational polyhedral cone*
 2696 *in its affine span, and it may have lineality when $\ker b_{\mathbb{R}} \neq 0$. On every relatively open cell*
 2697 *C and on every coset of one finite-index sublattice of L , the function $\varphi_A \circ b$ is the restriction*
 2698 *to C of a polynomial on $\text{aff}(C)$ of degree at most d .*

2699 *The complex may be chosen so that each cell either lies in $b_{\mathbb{R}}^{-1}(\text{relint cone}(A))$ or maps*
 2700 *into a proper face of $\text{cone}(A)$. We call a cell C strictly feasible in the first case. Equiva-*
 2701 *lently, for one, and hence for every, $\lambda \in C$, there is a vector $v \in \mathbb{R}_{> 0}^n$ such that*

$$Av = b_{\mathbb{R}}(\lambda). \quad (\text{B.49})$$

2702 *On a strictly feasible cell, the homogeneous degree- d term is independent of the period coset*
 2703 *and is*

$$T_A(b_{\mathbb{R}}(\lambda)), \quad T_A(u) = \text{vol}_{\ker A \cap \mathbb{Z}^n} \{v \in \mathbb{R}_{\geq 0}^n : Av = u\}. \quad (\text{B.50})$$

2704 *Here the measure on an affine fibre is normalized by its direction lattice $\ker A \cap \mathbb{Z}^n$. For*
 2705 *every strictly feasible relatively open cell C , every $K \subseteq C$, and every $B < \infty$, uniformly*
 2706 *for sequences*

$$\begin{aligned} \lambda_r &= ru_r + e_r \in L \cap C, \\ u_r &\in K, \quad e_r \in L_{\mathbb{R}}, \quad \|e_r\| \leq B, \\ r &\longrightarrow \infty \quad \text{through positive integers,} \end{aligned} \quad (\text{B.51})$$

2707 *one has*

$$\varphi_A(b(\lambda_r)) = T_A(b_{\mathbb{R}}(\lambda_r)) + O_{K,B}(r^{d-1}), \quad (\text{B.52})$$

2708 *with no remainder if $d = 0$. On every other cell, including a cell where the fibre dimension*
 2709 *drops, the conclusion is the polynomial statement and the degree- d upper bound. These*
 2710 *conclusions remain valid when $b_{\mathbb{R}}(L_{\mathbb{R}})$ is contained in an ambient chamber wall.*

2711 *Proof.* The chamber structure is that of [17]. We apply the chamber and enlarged-sector
 2712 formulas of [18] to the full configuration and to the configurations carried by its proper
 2713 faces. Pullback to $L_{\mathbb{R}}$ then requires a common period lattice and agreement of the formulas
 2714 on internal walls. We establish these before identifying the leading term.

2715 *Face reduction and the relative complex.* Let $A_1, \dots, A_n$ be the columns of A . Begin with
 2716 the face decomposition of $\text{cone}(A)$. For each face F , let

$$J(F) = \{j : A_j \in F\}, \quad (\text{B.53})$$

2717 and refine $\text{relint } F$ by the chamber fan of the subconfiguration $(A_j)_{j \in J(F)}$ in $\text{span}_{\mathbb{R}} F$. There
 2718 are only finitely many faces and hence only finitely many such chamber fans. Their facewise

common refinement, pulled back by $b_{\mathbb{R}}$, gives the complex $\mathcal{C}(A, b)$. This construction takes place in the parameter space $L_{\mathbb{R}}$ and allows the image of $b_{\mathbb{R}}$ to meet the ambient chamber walls in arbitrary relative position.

The passage to a smaller column configuration on an outer face has a simple reason. If $u \in \text{relint } F$ and $u = \sum_j v_j A_j$ with $v_j \geq 0$, choose a supporting functional $\ell_F \geq 0$ with $F = \{\ell_F = 0\}$. Then

$$0 = \ell_F(u) = \sum_j v_j \ell_F(A_j). \quad (\text{B.54})$$

Every summand is nonnegative, and consequently $v_j = 0$ whenever $A_j \notin F$. The term “active columns” will mean the columns which are allowed to occur, namely those indexed by $J(F)$; an individual representation of u may use a proper subset of them. For an active set $J = J(F)$, put

$$\Gamma_J = \sum_{j \in J} \mathbb{Z} A_j, \quad \bar{\Gamma}_J = \Gamma \cap \text{span}_{\mathbb{R}}\{A_j : j \in J\}. \quad (\text{B.55})$$

The first lattice is generated by the active columns, while the second is its saturation in the ambient lattice, so $\bar{\Gamma}_J/\Gamma_J$ is finite. On the face F , the partition function vanishes unless $u \in \Gamma_J$; when this condition holds it is the vector-partition function of the surjective map $A_J : \mathbb{Z}^J \rightarrow \Gamma_J$. Thus a lower face is treated in the real span and integral lattice of its active columns, rather than in the ambient target lattice. If $s_J = \text{rank } A_J$, adjoining the $n - |J|$ omitted columns can raise the rank by at most $n - |J|$. Hence

$$s \leq s_J + n - |J|, \quad |J| - s_J \leq n - s = d. \quad (\text{B.56})$$

The polynomial degree of every lower-face partition function is therefore at most d ; a proper subconfiguration can attain degree d only on the proper face on which it is active.

Chamber closures and a common period. For each active configuration A_J , the exact counting statement we use is Theorem 0.2 of [18], specialized to the constant function $f = 1$. If $\mathfrak{c}$ is a relatively open chamber of its chamber fan, put

$$\text{Box}(A_J) = \sum_{j \in J} [0, 1] A_j.$$

In the notation of that theorem, $\varphi_{A_J}(u) = \mathcal{P}[\mathfrak{c}, 1, A_J](u)$ for $u \in (\mathfrak{c} - \text{Box}(A_J)) \cap \Gamma_J$; the observation following the theorem records that this enlarged sector contains $\bar{\mathfrak{c}}$. The residue formula of Theorem 3.1 there expands $\mathcal{P}[\mathfrak{c}, 1, A_J]$ as follows.

$$\varphi_{A_J}(u) = \sum_{\chi \in \mathcal{X}_{\mathfrak{c}, J}} \chi(u) P_{\mathfrak{c}, J, \chi}(u), \quad u \in (\mathfrak{c} - \text{Box}(A_J)) \cap \Gamma_J, \quad (\text{B.57})$$

Here $\mathcal{X}_{\mathfrak{c}, J}$ is a finite set of finite-order characters of Γ_J , and each $P_{\mathfrak{c}, J, \chi}$ is an ordinary polynomial on $\text{span}_{\mathbb{R}}\{A_j : j \in J\}$. Thus Eq. (B.57) holds on the chamber closure as well as in its interior. There are finitely many active sets J and finitely many chambers $\mathfrak{c}$ for each active configuration. Hence

$$\mathcal{X}_J^{\text{all}} := \bigcup_{\mathfrak{c}} \mathcal{X}_{\mathfrak{c}, J} \quad (\text{B.58})$$

is finite. Choose one positive integer N_0 , simultaneously for all active sets and all their chambers, such that

$$N_0 \bar{\Gamma}_J \subset \Gamma_J, \quad \chi|_{N_0 \Gamma_J} = 1 \quad (\chi \in \mathcal{X}_J^{\text{all}}) \quad (\text{B.59})$$

for every active set J . One may take a common multiple of the exponents of $\bar{\Gamma}_J/\Gamma_J$ and the orders of all characters in $\mathcal{X}_J^{\text{all}}$. Put

$$L_0 = N_0^2 L. \quad (\text{B.60})$$

If λ and λ' lie in the same relative face and in one coset of L_0 , then, for the corresponding active set J ,

$$\begin{aligned} b(\lambda) - b(\lambda') &\in N_0^2 \Gamma \cap \text{span}_{\mathbb{R}}\{A_j : j \in J\} \\ &= N_0^2 \bar{\Gamma}_J \subset N_0 \Gamma_J \subset \Gamma_J. \end{aligned} \quad (\text{B.61})$$

The equality holds because the span of the active columns is a real vector space: if $N_0^2 \gamma$ lies in that span, then so does γ . The inclusion in Γ_J makes membership in the active-column lattice constant on the coset. If that membership holds, the stronger inclusion in $N_0 \Gamma_J$ makes every character in (B.57) constant there as well. Both conditions are ensured by the choice $L_0 = N_0^2 L$. Thus the same finite-index period lattice works after pullback on every relative face, including a cell contained in an ambient wall.

Agreement on internal walls. We spell out the internal-wall case, where no column need be omitted. Let σ be a relatively open cell of the common refinement such that $b_{\mathbb{R}}(\sigma)$ lies in an arbitrary intersection of internal chamber walls for A_J , and fix a coset D of L_0 meeting σ . Membership of $b(D \cap \sigma)$ in Γ_J is constant by the preceding calculation. If membership fails, $\varphi_A \circ b$ vanishes on $D \cap \sigma$. Otherwise, for every chamber $\mathfrak{c}$ with $b_{\mathbb{R}}(\sigma) \subset \bar{\mathfrak{c}}$, restrict Eq. (B.57) to $D \cap \sigma$ and compose with $b_{\mathbb{R}}$. The characters are constant on $b(D \cap \sigma)$, so this restriction is an ordinary polynomial on $\text{aff}(\sigma)$; the closure formula shows that it equals $\varphi_A \circ b$ at every lattice point of $D \cap \sigma$. If two chamber closures contain the same stratum, the resulting restricted polynomials agree. Both equal $\varphi_A \circ b$ on the lattice points of $D \cap \sigma$, and these points are Zariski dense in $\text{aff}(\sigma)$ whenever the intersection is nonempty.

Rationality also gives this density directly. If $q = \dim \sigma > 0$, the direction lattice

$$L_{\sigma} = L_0 \cap \text{lin}(\sigma) \quad (\text{B.62})$$

has full rank in $\text{lin}(\sigma)$. A relatively open rational cone contains q linearly independent lattice vectors $v_1, \dots, v_q \in L_{\sigma} \cap \sigma$. Starting from any $\lambda_0 \in D \cap \sigma$, the fact that σ is a convex cone gives

$$\lambda_0 + n_1 v_1 + \dots + n_q v_q \in D \cap \sigma, \quad n_1, \dots, n_q \in \mathbb{Z}_{\geq 0}. \quad (\text{B.63})$$

A polynomial which vanishes on this grid is zero: apply the one-variable fact successively in the q independent directions. The zero-dimensional case is immediate. The two chamber polynomials therefore have the same restriction to σ . This proves compatibility on internal strata of arbitrary codimension. On an outer face the same argument is applied after replacing A by its active subconfiguration A_J .

Leading term and uniform remainder. We next identify the degrees of the residue terms on a strictly feasible cell, where the active set consists of all n columns. Thus $J = \{1, \dots, n\}$ in Eq. (B.57); in the remainder of this paragraph we suppress that fixed index and write $P_{\mathbf{c}, \chi} = P_{\mathbf{c}, J, \chi}$. For a torsion character χ , let

$$I_\chi = \{j : \chi(A_j) = 1\}. \quad (\text{B.64})$$

Near the pole associated with χ , precisely the denominator factors indexed by I_χ have a zero. If these columns do not span $\Gamma_{\mathbb{R}}$, the corresponding total residue is zero. Otherwise the local denominator has $|I_\chi|$ simple polar factors, while total residue extracts degree $-s$. The proof of Lemma 2.1 in [18] therefore gives

$$\deg P_{\mathbf{c}, \chi} \leq |I_\chi| - s. \quad (\text{B.65})$$

The same estimate can be seen directly by expanding

$$(1 - e^{-\langle A_j, z \rangle})^{-1} = \langle A_j, z \rangle^{-1} + \frac{1}{2} + O(z), \quad j \in I_\chi. \quad (\text{B.66})$$

A term of polynomial degree m can enter the total residue only when $m - |I_\chi| \leq -s$.

For the identity character, $I_\chi = \{1, \dots, n\}$, and the degree- d part is the continuous partition function. More precisely, Eq. (3.2) of [18], equivalently the continuous residue formula of [16], gives

$$P_1^{[d]}(u) = T_A(u) = \text{vol}_{\ker A \cap \mathbb{Z}^n} \{v \in \mathbb{R}_{\geq 0}^n : Av = u\}. \quad (\text{B.67})$$

Because $A : \mathbb{Z}^n \rightarrow \Gamma$ is surjective, a nonidentity character is nontrivial on at least one column. Hence $|I_\chi| \leq n - 1$, and (B.65) gives

$$\deg P_{\mathbf{c}, \chi} \leq n - 1 - s = d - 1. \quad (\text{B.68})$$

Strict feasibility is used here to ensure that the full n -column configuration, rather than a proper face configuration, controls the fibre. Thus the degree- d term is independent of the period coset and is T_A .

After subtracting that common leading term, only finitely many polynomials of degree at most $d - 1$ remain, on finitely many period cosets. If $P = \sum_{j=0}^{d-1} P^{[j]}$ is one of them and $\lambda_r = ru_r + e_r$ is as in the statement, multilinearity of its homogeneous pieces gives

$$|P(ru_r + e_r)| \leq C_{P, K, B}(1 + r)^{d-1}. \quad (\text{B.69})$$

Taking the maximum over the finite character set, relative cells, and period cosets yields a constant depending only on K and B . This proves (B.52). If $d = 0$, surjectivity and equality of the two ranks make A a lattice isomorphism. On a strictly feasible cell the real fibre is one positive point, so the partition count and its normalized zero-dimensional volume are both one; the remainder is zero. A proper subconfiguration can have fibre dimension d and can therefore contribute a term of order r^d . Such a term is confined to a proper relative face and cannot enter a compact subset of a strictly feasible cell. On these faces the stated global degree bound survives, while the leading polynomial is the one supplied by the active face configuration and may differ from the restrictions of its neighbors. $\square$

Theorem B.6 (Polyhedral multiplicity theorem). *There is a finite rational polyhedral complex $\mathcal{C}_G$, supported on the real tensor cone, and a finite-index sublattice $\Lambda_0 \subset \mathcal{L}_G$ with the following properties. On every relatively open cell C and every coset of Λ_0 , the restriction of m_{ab}^c to $C \cap \mathcal{L}_G \cap (P_+)^3$ is a polynomial of degree at most d_G .*

A relatively open cell C is called strictly feasible if, for one and hence every $(a, b, c) \in C$, the reduced BZ fibre contains a point for which

$$t_j > 0 \quad (1 \leq j \leq |\Delta_+|) \quad (\text{B.70})$$

and every retained trail inequality is strict. Equivalently, in the reduced presentation described in the proof below, all Lusztig coordinates and all retained slack coordinates can be chosen strictly positive. On such a cell, the homogeneous degree- d_G term is independent of the Λ_0 -coset and equals the lattice-normalized Berenstein–Zelevinsky fibre volume V_C . More precisely, for every $K \Subset C$ and $B < \infty$, uniformly for compatible lattice parameters

$$\begin{aligned} \lambda_r &= (a_r, b_r, c_r) = ru_r + e_r \in \mathcal{L}_G \cap (P_+)^3 \cap C, \\ u_r &\in K, \quad \|e_r\| \leq B, \\ r &\longrightarrow \infty \quad \text{through positive integers,} \end{aligned} \quad (\text{B.71})$$

one has

$$m_{a_r b_r}^{c_r} = V_C(a_r, b_r; c_r) + O_{K,B}(r^{d_G-1}), \quad (\text{B.72})$$

with no remainder when $d_G = 0$. A strictly feasible cell may itself lie in an internal wall of the ambient vector-partition fan. On every other cell the conclusion is the polynomial degree bound stated above. Globally, for all $a, b, c \in P_+$,

$$m_{ab}^c \leq C_G(1 + \|a\| + \|b\| + \|c\|)^{d_G}. \quad (\text{B.73})$$

Proof. Reduced BZ presentation. Fix a reduced word $\mathbf{i} = (i_1, \dots, i_{|\Delta_+|})$ for the longest Weyl element w_0 . The associated Berenstein–Zelevinsky parametrization turns the tensor multiplicity into a pointed vector-partition function; we first describe this reduction and then identify the lattice that normalizes its leading volume. Put

$$\beta_j = s_{i_1} \cdots s_{i_{j-1}} \alpha_{i_j}, \quad 1 \leq j \leq |\Delta_+|. \quad (\text{B.74})$$

These are the positive roots, each occurring once. We use Theorem 2.2 of [15], with the source notation (λ, ν, μ) replaced by (a, b, c) . If $\kappa_j(\pi) \in \mathbb{Z}_{\geq 0}$ denotes the integer attached to the j th step of a $\mathbf{i}$ -trail π , that theorem says that m_{ab}^c is the number of $t = (t_1, \dots, t_{|\Delta_+|}) \in \mathbb{Z}_{\geq 0}^{|\Delta_+|}$ satisfying

$$R(t) := \sum_{j=1}^{|\Delta_+|} t_j \beta_j = a + b - c \quad (\text{B.75})$$

and the following two finite families of inequalities:

$$\sum_j \kappa_j(\pi) t_j \geq \langle s_i a + b - c, \omega_i^\vee \rangle, \quad (\text{B.76})$$

$$\sum_j \kappa_j(\pi') t_j \geq \langle a + s_i b - c, \omega_i^\vee \rangle. \quad (\text{B.77})$$

Here π runs through the $\mathbf{i}$ -trails from $s_i\omega_i^\vee$ to $w_0\omega_i^\vee$, while π' runs through the trails from $\omega_i^\vee$ to $w_0s_i\omega_i^\vee$, in the fundamental module of the Langlands dual Lie algebra. We regard the i th fundamental coweight $\omega_i^\vee$ as a fundamental weight of the Langlands dual root system. We enumerate these finitely many inequalities as

$$L_r(t) \geq \ell_r(a, b, c), \quad 1 \leq r \leq M. \quad (\text{B.78})$$

The balance equation (B.75) is the sole affine identity of the joint family. Substituting $c = a + b - R(t)$ into the two right-hand sides above gives

$$\langle s_i a + b - c, \omega_i^\vee \rangle = -\langle a, \alpha_i^\vee \rangle + \langle R(t), \omega_i^\vee \rangle, \quad (\text{B.79})$$

$$\langle a + s_i b - c, \omega_i^\vee \rangle = -\langle b, \alpha_i^\vee \rangle + \langle R(t), \omega_i^\vee \rangle. \quad (\text{B.80})$$

Choose an integral vector t° with every coordinate positive. There are only finitely many trails, so taking all Dynkin labels of a and b sufficiently large makes both trail families strict at $t = t^\circ$. The labels may be taken still larger so that $c = a + b - R(t^\circ)$ is strictly dominant. The resulting point has every t -coordinate and every trail slack positive. It therefore has a relative neighbourhood, inside the affine subspace cut out by (B.75), on which all these inequalities remain strict. No further affine equality can hold identically on the family.

The relevant joint rational polyhedral family is

$$\{((a, b, c), t) : R(t) = a + b - c, \quad t_j \geq 0, \quad L_r(t) \geq \ell_r(a, b, c)\} \subset \mathcal{L}_{G, \mathbb{R}} \times \mathbb{R}^{|\Delta_+|}. \quad (\text{B.81})$$

We retain all coordinate inequalities $t_j \geq 0$. Among the trail inequalities, choose an irredundant subfamily defining the same joint polyhedron in the affine space (B.75), and relabel its rows as in (B.78). Since the joint family has nonempty relative interior in that affine space, every retained row defines a relative facet. In particular, no row whose slack vanishes identically is retained. We continue to denote the number of retained trail rows by M .

Removing a redundant trail row does not alter any t -fibre or its intrinsic direction lattice

$$\left\{ t \in \mathbb{Z}^{|\Delta_+|} : \sum_j t_j \beta_j = 0 \right\}. \quad (\text{B.82})$$

If an omitted row is restored, its slack is the uniquely determined graph coordinate $L_r(t) - \ell_r(a, b, c)$. After the common denominator is cleared below, this coordinate is integral on the lattice under consideration. Projection to the t -variables therefore identifies the direction lattice of the enlarged slack presentation with (B.82). Passing to the irredundant trail list changes neither the lattice-point count nor the lattice normalization of the fibre volume.

For this reduced presentation, a parameter is *strictly feasible* when its fibre contains a point with $t_j > 0$ for every j and with every retained trail inequality strict. Equivalently, the fibre meets the relative interior of the total BZ family. This is stronger than merely requiring the fibre itself to have nonempty relative interior.

Here the L_r and ℓ_r are fixed rational linear forms. They are obtained from the finitely many $\mathbf{i}$ -trails in fundamental representations of the Langlands dual Lie algebra. Choose a positive integer q_0 that clears the denominators of all original trail forms, including the rows omitted from the reduced presentation above. In particular, for every retained row r ,

$$q_0 L_r(\mathbb{Z}^{|\Delta_+|}) \subset \mathbb{Z}, \quad q_0 \ell_r(P^3) \subset \mathbb{Z}, \quad (\text{B.83})$$

and set

$$C_r = q_0 L_r, \quad h_r = q_0 \ell_r, \quad C(t) = (C_1(t), \dots, C_M(t)), \quad h = (h_1, \dots, h_M), \quad (\text{B.84})$$

with $R(t)$ as in (B.75). Thus $C(t) \in \mathbb{Z}^M$ for integral t and $h(a, b, c) \in \mathbb{Z}^M$ for every weight triple, in particular for every compatible triple. Then

$$m_{ab}^c = \#\{t \in \mathbb{Z}_{\geq 0}^{|\Delta_+|} : R(t) = a + b - c, \ C(t) \geq h(a, b, c)\}. \quad (\text{B.85})$$

Pointed partition model. Introduce slack variables to express the BZ inequalities as a pointed vector-partition problem. The uniquely determined slack variable is $s = C(t) - h(a, b, c) \in \mathbb{Z}_{\geq 0}^M$. On $\Gamma := Q \oplus \mathbb{Z}^M$ define

$$A : \mathbb{Z}^{|\Delta_+|+M} \longrightarrow \Gamma, \quad A(t, s) = (R(t), C(t) - s), \quad \mathbf{b}(a, b, c) = (a + b - c, h(a, b, c)). \quad (\text{B.86})$$

The compatibility condition $(a, b, c) \in \mathcal{L}_G$ is exactly what makes the first component of $\mathbf{b}(a, b, c)$ lie in Q . Together with (B.83), this shows that $\mathbf{b} : \mathcal{L}_G \rightarrow Q \oplus \mathbb{Z}^M$ is a lattice homomorphism. Equation (B.85) becomes

$$m_{ab}^c = \varphi_A(\mathbf{b}(a, b, c)), \quad \varphi_A(u) = \#\{v \in \mathbb{Z}_{\geq 0}^{|\Delta_+|+M} : Av = u\}. \quad (\text{B.87})$$

The two kinds of columns of A are

$$(\beta_j, C(e_j)), \quad 1 \leq j \leq |\Delta_+|, \quad \text{and} \quad (0, -e_r), \quad 1 \leq r \leq M. \quad (\text{B.88})$$

Here e_j and e_r denote the standard basis vectors in the two respective coordinate factors. These columns generate the target lattice:

$$\mathbb{Z}A = \Gamma = Q \oplus \mathbb{Z}^M. \quad (\text{B.89})$$

Indeed, the slack columns $(0, -e_r)$ generate the second factor. Adding appropriate integral combinations of them to $(\beta_j, C(e_j))$ produces $(\beta_j, 0)$. The roots β_j run through all positive roots, so they include the simple roots and generate Q . It follows that

$$\text{rank } A = \ell + M, \quad \dim_{\mathbb{R}} \ker A = (|\Delta_+| + M) - (\ell + M) = d_G. \quad (\text{B.90})$$

The configuration is pointed. Namely, if $A(t, s) = 0$ with $t, s \geq 0$, then $R(t) = 0$. Pairing with a strictly dominant coweight gives $t = 0$, and then $s = 0$. Thus

$$\ker A \cap \mathbb{R}_{\geq 0}^{|\Delta_+|+M} = \{0\}. \quad (\text{B.91})$$

In particular, every partition fibre is bounded, as required by the chamber theorem.

For the present configuration, put $N = |\Delta_+| + M$. Then

$$A(\mathbb{R}_{\geq 0}^N) = \text{relint cone}(A).$$

Indeed, a strictly positive combination lies in the relative interior. In the other direction, if u is in that relative interior, then for all sufficiently small $\varepsilon > 0$ the vector $u - \varepsilon \sum_j A_j$ remains in the cone, which gives a representation of u with every coefficient positive. It follows that, if $C_A := \text{cone}(A)$ and

$$\tilde{\mathcal{T}}_G := \mathbf{b}_{\mathbb{R}}^{-1}(C_A) \tag{B.92}$$

is the feasible parameter cone before dominance is imposed, then its strictly feasible part is

$$\tilde{\mathcal{T}}_G^{\text{sf}} = \mathbf{b}_{\mathbb{R}}^{-1}(\text{relint } C_A), \tag{B.93}$$

and

$$\tilde{\mathcal{T}}_G \setminus \tilde{\mathcal{T}}_G^{\text{sf}} \subseteq \bigcup_{F \prec C_A} (\tilde{\mathcal{T}}_G \cap \mathbf{b}_{\mathbb{R}}^{-1}(F)). \tag{B.94}$$

Here F runs over the proper faces of C_A . The strictly feasible dominant example above shows that $\mathbf{b}_{\mathbb{R}}(\tilde{\mathcal{T}}_G)$ meets $\text{relint } C_A$. Hence every nonempty intersection on the right is a proper rational polyhedral face of $\tilde{\mathcal{T}}_G$. There are only finitely many such faces. Moreover, if $\mathbf{b}_{\mathbb{R}}(a, b, c) = Av_0$ with $v_0 > 0$, then a relative neighbourhood of v_0 in $v_0 + \ker A_{\mathbb{R}}$ remains in the positive orthant. The partition fibre therefore has its expected affine dimension

$$\dim \ker A_{\mathbb{R}} = d_G. \tag{B.95}$$

Consequently, the locus where the fibre dimension is smaller than d_G , or where a retained coordinate is forced to vanish, is contained in a finite union of proper rational faces. We include in this wall set the boundary faces whose fibres still have dimension d_G , since their leading residue need not restrict from the adjacent strictly feasible cell.

Relative chambers and leading volume. Apply Lemma B.5 with $L = \mathcal{L}_G$ and $b = \mathbf{b}$. Its hypotheses are exactly (B.89) and (B.91). The resulting rational complex lies in the parameter space and has one finite-index period lattice $\Lambda_0 \subset \mathcal{L}_G$. On every relative cell and every coset of Λ_0 , the restricted partition function is polynomial of degree at most

$$(|\Delta_+| + M) - (\ell + M) = d_G. \tag{B.96}$$

On a strictly feasible cell, its degree- d_G term is independent of the coset, including when the image of $\mathbf{b}_{\mathbb{R}}$ lies in an internal wall of the ambient fan. It is the continuous partition function

$$T_A(u) = \text{vol}_{\ker A \cap \mathbb{Z}^{|\Delta_+|+M}} \{v \in \mathbb{R}_{\geq 0}^{|\Delta_+|+M} : Av = u\}, \tag{B.97}$$

where the relative measure is normalized so that the direction lattice $\ker A \cap \mathbb{Z}^{|\Delta_+|+M}$ has covolume one.

For $u = (\gamma, h_0)$, projection $(t, s) \mapsto t$ maps the fibre in (B.97) bijectively onto

$$\{t \in \mathbb{R}_{\geq 0}^{|\Delta_+|} : R(t) = \gamma, C(t) \geq h_0\}, \quad s = C(t) - h_0. \quad (\text{B.98})$$

It also maps the direction lattices isomorphically:

$$\ker A \cap \mathbb{Z}^{|\Delta_+|+M} \xrightarrow{\sim} \ker R \cap \mathbb{Z}^{|\Delta_+|}, \quad (\dot{t}, \dot{s}) \mapsto \dot{t}, \quad \dot{s} = C(\dot{t}). \quad (\text{B.99})$$

Indeed, for every $\dot{t} \in \ker R \cap \mathbb{Z}^{|\Delta_+|}$, the vector $\dot{s} = C(\dot{t})$ is integral by (B.83), and it is the unique $\dot{s}$ for which $(\dot{t}, \dot{s}) \in \ker A$. Projection consequently has determinant one when the two affine fibres are equipped with their lattice-normalized measures.

This also explains why clearing denominators with q_0 introduces no factor in the leading volume. The Euclidean metric on the graph $s = C(t) - h_0$ does depend on q_0 , but the measure used here is normalized by the graph lattice

$$\{(\dot{t}, C(\dot{t})) : \dot{t} \in \ker R \cap \mathbb{Z}^{|\Delta_+|}\}. \quad (\text{B.100})$$

Projection carries this lattice bijectively to $\ker R \cap \mathbb{Z}^{|\Delta_+|}$. The same observation applies if one temporarily restores any redundant trail row removed above. Therefore projection identifies the two normalized lattice measures, and $T_A \circ \mathbf{b}$ is exactly the intrinsic lattice-normalized BZ fibre volume V_G .

The distinction between strictly feasible cells and proper faces remains essential. On a proper face the active subconfiguration can have a different top residue, and the relative lemma supplies only the degree- d_G bound asserted in the theorem.

Pullback and uniform estimates. Write $P_{+, \mathbb{R}} := \sum_{i=1}^{\ell} \mathbb{R}_{\geq 0} \omega_i$ for the real dominant cone, and define the real tensor cone by

$$\mathcal{T}_G^{\mathbb{R}} = \overline{\left\{ r^{-1}(a, b, c) : \begin{array}{l} r \in \mathbb{Z}_{>0}, \quad (a, b, c) \in \mathcal{L}_G \cap (P_+)^3, \\ m_{ab}^c > 0 \end{array} \right\}} \subset \mathcal{L}_{G, \mathbb{R}}. \quad (\text{B.101})$$

The BZ description gives

$$\mathcal{T}_G^{\mathbb{R}} = (P_{+, \mathbb{R}})^3 \cap \mathbf{b}_{\mathbb{R}}^{-1}(\text{cone}(A)). \quad (\text{B.102})$$

One inclusion follows immediately from an integral BZ pattern. For the converse, take a rational dominant compatible triple in the right-hand side. Its fibre is a nonempty rational polytope and therefore contains a rational point. Multiply first to clear the denominators of the triple and then to clear those of the fibre point. The result is an integral BZ pattern for a positive multiple of the original triple. Rational points are dense in the rational cone on the right-hand side, which proves (B.102) after closure.

This argument uses rational feasibility and produces a positive multiple of nonzero multiplicity. The original integral triple may still be a hole of the tensor semigroup; such a hole is recorded by the residue-class terms of the vector-partition formula.

The relative vector-partition complex supplied by Lemma B.5 already lies in the parameter space $\mathcal{L}_{G, \mathbb{R}}$. Take its common refinement with the face complex of $(P_{+, \mathbb{R}})^3$, and denote the resulting finite rational conical complex by $\mathcal{C}_G$. By (B.102), its support is precisely $\mathcal{T}_G^{\mathbb{R}}$.

On a strictly feasible relative cell C of the reduced presentation define

$$V_C(a, b, c) = T_A(\mathbf{b}_{\mathbb{R}}(a, b, c)). \quad (\text{B.103})$$

Fibre dilation makes V_C homogeneous of degree d_G . The period lattice Λ_0 is the finite-index lattice obtained in the relative lemma. It may instead be taken to be

$$\Lambda_0 = N_0^2 \mathcal{L}_G, \quad (\text{B.104})$$

where N_0 is the common annihilator chosen in Lemma B.5. This formulation does not require extending the torsion characters belonging to a lower-rank saturated span to the ambient target lattice. Pulling back the residue-class formulae proves the polynomial statement and identifies their common highest homogeneous part with (B.103).

The remainder estimate (B.72), including the zero remainder when $d_G = 0$, now follows from (B.52) and the identification of normalized fibre measures above.

For the global bound, there are finitely many relatively open cells and finitely many cosets of Λ_0 . On each cell and coset, the multiplicity is the restriction of a polynomial P of degree at most d_G . For the fixed norm on $\mathcal{L}_{G, \mathbb{R}}$, each such polynomial satisfies

$$|P(a, b, c)| \leq C_P(1 + \|a\| + \|b\| + \|c\|)^{d_G}. \quad (\text{B.105})$$

Taking the maximum of the finitely many constants C_P gives one constant C_G on $\mathcal{L}_G \cap (P_+)^3 \cap \mathcal{T}_G^{\mathbb{R}}$. If the triple is incompatible, its multiplicity vanishes by the central-character condition; if it lies outside $\mathcal{T}_G^{\mathbb{R}}$, it vanishes by the BZ feasibility criterion. This proves (B.73) for all dominant integral triples. The relation between the leading BZ volume and the usual volume function is discussed in [21]. $\square$

Corollary B.7 (Non-affine coefficient asymptotic). *Let C be a full-dimensional strictly feasible BZ cell, let $K \subseteq C \cap \mathcal{R}_G$, and let $E \subset P^3$ be a finite set of shifts $(\alpha, \beta, \varsigma)$ satisfying $\alpha + \beta - \varsigma \in Q$. Uniformly for*

$$(a, b, c) \in \mathcal{L}_G \cap (P_k^+)^3, \quad (z_a, z_b, z_c) \in K,$$

and for $(\alpha, \beta, \varsigma) \in E$ for which $a + \alpha, b + \beta, c + \varsigma \in P_k^+$, one has

$$r_k^{-d_G} N_{a+\alpha, b+\beta}^{(k)c+\varsigma} = V_C(z_a, z_b, z_c) + O_{K,E}(r_k^{-1}). \quad (\text{B.106})$$

Proof. For the shifted labels,

$$z_{a+\alpha} = z_a + \frac{\alpha}{r_k}, \quad z_{b+\beta} = z_b + \frac{\beta}{r_k}, \quad z_{c+\varsigma} = z_c + \frac{\varsigma}{r_k}.$$

Since $K \subseteq \mathcal{R}_G$ and E is finite, these points remain in a fixed compact subset of $\mathcal{R}_G$ for all sufficiently large k . Proposition B.4, applied to the shifted labels, therefore gives

$$N_{a+\alpha, b+\beta}^{(k)c+\varsigma} = m_{a+\alpha, b+\beta}^{c+\varsigma}.$$

2968 The normalized parameters entering the tensor-multiplicity theorem are

$$\left(\frac{a + \alpha}{r_k}, \frac{b + \beta}{r_k}, \frac{c + \varsigma}{r_k} \right) = (z_a, z_b, z_c) + \frac{1}{r_k}(\alpha - \rho, \beta - \rho, \varsigma - \rho).$$

2969 Their displacement from K is $O_E(r_k^{-1})$. Since C is full-dimensional and K has positive
 2970 distance from its walls, the displaced parameters remain in C for all large k . Because the BZ
 2971 complex is conical, multiplication by the positive number r_k shows that the corresponding
 2972 unscaled shifted triple also lies in C , as required in Theorem B.6. Moreover,

$$(a + \alpha) + (b + \beta) - (c + \varsigma) \in Q$$

2973 by the two compatibility assumptions. We may therefore apply Eq. (B.72). After division
 2974 by $r_k^{d_G}$, homogeneity and local Lipschitz continuity of V_C on a compact neighbourhood of
 2975 K give the stated estimate. $\square$

2976 To identify the normalization, extend the quantum-dimension formula from the sam-
 2977 pling grid by

$$D_k(u) = \prod_{\alpha \in \Delta_+} \frac{\sin \pi(\alpha, u)}{\sin(\pi(\rho, \alpha)/r_k)}, \quad D_k(z_c) = d_c^{(k)}. \quad (\text{B.107})$$

2978 Fix $x, y \in Y$ and sequences a_k, b_k with $z_{a_k} \rightarrow x$ and $z_{b_k} \rightarrow y$. For $f \in C_c(Y)$, Eq. (4.41)
 2979 gives the exact identity

$$r_k^{-|\Delta_+|} \sum_c N_{a_k b_k}^{(k)c} f(z_c) = \int_Y f(u) h_k(u) d\sigma_{a_k, b_k}^{(k)}(u), \quad h_k(u) = r_k^{-|\Delta_+|} \frac{d_{a_k}^{(k)} d_{b_k}^{(k)}}{D_k(u)}. \quad (\text{B.108})$$

2980 The sine product converges uniformly on compact regular sets and yields

$$h_k(u) \longrightarrow \frac{|\delta_G(x)| |\delta_G(y)|}{\Pi_\rho |\delta_G(u)|}. \quad (\text{B.109})$$

2981 Denote this limit by $h(u)$. For $f \in C_c(Y)$, extend fh_k and fh by zero to $\overline{Y}$. These func-
 2982 tions are continuous and $fh_k \rightarrow fh$ uniformly. Since the fusion measures are probability
 2983 measures,

$$\left| \int fh_k d\sigma_{a_k, b_k}^{(k)} - \int fh d\nu_{x, y} \right| \leq \|f(h_k - h)\|_\infty + \left| \int fh d\sigma_{a_k, b_k}^{(k)} - \int fh d\nu_{x, y} \right|.$$

2984 Both terms tend to zero by compact-uniform convergence and Proposition 4.5. Thus
 2985 (B.108) can be passed to the limit for every fixed regular input pair.

2986 We use the preceding identity locally. Let C be a full-dimensional strictly feasible BZ
 2987 cell, fix $(x, y, z_0) \in C \cap \mathcal{R}_G$, and choose relatively compact open sets $U_x, U_y, U_z \Subset Y$ such
 2988 that

$$\overline{U}_x \times \overline{U}_y \times \overline{U}_z \subset C \cap \mathcal{R}_G. \quad (\text{B.110})$$

Choose the input sequences above with $z_{a_k} \in U_x$ and $z_{b_k} \in U_y$ for all large k , and take $f \in C_c(U_z)$. Proposition 4.5, the exact change-of-weight identity, and Eq. (4.38) give

$$\lim_{k \rightarrow \infty} r_k^{-|\Delta_+|} \sum_{c \in P_k^+} N_{a_k b_k}^{(k)c} f(z_c) = \frac{1}{\text{covol}(Q)} \int_{U_z} f(u) V_{\text{BF}}(x, y; u) du. \quad (\text{B.111})$$

For fixed a_k, b_k , the central-character rule restricts the nonzero outputs to $c \in a_k + b_k + Q$. Their shifted coordinates form the translated lattice

$$\Lambda_k = \frac{a_k + b_k + \rho + Q}{r_k},$$

whose fundamental-cell volume is $\text{covol}(Q)r_k^{-\ell}$. Since $\text{supp } f \Subset Y$, every point of Λ_k in the support of f comes from a weight in P_k^+ once k is large, and there are $O(r_k^\ell)$ such points. Corollary B.7, applied uniformly on the product box, therefore gives

$$\begin{aligned} r_k^{-|\Delta_+|} \sum_{c \in P_k^+} N_{a_k b_k}^{(k)c} f(z_c) \\ = r_k^{-\ell} \sum_{z_c \in \Lambda_k} f(z_c) V_C(z_{a_k}, z_{b_k}; z_c) + O(r_k^{-1}). \end{aligned} \quad (\text{B.112})$$

Here $|\Delta_+| = d_G + \ell$: after the normalization, the uniform $O(r_k^{d_G-1})$ coefficient error is summed over $O(r_k^\ell)$ points and therefore contributes $O(r_k^{-1})$. Uniform continuity of V_C on the closure of the product box and the translated-lattice Riemann sum now give

$$\lim_{k \rightarrow \infty} r_k^{-|\Delta_+|} \sum_{c \in P_k^+} N_{a_k b_k}^{(k)c} f(z_c) = \frac{1}{\text{covol}(Q)} \int_{U_z} f(u) V_C(x, y; u) du. \quad (\text{B.113})$$

Comparison with (B.111) proves, for each fixed input pair (x, y) , that on every full-dimensional strictly feasible BZ cell C ,

$$V_C(x, y; z) = \frac{\text{covol}(Q)}{\text{covol}(Q^\vee) \Pi_\rho} |\delta_G(x) \delta_G(y) \delta_G(z)| \Phi_G(x, y; z) = V_{\text{BF}}(x, y; z) \quad (\text{B.114})$$

for almost every z in the section $\{z : (x, y, z) \in C \cap \mathcal{R}_G\}$. Fubini's theorem then gives the corresponding joint almost-everywhere identity. Thus the lattice normalization of the BZ fibre agrees with the independently normalized BF volume in $\mathcal{R}_G$. Outside this region, nonidentity affine Weyl images can contribute and must be retained.

B.5 Global affine completion and cellwise asymptotics

On every strictly feasible cell, let V_{add} be its leading BZ volume polynomial. For the off-wall antisymmetrization below, set $V_{\text{add}} = 0$ outside the closed real tensor cone and on cells that are not strictly feasible. This convention is used only in the off-wall antisymmetrization; the actual leading behavior on non-strict and lower-dimensional cells is left unspecified. Define the affine antisymmetrization by

$$V_{\text{aff}}(x, y; z) = \sum_{\bar{w} \in W} \sum_{\gamma^\vee \in Q^\vee} \epsilon(\bar{w}) V_{\text{add}}(x, y; \bar{w}z + \gamma^\vee). \quad (\text{B.115})$$

Lemma B.8 (Global finiteness of the affine set). *There is a finite set $\mathcal{A}_G \subset W \ltimes Q^\vee$ such that, if $(x, y, \bar{w}z + \gamma^\vee)$ belongs to the real additive tensor cone for some $(x, y, z) \in Y^3$, then $(\bar{w}, \gamma^\vee) \in \mathcal{A}_G$. Consequently only elements of $\mathcal{A}_G$ can contribute to (B.115) anywhere on Y^3 .*

For every finite set $E \subset P^3$ of shifts, the same finite set may be chosen so that, after increasing a number $k_0(E)$, every nonzero Kac–Walton term arising from sampling points in Y^3 and shifts in E belongs to $\mathcal{A}_G$ whenever $k \geq k_0(E)$. For the same finite set E , there is a constant $C_E > 0$ such that

$$0 \leq N_{a+\alpha, b+\beta}^{(k)c+\varsigma} \leq C_E r_k^{d_G} \quad (\text{B.116})$$

whenever $(\alpha, \beta, \varsigma) \in E$, $k \geq k_0(E)$, and the shifted labels are integrable.

Proof. If (x, y, ξ) lies in the real tensor cone, the highest-weight norm inequality gives $\|\xi\| \leq \|x + y\|$. Therefore

$$\|\gamma^\vee\| \leq \|\bar{w}z + \gamma^\vee\| + \|z\| \leq \|x + y\| + \|z\|. \quad (\text{B.117})$$

The closed alcove is compact, so the number

$$R_Y := \max_{(x, y, z) \in \bar{Y}^3} (\|x + y\| + \|z\|) \quad (\text{B.118})$$

is finite.

For the finite-level assertion, nonvanishing of $m_{a+\alpha, b+\beta}^{\hat{w} \cdot (c+\varsigma)}$ implies

$$\|\hat{w} \cdot (c + \varsigma)\| \leq \|a + b + \alpha + \beta\|. \quad (\text{B.119})$$

Writing $\hat{w} = (\bar{w}, \gamma^\vee)$ and dividing by r_k gives

$$\|\bar{w}z_c + \gamma^\vee\| \leq \|z_a + z_b\| + O_E(r_k^{-1}).$$

Consequently $\|\gamma^\vee\| \leq R_Y + O_E(r_k^{-1})$. For all sufficiently large k , every classical contribution and every shifted finite-level term therefore belongs to

$$\mathcal{A}_G = \{(\bar{w}, \gamma^\vee) \in W \ltimes Q^\vee : \|\gamma^\vee\| \leq R_Y + 1\}. \quad (\text{B.120})$$

This set is finite because W is finite and $Q^\vee$ is a lattice. For $k \geq k_0(E)$, the Kac–Walton formula contains no nonzero term outside $\mathcal{A}_G$. Since a fusion coefficient is nonnegative,

$$0 \leq N_{a+\alpha, b+\beta}^{(k)c+\varsigma} \leq \sum_{(\bar{w}, \gamma^\vee) \in \mathcal{A}_G} m_{a+\alpha, b+\beta}^{\bar{w}(c+\varsigma+\rho)+r_k\gamma^\vee-\rho}. \quad (\text{B.121})$$

All weights in this finite sum have norm $O_E(r_k)$. Applying Eq. (B.73) to each summand proves Eq. (B.116). $\square$

The fixed affine–BZ refinement. The family $\mathcal{H}_{\text{aff}}$ in Eq. (4.46) is finite. Let C be a connected component of $Y^3 \setminus \mathcal{C}_{\text{aff}}$. By Lemma 4.6, for every $\widehat{w} \in \mathcal{A}_G$ the image $\Psi_{\widehat{w}}(C)$ is either disjoint from the closed tensor cone or contained in one full-dimensional strictly feasible BZ cell.

If $K \Subset C$, then K has positive distance from every hyperplane in $\mathcal{H}_{\text{aff}}$. The $O_{K,E}(r_k^{-1})$ displacements in Eq. (B.123) therefore preserve these assignments for sufficiently large k . Hence $V_{\text{aff}}|_C$ is the single polynomial $V_{\text{aff},C}$.

Proposition B.9 (Uniform asymptotic on a polyhedral affine–BZ cell). *Let C be a polyhedral affine–BZ cell, let $K \Subset C$, and let $E \subset P^3$ be a finite set of compatible integral shifts as in Theorem 4.7. Uniformly for $(a, b, c) \in \mathcal{L}_G \cap (P_k^+)^3$ with $(z_a, z_b, z_c) \in K$ and for $(\alpha, \beta, \varsigma) \in E$ for which the shifted labels lie in P_k^+ ,*

$$r_k^{-d_G} N_{a+\alpha, b+\beta}^{(k)c+\varsigma} = V_{\text{aff},C}(z_a, z_b; z_c) + O_{K,E}(r_k^{-1}). \quad (\text{B.122})$$

Proof. Insert the Kac–Walton formula (B.45). By Lemma B.8, the sum reduces uniformly to one fixed finite set of affine Weyl elements. For $\widehat{w} = (\bar{w}, \gamma^\vee)$,

$$\frac{a + \alpha}{r_k} = z_a + \frac{\alpha - \rho}{r_k}, \quad \frac{b + \beta}{r_k} = z_b + \frac{\beta - \rho}{r_k}, \quad \frac{\widehat{w} \cdot (c + \varsigma)}{r_k} = \bar{w}z_c + \gamma^\vee + \frac{\bar{w}\varsigma - \rho}{r_k}. \quad (\text{B.123})$$

Before applying the polyhedral multiplicity theorem, we verify its lattice hypothesis for every affine image. Put $\lambda = c + \varsigma + \rho$. Then

$$\begin{aligned} & (a + \alpha) + (b + \beta) - \widehat{w} \cdot (c + \varsigma) \\ &= (a + b - c) + (\alpha + \beta - \varsigma) + (\lambda - \bar{w}\lambda) - r_k \gamma^\vee. \end{aligned} \quad (\text{B.124})$$

The first term lies in Q because $(a, b, c) \in \mathcal{L}_G$, and the second lies in Q by the assumption on the shifts. For every $\lambda \in P$ and $\bar{w} \in W$, one has $\lambda - \bar{w}\lambda \in Q$; this follows first for a simple reflection and then for every product of simple reflections. Finally, Eq. (2.8) gives $\gamma^\vee \in Q^\vee \subset Q$, and r_k is an integer. Concretely, with long roots normalized to have squared length two, the coroot of a long root is that root, while the coroot of a short root is two or three times the short root. Thus every transformed triple is compatible. For each affine element in this finite set, the transformed compact set is either outside the tensor cone or contained in one strictly feasible BZ cell. In the first case it has positive distance from the closed tensor cone. The $O_{K,E}(r_k^{-1})$ displacement in (B.123) therefore leaves the transformed triple outside the cone for all large k , so its tensor multiplicity is zero. In the second case, Theorem B.6, homogeneity, and local Lipschitz continuity give, uniformly over the finite affine and shift sets,

$$r_k^{-d_G} m_{a+\alpha, b+\beta}^{\widehat{w} \cdot (c+\varsigma)} = V_{\text{add}}(z_a, z_b; \bar{w}z_c + \gamma^\vee) + O_{K,E}(r_k^{-1}). \quad (\text{B.125})$$

The affine sign is $\epsilon(\widehat{w}) = \epsilon(\bar{w})$ because a translation is orientation preserving. Alternating these finitely many terms proves (B.122): the affine-image and shift sets are fixed and finite on K , so taking the maximum of their finitely many remainder constants and summing the errors preserves the uniform $O_{K,E}(r_k^{-1})$ bound. Theorem B.6 also shows that the leading term is independent of the Λ_0 -coset, so no further congruence condition occurs. $\square$

To identify the affine sum, we compute the local density of the same unweighted fusion measure in two ways: globally from the hypergroup limit and locally from the coefficient asymptotic.

Proposition B.10 (Identification of the affine leading polynomial). *Let C be a polyhedral affine-BZ cell and put $C_{x,y} = \{z \in Y : (x, y, z) \in C\}$. For every fixed input pair for which $C_{x,y}$ is nonempty,*

$$V_{\text{aff},C}(x, y; z) = V_{\text{BF}}(x, y; z) \quad \text{for } dz\text{-almost every } z \in C_{x,y}. \quad (\text{B.126})$$

If S is a principal regular-value chamber, then

$$V_{\text{aff},C} = V_{\text{geom}} \quad \text{on } S \cap C. \quad (\text{B.127})$$

Proof. Fix $x, y \in Y$ and $z_0 \in C_{x,y}$. Since C is open, there are relatively compact open neighbourhoods $U_x, U_y, U_z \Subset Y$ such that

$$x \in U_x, \quad y \in U_y, \quad z_0 \in U_z, \quad \overline{U}_x \times \overline{U}_y \times \overline{U}_z \subset C. \quad (\text{B.128})$$

Choose input sequences with $z_{a_k} \rightarrow x$ and $z_{b_k} \rightarrow y$. Such sequences exist in any prescribed cosets of Q : the finite covering radius of the full-rank lattice $Q \subset P_{\mathbb{R}}$ gives

$$a_k + \rho = r_k x + O(1), \quad b_k + \rho = r_k y + O(1). \quad (\text{B.129})$$

Regularity of x, y makes these weights dominant and level- k integrable for all large k . The compatible outputs form the translate $a_k + b_k + Q$. Moreover, every nonzero output coefficient satisfies $(a_k, b_k, c) \in \mathcal{L}_G$ by the central-character rule, so the cellwise estimate applies to every surviving term. For $f \in C_c(U_z)$, the exact change-of-weight identity (B.108), the global weak limit of the normalized fusion distributions, and the spectral-density formula (4.38) give

$$\lim_{k \rightarrow \infty} r_k^{-|\Delta_+|} \sum_{c \in P_k^+} N_{a_k b_k}^{(k)c} f(z_c) = \frac{1}{\text{covol}(Q)} \int_{U_z} f(u) V_{\text{BF}}(x, y; u) du. \quad (\text{B.130})$$

The passage with the varying test function $f h_k$ was justified after (B.108). Multiplying its limiting weight by the density in (4.38) gives $f V_{\text{BF}} / \text{covol}(Q)$.

On the other hand, Proposition B.9, with zero shifts, applies uniformly on the compact set in Eq. (B.128). The central-character rule restricts the nonzero terms to $c \in a_k + b_k + Q$. Their shifted coordinates form the translated lattice

$$\Lambda_k = \frac{a_k + b_k + \rho + Q}{r_k}, \quad (\text{B.131})$$

whose fundamental-cell volume is $\text{covol}(Q) r_k^{-\ell}$. Since $\text{supp } f \Subset Y$, every point of Λ_k in the support of f gives an integral weight $c = r_k u - \rho$ whose dominance and level inequalities hold for all sufficiently large k . There are $O(r_k^\ell)$ such points, uniformly in the translating coset.

Using $|\Delta_+| = d_G + \ell$, the uniform coefficient estimate gives

$$\begin{aligned}
& r_k^{-|\Delta_+|} \sum_{c \in P_k^+} N_{a_k b_k}^{(k)c} f(z_c) \\
&= r_k^{-\ell} \sum_{z_c \in \Lambda_k} f(z_c) V_{\text{aff}, C}(z_{a_k}, z_{b_k}; z_c) + O(r_k^{-1}).
\end{aligned} \tag{B.132}$$

3091 The total error is $O(r_k^{-1})$ by the same count used in (B.112). Set

$$g_k(u) := f(u) V_{\text{aff}, C}(z_{a_k}, z_{b_k}; u), \quad g(u) := f(u) V_{\text{aff}, C}(x, y; u),$$

3092 Their zero extensions are continuous with a common compact support, and polynomiality
3093 on the product box gives $\|g_k - g\|_\infty \rightarrow 0$. The lattice-point bound therefore implies

$$r_k^{-\ell} \sum_{\lambda \in \Lambda_k} |g_k(\lambda) - g(\lambda)| \leq C \|g_k - g\|_\infty \longrightarrow 0.$$

3094 For completeness, the fixed-function Riemann estimate is uniform in the translate as
3095 well. Choose a bounded half-open fundamental parallelepiped F for Q , and put

$$D_{k, \lambda} := \lambda + r_k^{-1} F, \quad \lambda \in \Lambda_k.$$

3096 These cells tile $P_{\mathbb{R}}$ and have volume $\text{covol}(Q) r_k^{-\ell}$. With $R_F = \sup_{v \in F} \|v\|$ and ω_g the
3097 modulus of continuity of g ,

$$\begin{aligned}
& \left| \text{covol}(Q) r_k^{-\ell} \sum_{\lambda \in \Lambda_k} g(\lambda) - \int_{P_{\mathbb{R}}} g(u) du \right| \\
& \leq \sum_{\lambda \in \Lambda_k} \int_{D_{k, \lambda}} |g(\lambda) - g(u)| du \leq \text{vol}(K') \omega_g(R_F / r_k).
\end{aligned} \tag{B.133}$$

3098 Here K' is a fixed compact neighbourhood of $\text{supp } g$ containing every contributing cell
3099 for all large k . The right-hand side tends to zero independently of the translating coset.
3100 Inserting this and the preceding estimate into (B.132) yields

$$\lim_{k \rightarrow \infty} r_k^{-|\Delta_+|} \sum_{c \in P_k^+} N_{a_k b_k}^{(k)c} f(z_c) = \frac{1}{\text{covol}(Q)} \int_{U_z} f(u) V_{\text{aff}, C}(x, y; u) du. \tag{B.134}$$

3101 Comparing Eqs. (B.130) and (B.134) for every $f \in C_c(U_z)$ identifies the two densities for
3102 dz -almost every point of U_z . For the fixed input pair (x, y) , a countable collection of such
3103 neighbourhoods covers $C_{x, y}$ and proves Eq. (B.126). Applying this assertion to every fixed
3104 input pair also gives $V_{\text{aff}, C} = V_{\text{BF}}$ for Lebesgue-almost every point of C by Fubini.

3105 Now let S be a principal regular-value chamber. On each nonempty connected com-
3106 ponent D of $S \cap C$, Eq. (4.33) and the preceding sectionwise identity give

$$V_{\text{aff}, C} = V_{\text{geom}} \quad \text{for Lebesgue-almost every point of } D.$$

3107 The Duistermaat–Heckman variation theorem makes V_{geom} polynomial on S , while $V_{\text{aff}, C}$
3108 is polynomial on C . Their difference is therefore a polynomial on D which vanishes almost
3109 everywhere, and hence vanishes identically. Applying this argument to every component
3110 of $S \cap C$ proves Eq. (B.127). $\square$

This agrees with the oriented affine-cobordism and volume formulae of Meinrenken–Woodward [46, Theorem 1.2 and Eq. (12)].

Combining Propositions B.9 and B.10 proves Theorem 4.7. In particular, outside the non-affine region the full affine Weyl antisymmetrization, rather than its identity term, equals the group-valued BF volume.

B.6 Uniform type- A asymptotics on the regular alcove

Korff’s formulas identify the WZW coefficients with the quantum-cohomology coefficients used by François–Tarrago. We first record this dictionary, including the scalar factor in its centered form. The proof of the compact-uniform limit then starts directly from the Kac–Peterson determinant. It uses the summability argument behind Theorem 3.8 of François–Tarrago [22], which we give below with bounds independent of the three labels. Throughout this subsection, $m \geq 3$. Put

$$N = r_k = k + m, \quad q_m = \frac{m(m-1)}{2}, \quad d_m = q_m - (m-1) = \frac{(m-1)(m-2)}{2}. \quad (\text{B.135})$$

B.6.1 The quantum-cohomology dictionary

For a dominant weight $a = (a_1, \dots, a_{m-1})$ in Dynkin coordinates, let

$$\lambda_s(a) = \sum_{j=s}^{m-1} a_j \quad (1 \leq s < m), \quad \lambda_m(a) = 0, \quad (\text{B.136})$$

so that $\lambda(a)$ lies in the $m \times k$ rectangle. For any partition $\Lambda = (\Lambda_1 \geq \dots \geq \Lambda_m \geq 0)$ in that rectangle, set

$$\Lambda' = (\Lambda_1 - \Lambda_m, \dots, \Lambda_{m-1} - \Lambda_m, 0), \quad (\text{B.137})$$

that is, delete every column of height m , and let $\widehat{\Lambda}' \in P_k^+$ be the corresponding affine weight. With $a_0 = k - \sum_{i=1}^{m-1} a_i$, fix the simple-current convention

$$\text{rot}(a_0, a_1, \dots, a_{m-1}) = (a_{m-1}, a_0, a_1, \dots, a_{m-2}). \quad (\text{B.138})$$

Let $c_{\Lambda, \mathbf{M}}^{\Upsilon, d}$ denote the degree- d quantum Littlewood–Richardson coefficient in the quantum cohomology of $\text{Gr}(m, N)$, with the convention that it is zero for $d < 0$. Its grading condition is

$$|\Lambda| + |\mathbf{M}| - |\Upsilon| = Nd. \quad (\text{B.139})$$

For an effective degree $d \geq 0$, Korff’s exact projection formula is

$$c_{\Lambda, \mathbf{M}}^{\Upsilon, d} = N_{\widehat{\Lambda'} \widehat{\mathbf{M}'}}^{(k) \text{rot}^d(\widehat{\Upsilon'})}. \quad (\text{B.140})$$

This is Proposition 1.5, Eq. (1.14), of [47]. In the transposed $\text{Gr}(k, N)$ presentation, the corresponding exact ring statement is

$$\text{Ver}_k(SU(m)) \simeq QH^\bullet(\text{Gr}(k, N))/(q - e_k, h_m - 1) \quad (\text{B.141})$$

[48, Theorem 6.20 and Corollary 8.2]. Thus the $SU(m)_k$ dictionary is not the specialization $q = 1$ alone: full height- m columns are forgotten and the quantum degree becomes the simple current rot^d .

For a desired coefficient $N_{ab}^{(k)c}$ put

$$\lambda = \lambda(a), \quad \mu = \lambda(b), \quad \nu = \lambda(c). \quad (\text{B.142})$$

If $a + b - c \in Q$, then

$$\widehat{d}_N = \frac{|\lambda| + |\mu| - |\nu|}{m} \in \mathbb{Z}. \quad (\text{B.143})$$

Let Rot be the order- N operation on partitions in the $m \times k$ rectangle obtained by the left cyclic shift $w_1 \cdots w_N \mapsto w_2 \cdots w_N w_1$ of the associated length- N 01-word, as in Eq. (4.6) of [47]. It is different from the order- m affine Dynkin automorphism rot fixed in Eq. (B.138). Define

$$\Sigma_N = \text{Rot}^{-\widehat{d}_N}(\nu), \quad d_N = \frac{|\lambda| + |\mu| - |\Sigma_N|}{N}. \quad (\text{B.144})$$

The partition Σ_N depends only on the residue class of $\widehat{d}_N$ modulo N , because Rot has order N . The residue class of $\widehat{d}_N$ therefore determines Σ_N , whereas the displayed cardinalities determine the integer d_N itself. This integer records the winding, or quantum degree, which is lost when one remembers only the rotated partition.

For the inverse construction, Corollary 4.4, Eqs. (4.16)–(4.17), of [47] shows that $d_N \in \mathbb{Z}$ and gives

$$N_{ab}^{(k)c} = c_{\lambda, \mu}^{\Sigma_N, d_N}, \quad \text{rot}^{d_N}(\widehat{\Sigma'_N}) = c. \quad (\text{B.145})$$

Here and below we extend the notation by $c_{\lambda, \mu}^{\Sigma, d} = 0$ for $d < 0$; this convention applies when a compatible but fusion-forbidden triple has negative canonical degree. Corollary 1.4, Eq. (1.7), of [47] expresses a quantum Littlewood–Richardson coefficient as a sum whose degree $d(\pi)$ counts a subset of m indices. Thus $0 \leq d(\pi) \leq m$, and the coefficient vanishes for $d > m$. Consequently the right-hand side of Eq. (B.145) vanishes whenever $d_N \notin \{0, \dots, m\}$. A nonzero lifted invariant therefore has $d_N \in \{0, \dots, m\}$. If $a + b - c \notin Q$, the WZW coefficient vanishes by the central selection rule.

Centering the canonical lift. Let $C(v) = v - m^{-1}(\sum_s v_s)\mathbf{1}$, put $\rho_s = (m + 1 - 2s)/2$, and, for a partition η in the rectangle, write $\tau_\eta = |\eta|/(mN)$. Directly from the definition of z_a ,

$$C(\lambda(a)/N) = z_a - \rho/N, \quad C(\lambda(b)/N) = z_b - \rho/N. \quad (\text{B.146})$$

Deleting height- m columns does not change a centered row vector. If $w_N = \widehat{\Sigma'_N}$, the second identity in Eq. (B.145) says $w_N = \text{rot}^{-d_N} c$. Hence the centering identity above gives the exact relation

$$C(\Sigma_N/N) = z_{\text{rot}^{-d_N} c} - \rho/N. \quad (\text{B.147})$$

The remaining datum is the scalar part of the $U(m)$ lift. The grading in Eq. (B.144) gives

$$\tau_\lambda + \tau_\mu - \tau_{\Sigma_N} = \frac{d_N}{m}, \quad \exp[-2\pi i(\tau_\lambda + \tau_\mu - \tau_{\Sigma_N})] = \zeta^{-d_N}, \quad \zeta = e^{2\pi i/m}. \quad (\text{B.148})$$

With the convention in Eq. (B.138), one application of rot multiplies the corresponding centered $SU(m)$ conjugacy class by ζ^{-1} . Multiplication by the scalar ζ^{-d_N} in Eq. (B.148) therefore carries the class indexed by $\text{rot}^{-d_N} c$ to the class indexed by c . After this scalar multiplication, the only discrepancy between the centered canonical lift and the class indexed by z_c is the Weyl shift ρ/N in Eq. (B.147). Consequently, whenever z_c converges, the classes of $\zeta^{-d_N} \exp(2\pi i C(\Sigma_N/N))$ converge to the class of the same limit. The scalar ζ^{-d_N} must be retained in this comparison; centering alone leaves the output indexed by $\text{rot}^{-d_N} c$.

Relation with the François–Tarrago limit. Korff’s lift reduces a compatible WZW coefficient to one quantum Littlewood–Richardson coefficient. If a subsequence has fixed canonical degree $d \in \{0, \dots, m\}$ and the three normalized partitions approach a regular triple, then the exact identity $|\lambda_N| + |\mu_N| = |\Sigma_N| + dN$ is precisely the fixed-degree grading hypothesis in Theorem 3.8 of [22]. That theorem gives the corresponding $U(m)$ orbital limit. Theorem 3.8 itself permits any fixed $d \geq 0$; the upper bound $d \leq m$ here comes from Korff’s canonical lift. When the canonical degree lies outside $\{0, \dots, m\}$, the lifted invariant vanishes by the preceding paragraph.

B.6.2 The centered determinant proof

We now work in centered WZW coordinates. The finite determinant formula will give both the limit and its uniformity under finitely many fixed compatible shifts.

Centered coordinates and the central phase. Set

$$A_s(a) = \lambda_s(a) + m - s, \quad \bar{A}(a) = \frac{1}{m} \sum_{s=1}^m A_s(a), \quad L_s(a) = A_s(a) - \bar{A}(a). \quad (\text{B.149})$$

Thus $L(a) = (L_1(a), \dots, L_m(a))$ is the standard-coordinate vector of $a + \rho$ in the trace-zero hyperplane. The uncentred normalized rows in the quantum-cohomology determinant differ from $L(a)/N$ by the scalar $\bar{A}(a)\mathbf{1}/N$; adding or removing a full height- m column changes only this scalar part. The grading retains this $U(1)$ datum, whereas column deletion and rot^{d_N} implement the $SU(m)$ quotient described above. The residue-translation phase in the determinant calculation below has a different role: it is trivial exactly when $a + b - c \in Q$, which is equivalent to the integrality of $\hat{d}_N$ in Eq. (B.143).

Exact finite-level formula. Integrable weights are in bijection with

$$\mathcal{I}_{m,N}^0 = \{I = (I_1 > \dots > I_m = 0) : I_1 < N\}, \quad I_s = A_s(\mu), \quad \mu \in P_k^+. \quad (\text{B.150})$$

Using the complex-conjugate Kac–Peterson convention, which leaves the real fusion coefficient unchanged, define

$$D_a^{(N)}(I) = \det \left[e^{2\pi i I_r L_s(a)/N} \right]_{r,s=1}^m, \quad F_N(I; a, b, c) = \frac{D_a^{(N)}(I) D_b^{(N)}(I) \overline{D_c^{(N)}(I)}}{D_0^{(N)}(I)}. \quad (\text{B.151})$$

Lemma B.11 (Exact compatible $SU(m)$ orbit formula). *For every $(a, b, c) \in \mathcal{L}_G \cap (P_k^+)^3$, the summand F_N is invariant under common translation of the residue tuple, including on orbits with nontrivial stabilizer, and the fusion coefficient is given exactly by Eq. (B.158). For an incompatible triple one must instead use Eq. (B.154).*

Proof. For the raw convention in (B.151), a direct Vandermonde calculation gives

$$D_0^{(N)}(I) = (2i)^{q_m} \prod_{r < s} \sin\left(\frac{\pi(I_r - I_s)}{N}\right). \quad (\text{B.152})$$

Its modulus is the positive sine Vandermonde. Let $\mu(I)$ be the integrable weight determined by $I_s = A_s(\mu(I))$ and $I_m = 0$. In the complex-conjugate Kac–Peterson convention used in Eq. (B.151), the specialization of the Weyl sum to type A_{m-1} is

$$S_{a, \mu(I)}^{(k)} = \frac{(-i)^{q_m}}{\sqrt{m} N^{(m-1)/2}} D_a^{(N)}(I). \quad (\text{B.153})$$

The alternating S_m -sum in [40, Eq. (3.3d)] is the determinant, while $|P/Q| = m$ supplies the factor $m^{-1/2}$. For $a = 0$, Eq. (B.152) shows that multiplication by $(-i)^{q_m}$ makes $S_{0, \mu(I)}^{(k)}$ positive.

In the Verlinde quotient $S_{a\mu} S_{b\mu} \overline{S_{c\mu}} / S_{0\mu}$, the constants in the last display reduce to $|(-i)^{q_m} / (\sqrt{m} N^{(m-1)/2})|^2 = 1/(mN^{m-1})$. The remaining quotient of determinants is exactly F_N , and hence

$$N_{ab}^{(k)c} = \frac{1}{mN^{m-1}} \sum_{I \in \mathcal{T}_{m,N}^0} F_N(I; a, b, c). \quad (\text{B.154})$$

We next remove the choice $I_m = 0$. Let $\mathcal{J}_{m,N}$ be the set of decreasing m -tuples of distinct residues in $\{0, \dots, N-1\}$, and let translation in $\mathbb{Z}/N\mathbb{Z}$ act by adding a common residue and reordering. When a translation crosses the cut, write w for the number of wrapped entries and ε for the sign of the reordering. The raw determinant in (B.151) transforms as

$$D_a^{(N)}(I) \mapsto \varepsilon e^{2\pi i w \overline{A}(a)} D_a^{(N)}(I). \quad (\text{B.155})$$

The signs cancel in F_N , and its residual phase is

$$\exp(2\pi i w [\overline{A}(a) + \overline{A}(b) - \overline{A}(c) - \overline{A}(0)]) = \exp\left(\frac{2\pi i w}{m} [|\lambda(a)| + |\lambda(b)| - |\lambda(c)|]\right). \quad (\text{B.156})$$

For a compatible output this is one, because $a + b - c \in Q$ is equivalent to the bracketed number being divisible by m . Thus F_N is constant on every translation orbit.

It remains to compare the full residue sum with the slice on which one residue is zero. Fix a residue set S in a translation orbit Ω and write

$$H = \{t \in \mathbb{Z}/N\mathbb{Z} : S + t = S\}, \quad |H| = s.$$

Then $|\Omega| = N/s$. The action of H on the m residues of S is free: a translation fixing one residue is the zero translation. Consequently $s \mid m$.

For each $u \in S$, the translate $S - u$ contains zero. Two residues u, u' give the same translated set if and only if $u - u' \in H$. Thus exactly m/s members of Ω contain zero. The ratio between the number of all orbit members and the number in the zero-containing slice is therefore

$$\frac{N/s}{m/s} = \frac{N}{m},$$

also for a nontrivial stabilizer. Since F_N is constant on Ω for $(a, b, c) \in \mathcal{L}_G$, summing this identity over the orbits gives

$$\sum_{I \in \mathcal{J}_{m,N}} F_N(I; a, b, c) = \frac{N}{m} \sum_{I \in \mathcal{I}_{m,N}^0} F_N(I; a, b, c). \quad (\text{B.157})$$

Combining (B.154) and (B.157) yields

$$N_{ab}^{(k)c} = N^{-m} \sum_{I \in \mathcal{J}_{m,N}} F_N(I; a, b, c), \quad (a, b, c) \in \mathcal{L}_G. \quad (\text{B.158})$$

For an incompatible triple the phase in Eq. (B.156) need not equal one, so we retain the zero-containing-slice formula (B.154). Equations (B.154)–(B.158) are the centered $SU(m)$ counterpart of the determinant formula in [22, Eqs. (3.27)–(3.30)]. Our exponent convention differs from the one printed there. For partitions written in nonincreasing order we use the bialternant exponents $\lambda_s + m - s$, as in $A_s(a)$ above. Reversing the determinant columns gives the equivalent exponents $\lambda_{m+1-s} + s - 1$ and the fixed column-reversal sign. This is the convention obtained directly from the Schur and Weyl character formulae. As a check, for $m = 2$ and $\lambda = (1, 0)$ it gives the distinct exponents 2 and 0.

François–Tarrago use uncentred $U(m)$ rows. Removing their common scalar parts produces $D_a^{(N)}$, while Eq. (B.148) converts the canonical lifted output back to c . The separate phase in Eq. (B.156) enforces the $SU(m)$ central selection rule. This choice of order does not alter their spacing or summability estimates: those estimates use the sine Vandermonde and uniform determinant bounds, and the bialternants above satisfy exactly the same bounds. $\square$

Orbit representatives and dominated convergence. Allow the labels $a = a_N$, $b = b_N$, and $c = c_N$ to vary with N . For an orbit Ω in $\mathcal{J}_{m,N}$ choose its lexicographically least member. It necessarily has last entry zero. With

$$\mathcal{I}_m^\infty = \{I = (I_1 > \cdots > I_{m-1} > I_m = 0) : I_r \in \mathbb{Z}\}, \quad \Delta(I) = \prod_{r < s} (I_r - I_s), \quad (\text{B.159})$$

write $\Omega_N(I)$ for the orbit of the residue set $\{I_1, \dots, I_m\} \bmod N$ under common translation by $\mathbb{Z}/N\mathbb{Z}$, with the translated residues reordered in decreasing order. The orbit decomposition of (B.158) reads

$$N^{-d_m} N_{a_N b_N}^{(k)c_N} = \sum_{I \in \mathcal{I}_m^\infty} \mathbf{1}_{\{I = \min \Omega_N(I)\}} \frac{|\Omega_N(I)|}{N} \times \left[N^{-m+1-d_m} F_N(I; a_N, b_N, c_N) \right], \quad (\text{B.160})$$

where a term is zero unless $I_1 < N$. For every fixed I , once N is sufficiently large, I is the least member of its orbit and $|\Omega_N(I)| = N$.

Lemma B.12 (Uniform summable majorant). *For $m \geq 3$ there is a constant $C_m > 0$ such that, for every $N \geq m$, every $I \in \mathcal{I}_m^\infty$ with $I_1 < N$, and all integrable labels a, b, c ,*

$$\mathbf{1}_{\{I=\min \Omega_N(I)\}} \frac{|\Omega_N(I)|}{N} N^{-m+1-d_m} |F_N(I; a, b, c)| \leq \frac{C_m}{\Delta(I)} =: C_I, \quad (B.161)$$

$$\sum_{I \in \mathcal{I}_m^\infty} C_I < \infty.$$

Proof. If $I \neq \min \Omega_N(I)$, the left-hand side of (B.161) vanishes. We may therefore assume that I is the lexicographically least representative of its translation orbit. The following spacing argument is the one used in [22, Lemmas 3.14–3.15]. Let the m cyclic gaps between the residues of I be ordered in increasing circular order. Lexicographic minimality places a largest gap at the wrap from I_1 back to $I_m = 0$: if a strictly larger gap occurred elsewhere, translating the residue immediately after that gap to zero would produce an orbit representative with smaller first entry. Since the largest of m gaps is at least N/m , we have

$$N - I_1 \geq \frac{N}{m}, \quad I_1 \leq N \left(1 - \frac{1}{m}\right). \quad (B.162)$$

Consequently, for $d_{rs} = I_r - I_s$ with $r < s$,

$$N - d_{rs} \geq N - I_1 \geq \frac{N}{m} \geq \frac{d_{rs}}{m-1}, \quad \min\{d_{rs}, N - d_{rs}\} \geq \frac{d_{rs}}{m-1}. \quad (B.163)$$

Every entry of a numerator matrix has modulus one. Hadamard's inequality therefore gives

$$|D_a^{(N)}(I)|, |D_b^{(N)}(I)|, |D_c^{(N)}(I)| \leq m^{m/2}, \quad (B.164)$$

uniformly in the labels. On the other hand,

$$|D_0^{(N)}(I)| = \prod_{r < s} 2 \left| \sin \frac{\pi(I_r - I_s)}{N} \right|. \quad (B.165)$$

For $1 \leq d < N$, the elementary inequality $|\sin(\pi d/N)| \geq 2 \min\{d, N-d\}/N$, together with Eq. (B.163), gives

$$|D_0^{(N)}(I)| \geq c_m N^{-q_m} \Delta(I) \quad (B.166)$$

for a constant $c_m > 0$. Since $m-1+d_m = q_m$, Eqs. (B.164) and (B.166) give a constant C_m , depending only on m , such that

$$\mathbf{1}_{\{I=\min \Omega_N(I)\}} \frac{|\Omega_N(I)|}{N} N^{-m+1-d_m} |F_N(I; a, b, c)| \leq \frac{C_m}{\Delta(I)} =: C_I,$$

uniformly in all integrable a, b, c ; here we also used $|\Omega_N(I)|/N \leq 1$.

For completeness, summability follows by writing the consecutive gaps as $n_j = I_j - I_{j+1} \geq 1$, $1 \leq j < m$. The Vandermonde contains the factors $n_1, n_2, n_1 + n_2$ and, for each $3 \leq j < m$, the two factors $n_j, n_{j-1} + n_j$. All remaining factors are positive integers. Hence

$$\Delta(I) \geq n_1 n_2 (n_1 + n_2) \prod_{j=3}^{m-1} n_j^2.$$

The first two gaps satisfy

$$\sum_{a,b \geq 1} \frac{1}{ab(a+b)} = 2 \sum_{s=2}^{\infty} \frac{H_{s-1}}{s^2} < \infty, \quad H_s = \sum_{j=1}^s \frac{1}{j},$$

because $H_s \leq 1 + \log s$. Summing separately over the other gaps gives

$$\sum_{I \in \mathcal{I}_m^\infty} C_I \leq 2C_m \left(\sum_{s=2}^{\infty} \frac{H_{s-1}}{s^2} \right) \left(\sum_{n=1}^{\infty} \frac{1}{n^2} \right)^{m-3} < \infty. \quad (\text{B.167})$$

This is the only step requiring $m \geq 3$. The estimate is unchanged by centering, which multiplies determinant rows by unit phases. $\square$

The orbit indicator in the lemma is essential. A nonminimal representative need not satisfy (B.162); for example, $I = (N-1, 1, 0)$ has wrap gap $N - I_1 = 1$. The lemma is therefore stated for the complete summand that actually occurs in (B.160), rather than for F_N in isolation.

For $x \in Y$ write its standard coordinates as $x = (x_1 > \cdots > x_m)$, with $\sum_s x_s = 0$ and $x_1 - x_m < 1$, and define

$$\mathcal{A}_I(x) = \det[e^{2\pi i I_r x_s}]_{r,s=1}^m. \quad (\text{B.168})$$

Equation (B.149) gives $L(a)/N = z_a$ exactly, so the three numerator determinants in F_N are already the continuum alternants $\mathcal{A}_I(z_a)$, $\mathcal{A}_I(z_b)$, and $\overline{\mathcal{A}_I(z_c)}$. For fixed I , $\sin(\pi(I_r - I_s)/N) = \pi(I_r - I_s)/N + O_I(N^{-3})$; hence Eq. (B.152) gives

$$D_0^{(N)}(I) = (2\pi i)^{q_m} N^{-q_m} \Delta(I) (1 + O_I(N^{-2})). \quad (\text{B.169})$$

Because $q_m = m - 1 + d_m$, substitution in Eq. (B.151) yields

$$N^{-m+1-d_m} F_N(I; a, b, c) = \frac{(2\pi i)^{-q_m}}{\Delta(I)} \mathcal{A}_I(z_a) \mathcal{A}_I(z_b) \overline{\mathcal{A}_I(z_c)} + o_I(1), \quad (\text{B.170})$$

where the error is uniform in all integrable labels: Hadamard's inequality bounds the numerator alternants independently of the labels, and the only approximation in a fixed- I term is the vacuum sine Vandermonde.

Define the trace-zero type- A volume series

$$J_m^A(x, y; z) := \sum_{I \in \mathcal{I}_m^\infty} \frac{(2\pi i)^{-q_m}}{\Delta(I)} \mathcal{A}_I(x) \mathcal{A}_I(y) \overline{\mathcal{A}_I(z)}. \quad (\text{B.171})$$

Proposition B.13 (Uniform dominated convergence in centered coordinates). *The series (B.171) is absolutely and uniformly convergent on Y^3 . For every $K \Subset Y^3$, the finite coefficients converge to it uniformly as in Eq. (B.173).*

Proof. By Hadamard's inequality, the absolute value of the I th continuum summand is at most $C_m/\Delta(I)$, independently of $(x, y, z) \in Y^3$. The Weierstrass test and Eq. (B.167) therefore give absolute and uniform convergence of the series, and hence continuity of J_m^A .

Compare the two series. Fix $\varepsilon > 0$. Choose a finite subset $E \subset \mathcal{I}_m^\infty$ so that

$$\sum_{I \notin E} C_I < \varepsilon. \quad (\text{B.172})$$

The same tail bound applies both to the continuum series and, by Lemma B.12, to the finite series in Eq. (B.160). For each $I \in E$, the wrap gap $N - I_1$ is eventually strictly larger than all fixed internal gaps. Thus I is the least element of its translation orbit and the orbit has trivial stabilizer once N is sufficiently large. The indicator and $|\Omega_N(I)|/N$ are then both equal to one. Finally, Eq. (B.169) is independent of the labels, so the convergence of the finitely many $I \in E$ terms is uniform in a, b, c . The two tails contribute at most 2ε , while the finite part tends uniformly to zero. Since ε was arbitrary, we obtain

$$\lim_{k \rightarrow \infty} \sup_{\substack{(a,b,c) \in \mathcal{L}_G \cap (P_k^+)^3 \\ (z_a, z_b, z_c) \in K}} \left| N^{-d_m} N_{ab}^{(k)c} - J_m^A(z_a, z_b; z_c) \right| = 0. \quad (\text{B.173})$$

This argument includes internal affine and polyhedral walls and points at which the limiting function vanishes. $\square$

For a fixed shift $(u, v, w) \in P^3$ with $u+v-w \in Q$, the shifted triple $(a+u, b+v, c+w)$ is compatible whenever its three labels are integrable, because $(a, b, c) \in \mathcal{L}_G$. Its normalized coordinates satisfy

$$z_{a+u} = z_a + \frac{u}{N}, \quad z_{b+v} = z_b + \frac{v}{N}, \quad z_{c+w} = z_c + \frac{w}{N}. \quad (\text{B.174})$$

If $K \Subset Y^3$ and the set of shifts is finite, then all shifted coordinate triples based in K lie, for sufficiently large N , in a single compact set $K' \Subset Y^3$. Apply Eq. (B.173) on K' . Uniform continuity of J_m^A there and Eq. (B.174) replace the shifted target by $J_m^A(z_a, z_b; z_c)$ with an error tending uniformly to zero. This proves the uniform part of Corollary 4.8 once the target is identified.

We now match the determinant and character series term by term. Put $\text{sf}(m-1) = \prod_{j=1}^{m-1} j!$. Every $I \in \mathcal{I}_m^\infty$ is $I_s = \lambda_s + m - s$ for one dominant $SU(m)$ weight λ , and the Weyl character and dimension formulas give

$$\mathcal{A}_I(x) = \delta_G(x) \chi_\lambda(g_x), \quad \dim V_\lambda = \frac{\Delta(I)}{\text{sf}(m-1)}. \quad (\text{B.175})$$

On the chosen trace-zero alcove all positive-root coordinates lie in $(0, 1)$, so the Weyl denominator has the fixed phase

$$\delta_G(x) = i^{q_m} |\delta_G(x)|. \quad (\text{B.176})$$

3317 Define the regular character representative

$$\Phi_G^{A,\text{reg}}(x, y; z) := \sum_{\lambda \in P_+} \frac{\chi_\lambda(g_x) \chi_\lambda(g_y) \overline{\chi_\lambda(g_z)}}{\dim V_\lambda}. \quad (\text{B.177})$$

3318 The majorant in Lemma B.12, together with the nonvanishing of the Weyl denominators
 3319 on Y , gives absolute convergence at every point of Y^3 and local uniform convergence there.
 3320 Substitution of Eq. (B.175) into Eq. (B.171) now gives

$$J_m^A(x, y; z) = \frac{|\delta_G(x) \delta_G(y) \delta_G(z)|}{(2\pi)^{q_m} \text{sf}(m-1)} \Phi_G^{A,\text{reg}}(x, y; z). \quad (\text{B.178})$$

3321 For fixed (x, y) , the partial sums in Eq. (B.177) are the Peter–Weyl partial sums
 3322 of Eq. (4.35). They converge in $L^2(G)^G$ to the class denoted by $\Phi_G(x, y; \cdot)$. Choose
 3323 a subsequence of the partial sums which converges almost everywhere to the L^2 limit.
 3324 On regular conjugacy classes the same subsequence converges to $\Phi_G^{A,\text{reg}}$ by local uniform
 3325 convergence. Weyl integration therefore gives

$$[\Phi_G^{A,\text{reg}}(x, y; \cdot)]_{L^2} = \Phi_G(x, y; \cdot). \quad (\text{B.179})$$

3326 For A_{m-1} , $Q = Q^\vee$ and

$$\Pi_\rho = (2\pi)^{q_m} \prod_{1 \leq i < j \leq m} (j - i) = (2\pi)^{q_m} \text{sf}(m-1). \quad (\text{B.180})$$

3327 Equations (4.37), (B.178), and (B.179) imply, for every fixed $(x, y) \in Y^2$,

$$J_m^A(x, y; \cdot) = V_{\text{BF}}(x, y; \cdot) \quad \text{for } dz\text{-almost every } z \in Y. \quad (\text{B.181})$$

3328 We define

$$V_{\text{BF}}^{A,\text{cont}}(x, y; z) := J_m^A(x, y; z), \quad (x, y, z) \in Y^3. \quad (\text{B.182})$$

3329 On every principal regular-value chamber this continuous function agrees almost every-
 3330 where with the polynomial V_{geom} , and hence agrees with it pointwise. Its continuity sup-
 3331 plies the common values on the internal critical walls. Combining this identification with
 3332 Eq. (B.173) proves Corollary 4.8.

3333 *Proof of Corollary 4.9.* In the conventions of Definitions 2.1, 2.3, and 2.4 of [22], for $0 \leq$
 3334 $d \leq m$ set

$$R_{d,m} := \{(v_1, v_2) \in \{0, \dots, m\}^2 : d \leq v_1 + v_2 \leq m + d\}. \quad (\text{B.183})$$

3335 This is a hexagonal subregion of the triangular lattice. Its north-west, east, and south-west
 3336 boundary paths are denoted by A, B, C , with vertices indexed by $0 \leq i \leq m$ as in the cited
 3337 definition.

3338 A map $g : R_{d,m} \rightarrow \mathbb{Z}/3\mathbb{Z}$ is a regular labeling when

$$g_i^A = m + i \pmod{3}, \quad g_i^B = i \pmod{3}, \quad g_i^C = i \pmod{3}, \quad (\text{B.184})$$

and, for each oriented lozenge $\ell = (v_1, v_2, v_3, v_4)$ ordered as in [22, Fig. 3],

$$g(v_2) = g(v_4) \implies \{g(v_1), g(v_3)\} = \{g(v_2) + 1, g(v_2) + 2\} \quad \text{in } \mathbb{Z}/3\mathbb{Z}. \quad (\text{B.185})$$

The order of the four vertices matters. A lozenge is rigid when its ordered labels are $(a, a + 1, a + 2, a + 1)$ for some $a \in \mathbb{Z}/3\mathbb{Z}$. The support $\text{Supp}(g)$ is obtained by removing every vertex v_4 of a rigid lozenge; no boundary vertex is removed. The set $\mathfrak{G}_{d,m}$ is precisely the finite set of these regular labelings.

A function $f : R_{d,m} \rightarrow \mathbb{R}$ is rhombus-concave relative to g when

$$f(v_2) + f(v_4) \geq f(v_1) + f(v_3) \quad (\text{B.186})$$

for every oriented lozenge, with equality on every rigid lozenge. The polytope $\mathcal{P}_{\alpha,\beta,\gamma}^g$ consists of the coordinates $f|_{\text{Supp}(g) \setminus \partial R_{d,m}}$ of such functions whose boundary lists are

$$\begin{aligned} f_i^A &= \sum_{s=1}^m \beta_s + \sum_{s=1}^i \alpha_s, \\ f_i^B &= (d - i)^+ + \sum_{s=1}^i \beta_s, \quad 0 \leq i \leq m. \\ f_i^C &= d + \sum_{s=1}^i \gamma_s, \end{aligned} \quad (\text{B.187})$$

The relation $|\alpha|_1 + |\beta|_1 = |\gamma|_1 + d$ makes the values agree at the three shared corners. Finally, Vol_g means ordinary coordinate Lebesgue measure on $\mathbb{R}^{\text{Supp}(g) \setminus \partial R_{d,m}}$. Empty polytopes have volume zero.

Let $d\gamma$ be the $(m - 1)$ -dimensional Lebesgue measure used in [22] on the ordered fixed-determinant eigenangle hyperplane, and let $p(\gamma \mid \alpha, \beta)$ be the corresponding density. For a strictly decreasing (hence regular) ordered list $\theta \in [0, 1]^m$ set

$$\Delta_{\mathbb{R}}(\theta) := 2^{qm} \prod_{i < j} \sin(\pi(\theta_i - \theta_j)) > 0. \quad (\text{B.188})$$

For a merely nonincreasing list the same product is nonnegative and may vanish; only regular lists occur in the division below. Their Proposition 3.1, equation (3.3), expresses the ordered eigenangle density as

$$p(\gamma \mid \alpha, \beta) = \frac{\Delta_{\mathbb{R}}(\gamma)^2}{(2\pi)^{m-1} m!} \Phi_G(x, y; z). \quad (\text{B.189})$$

Their Theorem 2.5 expresses the same density, for the canonical degree $d = |\alpha|_1 + |\beta|_1 - |\gamma|_1 \in \{0, \dots, m\}$, as

$$p(\gamma \mid \alpha, \beta) = \frac{\text{sf}(m - 1)(2\pi)^{d_m} \Delta_{\mathbb{R}}(\gamma)}{m! \Delta_{\mathbb{R}}(\alpha) \Delta_{\mathbb{R}}(\beta)} \sum_{g \in \mathfrak{G}_{d,m}} \text{Vol}_g(\mathcal{P}_{\alpha,\beta,\gamma}^g). \quad (\text{B.190})$$

Equating these two expressions and using $d_m + (m - 1) = q_m$ gives

$$\sum_{g \in \mathfrak{G}_{d,m}} \text{Vol}_g(\mathcal{P}_{\alpha,\beta,\gamma}^g) = \frac{\Delta_{\mathbb{R}}(\alpha)\Delta_{\mathbb{R}}(\beta)\Delta_{\mathbb{R}}(\gamma)}{(2\pi)^{q_m} \text{sf}(m-1)} \Phi_G(x, y; z). \quad (\text{B.191})$$

The real sine Vandermondes are the absolute Weyl denominators of the three conjugacy classes. Equations (B.178) and (B.180) identify the right-hand side with $V_{\text{BF}}^{A,\text{cont}}(x, y; z)$, proving the corollary. The degree d is the canonical François–Tarrago degree determined by the representatives in $[0, 1]^m$; this convention is used throughout. $\square$

B.7 A square-root-weighted corollary

For an orthonormal splitting of $A \otimes B$, the normalized spherical trace assigns each multiplicity channel with output C the probability $D_C/(D_A D_B)$, where $D_{(u^+, u^-)} = d_{u^+}^{(k)} d_{u^-}^{(k)}$. Summing the positive square roots of these channel probabilities gives the following ℓ^1 statistic. For doubled labels, define the positive measure

$$\mu_{A,B}^{(k),\text{sr}} := \sum_C \mathcal{N}_{AB}^{(k)C} \left(\frac{d_{c^+}^{(k)} d_{c^-}^{(k)}}{d_{a^+}^{(k)} d_{b^+}^{(k)} d_{a^-}^{(k)} d_{b^-}^{(k)}} \right)^{1/2} \delta_{(x_C, \eta_C)}. \quad (\text{B.192})$$

After multiplication by $r_k^{-d_G}$, it is obtained from the raw measure of Main Theorem C by the multiplier in Eq. (B.195). Its continuum limit is

$$\begin{aligned} \mu_{(x,\eta),(y,\omega)}^{\text{BF},\text{sr}} &:= V_{\text{BF}}(x, y; z) d\nu_{x,y}(z) \otimes \sum_{\tau \in \eta + \omega + Q} \delta_\tau \\ &= \frac{\Pi_\rho}{\text{covol}(Q)} \frac{|\delta_G(z)|}{|\delta_G(x)\delta_G(y)|} V_{\text{BF}}(x, y; z)^2 dz \otimes \sum_{\tau \in \eta + \omega + Q} \delta_\tau. \end{aligned} \quad (\text{B.193})$$

Corollary B.14 (Square-root measure). *Under the hypotheses of Theorem 5.3(ii),*

$$r_k^{-d_G} \mu_{A_k, B_k}^{(k),\text{sr}} \xrightarrow{\text{v}} \mu_{(x,\eta),(y,\omega)}^{\text{BF},\text{sr}} \quad \text{on } U \times P. \quad (\text{B.194})$$

The type-A strengthening of Main Theorem C gives the same conclusion on every relatively compact regular output window.

Proof. At a finite-level output atom put

$$W_k(x_C, \eta_C) := r_k^{|\Delta_+|} \left(\frac{d_{c^+}^{(k)} d_{c^-}^{(k)}}{d_{a^+}^{(k)} d_{b^+}^{(k)} d_{a^-}^{(k)} d_{b^-}^{(k)}} \right)^{1/2}. \quad (\text{B.195})$$

Since $|\Delta_+| = d_G + \text{rank } G$, the definitions give

$$r_k^{-d_G} \int f d\mu_{A_k, B_k}^{(k),\text{sr}} = \frac{1}{\text{covol}(Q)} \int W_k f d\mathfrak{N}_{A_k, B_k}^{(k)}. \quad (\text{B.196})$$

For $f \in C_c(U \times P)$, taking the positive square root of Eq. (5.39) gives

$$\left(\frac{d_{c^+}^{(k)} d_{c^-}^{(k)}}{d_{a^+}^{(k)} d_{b^+}^{(k)} d_{a^-}^{(k)} d_{b^-}^{(k)}} \right)^{1/2} = r_k^{-|\Delta_+|} \frac{\Pi_\rho |\delta_G(x_C)|}{|\delta_G(x_{A_k}) \delta_G(x_{B_k})|} (1 + O(r_k^{-1})). \quad (\text{B.197})$$

3376 The estimate is uniform over the finite-level output atoms $(x_C, \eta_C) \in \text{supp } f$ carrying
 3377 nonzero raw mass. Hence

$$W_k(x_C, \eta_C) \longrightarrow W(z) := \Pi_\rho \frac{|\delta_G(z)|}{|\delta_G(x)\delta_G(y)|} \quad (\text{B.198})$$

3378 uniformly over those atoms. Positivity and the vague convergence of $\mathfrak{N}_{A_k, B_k}^{(k)}$ bound their
 3379 masses on $\text{supp } f$. Therefore

$$\int (W_k - W) f d\mathfrak{N}_{A_k, B_k}^{(k)} \longrightarrow 0.$$

3380 Applying Main Theorem C to the fixed compactly supported function Wf in Eq. (B.196)
 3381 gives precisely Eq. (B.193). $\square$

3382 For $G = SU(2)$, write $t = \langle x, \alpha^\vee \rangle \in (0, 1)$. Then $d_G = 0$ and $V_{\text{BF}}(x, y; t) = \mathbf{1}_{I_{x,y}}(t)$ for
 3383 almost every t , where

$$I_{x,y} = [|t_x - t_y|, \min\{t_x + t_y, 2 - t_x - t_y\}], \quad (\text{B.199})$$

3384 so each compatible fixed charge has limiting measure

$$\frac{\pi \sin(\pi t)}{2 \sin(\pi t_x) \sin(\pi t_y)} \mathbf{1}_{I_{x,y}}(t) dt. \quad (\text{B.200})$$

3385 **B.8 The $SU(2)$ endpoint measure and the Jia–Tian kernel**

3386 For $SU(2)$ the double-coset coordinate gives the endpoint behavior directly. Let $0 <$
 3387 $t_x, t_y < 1$ and put

$$t_- = |t_x - t_y|, \quad t_+ = \min\{t_x + t_y, 2 - t_x - t_y\}. \quad (\text{B.201})$$

3388 The exact rank-one fusion rule [47, Eq. (4.21)] is

$$N_{ab}^{(k)c} = \begin{cases} 1, & a + b - c \in 2\mathbb{Z}, \quad |a - b| \leq c \leq \min\{a + b, 2k - a - b\}, \\ 0, & \text{otherwise,} \end{cases} \quad 0 \leq a, b, c \leq k.$$

3389 Here weights are written as integers in units of the fundamental weight, so $P = \mathbb{Z}$ and
 3390 $Q = 2\mathbb{Z}$. A fixed root-lattice shift can cross an endpoint inequality even when the shifted
 3391 and unshifted normalized triples have the same limit. For example, for each $n \geq 1$ take

$$k = 10n, \quad a = b = 2n, \quad c = 4n.$$

3392 Both c and $c + 2$ are integrable and both triples satisfy the central congruence. The upper
 3393 fusion inequality is $c \leq a + b = 4n$, however, so

$$N_{2n, 2n}^{(10n), 4n} = 1, \quad N_{2n, 2n}^{(10n), 4n+2} = 0.$$

3394 With $r_k = k + 2$ and $t_j = (j + 1)/r_k$, the two normalized triples converge to $(1/5, 1/5, 2/5)$.
 3395 More precisely,

$$t_c = t_a + t_b - r_k^{-1}, \quad t_{c+2} = t_a + t_b + r_k^{-1}.$$

Thus the compatible shift $(0, 0, 2)$ crosses the internal upper endpoint. Since $d_{SU(2)} = 0$, the difference between the two coefficients survives their asymptotic normalization. This explains the restriction $m \geq 3$ in the wall-crossing statement with fixed shifts in Appendix B.6. It is consistent with the measure limits: the endpoint cells have length $O(r_k^{-1})$, and bounded $SU(2)$ coefficients give a vanishing contribution from these cells in the normalized Riemann sums. In particular, the example creates no atom at the limiting endpoint.

Proposition B.15 (Endpoint measure and endpoint charge rule). *Assume*

$$0 < t_- < t_+ < 1, \quad (\text{B.202})$$

so both endpoints remain in the regular output alcove. Then

$$d\nu_{x,y}(t) = \frac{\pi \sin(\pi t)}{2 \sin(\pi t_x) \sin(\pi t_y)} \mathbf{1}_{[t_-, t_+]}(t) dt, \quad \nu_{x,y}(\{t_-\}) = \nu_{x,y}(\{t_+\}) = 0. \quad (\text{B.203})$$

At the aligned and anti-aligned endpoint double cosets, respectively, the full stabilizer rule gives, before the Weyl return to the open alcove,

$$(g_{t_x+t_y}, \eta + \omega), \quad (g_{t_x-t_y}, \eta - \omega). \quad (\text{B.204})$$

If $t_x + t_y > 1$, the first pair is represented by $(g_{2-t_x-t_y}, -\eta - \omega)$; if $t_x < t_y$, the second is represented by $(g_{t_y-t_x}, \omega - \eta)$. These endpoint labels lie on two null strata and do not give discrete summands of the Hilbert direct integral.

Proof. Identify $T \backslash SU(2)/T$ with $\vartheta \in [0, \pi]$, carrying the probability measure $\frac{1}{2} \sin \vartheta d\vartheta$. The product-class coordinate satisfies

$$\cos(\pi t) = \cos(\pi t_x) \cos(\pi t_y) - \cos \vartheta \sin(\pi t_x) \sin(\pi t_y).$$

The variable t decreases from t_+ to t_- as ϑ increases from 0 to π . In the interior, differentiation gives

$$\frac{1}{2} \sin \vartheta |d\vartheta| = \frac{\pi \sin(\pi t)}{2 \sin(\pi t_x) \sin(\pi t_y)} |dt|.$$

This proves Eq. (B.203). Its total mass is

$$\frac{\cos(\pi t_-) - \cos(\pi t_+)}{2 \sin(\pi t_x) \sin(\pi t_y)} = 1,$$

by the cosine addition formulas, including when $t_x + t_y > 1$. The endpoints are the images of $\vartheta = 0, \pi$, hence have zero measure. At these double cosets the two maximal tori coincide, so $H_q = T$; the one-dimensional Hom space in Eq. (B.8) adds or subtracts the two torus characters. Weyl transport back to the open alcove gives the two cases following Eq. (B.204). $\square$

Remark B.16 (Comparison with the Jia–Tian kernel). Jia and Tian use $g_\psi = \exp(i\psi H)$ with $H = \text{diag}(1, -1)/2$. If $\eta = m\varpi_1$, their variables are

$$\psi = 2\pi t, \quad \varrho = \frac{m}{2}.$$

Their interior kernel coefficient therefore becomes

$$\frac{\sin(\psi/2)}{2 \sin(\psi_g/2) \sin(\psi_h/2)} d\psi = \frac{\pi \sin(\pi t)}{\sin(\pi t_x) \sin(\pi t_y)} dt = 2 d\nu_{x,y}(t). \quad (\text{B.205})$$

The factor two reflects their conjugacy-class and kernel normalization, whereas Eq. (B.203) uses the probability measure $\frac{1}{2} \sin \vartheta d\vartheta$.

In their fixed- v kernel identity, summing the interior charge tower produces the periodic delta distribution in the Fourier-dual holonomy variable. The two endpoint summands in Eqs. (3.64)–(3.68) are separate terms. This distributional identity is different from the positive orbital spectral measure, so its endpoint coefficients are consistent with $\nu_{x,y}(\{t_-\}) = \nu_{x,y}(\{t_+\}) = 0$ and with the pointwise charges in Eq. (B.204) [9, Sec. 3.3]. The unitary direct integral of [6, Sec. 7.3, especially Eqs. (7.23)–(7.31)] likewise has an absolutely continuous double-coset measure and null endpoints.

References

- [1] E. Witten, *Quantum field theory and the Jones polynomial*, *Communications in Mathematical Physics* **121** (1989) 351.
- [2] S. Elitzur, G.W. Moore, A. Schwimmer and N. Seiberg, *Remarks on the canonical quantization of the Chern–Simons–Witten theory*, *Nuclear Physics B* **326** (1989) 108.
- [3] M. Müger, *From subfactors to categories and topology II: The quantum double of tensor categories and subfactors*, *Journal of Pure and Applied Algebra* **180** (2003) 159 [[math/0111205](#)].
- [4] C. Meusburger and B.J. Schroers, *The quantisation of Poisson structures arising in Chern–Simons theory with gauge group $G \ltimes \mathfrak{g}^*$* , *Advances in Theoretical and Mathematical Physics* **7** (2003) 1003 [[hep-th/0310218](#)].
- [5] T.H. Koornwinder and N.M. Muller, *The quantum double of a (locally) compact group*, *Journal of Lie Theory* **7** (1997) 101 [[q-alg/9605044](#)].
- [6] T.H. Koornwinder, F.A. Bais and N.M. Muller, *Tensor product representations of the quantum double of a compact group*, *Communications in Mathematical Physics* **198** (1998) 157 [[q-alg/9712042](#)].
- [7] Q. Jia, R. Luo, J. Tian, Y.-N. Wang and Y. Zhang, *Categorical symmetries via operator algebras*, [2604.25821](#).
- [8] L. Charles, *Asymptotic properties of the quantum representations of the modular group*, *Transactions of the American Mathematical Society* **364** (2012) 5829 [[1005.3447](#)].
- [9] Q. Jia and J. Tian, *Hilbert space and defect Hilbert spaces associated with categorical symmetries*, [2605.28485v1](#).
- [10] Q. Jia, C. Ma and J. Tian, *Candidate gaugings of categorical continuous symmetry*, [2604.25820](#).
- [11] S. Agnihotri and C. Woodward, *Eigenvalues of products of unitary matrices and quantum schubert calculus*, *Mathematical Research Letters* **5** (1998) 817 [[alg-geom/9712013](#)].
- [12] P. Belkale, *Quantum generalization of the Horn conjecture*, *Journal of the American Mathematical Society* **21** (2008) 365 [[math/0303013](#)].

- [13] C. Teleman and C. Woodward, *Parabolic bundles, products of conjugacy classes, and Gromov–Witten invariants*, *Annales de l’Institut Fourier* **53** (2003) 713 [[math/0012241](#)].
- [14] M. Defossez, *Fusion coefficients and random walks in alcoves*, *Annales de l’Institut Henri Poincaré, Probabilités et Statistiques* **52** (2016) 1515 [[1307.3830](#)].
- [15] A. Berenstein and A. Zelevinsky, *Tensor product multiplicities, canonical bases and totally positive varieties*, *Inventiones Mathematicae* **143** (2001) 77 [[math/9912012](#)].
- [16] M. Brion and M. Vergne, *Residue formulae, vector partition functions and lattice points in rational polytopes*, *Journal of the American Mathematical Society* **10** (1997) 797.
- [17] B. Sturmfels, *On vector partition functions*, *Journal of Combinatorial Theory, Series A* **72** (1995) 302.
- [18] A. Szenes and M. Vergne, *Residue formulae for vector partitions and Euler–Maclaurin sums*, *Advances in Applied Mathematics* **30** (2003) 295 [[math/0202253](#)].
- [19] E. Meinrenken and R. Sjamaar, *Singular reduction and quantization*, *Topology* **38** (1999) 699 [[dg-ga/9707023](#)].
- [20] Y. Loizides, P.-E. Paradan and M. Vergne, *Semi-classical analysis of piecewise quasi-polynomial functions and applications to geometric quantization*, *Indagationes Mathematicae* **32** (2021) 151 [[1907.01428](#)].
- [21] C. McSwiggen, *Box splines, tensor product multiplicities and the volume function*, *Algebraic Combinatorics* **4** (2021) 435 [[1909.12278](#)].
- [22] Q. François and P. Tarrago, *Positive formula for the product of conjugacy classes on the unitary group*, [2405.06723v2](#).
- [23] N. Bourbaki, *Lie Groups and Lie Algebras: Chapters 4–6*, Elements of Mathematics, Springer, Berlin (2002).
- [24] V.G. Kac, *Infinite-Dimensional Lie Algebras*, Cambridge University Press, 3 ed. (1990), [10.1017/CBO9780511626234](#).
- [25] D. Bump, *Lie Groups*, vol. 225 of *Graduate Texts in Mathematics*, Springer, New York, 2 ed. (2013), [10.1007/978-1-4614-8024-2](#).
- [26] D.S. Freed, *Classical Chern–Simons theory, part 1*, *Advances in Mathematics* **113** (1995) 237 [[hep-th/9206021](#)].
- [27] E. İnönü and E.P. Wigner, *On the contraction of groups and their representations*, *Proceedings of the National Academy of Sciences* **39** (1953) 510.
- [28] E.J. Saletan, *Contraction of Lie groups*, *Journal of Mathematical Physics* **2** (1961) 1.
- [29] A.A. Tseytlin, *On gauge theories for non-semisimple groups*, *Nuclear Physics B* **450** (1995) 231 [[hep-th/9505129](#)].
- [30] D.P. Williams, *Crossed Products of C^* -Algebras*, vol. 134 of *Mathematical Surveys and Monographs*, American Mathematical Society, Providence, RI (2007), [10.1090/surv/134](#).
- [31] M.A. Rieffel, *Induced representations of C^* -algebras*, *Advances in Mathematics* **13** (1974) 176.
- [32] M.A. Walton, *Fusion rules in Wess–Zumino–Witten models*, *Nuclear Physics B* **340** (1990) 777.
- [33] M.A. Walton, *Algorithm for WZW fusion rules: A proof*, *Physics Letters B* **241** (1990) 365.

- [34] B. Bakalov and A. Kirillov, Jr., *Lectures on Tensor Categories and Modular Functors*, vol. 21 of *University Lecture Series*, American Mathematical Society, Providence, RI (2001), [10.1090/ulect/021](https://doi.org/10.1090/ulect/021).
- [35] L.C. Jeffrey, *Flat connections on oriented 2-manifolds*, *Bulletin of the London Mathematical Society* **37** (2005) 1.
- [36] G.B. Folland, *A Course in Abstract Harmonic Analysis*, Studies in Advanced Mathematics, CRC Press, Boca Raton, FL (1995).
- [37] J.C. Baez, *An introduction to spin foam models of BF theory and quantum gravity*, in *Geometry and Quantum Physics*, H. Gausterer, L. Pittner and H. Grosse, eds., vol. 543 of *Lecture Notes in Physics*, (Berlin, Heidelberg), pp. 25–93, Springer (2000), DOI [\[gr-qc/9905087\]](https://doi.org/10.1007/978-3-540-69050-7).
- [38] P. Etingof, S. Gelaki, D. Nikshych and V. Ostrik, *Tensor Categories*, vol. 205 of *Mathematical Surveys and Monographs*, American Mathematical Society, Providence, RI (2015), [10.1090/surv/205](https://doi.org/10.1090/surv/205).
- [39] V.G. Kac and D.H. Peterson, *Infinite-dimensional Lie algebras, theta functions and modular forms*, *Advances in Mathematics* **53** (1984) 125.
- [40] T. Gannon, *Kac–Peterson, Perron–Frobenius, and the classification of conformal field theories*, [q-alg/9510026](https://arxiv.org/abs/q-alg/9510026).
- [41] E.M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*, vol. 1 of *Princeton Lectures in Analysis*, Princeton University Press, Princeton (2003).
- [42] D. Novikov, *Hahn decomposition and Radon–Nikodym theorem with a parameter*, [math/0501215](https://arxiv.org/abs/math/0501215).
- [43] A. Alekseev, A. Malkin and E. Meinrenken, *Lie group valued moment maps*, *Journal of Differential Geometry* **48** (1998) 445 [[dg-ga/9707021](https://arxiv.org/abs/dg-ga/9707021)].
- [44] R. Sjamaar and E. Lerman, *Stratified symplectic spaces and reduction*, *Annals of Mathematics* **134** (1991) 375.
- [45] A. Alekseev, E. Meinrenken and C. Woodward, *Duistermaat–heckman measures and moduli spaces of flat bundles over surfaces*, *Geometric and Functional Analysis* **12** (2002) 1 [[math/9903087](https://arxiv.org/abs/math/9903087)].
- [46] E. Meinrenken and C. Woodward, *Moduli spaces of flat connections on 2-manifolds, cobordism, and Witten’s volume formulas*, in *Advances in Geometry*, vol. 172 of *Progress in Mathematics*, pp. 271–295, Birkhäuser (1999), DOI [[dg-ga/9707018](https://arxiv.org/abs/dg-ga/9707018)].
- [47] C. Korff, *A combinatorial derivation of the Racah–Speiser algorithm for Gromov–Witten invariants*, [0910.3395](https://arxiv.org/abs/0910.3395).
- [48] C. Korff and C. Stroppel, *The $\widehat{\mathfrak{sl}}(n)_k$ -WZNW fusion ring: A combinatorial construction and a realisation as quotient of quantum cohomology*, *Advances in Mathematics* **225** (2010) 200 [[0909.2347](https://arxiv.org/abs/0909.2347)].
- [49] E. Verlinde, *Fusion rules and modular transformations in 2d conformal field theory*, *Nuclear Physics B* **300** (1988) 360.
- [50] V.G. Knizhnik and A.B. Zamolodchikov, *Current algebra and Wess–Zumino model in two dimensions*, *Nuclear Physics B* **247** (1984) 83.

- [51] C.R. Nappi and E. Witten, *A WZW model based on a nonsemisimple group*, *Physical Review Letters* **71** (1993) 3751 [[hep-th/9310112](#)].
- [52] D.I. Olive, E. Rabinovici and A. Schwimmer, *A class of string backgrounds as a semiclassical limit of WZW models*, *Physics Letters B* **321** (1994) 361 [[hep-th/9311081](#)].
- [53] K. Sfetsos, *Exact string backgrounds from WZW models based on non-semisimple groups*, *International Journal of Modern Physics A* **9** (1994) 4759 [[hep-th/9311093](#)].
- [54] J.M. Figueroa-O'Farrill and S. Stanciu, *Nonreductive WZW models and their CFTs*, *Nuclear Physics B* **458** (1996) 137 [[hep-th/9506151](#)].
- [55] F.A. Bais and N.M. Muller, *Topological field theory and the quantum double of $SU(2)$* , *Nuclear Physics B* **530** (1998) 349 [[hep-th/9804130](#)].
- [56] M.R. Gaberdiel and P. Suchanek, *Limits of minimal models and continuous orbifolds*, *Journal of High Energy Physics* **03** (2012) 104 [[1112.1708](#)].
- [57] I. Runkel and G.M.T. Watts, *A nonrational CFT with $c = 1$ as a limit of minimal models*, *Journal of High Energy Physics* **09** (2001) 006 [[hep-th/0107118](#)].
- [58] A. Kitaev and L. Kong, *Models for gapped boundaries and domain walls*, *Communications in Mathematical Physics* **313** (2012) 351 [[1104.5047](#)].
- [59] J. Fuchs, C. Schweigert and A. Valentino, *Bicategories for boundary conditions and for surface defects in 3-d TFT*, *Communications in Mathematical Physics* **321** (2013) 543 [[1203.4568](#)].
- [60] A. Kapustin and N. Saulina, *Surface operators in 3d topological field theory and 2d rational conformal field theory*, [1012.0911](#).
- [61] C. Bachas, J. de Boer, R. Dijkgraaf and H. Ooguri, *Permeable conformal walls and holography*, *Journal of High Energy Physics* **06** (2002) 027 [[hep-th/0111210](#)].
- [62] A. Davydov, M. Müger, D. Nikshych and V. Ostrik, *The Witt group of non-degenerate braided fusion categories*, *Journal für die reine und angewandte Mathematik* **677** (2013) 135 [[1009.2117](#)].
- [63] T. Lan, J.C. Wang and X.-G. Wen, *Gapped domain walls, gapped boundaries and topological degeneracy*, *Physical Review Letters* **114** (2015) 076402 [[1408.6514](#)].
- [64] P. Goddard, A. Kent and D.I. Olive, *Virasoro algebras and coset space models*, *Physics Letters B* **152** (1985) 88.
- [65] D. Gaiotto and K. Zeng, *Interface minimal model holography and topological string theory*, [2603.08783](#).
- [66] D. Friedan, Z. Qiu and S.H. Shenker, *Conformal invariance, unitarity, and critical exponents in two dimensions*, *Physical Review Letters* **52** (1984) 1575.
- [67] A.A. Belavin, A.M. Polyakov and A.B. Zamolodchikov, *Infinite conformal symmetry in two-dimensional quantum field theory*, *Nuclear Physics B* **241** (1984) 333.
- [68] P. Goddard, A. Kent and D.I. Olive, *Unitary representations of the Virasoro and Super-Virasoro algebras*, *Communications in Mathematical Physics* **103** (1986) 105.
- [69] D. Roggenkamp and K. Wendland, *Limits and degenerations of unitary conformal field theories*, *Communications in Mathematical Physics* **251** (2004) 589 [[hep-th/0308143](#)].

- 3581 [70] D. Roggenkamp and K. Wendland, *Decoding the geometry of conformal field theories*,
3582 *Bulgarian Journal of Physics* **35** (2008) 139 [[0803.0657](#)].
- 3583 [71] S. Fredenhagen and C. Restuccia, *The geometry of the limit of $N = 2$ minimal models*,
3584 *Journal of Physics A* **46** (2013) 045402 [[1208.6136](#)].
- 3585 [72] S. Fredenhagen, C. Restuccia and R. Sun, *The limit of $N = (2, 2)$ superconformal minimal*
3586 *models*, *Journal of High Energy Physics* **10** (2012) 141 [[1204.0446](#)].
- 3587 [73] C. Restuccia, *Limit Theories and Continuous Orbifolds*, Ph.D. thesis, Humboldt University
3588 of Berlin, 2013. [1310.6857](#).
- 3589 [74] I.G. Macdonald, *The volume of a compact Lie group*, *Inventiones Mathematicae* **56** (1980)
3590 [93](#).
- 3591 [75] J.J. Duistermaat and G.J. Heckman, *On the variation in the cohomology of the symplectic*
3592 *form of the reduced phase space*, *Inventiones Mathematicae* **69** (1982) 259.
- 3593 [76] A. Alekseev, E. Meinrenken and C. Woodward, “Formulas of Verlinde type for non-simply
3594 connected groups.” [arXiv:math/0005047](#), 2000.